



OPERATING INSTRUCTIONS

picoScan150

2D LiDAR sensors

Described product

picoScan150



NOTE

The functional scope of the picoScan150 depends on the selected configuration. Certain functions are supported or not supported, depending on the configured variant. The operating instructions describe the full functional scope of the picoScan.

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Original document

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1 About this document

1.1 Information on the operating instructions

Read these operating instructions carefully to familiarize yourself with the product and its functions before commencing any work.

The operating instructions are an integral part of the product. Store the instructions so they remain accessible to staff at all times. If the product is passed on to a third party, these operating instructions should be handed over with it.

These operating instructions do not provide information on the handling and safe operation of the machine or system in which the product is integrated. Information on this can be found in the operating instructions for the machine or system.

1.2 Symbols and document conventions

Warnings and other notes



DANGER

Indicates a situation presenting imminent danger, which will lead to death or serious injuries if not prevented.



WARNING

Indicates a situation presenting possible danger, which may lead to death or serious injuries if not prevented.



CAUTION

Indicates a situation presenting possible danger, which may lead to moderate or minor injuries if not prevented.



NOTICE

Indicates a situation presenting possible danger, which may lead to property damage if not prevented.



NOTE

Highlights useful tips and recommendations as well as information for efficient and trouble-free operation.

Instructions to action

- The arrow denotes instructions to action.
- 1. The sequence of instructions is numbered.
- 2. Follow the order in which the numbered instructions are given.
- ✓ The tick denotes the results of an action.

1.3 Further information

More information can be found on the product page.

The call is made via the **SICK Product ID: pid.sick.com/{P/N}/{S/N}**

{P/N} corresponds to the part number of the product, see type label.

{S/N} corresponds to the serial number of the product, see type label (if indicated).

The following information is available depending on the product:

- Data sheets
- This document in all available language versions
- CAD files and dimensional drawings
- Certificates (e.g., declaration of conformity)
- Other publications
- Software
- Accessories

2 Safety information

2.1 Intended use

Important notes



NOTE

The functional scope of the picoScan150 depends on the selected configuration. Certain functions are supported or not supported, depending on the configured variant. The operating instructions describe the full functional scope of the picoScan150.

The picoScan150 2D LiDAR sensor is a non-contact distance measuring sensor with one scan plane. It has been designed for indoor or outdoor and mobile and stationary use in stand-alone operation.

Depending on the configuration and application software, the following usage scenarios can be solved:

- Detection of objects during continuous output of measurement data as required.
- Field monitoring of freely defined areas with signaling of object detection via digital outputs or telegrams.

The picoScan150 is suitable for applications that demand precise, non-contact optical contour measurements and environment perception. Typical fields of application are, for example, stationary field protection, area monitoring, access control, mobile applications (navigation and anti-collision of mobile platforms) as well as profile detection. It can also be used, for example, to implement systems for collision protection, object protection or access monitoring.

The product is designed for use in industrial and logistics areas, and meets the requirements for industrial ruggedness, interfaces and data processing.

The product was developed for use in industrial environments (EN 61000-6-4).

Incorrect use, improper modification, or tampering with the product will invalidate any warranty offered by SICK AG. Furthermore, SICK AG shall not accept any responsibility or liability for any resulting damage and consequential damage.

2.2 Improper use

Impermissible use

- As a safety component as defined in the relevant applicable safety standards for machines, e.g. Machinery Directive.

Impermissible ambient conditions

- Explosion-hazardous area
- Corrosive environment

2.3 Cybersecurity

Overview

To protect against cybersecurity threats, the operator must have a comprehensive cybersecurity concept, which must be continuously monitored and maintained. A suitable concept consists of organizational, technical, procedural, electronic, and physical levels of defense and considers suitable measures for different types of risks. The measures implemented in this product can only support protection against cybersecurity threats if the product is used as part of such a concept.

You will find further information at www.sick.com/psirt, e.g.:

- General information on cybersecurity
- Contact option for reporting vulnerabilities
- Information on known vulnerabilities (security advisories)

2.4 Limitation of liability

Relevant standards and regulations, the latest technological developments, and our many years of knowledge and experience have all been taken into account when compiling the data and information contained in these operating instructions. The manufacturer accepts no liability for damage caused by:

- Non-adherence to the product documentation (e.g., operating instructions)
- Incorrect use
- Use of untrained staff
- Unauthorized conversions or repair
- Technical modifications
- Use of unauthorized spare parts, consumables, and accessories

2.5 Qualification of personnel

Any work on the product may only be carried out by personnel qualified and authorized to do so.

Qualified personnel are able to perform tasks assigned to them and can independently recognize and avoid any potential hazards. This requires, for example:

- technical training
- experience
- knowledge of the applicable regulations and standards

2.6 Basic safety notes

Please observe the safety notes and the warnings listed here and in other sections of this product documentation to reduce the possibility of risks to health and avoid dangerous situations.

Danger due to visible radiation is product-specific. See the technical data for more information.

Laser notes



CAUTION

Optical radiation: Class 1 Laser Product

The accessible radiation does not pose a danger when viewed directly for up to 100 seconds. It may pose a danger to the eyes and skin in the event of incorrect use.

- Do not open the housing. Opening the housing may increase the level of risk.
- Current national regulations regarding laser protection must be observed.



This laser product complies with laser class 1 according to IEC 60825-1:2014 / EN 60825-1:2014+A11:2021.

Complies with 21 CFR 1040.10 and 1040.11 except for conformance with IEC 60825-1 Ed. 3., as described in Laser Notice No. 56, dated May 8, 2019.

Caution – Use of controls or adjustments or performance of procedures other than those specified herein may result in hazardous radiation exposure.

It is not possible to entirely rule out temporary disorienting optical effects, particularly in conditions of dim lighting. Disorienting optical effects may come in the form of dazzle, flash blindness, afterimages, photosensitive epilepsy, or impairment of color vision, for example.

Mounting and electrical installation



CAUTION

Risk of injury due to hot device surface!

The surface of the product can become hot.

- Before performing work on the product (e.g. mounting, cleaning, disassembly), switch off the product and allow it to cool down.
 - Ensure good dissipation of excess heat from the product to the surroundings.
-



WARNING

Electrical voltage!

Electrical voltage can cause severe injury or death.

- Work on electrical systems must only be performed by qualified electricians.
 - The power supply must be disconnected when attaching and detaching electrical connections.
 - The product must only be connected to a voltage supply as set out in the requirements in the operating instructions.
 - National and regional regulations must be complied with.
 - Safety requirements relating to work on electrical systems must be complied with.
-



WARNING

Risk of injury and damage caused by potential equalization currents!

Improper grounding can lead to dangerous equipotential bonding currents, which may in turn lead to dangerous voltages on metallic surfaces, such as the housing. Electrical voltage can cause severe injury or death.

- Work on electrical systems must only be performed by qualified electricians.
 - Follow the notes in the operating instructions.
 - Install the grounding for the product and the system in accordance with national and regional regulations.
-

Repairs and modifications



WARNING

Electric shock!

Non-insulated electrical conductors are located in the housing. Electrical voltage can cause severe injury or death.

- Do not open the housing.
 - Protect the housing from damage.
 - If the housing is damaged, disconnect the product from the voltage supply and do not put it into operation.
-

3 Product description

3.1 Scope of delivery

Depending on the chosen product version, the scope of delivery of a product will include the following components:

Table 1: Scope of delivery

No. of units	Component	Note
1	Product in the ordered type (complete device or basic device). The functional scope of the product depends on the ordered configuration.	Complete device: • Components are mounted at the factory (housing and system plug). Basic device: • Mount the housing and system plug separately.
1	Printed safety notes, multilingual	Brief information and general safety notes

The actual scope of delivery may differ for special designs, additional orders or due to the latest technical changes.

Further topics

- [Mounting the system plug on the product](#)

3.2 Product overview

Product overview



- ① Status indicator LED 1
- ② Status indicator LED 2
- ③ Optics cover



- ① Marking for the position of the light emission level
- ② Ventilation element (do not remove/paint over)
- ③ System plug (mounted at the back)

Further topics

- [Dimensional drawing](#)

3.3 Display and control elements

Overview



Figure 1: Position of the two status LEDs

- ① LED1
- ② LED2

Status LEDs

LED1 (color)	LED2 (color)	Description
● (Red)	● (Red)	Start up, parameterization, firmware update
-	-	Off
⚡ Fast (red)	⚡ Fast (red)	Fatal error

3 PRODUCT DESCRIPTION

LED1 (color)	LED2 (color)	Description
⦿ Slow (red)	⦿ Slow (red)	Remediable error
● (Green)	● (Green)	No object detection in configured fields/ready for operation
–	● (Green)	No fields configured/ready for operation
● (Green)	⦿ (Yellow)	Warning
⦿ (Yellow)	⦿ (Green)	Restart after time or digital input
● (Green)	⦿ (Yellow)	Contamination warning
⦿ (Red)	⦿ (Yellow)	Contamination error
● (Green) ⦿ (Yellow) ⦿ (Red)	● (Green) ⦿ (Yellow) ⦿ (Red)	Identify product
⦿ (Red)	⦿ (Red) ⦿ (Yellow)	Teach-in environment
⦿ (Yellow)	● (Green)	Object detected

● = illuminated; ⦿ = flashing

Further topics

- [Enable/ disable LEDs \[sWN LEDEnable\]](#)
- [Read state of LEDs \[sRN LEDState\]](#)

3.4 Type label

Product

Information for identifying the product is located on the left side of the product.

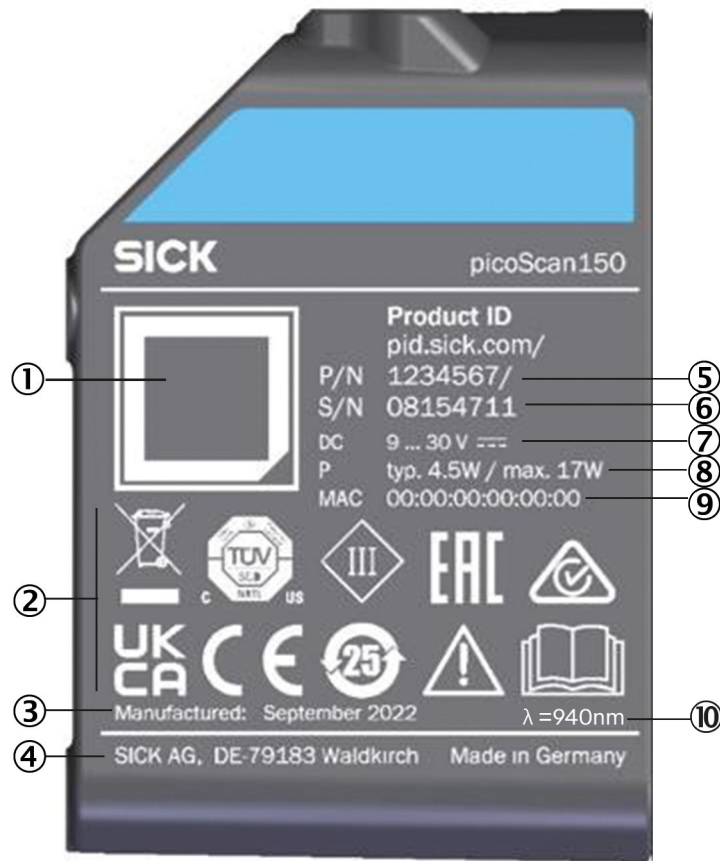


Figure 2: picoScan150 type label (example)

- ① Data Matrix code with product data and link to product page
- ② Conformity mark/certification mark, protection class, symbol: Observe the operating instructions!
- ③ Production date
- ④ Manufacturer, place of production
- ⑤ Part number
- ⑥ Serial number
- ⑦ Voltage supply
- ⑧ Typical power, max. power
- ⑨ MAC address
- ⑩ $\lambda=940\text{ nm}$: The values for the light source in the Technical Data section apply.

Not specified: The following light source values apply: Infrared (wavelength 905 nm, maximum output power < 5 W, maximum output power per area 0.2 W/mm², pulse duration 1.2 ns, average power < 10 mW)

Male connector

Information for identifying the male connector is located on the connector.

3 PRODUCT DESCRIPTION

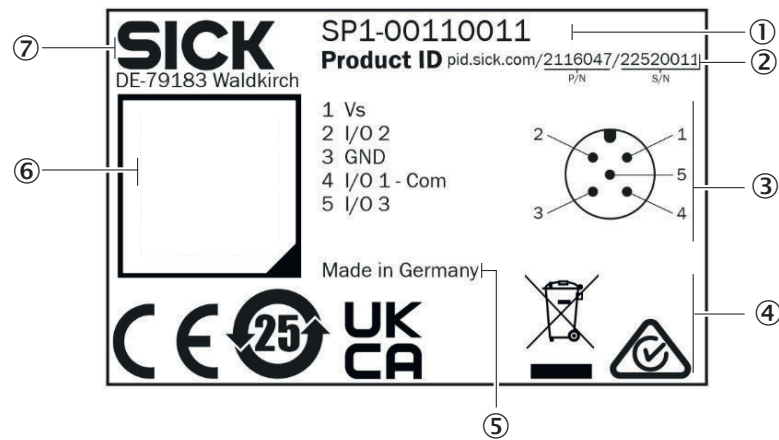


Figure 3: Type label for system plug (example)

- ① Type code
- ② Product ID with part number (P/N) and serial number (S/N)
- ③ Pin assignment or wire colors
- ④ Conformity mark/certification mark
- ⑤ Production site
- ⑥ Data Matrix code with product data and link to product page
- ⑦ Manufacturer

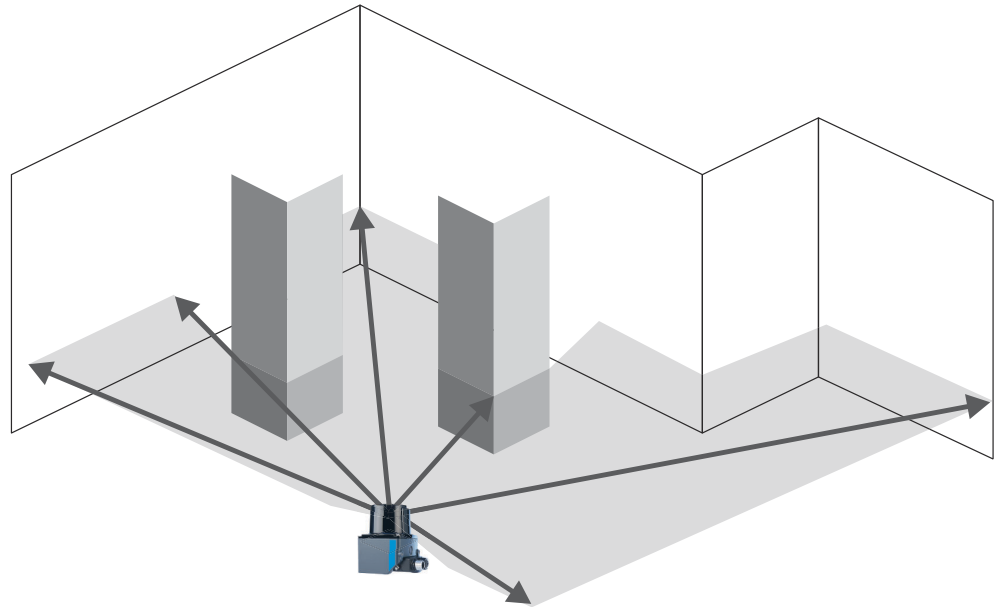
Further topics

- [Read serial number \[sRN SerialNumber\]](#)
- [Read laser type \[sRN laserType\]](#)

3.5 Principle of operation

3.5.1 Measurement principle

The product is an opto-electronic LiDAR sensor (laser scanner) that uses laser beams for non-contact scanning of the contour of its surroundings on a plane. The product measures its surroundings in two-dimensional polar coordinates, relative to its measurement origin. This is marked by a circular indentation in the center of the optics cover. If a laser beam strikes an object, the position of that object is determined in terms of distance and direction.



3.5.2 Distance measurement with HDDM+

Overview

This section describes the principle of distance measurement using pulsed laser beams and the use of SICK's own HDDM+ technology.

Principle of distance measurement

The product emits beams pulsed by a laser diode. If these hit an object and are reflected, the product recognizes the returning signal and uses it to determine a measurement point. The distance to the object is calculated using the time-of-flight of the laser pulse – i.e. the time it takes for the light pulse to travel from the product to the object and back again.



- ① Emitted pulse
- ② Receive pulse

HDDM+ technology

The product uses SICK's own HDDM+ (High Definition Distance Measurement plus) technology. With this measurement procedure, a measured value is formed from the statistical evaluation of multiple single pulses.

This ensures a significantly more stable time and distance measurement. This gap-free scanning for objects leads to a high sensitivity even for small objects. The individual laser pulses overlap as the distance increases. The number of subpulses for a measurement point differs depending on the sensor profile.

A rotating mirror deflects the emitted laser beams and uses them to scan the environment in a circular pattern. This creates a two-dimensional measuring plane. The angle information of the measurement point is provided by an internal encoder of the rotating mirror.

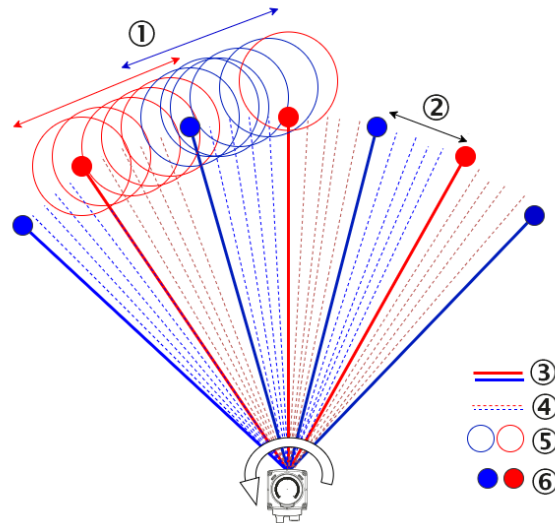


Figure 4: Gap-free scanning using the HDDM+ measurement procedure (schematic illustration)

- ① Optically covered area for the respective measurement point
- ② Angular resolution
- ③ Angle of an output measurement point with specified angular resolution
- ④ Angle of an internally used HDDM+ subpulse
- ⑤ Laser spot of a subpulse
- ⑥ Output measurement point in the polar coordinate system

3.5.3 Multi-echo analysis

The distance between the product and an object is calculated via the time-of-flight of the emitted and then received again pulse. The product can evaluate up to three echo signals for each measuring beam to deliver reliable measurement results, even under adverse ambient conditions.

To be able to display echoes as separate measured values, there must be a minimum distance of approx. 0.5 m between the two objects.

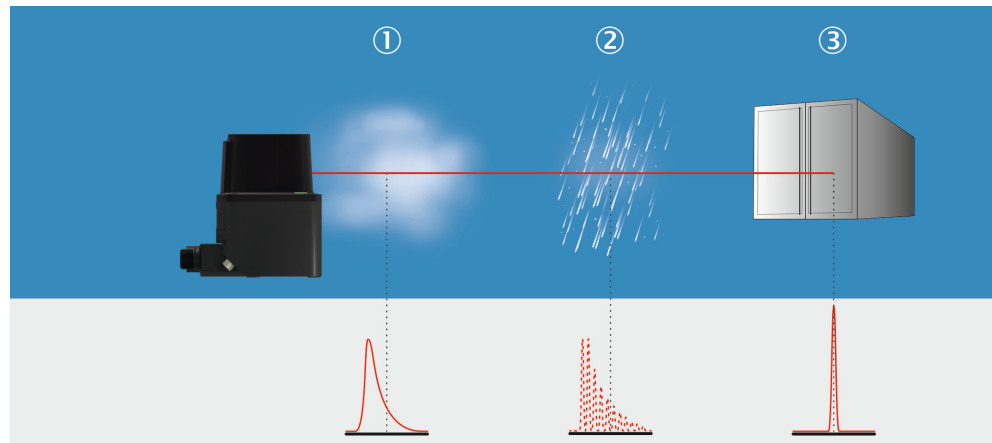


Figure 5: Multi-echo analysis using the example of building management

- ① Fog
- ② Rain
- ③ Measuring object

3.5.4 Dynamic Sensing Profiles

LiDAR sensors are used for a variety of sub-applications. Within these sub-applications, the requirements can change dynamically. When used in an AMR (Autonomous Mobile Robot), for example, different sensor characteristics are required depending on the area in which the AMR is employed. For example, collision protection in an area with a lot of traffic requires the shortest possible response time. If the AMR is traveling in a wide area of a warehouse, however, a large scanning range and good angular resolution is needed. When an AMR docks with a roller conveyor or a loading station, for example, the response time is no longer of such high priority and the high angular resolution to precisely detect the close range becomes more important.

If an application requires a very high angular resolution, this necessitates a low scanning frequency and vice versa. Which functions have priority should be prioritized for each specific application.

The product therefore offers a Dynamic Sensing Profiles feature, which enables the measurement core parameters to be dynamically adjusted to the specific requirements in subtasks of the LiDAR sensor. This is done during operation and without the need to restart the product.

The product therefore comes with a Scan Data Configuration function that allows quick switching between profiles.

The following profiles are available:

Profile number	Frequency in Hz	Angular resolution in °	Possible use
Profile 1	15	0.5	Object detection with large scanning ranges
Profile 2	15	0,33	TiM series retrofit
Profile 3	20	0.1	Fine positioning or recording of environment maps
Profile 4	20	0.25	Low-noise measurements, recommended when using LiDAR-LOC
Profile 5	25	0.25	LMS100 series retrofit
Profile 6	30	0.1	Fine positioning or recording of environment maps

3 PRODUCT DESCRIPTION

Profile number	Frequency in Hz	Angular resolution in °	Possible use
Profile 7	40	0.25	Fast object detection at a large scanning range
Profile 8	50	0.25	Very fast object detection
Profile 9	15	0.05	Highest angular resolution for static measurement tasks
Profile 10	40	0.125	High angular resolution and high scan rate
Profile 11	15	1°	TiM series retrofit

Further topics

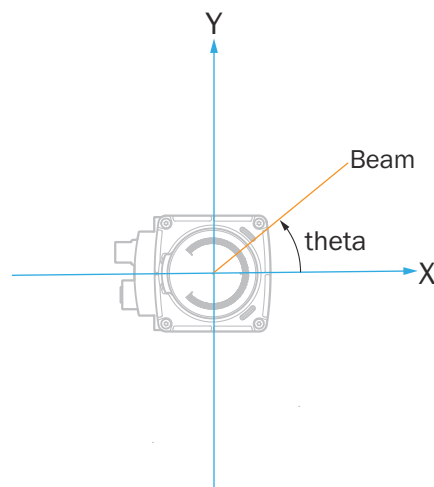
- [Set Performance Profile \[sWN PerformanceProfileNumber\]](#)

3.5.5 Coordinate system

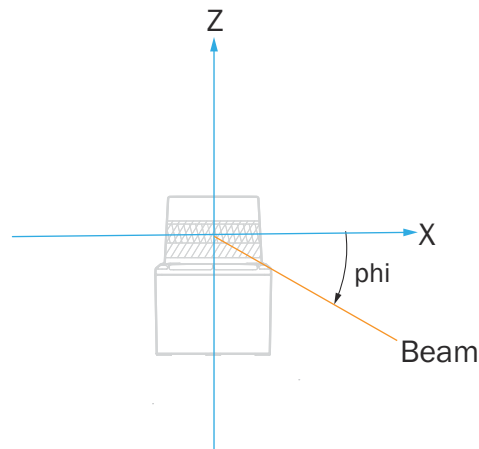
Device coordinate system

The origin of the device coordinate system ($X=0$, $Y=0$, $Z=0$) is a single point that serves as the origin and reference for all laser beams and the distance measurement of the product. When no translation is applied to the product in the world coordinate system, this point coincides with the origin of the world coordinate system.

The azimuthal (horizontal) angle of a beam is called theta. The beam at zero azimuth angle lies in the middle of the main viewing direction of the product so the scan is symmetrical.



The elevation angle (vertical angle) of a beam is designated phi and is measured relative to the x-y plane. For single plane sensors like the picoScan, the elevation angle is 0° when ideally mounted. Note the scan field flatness in regard to this.



The data is always output in the device coordinate system, not in the world coordinate system.

3.5.6 Filter

By using digital filters to pre-process and optimize the measured distance values, the product can be tailored to the specific requirements of the respective application. This makes it possible to prevent virtually all faults.

The active filter functions affect the output measured values. It is not possible to recalculate the original measured values from the filtered output values.

The filters – if present – are applied in the following order:

- Fog filter
- Echo filter
- Particle filter
- Moving average filter
- The filters for data reduction as well as the region of interest filter are then used.

3.5.6.1 Fog filter

The fog filter enables the product to eliminate unwanted echoes at close range (approx. 3 m). This considerably lowers the probability of false activations at close range in fog.

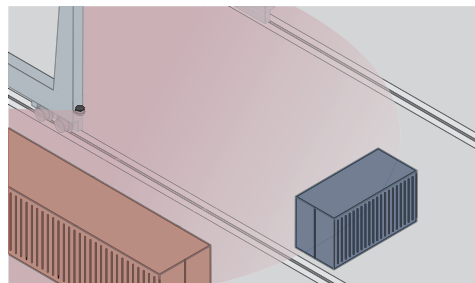


Figure 6: Without the fog filter: objects are difficult to detect through the fog due to reflections.

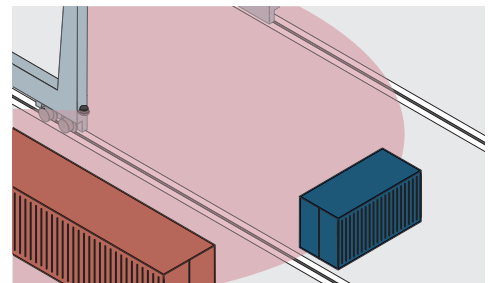


Figure 7: Using the fog filter: objects can be detected reliably because unwanted echoes are screened out.

Further topics

- [Set sensitivity fog filter \[sWN MCSenseLevel\]](#)

3.5.6.2 Echo filter

The echo filter screens out unwanted measurement data caused by edge hits, rain, dust, snow and other ambient conditions.

You can set whether the first, the last, or all three echoes are output.

The other pulses triggered by undesirable ambient conditions are not taken into account or output.

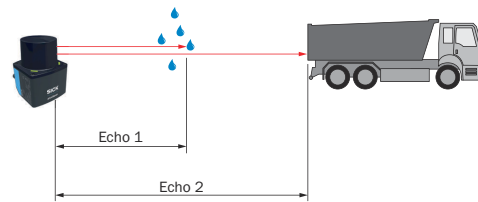


Figure 8: Without the echo filter: The product receives unwanted echoes from ambient conditions such as rain.

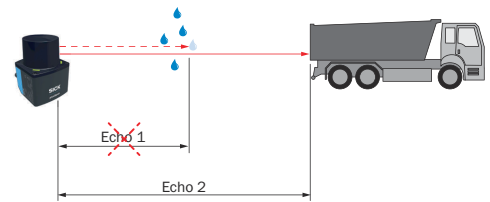


Figure 9: Using the echo filter (setting: last echo): the product screens out unwanted echoes from ambient conditions as per the settings chosen.

Further topics

- [Set echo filter \[sWN FREchoFilter\]](#)

3.5.6.3 Particle filter

The particle filter blanks small, irrelevant reflection pulses in dusty environments and in rain or snow which are caused by dust particles, raindrops, snowflakes or the like.

In doing so, successive scans are continuously evaluated in order to detect static objects. If the distance between a measured value and its temporal spatial neighbors is greater than a defined threshold value, this measured value is discarded as faulty and replaced by the last valid measured value.

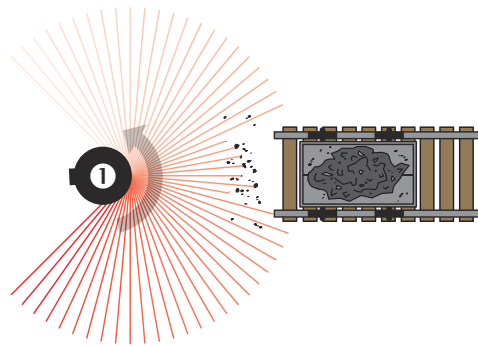


Figure 10: Without the particle filter: Violation of the contour due to dust particles in the vicinity of the object.

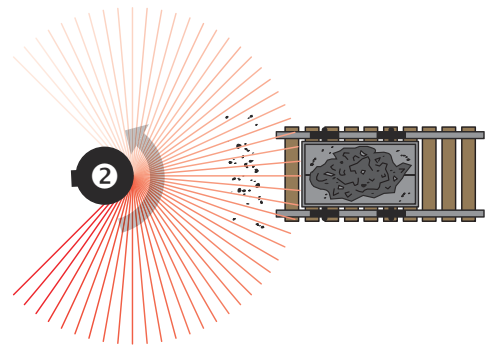


Figure 11: Using the particle filter: The response to dust particles in the detection field is delayed by one scan. Particles can thereby be blanked.

Important notes



NOTE

If the particle filter is activated, measurement data output is delayed by one scan.

Further topics

- [Set particle filter \[sWN LFPparticle\]](#)

3.5.6.4 Moving average filter

The sliding average filter smooths the distance value. It does this by calculating the arithmetic mean from several scans of the same point. The number of scans is configurable (2 ... 4 scans).

Table 2: Example: Moving average filter over 4 scans

Scan	Angle (distance values in mm)									
	1	2	3	4	5	6	7	8	9	...
1	0	0	1100	1100	1150	1150	1380	1380	0	...
2	0	0	1200	1200	1190	950	1500	1500	0	...
3	0	0	1150	1450	1200	1200	1450	1450	0	...
4	0	0	1170	1170	1220	1220	1470	1150	0	...
1. Output value (scan 1-4)	0	0	1155	1230	1190	1130	1450	1370	0	...
5	0	0	0	1110	1150	1150	1380	1380	0	...
2. Output value (scan 2-5)	0	0	1173	1233	1190	1130	1450	1370	0	...
6	0	0	1200	1210	1190	0	1500	1500	0	...
3. Output value (scan 3-6)	0	0	1173	1235	1190	1190	1450	1370	0	...
7	0	730	1150	0	1200	1200	1450	1450	0	...
4. Output value (4-7)	0	730	1173	1163	1190	1190	1450	1370	0	...
...	...	...	...	...	...	...	...	...	...	...

Individual outliers (shown in **bold** in the table) influence the average value.

After activating the filter, the first measured value is not output until after the configured number of scans. Therefore, there is always a time delay equivalent to the number of scans configured for averaging. The scan counter is taken from the latest scan included in the averaging process. Invalid distance values (= 0) are not included in the averaging calculation, so that in these places a smaller number of scans is used in the division calculation.

Further topics

- [Set moving averaging filter \[sWN LFPmovingAveragingFilter\]](#)

3.5.6.5 Data reduction filter

A data reduction filter is an algorithm that selects, based on various criteria, the relevant measurement data that should be excluded from the further processing.

3.5.6.5.1 Interval filter

The interval filter reduces the scan output rate by a configurable factor (reduction factor). When the reduction factor is set to three, for example, the output rate is reduced to one third. In this case only every third scan is output.

Further topics

- [Set interval filter \[sWN LFPintervalFilter\]](#)

3.5.6.5.2 Angular range filter

The horizontal angle range that is output per scan can be restricted by increasing the start angle (theta start) or reducing the stop angle (theta stop).

It can be set within a range of -138° to $+138^{\circ}$. Please note that the angle beam orthogonal to the front side is defined as 0° , and the direction of rotation of the product is set to counterclockwise.

For the compact format: If a complete segment of the output data is outside the angle range, it is not output. If a segment is partially within the specified angle range, it is filled with 0 values.

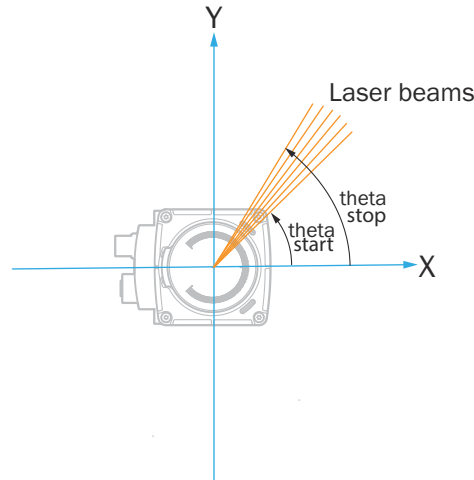


Figure 12: Definition of the theta start and theta stop angles (top view)

3.5.6.6 Range filter

A range filter is an algorithm that uses various criteria to select relevant measurement data that are set to a distance value and RSSI value of zero for further processing. Angle values are retained.

3.5.6.6.1 Rectangular filter

When the rectangular filter is activated, it cuts out everything except for the parts of the scan within a rectangle. Note that this filter does not reduce the data, it sets the data points outside the rectangle to zero. The rectangle can be adjusted by setting the minimum and maximum values [mm] for the X and Y axis.

Further topics

- [Set rectangular filter \[sWN LFPcubicareafilter\]](#)

3.5.6.6.2 Distance filter

The distance filter affects the display of a circular area around the product by limiting the minimum and maximum radial distance that is measured.

The distance filter does not reduce the data, but sets the data points outside the radial distance to zero.

To keep a circular area, set the max area to a specific radius [mm].

To keep a ring, set the max range to a specific radius [mm] and the min range to >0 mm.

Further topics

- [Set sensitivity fog filter \[sWN MCSenseLevel\]](#)

3.5.7 Measurement data output

3.5.7.1 Data formats

The product supports two data output formats: Compact (recommended) and MSGPACK. In both data formats, the data can be output segment by segment via UDP/IP (singlecast/multicast). Both data formats contain information such as serial number and time stamp.

MSGPACK can be easily integrated and parsed using existing libraries, but requires more bandwidth than the Compact data format due to the data types used. Compact is more efficient and requires less bandwidth.

When changing from TiM series or LMS series sensors to picoScan150, you can easily add the `LMDscandata` data format. This format is transmitted via TCP/IP and supplements the Compact and MSGPACK formats included as standard in the picoScan150.

The product has native ROS support for integration in a ROS environment.

Further topics

- [Data format description \(EN\)](#)
- [Native ROS 2 \(DDS\)](#)

3.5.7.2 Segmented data output

The product records data over a scan range of 276°. The data acquired within a 276° rotation for the scan layer are referred to below as a scan. For data output, a scan is divided into segments each of which contain the scan layer data for a smaller angular range.

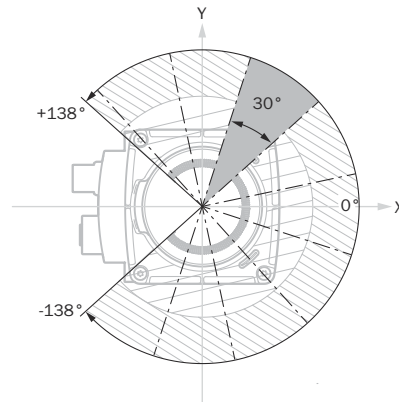


Figure 13: Example segment output at 276° scan range at 25 Hz

The segment size depends on the scanning frequency

- 30° at a scanning frequency ≤ 25 Hz
- 60° at a scanning frequency ≥ 30 Hz

The number of segments depends on the configured horizontal scan range.

Examples:

- 276° scan range at 25 Hz = 10 segments
- 276° scan range at 40 Hz = 5 segments
- 180° scan range (-90° ... +90°) at 40 Hz = 4 segments

If the scan range is adjusted, the segment boundaries remain.

In the Compact format, the measured values in hidden angle ranges are filled with zeros. If a complete segment is hidden, it is omitted from the data output.

Complementary information

- A code example can be found at: <https://github.com/SICKAG/ScanSegmentAPI>

3.5.7.3 Latency of the measurement data output

The latency value describes the data output latency between an HDDM+ laser pulse packet (measured value) and the transmission of the processed segment to a client system.

The specified latency value applies to a measurement scenario with many echoes and without the use of internal filters. In addition to this delay time, network delays, e.g. due to cables or Ethernet switches, must also be taken into account.

3.5.7.4 Native ROS 2 (DDS)

Overview

The product supports native ROS 2 communication via the DDS (Data Distribution Service) protocol. This enables efficient real-time transmission of measured data without additional drivers and with minimum delay time. The data is output directly in ROS 2-compatible formats.

Prerequisites

- Data output for **Native ROS 2 (DDS)** is activated
- CycloneDDS is used as ROS middleware (RMW)

ROS 2 topics

The following ROS 2 topics are streamed directly via DDS:

- PointCloud2
- LaserScan
- MultiEchoLaserScan
- tf2

Table 3: Available PointCloud2 topics

PointCloud2 topic	Contents ¹⁾	Output frequency	Description
rt<namespace>/cloud/all_points	x,y,z,i,p	Scan rate	Valid scan points of all segments and all echoes
rt<namespace>/cloud/all_points_echo0	x,y,z,i,p	Scan rate	Valid scan points of echo 0
rt<namespace>/cloud/all_points_echo1	x,y,z,i,p	Scan rate	Valid scan points of echo 1 (only available if multi-echo is configured)
rt<namespace>/cloud/all_points_echo2	x,y,z,i,p	Scan rate	Valid scan points of echo 2 (only available if multi-echo is configured)
rt<namespace>/cloud/reflector_points	x,y,z,i,p	Scan rate	Valid scan points that were recognized as reflectors
rt<namespace>/cloud/segment[s]	x,y,z,i,p	Segment frequency	Valid scan points of a selected segment s (only available for segments that are available after application of the angle range filter)

PointCloud2 topic	Con- tents ¹⁾	Output fre- quency	Description
rt<namespace>/cloud/ segment[s]_echo[e]	x,y,z,i,p	Segment frequency	Valid scan points of the selected echo e in a specific segment s (only available for segments that are avail- able after application of the angular range filter and depending on the configuration of the echo filter)

- ¹⁾
- intensity (i) : RSSI with value range: 0.0 ... 1.0
 - properties (p): Bit mask 0b00000001 indicates whether this measurement point was recognized as a reflector

Table 4: Available LaserScan / MultiEchoLaserScan topics

LaserScan / MultiEchoLaser- Scan topic	Data for- mat	Con- tents ¹⁾	Output fre- quency	Description
rt<namespace>/scan/ all_points	MultiEcho- LaserScan	x,y,z,i	Scan rate	Valid scan points of all seg- ments and all echoes
rt<namespace>/scan/ all_points_echo0	LaserScan	x,y,z,i	Scan rate	Valid scan points of echo 0
rt<namespace>/scan/ all_points_echo1	LaserScan	x,y,z,i	Scan rate	Valid scan points of echo 1 (only available if multi-echo is configured)
rt<namespace>/scan/ all_points_echo2	LaserScan	x,y,z,i	Scan rate	Valid scan points of echo 2 (only available if multi-echo is configured)
rt<namespace>/scan/ segment[s]	MultiEcho- LaserScan	x,y,z,i	Segment fre- quency	Only for available segment s (only available for seg- ments that are available after application of the angle range filter)
rt<namespace>/scan/ segment[s]_echo[e]	LaserScan	x,y,z,i	Segment fre- quency	Only for available segment s and echo e (only availa- ble for segments that are available after application of the angular range filter and depending on the con- figuration of the echo filter)

- ¹⁾ intensity (i) : Value range 0.0-1.0 for natural targets, 1.05 for detected reflector

Important information

- Filter settings in the product have a direct effect on the published data.
- Subscribed topics are published even if no measurement points are available. It is therefore recommended that you only subscribe to required topics.
- If the static IP address is changed, a restart of the device is required to output the topics correctly.

Complementary information

The Native ROS 2 interface is used solely as a data output interface. Parameterization is still carried out via the CoLa or REST interface.

The following functions are useful in the context of this feature:

- Switching on Native ROS 2: `ScanDataFormat`
- Setting the FrameID: `rosFrameID`
- Setting the namespace: `rosNamespace`

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- Setting the ParentFrameID: `rosParentFrameID`
- Setting the sensor position (`/tf Topic`): `SensorPosition`

Further topics

- [Communication interfaces](#)
- [Telegram listing \(EN\)](#)

3.5.7.5 ROS driver

Suitable drivers for integrating the product into the ROS (Robot Operating System) are available for download on the product page.

The product page can be accessed via the **SICK Product ID**: `pid.sick.com/{P/N}/{S/N}`

{P/N} corresponds to the part number of the product, see type label.

{S/N} corresponds to the serial number of the product, see type label (if indicated).

3.5.8 Object sizes

As the distance from the product increases, the laser beam expands. As a result, the diameter of the light spot on the surface of the object increases.

The more laser pulses (HDDM+ subpulses) that hit an object, the more reliably it will be detected. Other factors that can affect the object detection are the angle of incidence of the laser beam, and the remission and surface properties of the object. Ambient conditions (e.g. ambient light) as well as movement of the product or the object can also affect the object detection.



Figure 14: Beam expansion

- ① Expanded laser beam
- ② Optical axis

Formula

The formula applies to all result values ≥ 2 mm. If the value is less than this, 2 mm is used as the result.

$$\text{object distance [mm]} \cdot \arctan \left(\left(\text{laser divergence [}^\circ\text{]} + \text{angular resolution [}^\circ\text{]} \right) \cdot \frac{\pi}{180} \right)$$

Figure 15: Formula for the minimum object size at a certain distance

- ① Object distance in mm
- ② Laser divergence in $^\circ$
- ③ Angular resolution in $^\circ$

The angular resolution can be adjusted to suit the product variant. As a general rule, the lower the configured angular resolution, the smaller the objects that can be detected.

$$4000 \text{ mm} * \arctan\left((0.27^\circ + 0.1^\circ) * \frac{\pi}{180}\right) = 26 \text{ mm}$$

Figure 16: Example calculation: minimum object size at 4 m distance for an angular resolution of 0.1°

$$100 \text{ mm} * \arctan\left((0.27^\circ + 0.1^\circ) * \frac{\pi}{180}\right) = 0,65 \text{ mm} < 2 \text{ mm} \longrightarrow 2 \text{ mm}$$

Figure 17: Example calculation: minimum object size at 0.1 m distance for an angular resolution of 0.1°. According to the formula, the result is less than 2 mm, so the minimum result of 2 mm remains.

Diagrams

The following diagrams show typical minimum object sizes at various distances.

For reasons of improved readability, the near range up to 4 m is shown in a separate diagram.

When using the diagrams, note the specified max. scanning ranges depending on the device variant.

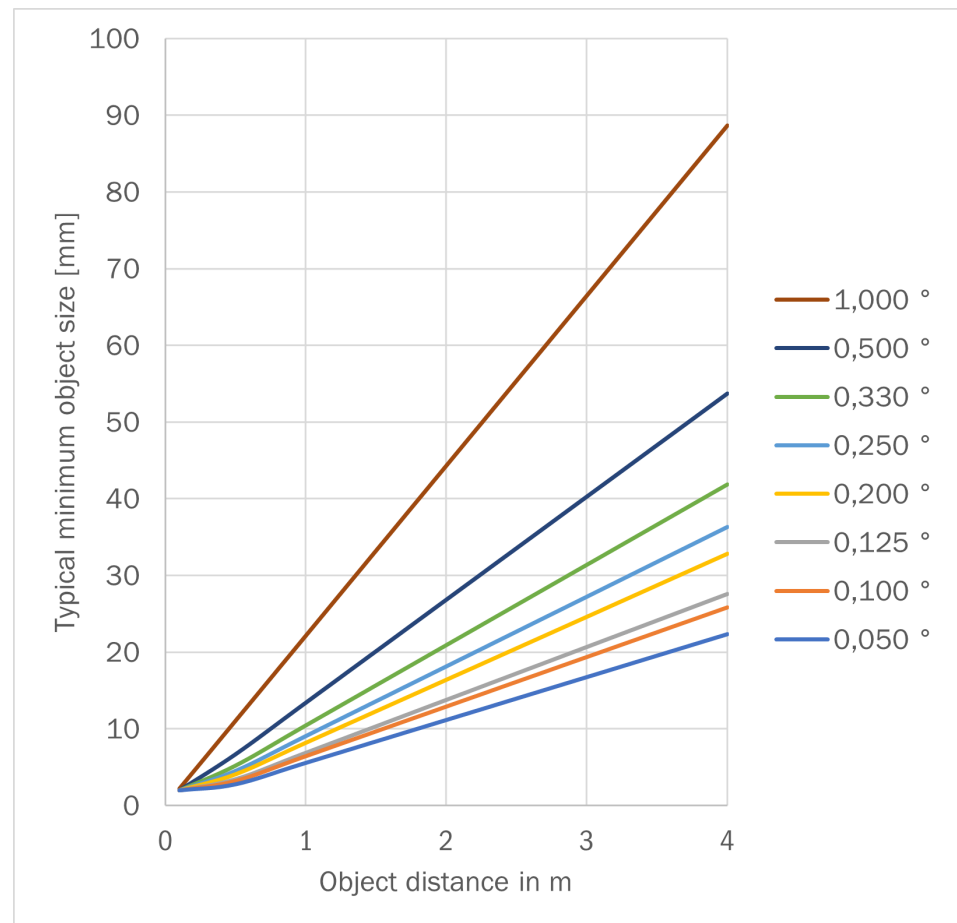


Figure 18: Typical minimum object size up to 4 m

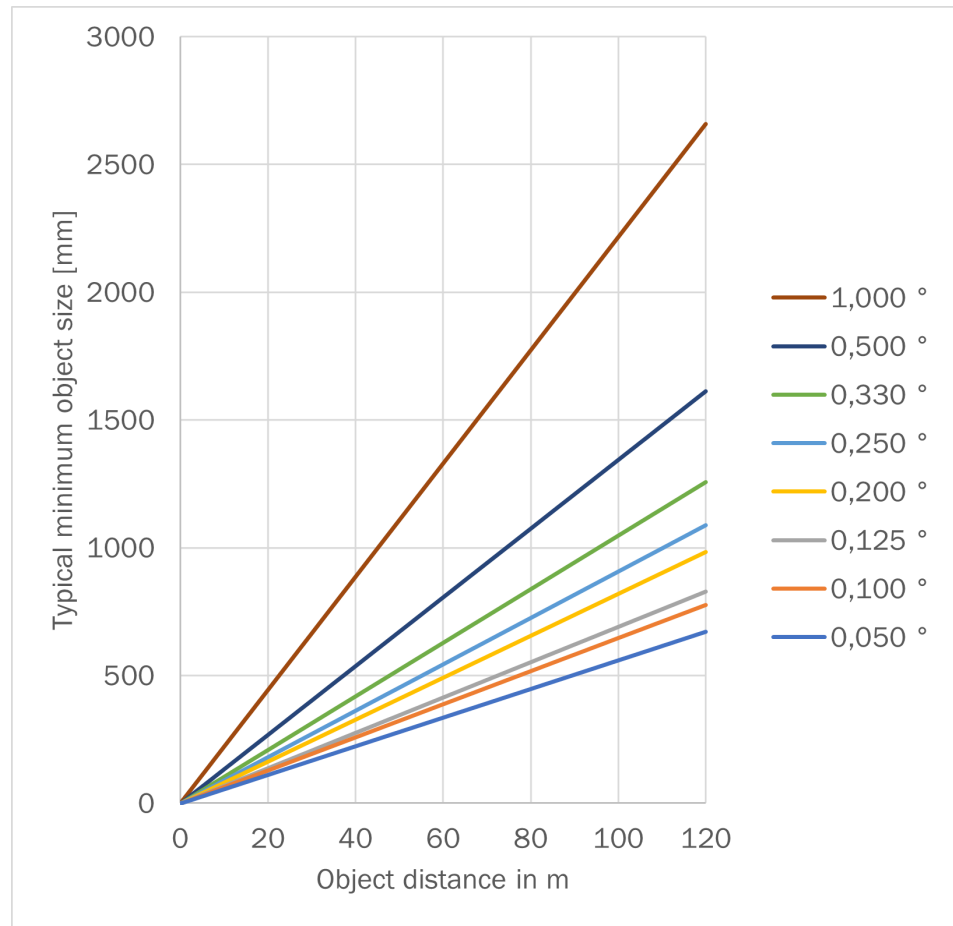


Figure 19: Typical minimum object size up to 120 m

Further topics

- [Data sheet](#)
- [Dynamic Sensing Profiles](#)

3.5.9 Impact of object surfaces on the measurement

Reflectance

Most surfaces produce a diffuse reflection of the laser beam in all directions. The structure (smooth or rough), shape (flat or curved), and color (light or dark) of the surface determine how well the laser beam is reflected.

On very rough surfaces, a large proportion of the energy is lost due to absorption. Curved surfaces produce a higher diffusion. Dark surfaces reflect the laser beam worse than light ones (brilliant white plaster reflects approx. 100% of the light, while black foam rubber reflects approx. 2.4%). The aforementioned surface characteristics can reduce the scanning range of the device, in particular for surfaces with low remission values.

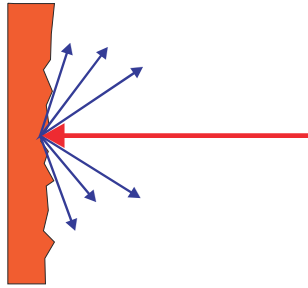


Figure 20: Reflection of light on the surface of the object

Angle of reflection

The angle of reflection corresponds to the angle of incidence. If the laser beam hits a surface at right angles, the energy is optimally reflected. If the laser beam hits a surface at an oblique angle, energy and range are lost accordingly.

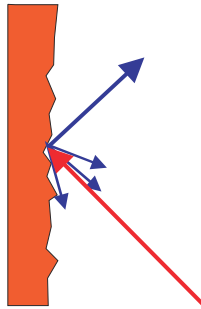


Figure 21: Angle of reflection

Retroreflection

If the reflective energy is greater than 100 %, the beam is not reflected diffusely in all directions; instead it is reflected in a targeted way (retroreflection). Thus a large part of the emitted energy can be received by the LiDAR sensor. Plastic reflectors (cat's eyes), reflective tape, and triple prisms have these properties.

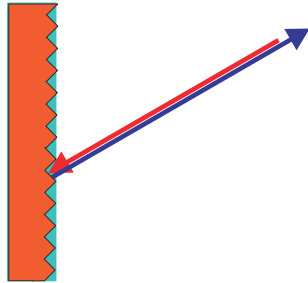


Figure 22: Retroreflection

Specular surfaces

The laser beam is almost completely deflected on reflective surfaces. This means that an object hit by the deflected beam may be detected instead of the reflective surface.

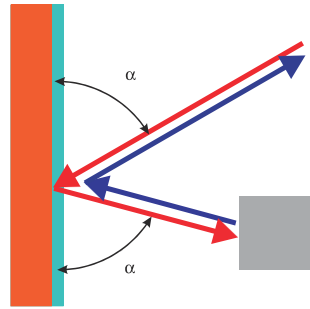


Figure 23: Specular surfaces

Small objects

Objects that are smaller than the diameter of the laser beam cannot reflect the laser light's full energy. The portion of the light beam that does not reach the object is lost. If all of the light reflected to the sensor is insufficient, the object may not be detected.

The portion of the light that does not reach the front object can be reflected by a larger object in the background. If all of the light reflected to the sensor is sufficient, this object is detected. This can lead to a corruption of the measured value.

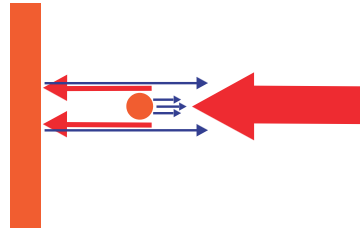


Figure 24: Object smaller than the laser beam diameter

3.5.10 Reflector detection

To determine the presence of a reflector, a value based on an evaluation of the energy reflected from the target (Received Signal Strength Indicator, RSSI) is often used. Several difficulties arise with this approach, especially in distinguishing between bright targets and a reflector in the vicinity of the product.

The product therefore has a special reflector detection integrated into it. This function uses not only the RSSI value, but also other internally available information (e.g. signal characteristics, distances, echoes or signal delays) to ensure a near error-free reflector detection.

A reflector marking is assigned to each distance measurement value generated by a reflector. This reflector marking is available in the measurement data telegram (Compact/MSGPACK) or can be displayed in the SOPASair web server. When using LMDscandata, this reflector information can be taken from the RSSI data. If a laser beam hits a reflector, the corresponding RSSI value is set to 254.

Important notes

At close range, reflectors may appear wider than they really are.

Adjustments

Different settings can be selected to ensure a reflector detection with high availability. In the product under **Defaults > Sensitivity**, you can select various modes that offer the following optimizations:

- **Standard** (recommended): Setting that delivers good results in a wide range of applications. The reflector size should be 1.5 times the angular resolution depending on the distance.
- **Optimized for reliability**: This setting can be used to suppress or reduce unwanted measurement artefacts (e.g. widening of the reflectors at close range). This makes the sensor less sensitive. Caution: Targets with a very low reflectivity are only detected with difficulty.
- **Optimized for low remission**: This setting makes it possible to also detect objects with a low reflectivity. This will also lead to more frequent detection of interfering factors such as ambient light or particles, and undesired measurement artefacts can be created. This setting is not suitable for applications focusing on reflectors because it has been optimized for targets with a low remission and reflectors have a very high remission.

Further topics

- [Data sheet](#)
- [Set Sensitivity Mode \[sWN SensitivityMode\]](#)

3.5.11 RSSI

RSSI (Received Signal Strength Indicator) is the measure of the signal strength that the device receives. This value is calculated for every measurement. The device therefore provides, for every echo signal, an associated RSSI value for the signal strength.

The value 0 (zero) means that the received energy was too low to produce a valid measured value and also represents the lowest possible RSSI value. An RSSI value of 1 represents the highest possible measured value. A linear interpolation is applied between the values 0 and 1 using a resolution and scaling specific to the data format.

The Compact and MSGPACK data formats represent the RSSI data in the range from 0 ... 65535. The LMDscandata data format represents the RSSI data in the range from 0 ... 255.

If the RSSI value is 0, then no distance measurement is possible. There can be two reasons for this:

- The target object lies outside the sensing range.
- The target object has an very low remission.

Note that objects like white paper can have similar RSSI values as a reflector at a short distance.

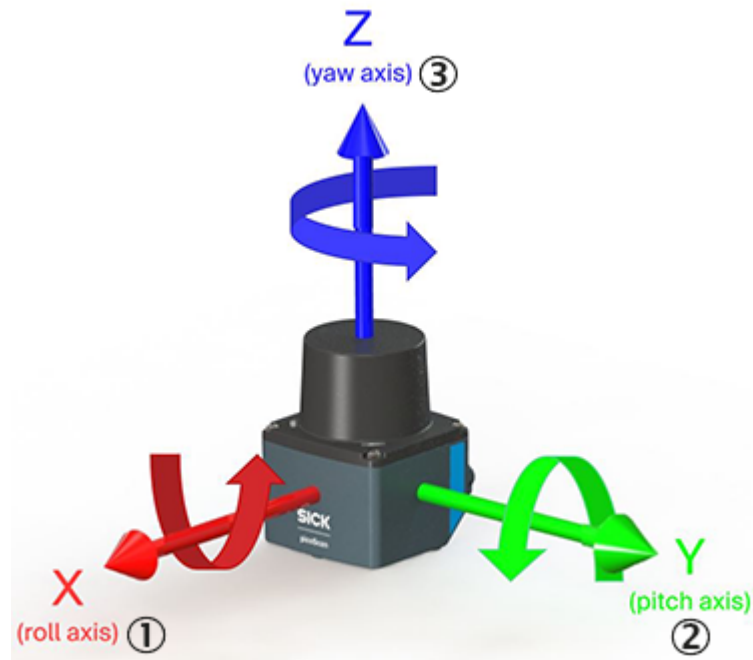
The RSSI values are sensor-specific, relative values that can vary slightly between different devices and during the service life of the device.

Further topics

- [Reflector detection](#)

3.5.12 Inertial measuring unit (IMU)

The product is equipped with an inertial measuring unit (IMU). This can be used to identify movements of the product. The IMU can output accelerations in X, Y and Z as well as the position angle in yaw, pitch and roll. The orientation of the IMU data is based on the coordinate system of the product.



- ① Roll angle
- ② pitch angle
- ③ Yaw angle

Further topics

- [Data format description \(EN\)](#)
- [Set IMU data enable \[sWN ImuDataEnable\]](#)
- [Set IMU data streaming ethernet settings \[sWN ImuDataEthSettings\]](#)

3.5.13 Scanning range

Overview

The scanning range of the device depends on the remission of the object to be detected. The better a surface reflects the incident beam back to the device, the greater the scanning range of the device.

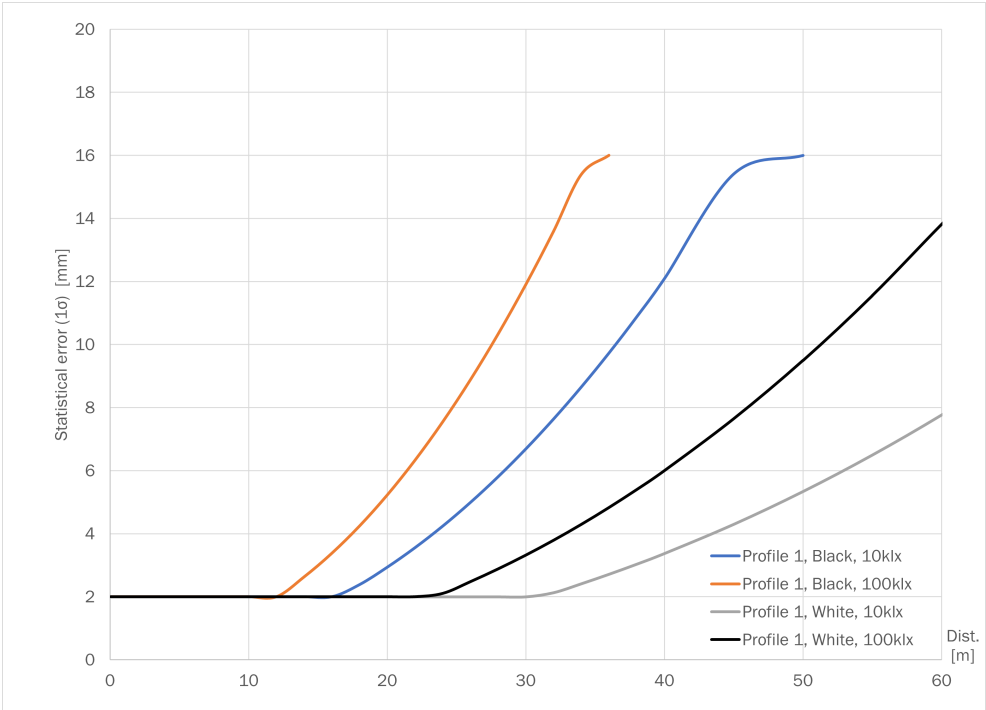


Figure 25: Profile 1 (15 Hz; 0.5°)

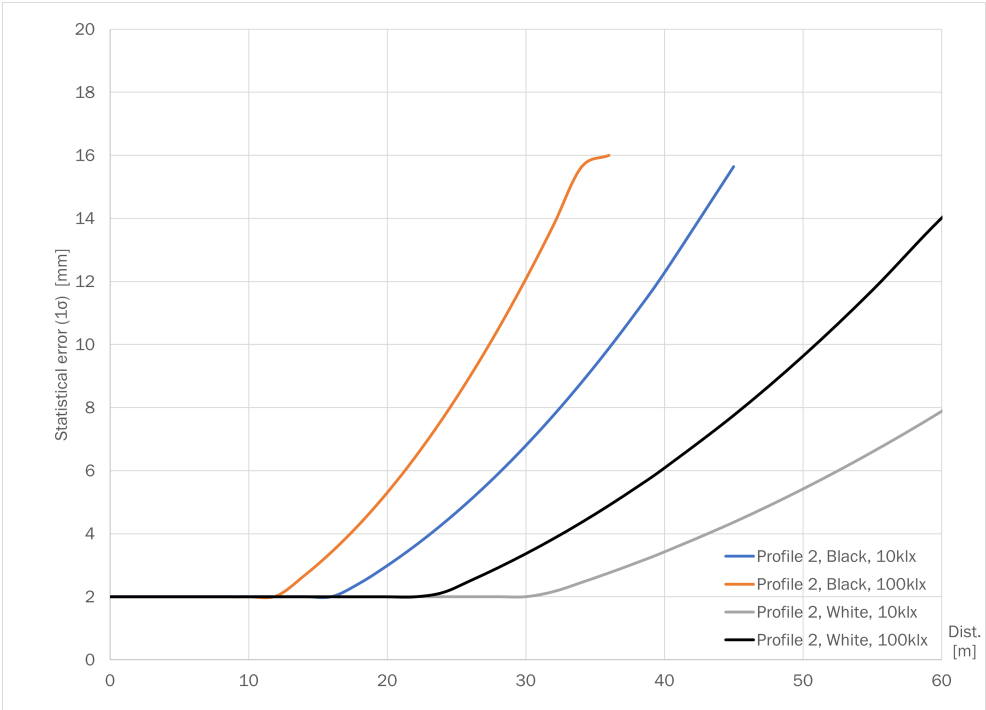


Figure 26: Profile 2 (15Hz; 0.33°)

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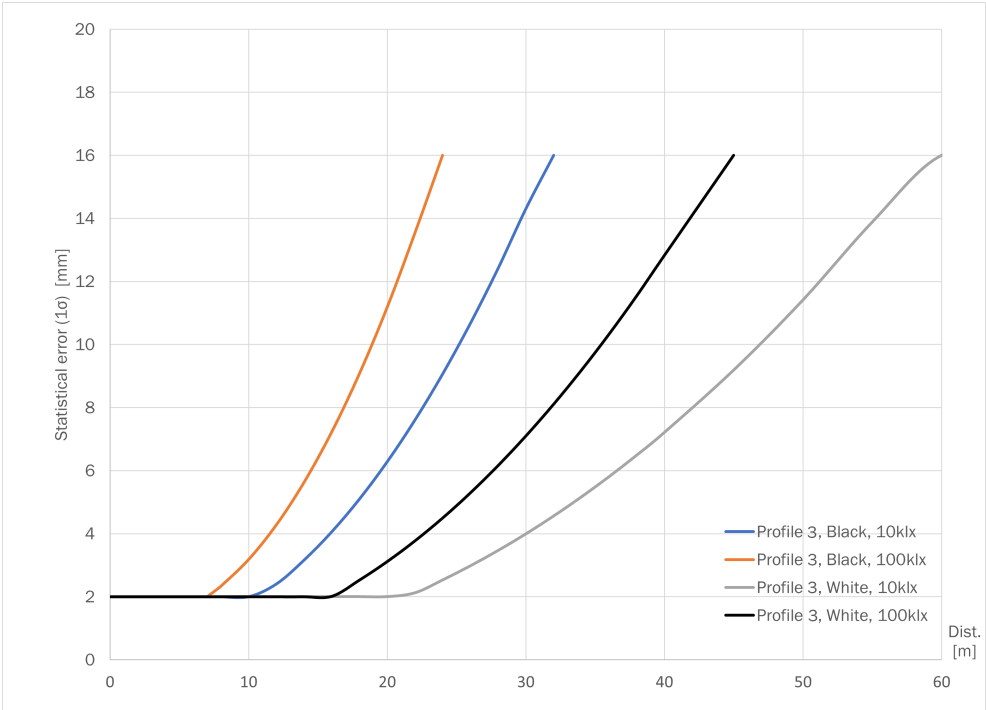


Figure 27: Profile 3 (20 Hz; 0.1°)

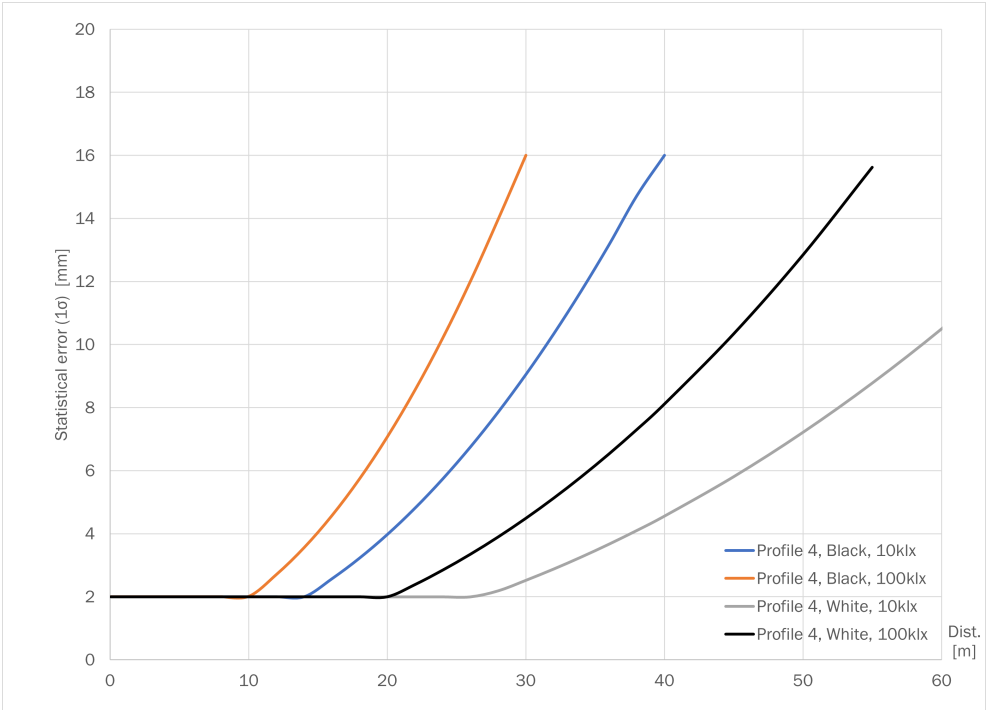


Figure 28: Profile 4 (20 Hz; 0.25°)

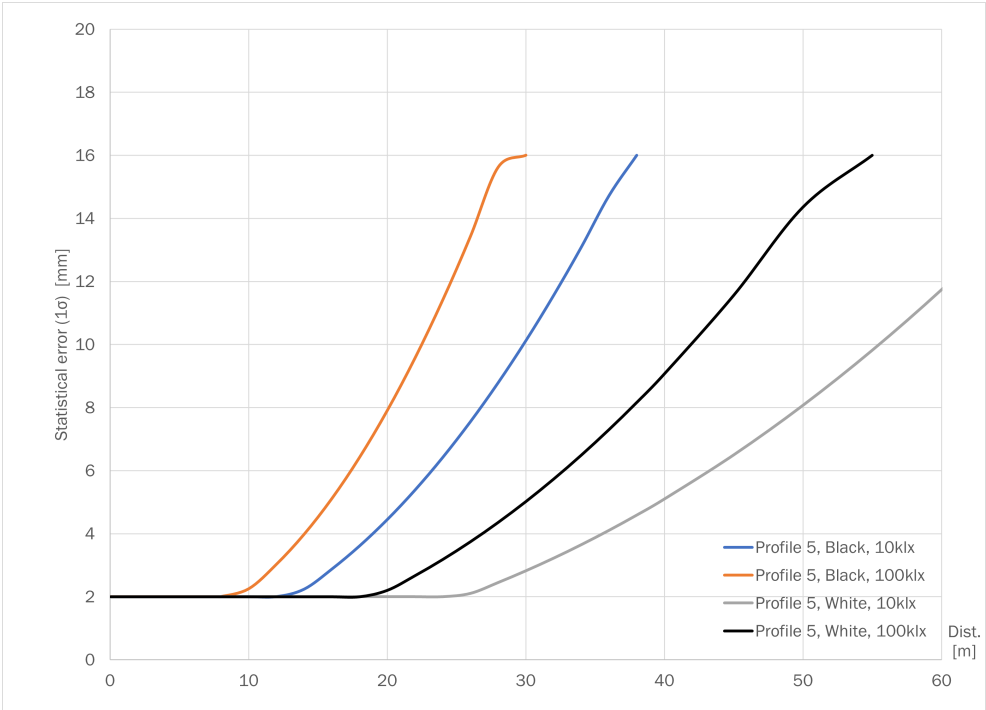


Figure 29: Profile 5 (25 Hz; 0.25°)

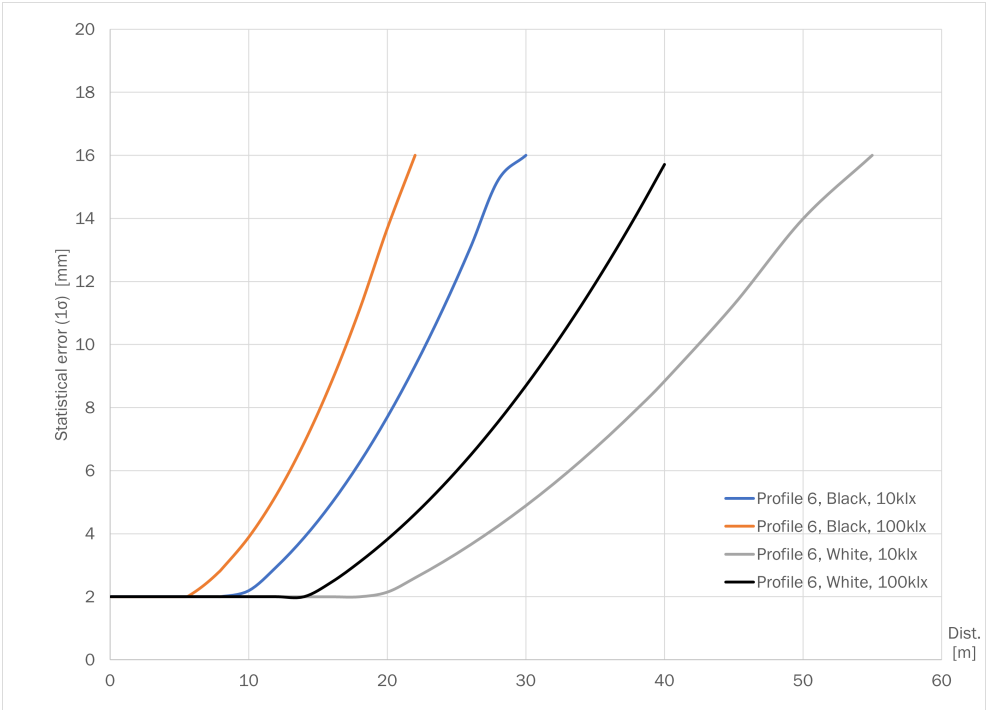


Figure 30: Profile 6 (30 Hz; 0.1°)

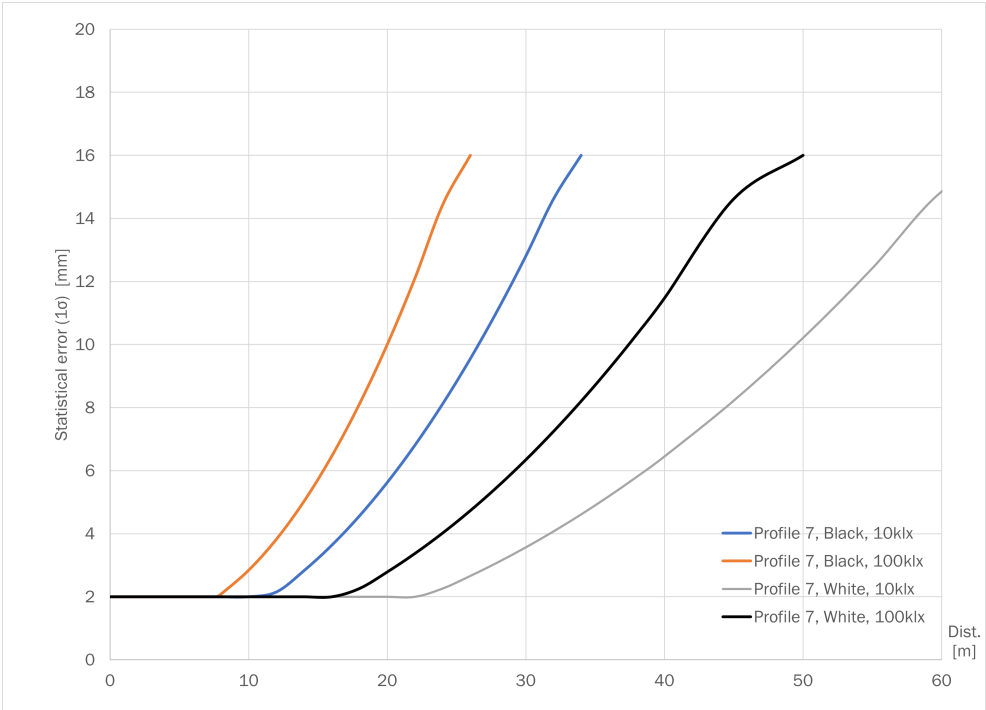


Figure 31: Profile 7 (40 Hz; 0.25°)

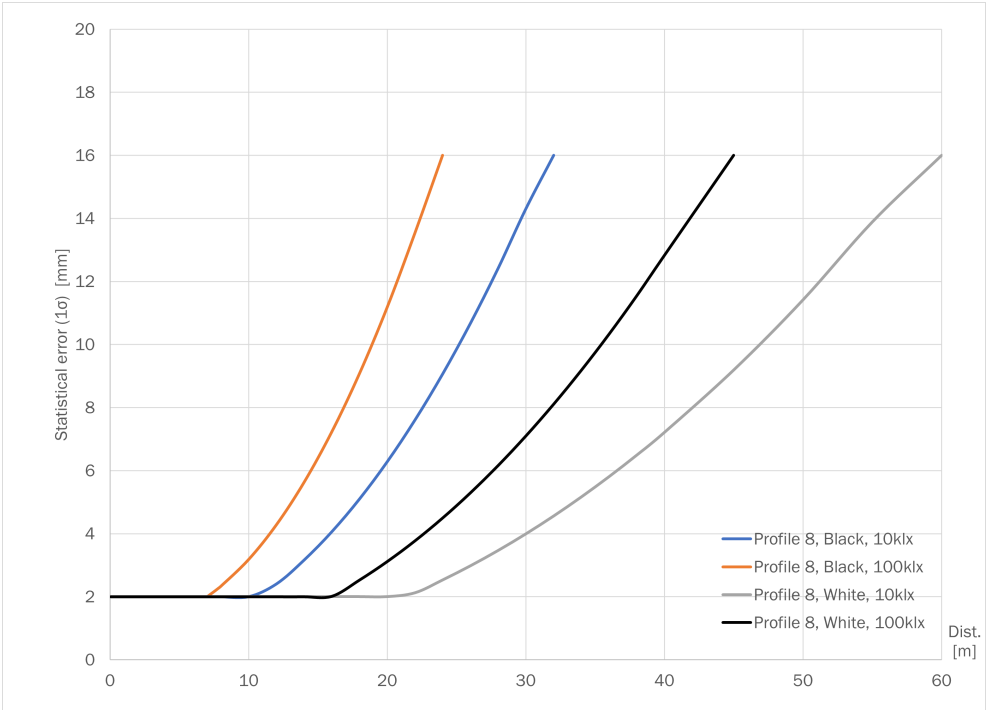


Figure 32: Profile 8 (50 Hz; 0.25°)

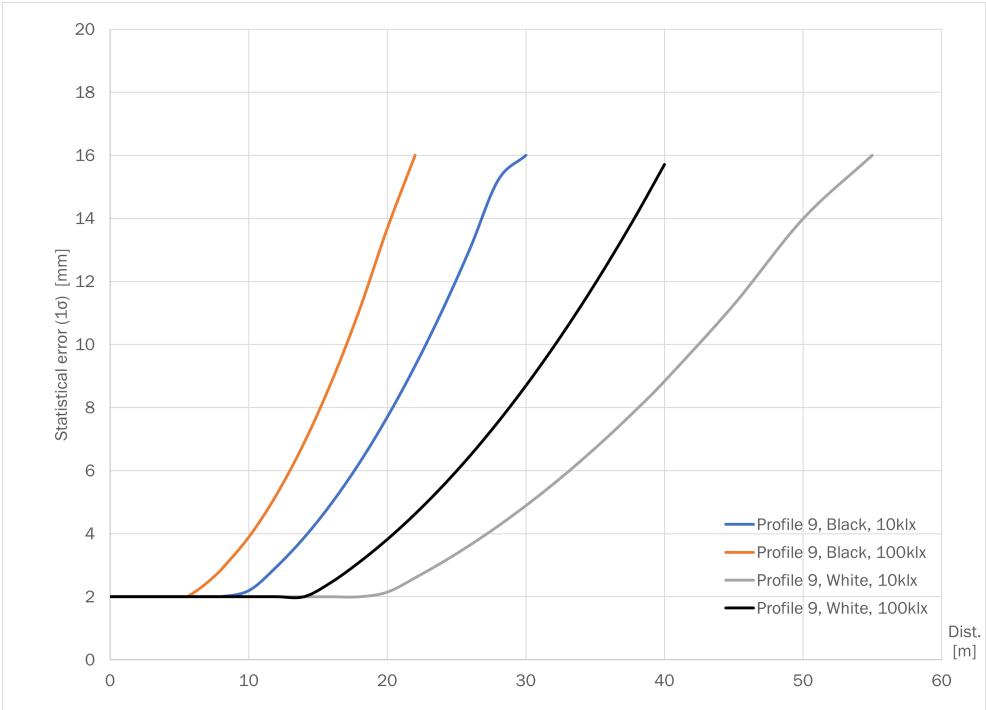


Figure 33: Profile 9 (15 Hz; 0.05°)

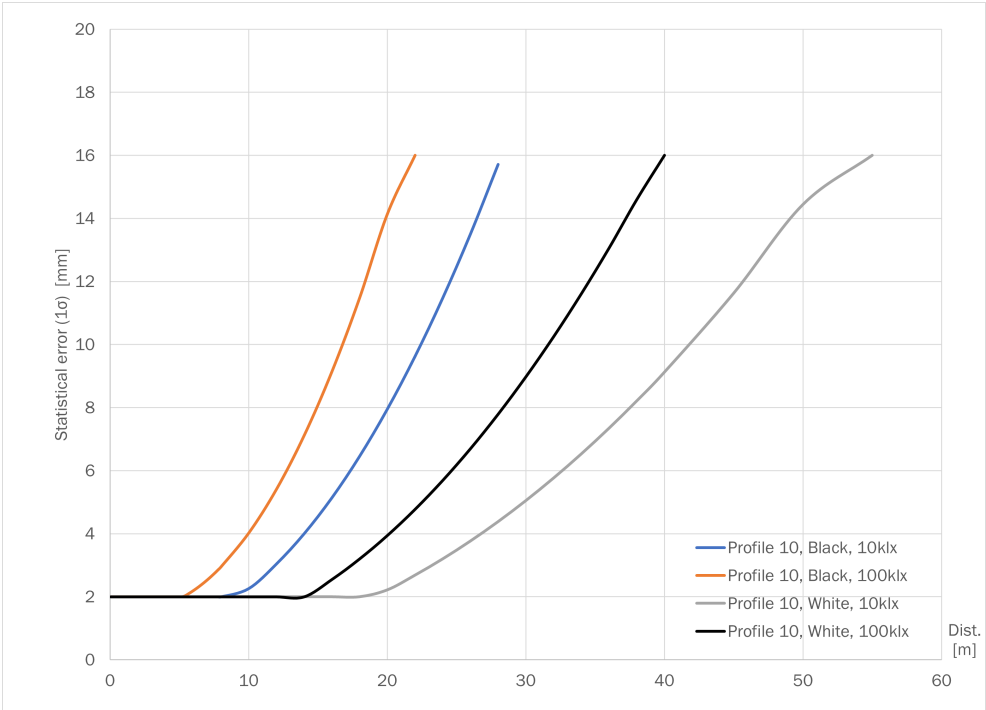


Figure 34: Profile 10 (40 Hz; 0.125°)

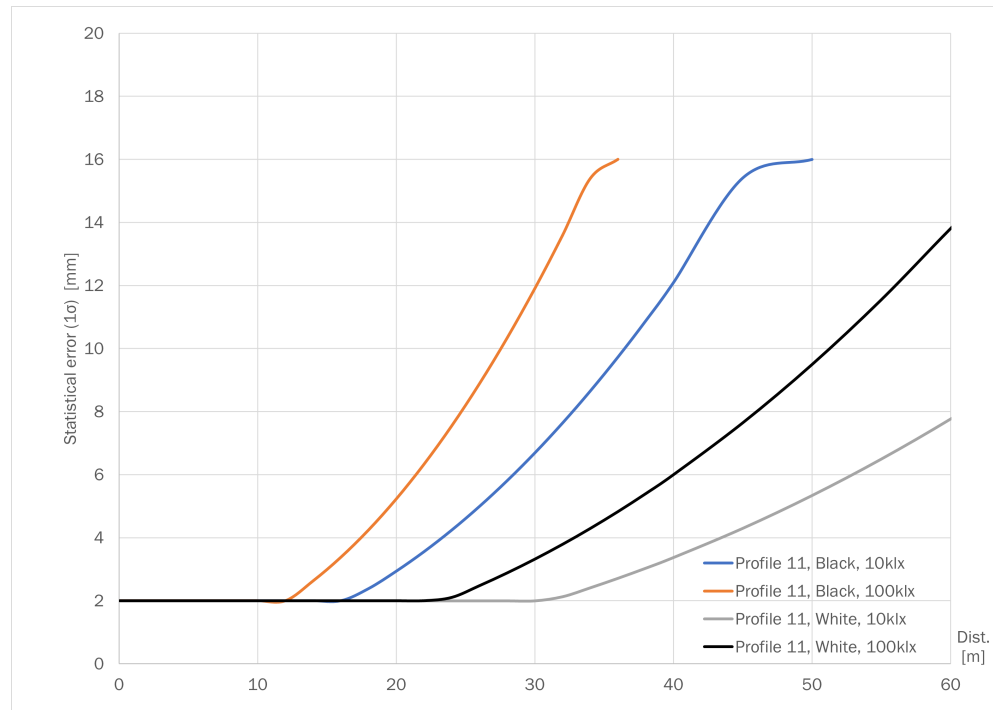


Figure 35: Profile 11 (15 Hz; 1°)

3.5.14 Extended scanning range

The picoScan150 Pro Performance variant offers the option of switching to high reflectivity targets such as reflectors for scanning ranges of up to 120 m. This switching changes the measuring core parameters so they are optimized for large scanning ranges.

Extended scanning range should only be activated if measuring objects will actually be measured at a distance of more than 60 m. Activating it will slightly worsen the noise characteristics at close range and increase the susceptibility to mutual interference when using multiple devices. Both effects are only at a low level, but increased compared to normal operation.

For the Compact and LMDscandata data formats, the distance resolution is changed from 1 mm to 2 mm when extended scanning range is active since these data formats transmit the measurement data as a 16-bit value. When using MSGPACK, the distance resolution remains at 1 mm even with the extended scanning range activated since MSGPACK works with 32 bits.

Further topics

- [Activate extended range mode \[sWN EnableLongRangeMode\]](#)

3.5.15 Time synchronization

You can synchronize the time and date of the devices in the network. This is important, among other things, for ensuring that diagnostics and reports have synchronized and correct time stamps.

The time base can be set using various methods:

- Explicitly setting the time with date, time
- Transferring the time from the computer used
- Via an external NTP server (Network Time Protocol) that is accessible in the network. The technology uses UDP on port 123. This setting is recommended to ensure a correct time stamp in the detection history or error entries.
- Via an external PTP (Precision Time Protocol) master that is accessible in the network. The technology uses UDP via port 319/320. This setting is recommended for precise synchronization requirements (e.g. between several devices).

3.5.16 Contamination indication

The device has an optics cover to protect it. This optics cover can get dirty. Contamination reduces the energy emitted and received by the laser beam. As a result, scanned objects appear to have a lower remission factor than they actually have and, from a certain degree of contamination, it will no longer be possible to perform measurements.

The contamination is constantly measured by a separate system during operation. A contamination warning is output first for the different degrees of contamination. If the optics cover is not cleaned and contamination increases, then a contamination error is output.

The device supports a contamination indication over the full 276°. 7 sectors are monitored independently. Each sector can be individually enabled for the contamination indication.

You can select different settings, depending on the application in which the device is used.

Warning/error output

- **All sectors contaminated:** If all active sectors have the same or a higher value, the device status “Contamination warning” or “Contamination error” is displayed.
- **One sector contaminated:** At least one sector must have a level of “Warning” or “Error” for a contamination warning or contamination error to be displayed.
- **No output:** Warning and error device status display is deactivated. The contamination measurement for the individual sectors continues to be performed but has no effect on the device status.

Sensitivity: Low, medium, high: Threshold for triggering contamination warnings and errors. The parameter makes it possible to tailor the display to the specific requirements of the application.

Sector preset: If an accessory is used that limits the angular range used, this can be specified in the contamination indication.

Response time: This can be used to define how quickly the contamination should result in an error or warning.

Default setting: In the default setting, the contamination display is deactivated. The monitoring of individual sectors is active and can be seen for each sector in the user interface and the command interface to assist with integrating the device into the application.

Strategy and evaluation are deactivated so no device warnings or errors are triggered. All associated parameters can be permanently changed.

Important information**NOTE**

- The cleaner the application environment is, the lower you can set the contamination indication sensitivity. If a high precision of the measured values is required, the contamination indication must be set to the most sensitive level.
- Sectors that are not relevant should be deactivated to ensure a higher availability.
- High illumination reduces the contamination level.
- Moisture in the form of drops creates a high contamination value.
- Contamination warnings and contamination errors are indicated on the display elements of the device
- If there are objects close to the product, a contamination indication in this segment is not possible. In this case, the relevant segment is evaluated as a blind zone.

Further topics

- [Display and control elements](#)

3.5.17 Field evaluation

The device uses the integrated field evaluation to evaluate the fields within its scan area. You can use the field evaluation, for example, to implement systems for collision protection, object protection or access monitoring.

Up to 48 fields can be defined, of which 20 can be active at the same time.

Limitations of simultaneously active fields:

- More active fields results in a greater computing effort and therefore a higher system load.
- Only the measuring points located within an active field result in an additional system load.
- A too high system load can lead to performance problems or system errors.
- The system load can be reduced by deactivating other functions, for example GUI, filter, and measurement data output, which may allow more fields to be active at the same time.
- We recommend testing the exact application requirements.

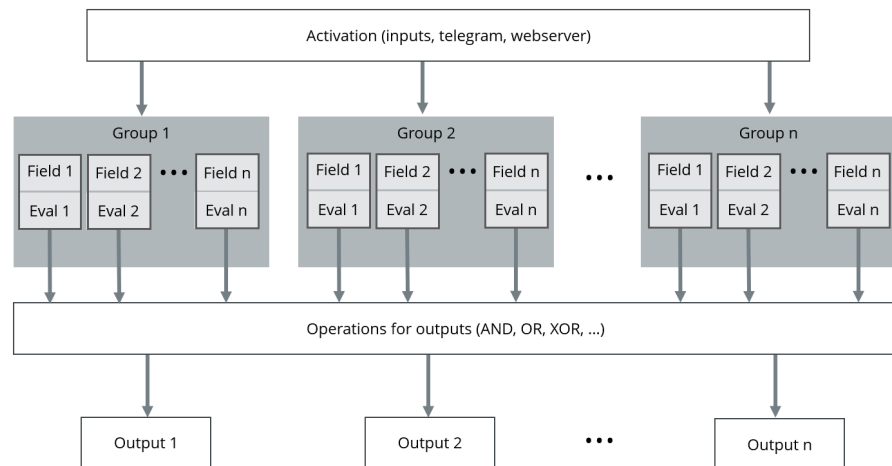


Figure 36: Principle of field evaluation

Multiple groups can be created, depending on application requirements. A group can be switched to active either by digital inputs of the device, telegrams, or permanently. A group can comprise several fields, each of which can be configured for the particular situation via its own evaluation.

The results of the evaluation can be linked to a digital output. It is also possible to link the respective evaluation to other signals using logical operands (AND, OR, XOR, ...).

Further topics

- For more information, see the online help in SOPASair.

3.5.17.1 Delay time of field evaluation

The latency of the field evaluation depends on several parameters. These include the location and time at which an object enters the sensor detection range in relation to the laser position. An evaluation is performed after each scan cycle.

The latency with which the field status is output is also influenced by:

- Number of activated fields
- Digital outputs
- Filters
- Measurement data streaming function

Typical latency of field evaluation:

Typ. latency of field evaluation = $1.5 \cdot 1/\text{scan rate} + 20 \text{ ms}$

Maximum latency of the field evaluation (3 σ):

Max. latency of field evaluation (3 σ) = $1.8 \cdot 1/\text{scan rate} + 50 \text{ ms}$

The 3-sigma value also takes into account rarely occurring extreme values. These can be caused by special conditions, such as late detection of an object due to an unfavorable laser position while at the same time using the maximum number of digital inputs and outputs, and the use of the measurement data streaming function in combination with several filter options.

3.5.18 Perpendicular distance measurement

The perpendicular distance measurement allows the detection fields and reference lines to be freely defined. When an object is located in the detection field, the minimum and maximum perpendicular distance between the object and the corresponding reference line is output.

4 **Transport and storage**

4.1 **Unpacking**

Procedure

- Check the components for completeness and integrity for all parts.
- In the event of complaints, contact the responsible SICK subsidiary.

5 Mounting

5.1 Installation site

The following information and recommendations are to be observed when positioning and installing the product:

General

- It can be mounted in any position.
- The mounting site is suitable for the weight of the device.
- The base is considered the reference surface for the scan field plane.
- When mounting the device to the rear, it is recommended to use the SICK mounting bracket (part number 2134874).
- During mounting, make sure there is no reflective surface behind the reference target.
- Make sure that the status indicator is clearly visible.
- Do not affix any labels or stickers to the optics cover.
- To avoid inaccurate measurements when installing multiple devices: Make sure that the laser spot of one device is not in the visible range of another device.

Vibrations

- Mount the device in a shock and vibration insulated manner.
- In applications with severe vibrations, shocks, jolts or abrupt changes in directions (e.g., when mounted to a manned forklift truck), mounting with vibration dampers is to be carried out. Mount the device in a freely suspended manner.
- In application areas with strong vibrations, use all five mounting points.

Temperature and humidity

- Protect the sensor from direct and indirect sunlight.
- To prevent condensation, avoid exposing the device to rapid changes in temperature.
- Protect the device from moisture, contamination, and damage.
- The ventilation element must not be sealed off during installation.
- The device must be mounted in such a way that no water can pool on the ventilation element. When using a mounting bracket, it is recommended to allow a narrow gap between the ventilation element and the mounting bracket. This is already appropriately provided for when using the SICK accessories.

Further topics

- [Ventilation element](#)
- [Dimensional drawing](#)
- [Mounting multiple products](#)

5.2 Ventilation element

Overview

The ventilation element ensures an improved pressure equalization and allows the exchange of air and heat between the housing and surroundings.

Prerequisites

- Do not affix any labels or stickers to the ventilation element.
- Do not paint over the ventilation element.
- Products that have been subjected to a long period of moisture or very rapid temperature changes need to first equilibrate after being switched on. In some circumstances, therefore, a period of time should be allowed before measurement readiness of the product because any moisture in the housing must first be taken up by the air in the housing, which is heated up through the operation of the product, so that it can then escape via the ventilation element. Depending on the nature of the precipitated moisture, this time period might be several minutes or even up to hours.

Ventilation element

The breathable membrane allows ambient air to either penetrate into the product, or escape again depending on the prevailing ambient conditions. In particular for applications with frequently changing environmental conditions (e.g., large temperature fluctuations or rapid temperature changes) or with standing water, the ventilation element ensures a reliable pressure equalization and thereby relieves the seals and adhesive joints of the housing. This can improve the expected service life of the product in the application.

Further topics

- [Product overview](#)
- [Installation site](#)

5.3 Mounting multiple products**Overview**

The product has been designed to minimize the probability of mutual interference, including between different LiDAR sensors. For applications where it is not possible to separate the scan planes, the product exhibits an extremely reliable measurement behavior on account of the HDDM+ process.

To rule out even the slightest effects on the measurement accuracy, the products should be arranged in such a way that as few laser beams as possible are received from other products.

Important notes**NOTICE RISK OF INTERFERENCE FROM OTHER DEVICES!**

Radiation sources with a wavelength of 905 nm can cause interference if they affect the device directly.

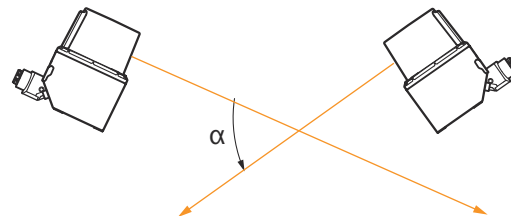
Recommended minimum distances

Figure 37: Angle $\geq 6^\circ$

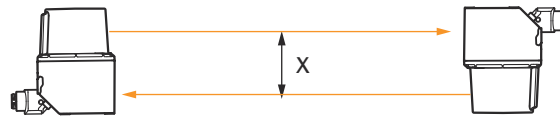


Figure 38: Distance ≥ 200 mm

5.4 Mounting the system plug on the product

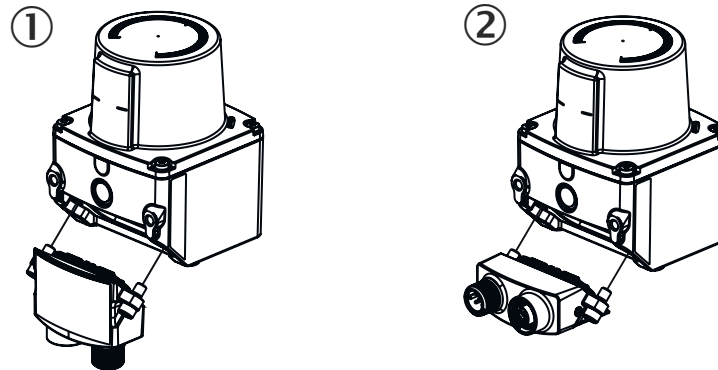
Overview

This mounting step is only required for the basic device product variant.

Prerequisites

- The system plug, the seal on the product, and the entire connection area are free of contamination and moisture and show no signs of damage.

Fitting the system plug



- ① System plug, installation at bottom
- ② System plug, installation at rear

1. Ensure the voltage supply is not connected.
2. Press the system plug flush against the product in the desired installation position.
3. Tighten the screws (tightening torque: max. 2 Nm).

Further topics

- [Mounting the product](#)

5.5 Mounting the product

Important notes



NOTICE

Risk of damage to the device

the device will be damaged if the tightening torque of the mounting screws is too high or if the maximum screw-in depth of the blind hole threads is exceeded.

- Observe maximum tightening torque.
- Use suitable mounting screws for the blind hole threads of the device. Observe the maximum screw-in depth.

Prerequisites

- The connector is mounted on the product.

Procedure

1. Mount the product in a suitably prepared bracket using the fixing holes provided. Mounting brackets are available as accessories.
2. Make the electrical connection. Attach and tighten a voltage-free cable.
3. Align the vertical center line of the viewing range of the product with the center of the area to be monitored. The marking on the upper side of the optics cover serves as a bearing alignment aid.
4. Switch on the supply voltage.
- ✓ The status LED 2 lights up green after successful initialization. The product is ready for operation.
5. Align the vertical center line of the viewing range of the product with the center of the area to be monitored. The marking (0° axis) on the upper side of the optics cover serves as a alignment aid.

Further topics

- [Mounting the system plug on the product](#)
- [Dimensional drawing](#)
- [Connecting the product electrically](#)

6 Electrical installation

6.1 Prerequisites for safe operation of the device



WARNING

Risk of injury and damage caused by electrical current!

As a result of equipotential bonding currents between the device and other grounded devices in the system, faulty grounding of the device can give rise to the following dangers and faults:

- Dangerous voltages are applied to the metal housings.
- Devices will behave incorrectly or be destroyed.
- Cable shielding will be damaged by overheating and cause cable fires.

Remedial measures

- Only skilled electricians should be permitted to carry out work on the electrical system.
- If the cable insulation is damaged, disconnect the voltage supply immediately and have the damage repaired.
- Ensure that the ground potential is the same at all grounding points.
- Where local conditions do not meet the requirements for a safe earthing method, take appropriate measures. For example, ensure low-impedance and current-carrying equipotential bonding.

The device is connected to the peripheral devices (any local trigger sensor(s), system controller) via shielded cables. The cable shield – for the data cable, for example – rests against the metal housing of the device.

The device can be grounded through the cable shield or through a blind tapped hole in the housing, for example.

If the peripheral devices have metal housings and the cable shields are also in contact with their housings, it is assumed that all devices involved in the installation have the **same ground potential**.

This is achieved by complying with the following conditions:

- Mounting the devices on conductive metal surfaces
- Correctly grounding the devices and metal surfaces in the system
- If necessary: low-impedance and current-carrying equipotential bonding between areas with different ground potentials

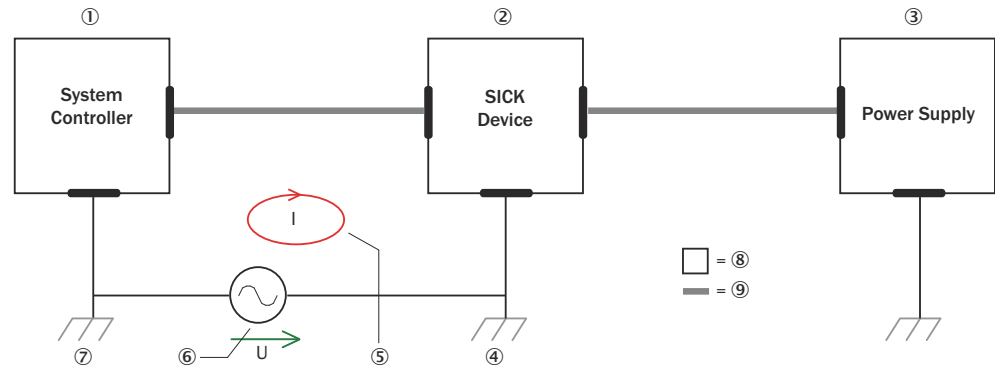


Figure 39: Example: Occurrence of equipotential bonding currents in the system configuration

- ① System controller
- ② Device
- ③ Voltage supply
- ④ Grounding point 2
- ⑤ Closed current loop with equalizing currents via cable shield
- ⑥ Ground potential difference
- ⑦ Grounding point 1
- ⑧ Metal housing
- ⑨ Shielded electrical cable

If these conditions are not fulfilled, equipotential bonding currents can flow along the cable shielding between the devices due to differing ground potentials and cause the hazards specified. This is, for example, possible in cases where there are devices within a widely distributed system covering several buildings.

Remedial measures

The most common solution to prevent equipotential bonding currents on cable shields is to ensure low-impedance and current-carrying equipotential bonding. If this equipotential bonding is not possible, the following solution approaches serve as a suggestion.



NOTICE

We expressly advise against opening up the cable shields. This would mean that the EMC limit values can no longer be complied with and that the safe operation of the device data interfaces can no longer be guaranteed.

Measures for widely distributed system installations

On widely distributed system installations with correspondingly large potential differences, the setting up of local islands and connecting them using commercially available **electro-optical signal isolators** is recommended. This measure achieves a high degree of resistance to electromagnetic interference.

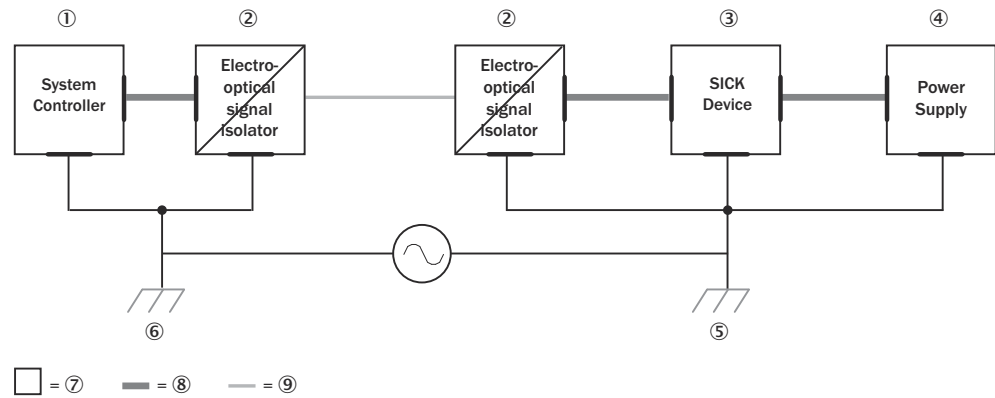


Figure 40: Example: Prevention of equipotential bonding currents in the system configuration by the use of electro-optical signal isolators

- ① System controller
- ② Electro-optical signal isolator
- ③ Device
- ④ Voltage supply
- ⑤ Grounding point 2
- ⑥ Grounding point 1
- ⑦ Metal housing
- ⑧ Shielded electrical cable
- ⑨ Fibers

The use of electro-optical signal isolators between the islands isolates the ground loop. Within the islands, a stable equipotential bonding prevents equalizing currents on the cable shields.

Measures for small system installations

For smaller installations with only slight potential differences, insulated mounting of the device and peripheral devices may be an adequate solution.

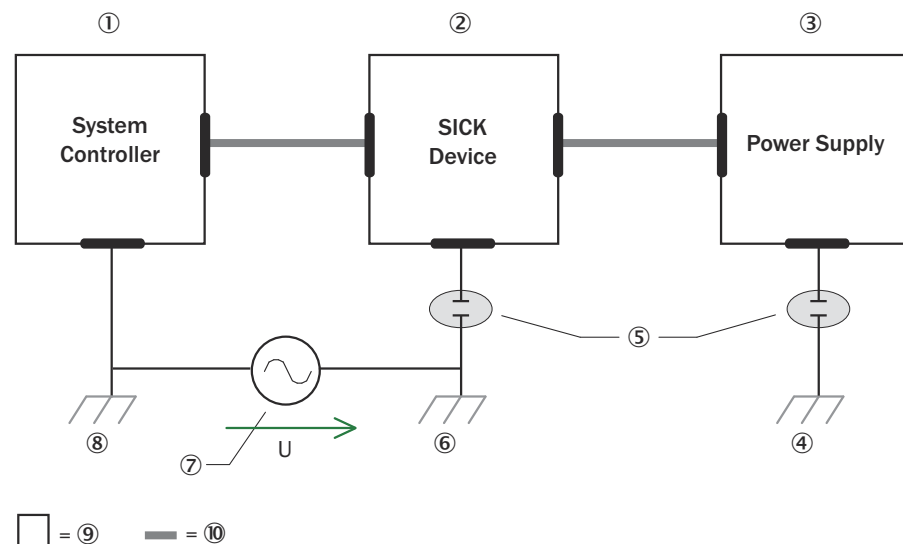


Figure 41: Example: Prevention of equipotential bonding currents in the system configuration by the insulated mounting of the device

- ① System controller
- ② Device
- ③ Voltage supply
- ④ Grounding point 3
- ⑤ Insulated mounting
- ⑥ Grounding point 2
- ⑦ Ground potential difference
- ⑧ Grounding point 1
- ⑨ Metal housing
- ⑩ Shielded electrical cable

Even in the event of large differences in the ground potential, ground loops are effectively prevented. As a result, equalizing currents can no longer flow via the cable shields and metal housing.



NOTICE

The voltage supply for the device and the connected peripheral devices must also guarantee the required level of insulation.

Under certain circumstances, a tangible potential can develop between the insulated metal housings and the local ground potential.

6.2 Calculation instructions for voltage drop and length of cable

Overview

The following formulas can be used to estimate the required cable lengths or supply voltages. Other conditions of the system must be considered in detail.

Formula for the voltage drop to be considered

$$\Delta V = \frac{I \cdot 2 \cdot L}{A} \cdot \rho \cdot (1 + \alpha \cdot (T - T_0))$$

Formula for permissible length of cable

$$L = \frac{\Delta V \cdot A}{2 \cdot I \cdot \rho \cdot (1 + \alpha \cdot (T - T_0))}$$

Example**Prerequisites:**

- Steady state of the voltage supply
- Only applies for copper cable material

Table 5: Values used in both example calculations

Cable properties	
$A = 0.34 \cdot 10^{-6} \text{ m}^2$	Cross-section of the cable surface [m ²]
$\rho = 1.72 \cdot 10^{-8} \Omega\text{m}$	Specific resistance of copper [Ωm]
$\alpha = 3.9 \cdot 10^{-3} \text{ K}^{-1}$	Temperature coefficient of copper [1/K]
Ambient conditions	
$T_0 = 20 \text{ }^\circ\text{C}$	Reference temperature [$^\circ\text{C}$]
$T = 80 \text{ }^\circ\text{C}$	Cable temperature [$^\circ\text{C}$]
Cable load	
$I = P/U = 1.46 \text{ A}$	Load current I [A]
$P = 35 \text{ W}$	Supply voltage U [V]
$U = 24 \text{ V}$	Maximum expected power consumption P

Table 6: Example: voltage drop to be considered for cable part no. 2096241

$L = 10 \text{ m}$	Cable length [m]
$\Delta V = \frac{I \cdot 2 \cdot L}{A} \cdot \rho \cdot (1 + \alpha \cdot (T - T_0)) = 1.82 \text{ V}$	Voltage drop ΔV [V]

Table 7: Calculation of the cable length for allowed voltage drop of 1.82 V

$\Delta V = 1.82 \text{ V}$	Voltage drop on the cable [V]
$L = \frac{\Delta V \cdot A}{2 \cdot I \cdot \rho \cdot (1 + \alpha \cdot (T - T_0))} = 10 \text{ m}$	Permissible length of cable [m]

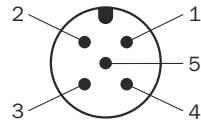
6.3 Connections and pin assignment

Overview

The connections depend on the mounted system plug.

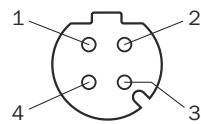
Prerequisites

- Connect the connecting cables in a de-energized state. Do not switch on the supply voltage until installation is complete and all connecting cables are connected to the product and controller.
- The voltage supply and digital inputs/outputs are not galvanically isolated.
- Always assign a fixed potential to digital inputs.
- Wire cross-sections in the supply cable from the user's power system must be implemented in accordance with the applicable standards.
- Use shielded cables.
- In the case of open end cables, make sure that bare wire ends do not touch. Wires must be properly insulated from each other.
- Do not lay cables over long distances in parallel with voltage supply cables and motor cables in cable ducts.

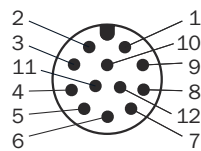
System plug PWR & 3 I/Os (Part No. 2116047)**Figure 42:** Male connector, M12, 5-pin, A-coded**Table 8:** Pin assignment for PWR & 3 I/Os connection

Contact	Signs	Description	Wire color, part number 2095733 ¹⁾
1	Vs	Supply voltage: +9 ... +30 V DC	Brown
2	IN2 / OUT2	Digital input 2 / digital output 2	White
3	GND	Supply voltage: 0 V	Blue
4	IN1 / OUT1	Digital input 1 / digital output 1	Black
5	IN3 / OUT3	Digital input 3 / digital output 3	Gray

¹⁾ Data only valid when using the specified connecting cable with flying leads, which is available as an accessory

**Figure 43:** M12 socket, 4-pin, D-coded**Table 9:** Pin assignment for Ethernet connection

Contact	Marking	Description
1	TX+	Sender+
2	RX+	Receiver+
3	TX-	Sender-
4	RX-	Receiver-

System plug PWR & 6 I/Os (Part No. 2130754)**Figure 44:** M12 plug, 12-pin, A-coded**Table 10:** Pin assignment for PWR & 6 I/Os connection

Contact	Marking	Description	Wire color part no. 2119019 ¹⁾
1	IN1/OUT1	Digital input 1 / digital output 1	Brown
2	GND	Supply voltage: 0 V	Blue
3	IN2/OUT2	Digital input 2 / digital output 2	White
4	-	-	Green
5	-	-	Pink
6	IN3/OUT3	Digital input 3 / digital output 3	Yellow
7	IN4/OUT4	Digital input 4 / digital output 4	Black
8	IN6/OUT6	Digital input 6 / digital output 6	Gray
9	Vs	Supply voltage: +9 ... +30 V DC	Red
10	IN5/OUT5	Digital input 5 / digital output 5	Violet

Contact	Marking	Description	Wire color part no. 2119019 ¹⁾
11	-	-	Gray/pink
12	-	-	Red/blue

¹⁾ Data only valid when using the specified connecting cable with flying leads, which is available as an accessory

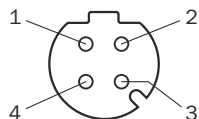


Figure 45: M12 socket, 4-pin, D-coded

Table 11: Pin assignment for Ethernet connection

Contact	Marking	Description
1	TX+	Sender+
2	RX+	Receiver+
3	TX-	Sender-
4	RX-	Receiver-

System plug PWR & 6 I/Os (Part No. 2130871)

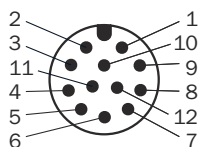


Figure 46: M12 plug, 12-pin, A-coded

Table 12: Pin assignment for PWR & 6 I/Os connection

Contact	Marking	Description
1	GND	Supply voltage: 0 V
2	Vs	Supply voltage: +9 ... +30 V DC
3	IN1/OUT1	Digital input 1 / digital output 1
4	IN2/OUT2	Digital input 2 / digital output 2
5	IN3/OUT3	Digital input 3 / digital output 3
6	IN4/OUT4	Digital input 4 / digital output 4
7	IN5/OUT5	Digital input 5 / digital output 5
8	IN6/OUT6	Digital input 6 / digital output 6
9	-	-
10	-	-
11	-	-
12	-	-

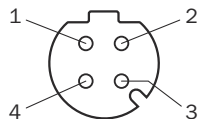


Figure 47: M12 socket, 4-pin, D-coded

Table 13: Pin assignment for Ethernet connection

Contact	Marking	Description
1	TX+	Sender+

Contact	Marking	Description
2	RX+	Receiver+
3	TX-	Sender-
4	RX-	Receiver-

System plug with integrated cables (part no. 2146513)

Table 14: Pin assignment for PWR connection & 3 I/Os with flying leads (0.8 m)

Wire color	Marking	Description
Blue	GND	Supply voltage: 0 V
Brown	Vs	Supply voltage: +9 ... +30 V DC
Black	IN1 / OUT1	Digital input 1 / digital output 1
White	IN2 / OUT2	Digital input 2 / digital output 2
Gray	IN3 / OUT3	Digital input 3 / digital output 3

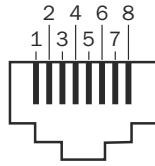


Figure 48: RJ45 plug

Table 15: Pin assignment for Ethernet connection with RJ45 plug (0.6 m)¹⁾

Wire color	Marking	Description
1	TX+	Sender+
2	TX-	Sender-
3	RX+	Receiver+
4	-	-
5	-	-
6	RX-	Receiver-
7	-	-
8	-	-

Complementary information

Pre-assembled cables can be found on the product page.

The product page can be accessed via the **SICK Product ID: pid.sick.com/{P/N}/{S/N}**

{P/N} corresponds to the part number of the product, see type label.

{S/N} corresponds to the serial number of the product, see type label (if indicated).

Further topics

- Information on interfaces: [Data sheet](#)

6.4 Encoder input

The device has a digital encoder input (port 1-port 4) for recording incremental encoder signals. Configuration can be carried out either via the user interface or via telegram using the communication interface.

¹⁾ When extending the data cable, ensure that the cable shield is sufficiently well connected to avoid EMC interference.

The encoder input supports single-channel encoders (A) and dual-channel encoders (A/B) for determining the direction. The recorded pulses are provided with a high-resolution time stamp (NTP / PTP) and output in the *Compact* data format. The time stamps are based on the internal system time of the device and enable the movement data of the device to be precisely assigned temporally to the measured data of the device.

Reference signals can also be fed in, for example to process an encoder Z signal or limit switch signals.

6.5 Voltage supply

Prerequisites

- All circuits connected to the product must be configured as SELV or PELV circuits. SELV = safety extra-low voltage, PELV = protective extra-low voltage.
- Connect the connecting cables in a de-energized state. Do not switch on the supply voltage until installation is complete and all connecting cables are connected to the product and controller.
- Protect the product with an external slow-blow fuse rated at min. 1 A/max. 3 A at the start of the supply cable.
- To ensure protection against short-circuits/overload in the customer's supply cables, choose and implement wire cross-sections in accordance with the applicable standards.
- When using the Power over Ethernet system plug (part number: 2149035), the device must be supplied via a Power Sourcing Equipment (PSE) that meets the Power over Ethernet standard IEEE 802.3af or higher.

6.6 Output load and power consumption

The maximum power consumption depends on how the product is employed. If no outputs are used, no additional power consumption by them needs to be considered. The typical power consumption is then as follows: $P_{typ} = 4.5 \text{ W}$ (at 24 V supply voltage, 25 °C ambient temperature, continuous scan data output, no output load)

If outputs are used, the power consumption increases according to the external sinks. This can be up to 6 configurable I/Os, depending on the system plug used.

Number of outputs used	Output current per output	P_{output}
2	200 mA	$P_{output} = 2 \times 30 \text{ V} \times 200 \text{ mA} = 12 \text{ W}$
3	133 mA	$P_{output} = 3 \times 30 \text{ V} \times 133 \text{ mA} = 12 \text{ W}$
4	100 mA	$P_{output} = 4 \times 30 \text{ V} \times 100 \text{ mA} = 12 \text{ W}$
6	67 mA	$P_{output} = 6 \times 30 \text{ V} \times 67 \text{ mA} = 12 \text{ W}$

The maximum power consumption is the sum of the typical power consumption and the power consumption of loaded outputs: $P_{max} = P_{typ} + P_{output} + P_{temp_res}$

For the example with 6 loaded outputs, this gives:

- $P_{typ} = 4.5 \text{ W}$
- $P_{output} = 12 \text{ W}$
- $P_{temp_res} = 0.5 \text{ W}$
- $P_{max} = 4.5 \text{ W} + 12 \text{ W} + 0.5 \text{ W} = 17 \text{ W}$

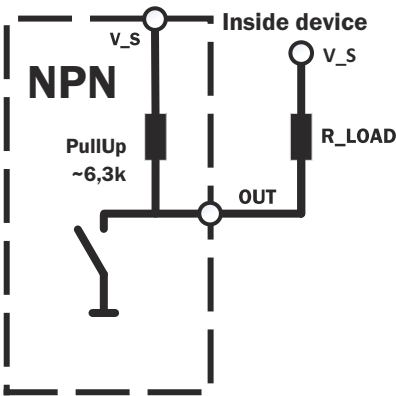


Figure 49: Wiring of the digital output for NPN configuration

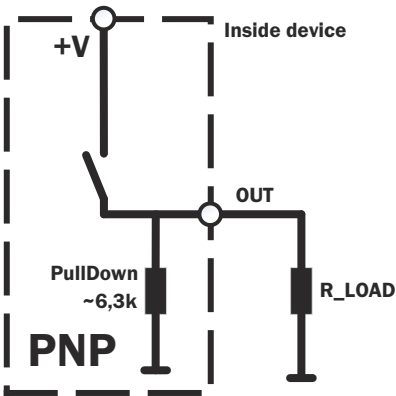


Figure 50: Wiring of the digital output for PNP configuration

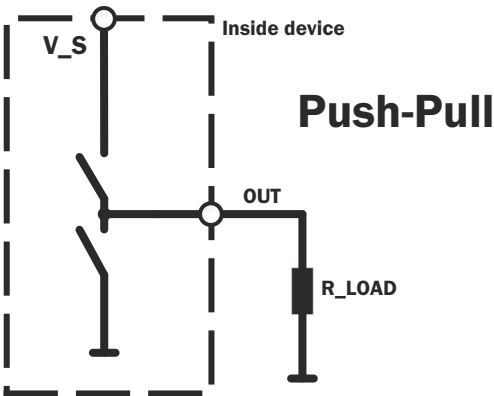


Figure 51: Wiring of the digital output for push-pull configuration

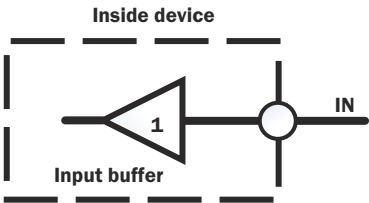
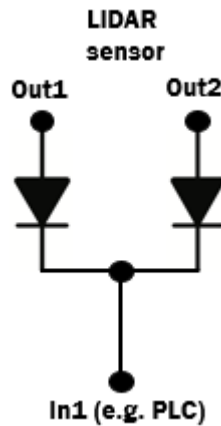


Figure 52: Wiring of the digital input

Important information

Parallel connection of the digital outputs of several identical devices – for example in PNP mode – to a common input (e.g. a PLC) is not recommended. For safe and reliable signal processing, an external logic or decoupling circuit, for example with diodes, should be used.



6.7 Connecting the product electrically

Prerequisites

- Ensure that the power supply unit can provide the necessary voltage and current for operating the product. Particular attention must be given to the voltage drop across the supply cable, and for digital outputs the additional voltage drop in the opposite direction and the required start-up power, without which the product cannot start reliably.

Procedure

1. Ensure the voltage supply is not connected.
2. Connect the product according to the connection example.
3. Any unused electrical M12 connections of the system plug are closed with a protective element of the corresponding type (as in the delivery state).

Further topics

- [Calculation instructions for voltage drop and length of cable](#)
- [Data sheet](#)

7 Commissioning

7.1 Virtual Sensor Lab

The Virtual Sensor Lab is a cloud-based service for displaying the user interface of selected sensors without a physical device.

It provides a visual representation of the functions and settings of a device and is used, in particular, to prepare for parameterization as well as for training and demonstration purposes.

The Virtual Sensor Lab does not replace commissioning or a function check-out on the real device.

Access: <https://devices.cloud.sick.com>

7.2 Starting SOPASair

Overview

SOPASair is a web server user interface that is used for operation, parameterization and for servicing purposes (e.g., diagnostics, data logger).

Prerequisites

- Computer with Ethernet port
- Product is connected to the computer via Ethernet.
- Voltage supply is connected.
- Computer and product are located on the same network.

Procedure

1. Open web browser (recommendation: Google Chrome).
2. Enter the IP address of the product in the address line.

Default IP address:

 - 192.168.0.1 (for parameterization)
 - 192.168.0.1/#/?fullscreen= (Full-screen display, e.g. on an external display, no parameterization possible)
- ✓ The user interface is displayed.
3. To make changes, log in to the product.

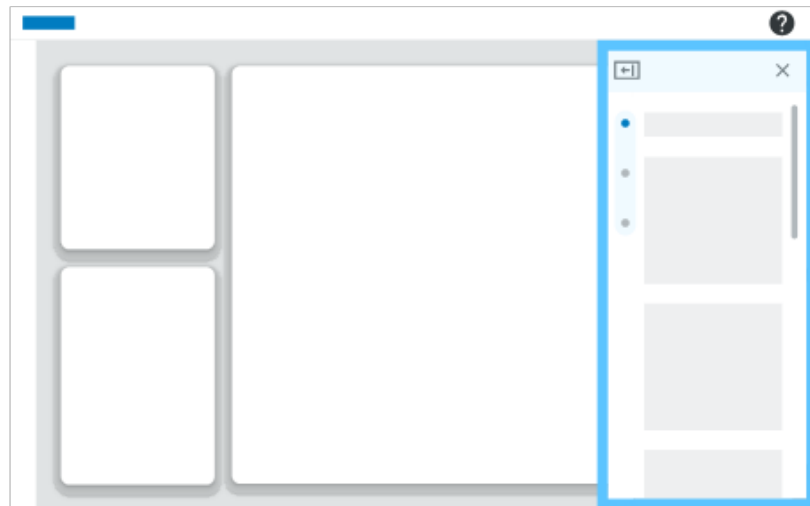
Complementary information

The most important commissioning steps can be found in the “Initial steps” tour of the SOPASair help. You can open the help at any time using the question mark in the toolbar.

7.2.1 SOPASair help

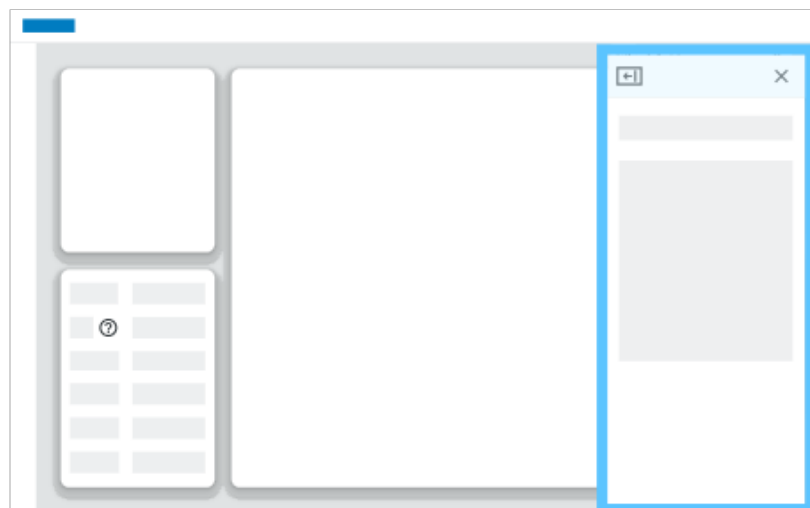
The software help is integrated into the SOPASair user interface.

The question mark in the toolbar opens the global help with step-by-step tours and general information.

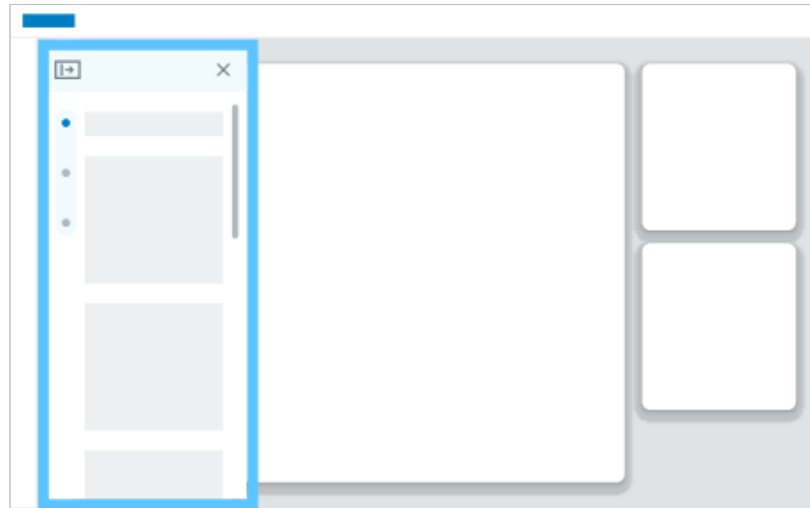


The **Initial steps** tour will help you get started with the user interface.

The question mark next to individual functions opens specific information about the function.



The help window can be moved: 



7.2.2 Logging on and off the device

Overview

The device has different user levels. Only the **Service** user level is activated in the delivered state.

The user levels have different authorizations for configuring the device.

The current user level is displayed in the **Log in** panel.

Procedure

Activate the user level during initial commissioning of the device

1. Click on the  button.
- ✓ The **Log in to device** input screen opens. The **Service** user level is selected.
2. Enter the password servicelevel and click on **Log in**.
3. Activate the desired user level.

Logging in to the device

1. User levels have been activated.
2. Click on the  button.
- ✓ The **Log in to device** input screen opens.
3. Select user level(**User level**), enter a password(**Password**) then click **Login in**.

User levels	Password
Maintenance (deactivated in the delivery state)	main
Authorized Client (deactivated in the delivery state)	client
Service	servicelevel

Logging out of the device

1. Click on the  button.
2. Click **Log out**.

Important information

- Change the passwords during initial commissioning to protect your device.
- A higher user level can change the password of a lower user level.
- If you leave the computer unattended, you must log off to prevent unwanted access to the device.

7.2.3 Managing passwords**7.2.3.1 Change password****Procedure**

1. Open SOPASair.
 2. Select .
 3. Log in with the last assigned password.
 4. Select **Change password**.
 5. Assign a new password.
- ✓ The new password is valid immediately.

Important information

The password can only be changed via the web browser (REST interface). It is not possible to change it via SOPAS ET (CoLa A/B interface).

7.2.3.2 Resetting the password**Procedure**

1. Establish a connection to the product in the web browser.
 2. Select .
 3. Select **Password forgotten?**.
 4. Send **Device key**, **Serial number** and **Part number** in an e-mail to the responsible SICK sales company or the responsible SICK service partner, see www.sick.com/worldwide.
 5. Click **Next** to confirm.
- ✓ The window for entering a code appears.
You can get the code from the responsible SICK sales company or the responsible SICK service partner. The code is only valid once for the reset process. You can close the window by clicking on the **x** without interrupting the reset process. If you select **Cancel** or enter an incorrect code several times, the current reset process is terminated. The requested code is no longer valid. The process must be restarted.
6. Optional: close the window by clicking on **x**. At a later time, open the window for entering the code via  and **Password forgotten?**.
 7. Enter the code.
 8. Click **Reset** to confirm.
- ✓ The password is reset to the default password **servicelevel**. Parameters are not changed.

7.2.4 Data backup

Manual data backup using project file

The parameter set can be manually saved as a project file (*.json) on the computer

(In SOPASair:  > **Download/Import Configuration File**). This is the generally recommended procedure. Using the project file, the parameter set can be transferred to a replacement product via download.

Parameter cloning

As the existing system plug continues to be in use, the product make it possible for the last parameter values used in the system plug to be automatically passed on to a product of the same type (cloning). Accordingly, the process overwrites all the existing parameter values in the product.

During cloning, the replacement product accepts the following configurable parameters:

- Addressing mode
- IP address
- Subnet mask
- Default gateway
- Device name

The following parameters are not included when cloning:

- Parameter settings such as filter, field geometries
- Device-specific parameters:
 - Serial number
 - MAC address
 - Licenses
 - Operating hours counter
 - Error memory

After being switched on, the product automatically checks the system plug memory. The subsequent product behavior depends on the content of the system plug memory. The goal is for the internal parameter set and the parameter set saved on the system plug to always be identical.

Content of the system plug memory	Device behavior
Empty	Once the parameter set is permanently saved on the product, the product also saves the internal parameter set on the system plug. The prerequisite is that the Save permanently function has been activated via SOPASair or SOPAS ET.
No parameter set possible to interpret	
Parameter set possible to interpret	After being switched on, the product automatically loads the compatible parameter set from the system plug into the temporary parameter memory of the product. The product then starts with its new valid parameter set. To permanently transfer the parameter set to the product, activate the Save permanently function via SOPASair or SOPAS ET.

Further topics

- [Saving the parameter set](#)

7.2.5 Saving the parameter set

Overview

The product is configured for the application using SOPASair. The parameter set can be permanently saved in SOPASair. To be able, for example in the event of a product failure, to restore the parameter set to a replacement product or to also save the parameter set in the cloning parameter memory of the system plug.

Procedure

1. Permanently saving the parameters in SOPASair:



- ✓ The product stores the parameter set internally in the permanent parameter memory and in the cloning parameter memory of the system plug.

Further topics

- [Data backup](#)

7.2.6 Install firmware updates

Procedure

1. Establish a connection to the product in the web browser.
- ✓ The firmware version is displayed in the **device information** tile on the start page.
2. Click on the firmware version.
- ✓ A dialog box opens.
3. Follow the further steps in the dialog box.

7.3 Starting SOPAS ET

Overview

The saved parameters can be manually saved, imported, and exported as a project file on the computer in the SOPAS ET configuration software.

Prerequisites

- Computer with the SOPAS ET software installed
The most up-to-date version of the SOPAS ET software can be downloaded from www.sick.com/SOPAS_ET. The respective system requirements for installing SOPAS ET are also specified there.

- Ethernet connection

- SDD file (device description file)

You can install the SDD file using the device catalog in SOPAS ET. Use the wizard in SOPAS ET to do this. The SDD file can be installed from the device or the SICK website. To install it from the SICK website, you need an Internet connection.

- Access data

User levels	Password
Maintenance (deactivated in the delivery state)	main
Authorized Client (deactivated in the delivery state)	client
Service	servicelevel

- Device is ready for use.

Procedure

1. Install the latest version of the SOPAS ET configuration software. In this case, select the “Complete” option as suggested by the installation wizard. Administrator rights may be required on the computer to perform the installation.
2. Start “SOPAS ET” after completing the installation.
- ✓ SOPAS ET automatically starts the search for connected devices. Connected devices are displayed in the **Device Search** window.
3. Select the desired device in the list of available devices.
The following IP addresses are configured by default on the device:
 - IP address: 192.168.0.1
 - Subnet mask: 255.255.255.0If necessary, install an updated device description file for the device.
4. Click on  **Add** to establish communication.
- ✓ SOPAS ET establishes communication with the device, loads its current device description (parameters), and displays the device in the **New Project** window.
5. Log into the device.

8 Maintenance

8.1 Maintenance

Table 16: Maintenance schedule

Maintenance work	Interval
Check device and connecting cables for damage at regular intervals.	Depends on ambient conditions and climate.
Check housing and optics cover for contamination and clean if necessary, .	Depends on ambient conditions and climate.
Check the screw connections and plug connectors.	Depends on the place of use, ambient conditions or operating requirements. Recommended: At least every 6 months.
Check the assembly accessories used.	Depends on the place of use, ambient conditions or operating requirements. Recommended: At least every 6 months.
Check that all unused connections are sealed with protective caps.	Depends on ambient conditions and climate. Recommended: At least every 6 months.

8.2 Cleaning the product

Important information



NOTICE

- Never use sharp objects for cleaning.
- Recommendation: Use anti-static cleaning agents.
- Recommendation: Use anti-static plastic cleaners and lens cloths from SICK.



NOTICE

If the optics cover is scratched or damaged, take the product out of operation and have it repaired by SICK.

Procedure

1. If possible, switch off the voltage supply to the product.
2. Use only a clean, damp, lint-free cloth and a mild anti-static plastic cleaner to clean the optics cover.
3. Remove any dust from the housing using a soft brush.

9 Troubleshooting

9.1 Troubleshooting

Faults, warnings and errors

Table 17: Troubleshooting Q&A

Question / status	Response / remedial actions
Both LEDs flash red.	Device error: Read out the error via the SOPASair or SOPAS ET PC software. Eliminate the cause of the error.
LEDs indicate an undefined status.	Check device status. If necessary, contact SICK Service.
All LEDs are off.	Check the voltage supply to the device. In SOPASair, check whether the LEDs were switched off.
All LEDs of the device light up red at startup and do not change to green.	Check the voltage supply to the device. The power supply unit may not be supplying the required current or the required voltage to start the device.
Measurement data show anomalies.	Optics cover contaminated: Clean the optics cover.
When accessing the device via a web browser, the SOPASair user interface is not loaded, the SOPASair loading screen is permanently displayed.	Try connecting again. If this does not work: Restart the device. Check the network settings. Standard IP address: 192.168.0.1 The IP address range 169.254.1.{1-15} is locked for device-internal communication. The device cannot be assigned any of these 15 IP addresses.
SOPASair is not started in the browser.	Check the IP address of device and network adapter (e.g., using device search in SOPAS ET) and adjust if necessary. Then try to establish the connection again.
Measurement data dropouts when using LMDscandata via TCP	Check whether there is sufficient bandwidth in the network and whether the receiving end can process the measurement data fast enough to receive further data.
Basic device after mounting the system plug: All LEDs are off, measurement data show anomalies, SOPASair is not started in the browser.	Remove the system plug and mount it again. Observe the mounting instructions.

Diagnostics using SOPASair

The error log contains current error messages.

In the event of faults, you can use the configuration software to create and download an encrypted diagnostic file. This contains important analysis data and can only be opened by SICK Service. In the event of an error, do not restart the device until the diagnostic file has been saved.

Complementary information

For faults that cannot be rectified based on the error description, please contact SICK Service. To help us to resolve the matter quickly, please note down the details on the type label.

9.2 Repair

Repairs on the device may only be performed by qualified and authorized personnel from SICK AG. Interference with or modifications to the device on the part of the customer will invalidate any warranty claims against SICK AG.

10 Decommissioning

10.1 Reset the device to factory settings

Procedure

1. In SOPASair, click  to open the menu.
2. **Device > Reset to factory settings** or **Reset and maintain communication interfaces**.



NOTE

- **Reset to factory settings:** All settings are deleted. Existing remote connections are disconnected.
- **Reset and maintain communication interfaces:** All settings, except those of the communication interfaces, are deleted. Existing remote connections are thus maintained.

10.2 Removing the product

Procedure

1. Reset device to factory settings.
2. Switch off the supply voltage.
3. Mark the position and alignment of the device on the bracket or surroundings.
4. Disconnect and remove the connecting cables of the device.
5. Remove the device from the bracket.

Complementary information

If you replace the product, you can transfer the parameter values to the replacement product by downloading them.

Further topics

- [Reset the device to factory settings](#)

10.3 Disposal of the product

Procedure

- Always dispose of unusable products in accordance with national waste disposal regulations.



Complementary information

SICK will be glad to help you dispose of these products on request.

11 Technical data



NOTE

The relevant online product page for your product, including technical data, dimensional drawing, and connection diagrams, can be downloaded, saved, and printed from the Internet.

The product page can be accessed via the **SICK Product ID: pid.sick.com/{P/N}/{S/N}**. **{P/N}** corresponds to the part number of the product, see type label.

{S/N} corresponds to the serial number of the product, see type label (if indicated).

Please note: This documentation may contain further technical data.

11.1 Data sheet

Features

Measurement principle	HDDM+ statistical measurement procedure
Application	Indoor and outdoor
Light source ¹⁾	Infrared (wavelength 940 nm, maximum output power < 8 W, maximum output power per area 0.3 W/mm ² , pulse duration 1.2 ns, average power < 14 mW)
Laser class	Laser class 1 (IEC 60825-1:2014, EN 60825-1:2014+A11:2021) Complies with 21 CFR 1040.10 and 1040.11 except for conformance with IEC 60825-1 Ed. 3 as described in Laser Notice No. 56, dated May 8, 2019.
Horizontal aperture angle	276°
Scanning frequency ²⁾	Core: 15 Hz, 25 Hz Prime: 15 Hz, 20 Hz, 40 Hz ³⁾ Pro: 15 Hz ... 50 Hz
Angular resolution ²⁾	Core: 0.33°; 0.25°; 1° Prime: 0.10°; 0.25°; 0.33°; 1° ³⁾ Pro: 0.05° ... 1°
Working range	Core: 0.05 m ... 25 m Prime: 0.05 m ... 60 m Pro: 0.05 m ... 120 m
Blind zone	0 m ... 0.05 m
Spot size	8 mm (at the optics cover) In 5 m: 32 mm In 10 m: 55 mm In 25 m: 126 mm Prime, Pro: In 50 m: 244 mm
Laser divergence	typ. 0.27 [°] / 4.8 [mrad] max. 0.3 [°] / 5.3 [mrad]
Number of echoes evaluated	3 (output options: first echo, last echo or all echoes)
Minimum distance between echoes	0.5 m
Scanning range at 10 % remission, > 99 % detection probability and 10 klx	Core: typ. 12 m Prime: typ. 25 m Pro: typ. 40 m For more details, see "Scanning range and working range diagrams", page 74
Scanning range at 90 % remission, > 99 % detection probability and 10 klx	Core: typ. 25 m Prime: typ. 47 m Pro: typ. 75 m For more details, see "Scanning range and working range diagrams", page 74

Reflector detection working range ⁴⁾	Core: typ. 0.3 m ... 25 m Prime: typ. 0.3 m ... 60 m Pro: typ. 0.3 m ... 120 m
Scan field flatness	- combined $\pm 1^\circ$ - Conical error: typ. $\pm 0.4^\circ$ - Tilt: typ. $\pm 0.6^\circ$ ⁵⁾

- 1) Data may vary depending on the hardware version. For devices with a wavelength of 940 nm specified on the type label, the values listed here apply. The values for devices without a wavelength specification see "Type label", page 14
- 2) Scan rate and angular resolution are not separately configurable, but are determined by the selected Dynamic Sensing Profile.
- 3) Extendable with Dynamic Sensing Profile
- 4) On a sufficiently large Diamond Grade reflector (part no. 5320565) and orthogonal angle of incidence
- 5) Correctable using mounting bracket with integrated fine alignment (part no. 2136134)

Mechanics and electronics

Connection type	Depending on the mounted system plug, 2 x M12 round connectors
Supply voltage (V_S)	9 V DC ... 30 V DC, reverse polarity protected
Permissible residual ripple	$\pm 5\%$
Power consumption	Typ. 4.5 W max. 17 W with loaded digital outputs, depending on the mounted system plug For more details, see "Output load and power consumption", page 57
Digital inputs	I/O (6 multiports) depending on the mounted system plug Voltage range: - low: $-3\text{ V} \dots 0.45 \times V_S$ - high: $0.72 \times V_S \dots V_S$ Switching frequency range: $\leq 100\text{ Hz}$ Encoder input (IN_x): - HTL / dual-channel encoder with 90° phase offset - Max. input frequency: 50 kHz - Recommended length of cable of the incremental encoder: max. 10 m
Digital outputs	Output mode (configurable): - Push/pull - NPN - PNP Voltage range: - low: $0\text{ V} \dots 1\text{ V}$ - high: $(V_S - 1\text{ V}) \dots V_S$ Output current per output, short-circuit protected: 2x max. 200 mA 4x max. 100 mA > 4x 50 mA
Material	Housing: Aluminum with Suretec650 coating Optics cover: Polycarbonate, scratch-resistant coating System plug: passivated zinc, black lacquered
Housing color	Anthracite gray (RAL 7016)

Enclosure rating ¹⁾	IP65, IP67 (IEC 60529:1989+AMD1:1999+AMD2:2013) Test conditions: • Water spray volume: 14 l/min ... 16 l/min • Water pressure/temperature: 10000 KPa (100 bar) / 80 °C • Flat jet nozzle distance: 100 mm ... 150 mm • Spray angle: 0°, 30°, 60°, 90° • Cycle: 30 seconds per position • Rotational speed of test specimen: 5 rpm
Protection class	III (IEC 61140:2016-11)
Electrical safety	IEC 61010-1:2010-06+AMD1:2016
Weight	220 g, without system plug
Dimensions (L x W x H)	60 mm x 60 mm x 82 mm
Mission time	Typ. 12 years ²⁾
MTBF	> 100 years
MTTF_D	> 100 years, at 25 °C ambient temperature (EN ISO 13849-1:2015)

1) Prerequisites:

- The system plug is mounted.
- The cables plugged into the electrical connections must be screwed tight.
- Unused electrical connections are sealed off with a protective cap.

2) Based on IEC/EN 61508-2 and continuous usage at 25 °C

Performance

Data output per scan segment	Core, Prime, Pro: 30° at a scanning frequency ≤ 25 Hz Prime, Pro: 60° at a scanning frequency ≥ 30 Hz
Scan/frame rate ¹⁾	Up to 265808 measuring point(s)
Latency of the measurement data output ²⁾	• Core, Prime, Pro: Segment size 30° at < 25 Hz: ≤ 10 ms (3 σ) • Prime, Pro: Segment size 60° at ≥ 30 Hz: ≤ 15 ms (3 σ) For more details, see "Latency of the measurement data output", page 26
Profile switching delay time	With equal motor frequencies: < 1 s With different motor frequencies: < 3 s
Delay time of field evaluation	typ. = $1.5 \cdot 1 / \text{scan rate} + 20$ ms max. (3 σ) = $1.8 \cdot 1 / \text{scan rate} + 50$ ms For more details, see "Delay time of field evaluation", page 43
Power-up time ³⁾	typ. 9.5 s
Systematic error	typ. ± 20 mm ⁴⁾ Typ. ± 10 mm (0.1 m ... 120 m) on Diamond Grade reflectors (part number 5320565) max. ± 30 mm Temperature drift: Typ. ± 0.5 mm/K
Statistical error (1 σ) ⁴⁾	≤ 2 mm (0.05 m ... 5 m) ≤ 16 mm (5 m ... 60 m) For more details, see "Scanning range", page 34
Zero point error (0° beam)	typ. $\pm 0.2^\circ$
Statistical angle error	$\leq 0.025^\circ$ (1 σ)
Integrated application	Measurement data output (based on ordered configuration), field evaluation (based on ordered configuration)
Filter	Fog filter, echo filter, particle filter, moving average filter

1) Depending on the selected Dynamic Sensing Profile and the number of echoes.

2) Typical value; real value depends on ambient conditions and the selected Dynamic Sensing Profile.

3) Time until the device can be reached in the network. The time to full operational readiness ("device ready") is typically a further 2 seconds longer.

4) 10 and 100 klx

Interfaces

Ethernet	✓ Functions: • Measurement data output (distance, RSSI, time stamp, reflector mark) • Field status information • DHCP • Time synchronization ◦ NTP (Network Time Protocol) ◦ PTP (Precision Time Protocol, PTPv2 per IEEE 1588-2008, operation as PTP slave, end-to-end (E2E) delay measurement) Protocols and data formats: • UDP/IP Singlecast (Compact (recommended), MSGPACK) • UDP/IP Multicast (Compact (recommended), MSGPACK) • TCP/IP (LMDscandata, HTTP/REST, maximum number of simultaneous connections to the device: 50) ¹⁾ • Native ROS2 (Data Distribution Service, CycloneDDS) Data transmission rate: • 10/100 Mbit/s half/full-duplex
Digital inputs/outputs	I/O (6 multiports) depending on the mounted system plug
Inertial measuring unit (IMU)	BOSCH BHI160B with 3-axis gyroscope and 3-axis accelerometer Sampling rate: 100 Hz Relative position of the IMU to the optical origin: • X: +12.4 mm • Y: +18.5 mm • Z: -48.4 mm
Optical displays	2 LEDs
Configuration software and interfaces	• SOPASair (web server) • REST API (HTTP/HTTPS) • CoLa A/B (SICK TCP-based proprietary protocol)
Driver	ROS1, ROS2, C++, Python

¹⁾ The number of simultaneous connections should be limited to avoid transmission delays.

Ambient data

Remission factor	1.8% ... > 1,000% (reflector)
Electromagnetic compatibility (EMC)	Radiation emitted: • Industrial environment (IEC 61000-6-4:2018 / EN IEC 61000-6-4:2019 IEC 61000-6-4:2006+A1:2010 / EN 61000-6-4:2007+A1:2011) • Business and commercial areas as well as small enterprises (IEC 61000-6-8:2020 / EN IEC 61000-6-8:2020) Electromagnetic immunity: • Industrial environment (IEC 61000-6-2:2016 / EN IEC 61000-6-2:2019 / IEC 61000-6-2:2005 / EN 61000-6-2:2005 / EN 61000-6-2:2005 / AC:2005) Application areas: • Automotive (UN ECE R10) ¹⁾ • Agricultural and forestry machinery (ISO 14982-1, ISO 14982-2) ^{1) 2)} • Earthmoving and construction machinery (ISO 13766-1) ^{1) 2)}

Vibration resistance	Test severity level 1 ³⁾: • Sine resonance scan: 10 Hz ... 1,000 Hz; 1 g (IEC 60068-2-6:2007-12) • Sine test: 10 Hz ... 500 Hz, 5 g, 10 cycles (IEC 60068-2-6:2007-12) • Noise test: 10 Hz ... 250 Hz, 4.24 g RMS, 5 h (IEC 60068-2-64:2008) Test severity level 2 ⁴⁾: • Sine resonance scan: 10 Hz ... 1,000 Hz; 1 g (IEC 60068-2-6:2007-12) • Sine test: 10 Hz ... 500 Hz, 10 g, 10 cycles (IEC 60068-2-6:2007-12) • Noise test: 10 Hz ... 500 Hz, 13.5 g RMS, 5 h (IEC 60068-2-64:2008)
Shock resistance	Test severity level 1 ³⁾: • Single shock according to IEC 60068-2-27:2008-02: 50 g, 11 ms, 3 shocks per axis • Continuous shock according to IEC 60068-2-27:2008-02: 25 g, 6 ms, 1,000 shocks per axis • Continuous shock according to IEC 60068-2-27:2008-02: 50 g, 3 ms, 5,000 shocks per axis Test severity level 2 ⁴⁾: • Single shock according to IEC 60068-2-27:2008-02: 100 g, 6 ms, 3 shocks per axis • Continuous shock according to IEC 60068-2-27:2008-02: 40 g, 6 ms, 4000 shocks per axis • Continuous shock according to IEC 60068-2-27:2008-02: 50 g, 3 ms, 5,000 shocks per axis
Impact resistance	IEC 60068-2-75: 2014, Test Ehc (0.5 J, vertical strike) IEC 61010-1: 2017, based on IK08 (5 J, vertical strike)
Ambient operating temperature	-33 °C ... +50 °C
Storage temperature	- 40 °C ... + 70 °C
switch-on temperature	-33 °C ... +50 °C (permissible switch-on temperature) -5 °C ... +50 °C (immediately ready for operation)
damp heat	Cyclical: +25 °C ... +55 °C, 95% rH (operation/storage/transport) (EN 60068-2-30:2005) Static: +40° C, 93% rF (operation) (EN 60068-2-78:2005)
Permissible relative humidity	Operation: max. 80% rH, non-condensing Storage: max. 90% rH, non-condensing
Ambient light immunity	100 klx (indirect)
Altitude	< 5,000 m above sea level

- 1) Load dump: from ISO 16750-2 Test B Severity Level 4 passed for 12 V systems. Required in case of transient disturbances on the input filtering signal lines (debounce > 10 ms).
- 2) The requirements of ISO 13766-1 and DIN EN ISO 14982-1 for immunity to electrostatic discharge (ESD) are only met in areas that can be easily touched from the outside.
- 3) Short restrictions in measurement data availability possible at low scan rates
- 4) Restrictions in measurement data availability possible at all scan rates

11.1.1 Scanning range and working range diagrams

Table 18: Scanning range of picoScan150 Core

Dynamic Sensing Profile	Minimum		Typical			
	100 klx		10 klx		100 klx	
	10 %	90 %	10 %	90 %	10 %	90 %
15 Hz and 0.33°	10 m	25 m	12 m	25 m	10 m	25 m
15 Hz and 1°	10 m	25 m	12 m	25 m	10 m	25 m

Dynamic Sensing Profile	Minimum		Typical			
	100 klx		10 klx		100 klx	
	10 %	90 %	10 %	90 %	10 %	90 %
25 Hz and 0.25° (default)	10 m	25 m	12 m	25 m	10 m	25 m

Table 19: Scanning range of picoScan150 Prime

Dynamic Sensing Profile	Minimum		Typical				On highly reflective targets and reflectors ¹⁾
	100 klx		10 klx		100 klx		100 klx
	10 %	90 %	10 %	90 %	10 %	90 %	
15 Hz and 0.33°	-	-	34 m	51 m	23 m	44 m	60 m
15 Hz and 0.5°	-	-	34 m	51 m	23 m	44 m	60 m
15 Hz and 1°	-	-	34 m	51 m	23 m	44 m	60 m
20 Hz and 0.1° (default)	14 m	27 m	23 m	38 m	16 m	31 m	45 m
20 Hz and 0.25°	-	-	29 m	51 m	20 m	38 m	60 m
25 Hz and 0.25°	17 m	33 m	28 m	51 m	19 m	36 m	60 m
30 Hz and 0.1°	-	-	21 m	26 m	15 m	26 m	30 m
40 Hz and 0.25°	15 m	29 m	25 m	47 m	17 m	32 m	60 m
50 Hz and 0.25°	-	-	23 m	44 m	16 m	31 m	55 m
15 Hz and 0.05°	-	-	21 m	21 m	15 m	21 m	25 m
40 Hz and 0.125°	-	-	21 m	26 m	14 m	26 m	30 m

¹⁾ On a sufficiently large Diamond Grade reflector (part no. 5320565) and orthogonal angle of incidence

Table 20: Scanning range of picoScan150 Pro

Dynamic Sensing Profile	Minimum		Typical				On highly reflective targets and reflectors ¹⁾	
	100 klx		10 klx		100 klx		100 klx	
	10 %	90 %	10 %	90 %	10 %	90 %	without extended scanning range	with extended scanning range
15 Hz and 0.33°	-	-	40 m	75 m	27 m	52 m	60 m	120 m
15 Hz and 0.5°	-	-	40 m	75 m	27 m	52 m	60 m	120 m
15 Hz and 1°	-	-	40 m	75 m	27 m	52 m	60 m	120 m
20 Hz and 0.1° (default)	17 m	32 m	27 m	45 m	19 m	36 m	45 m	-
20 Hz and 0.25°	-	-	34 m	65 m	24 m	45 m	60 m	120 m
25 Hz and 0.25°	20 m	39 m	33 m	62 m	22 m	43 m	60 m	110 m
30 Hz and 0.1°	-	-	25 m	30 m	17 m	30 m	30 m	-
40 Hz and 0.25°	18 m	34 m	29 m	55 m	20 m	38 m	60 m	70 m
50 Hz and 0.25°	-	-	27 m	52 m	19 m	36 m	55 m	-
15 Hz and 0.05°	-	-	25 m	25 m	17 m	25 m	25 m	-
40 Hz and 0.125°	-	-	24 m	30 m	17 m	30 m	30 m	-

¹⁾ On a sufficiently large Diamond Grade reflector (part no. 5320565) and orthogonal angle of incidence

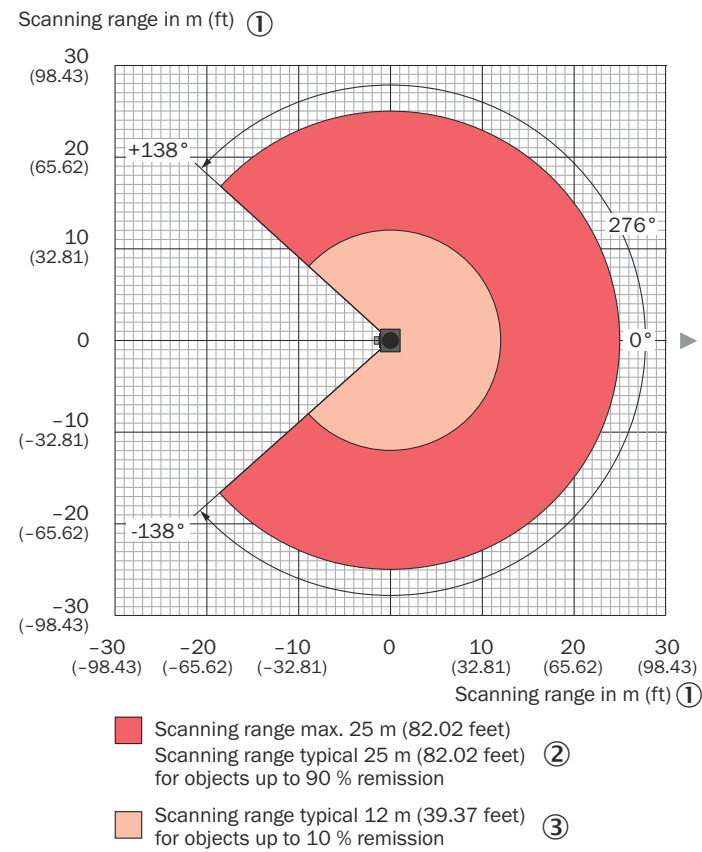


Figure 53: Working range diagram for picoScan150 Core

- ① Scanning range in m
- ② Max. scanning range 25 m/scanning range typically 25 m for objects with 90 % remission
- ③ Scanning range typically 12 m for objects with 10 % remission

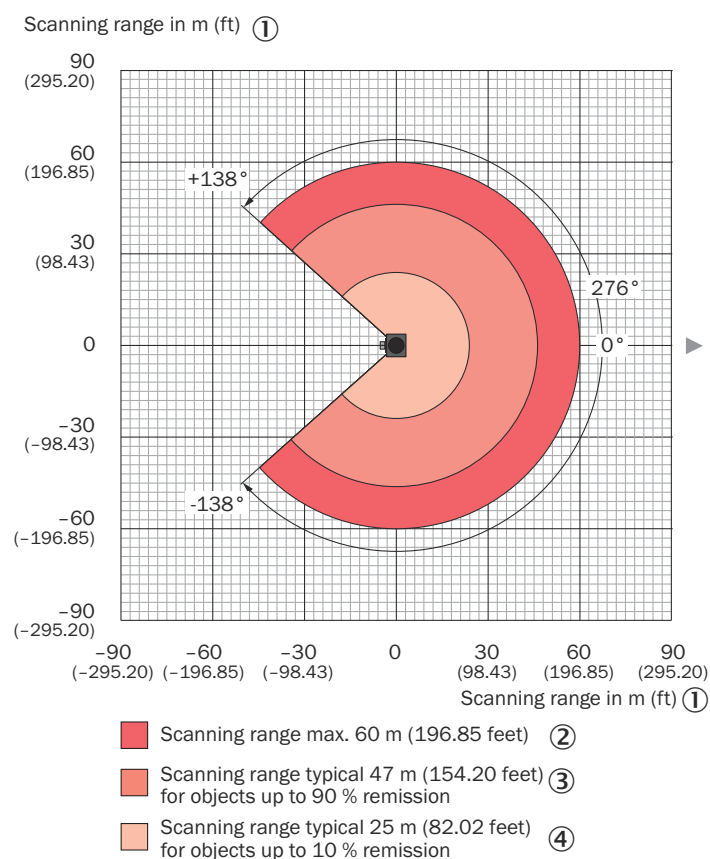


Figure 54: Working range diagram for picoScan150 Prime

- ① Scanning range in m
- ② Max. scanning range 60 m
- ③ Scanning range typically 47 m for objects with 90 % remission
- ④ Scanning range typically 25 m for objects with 10 % remission

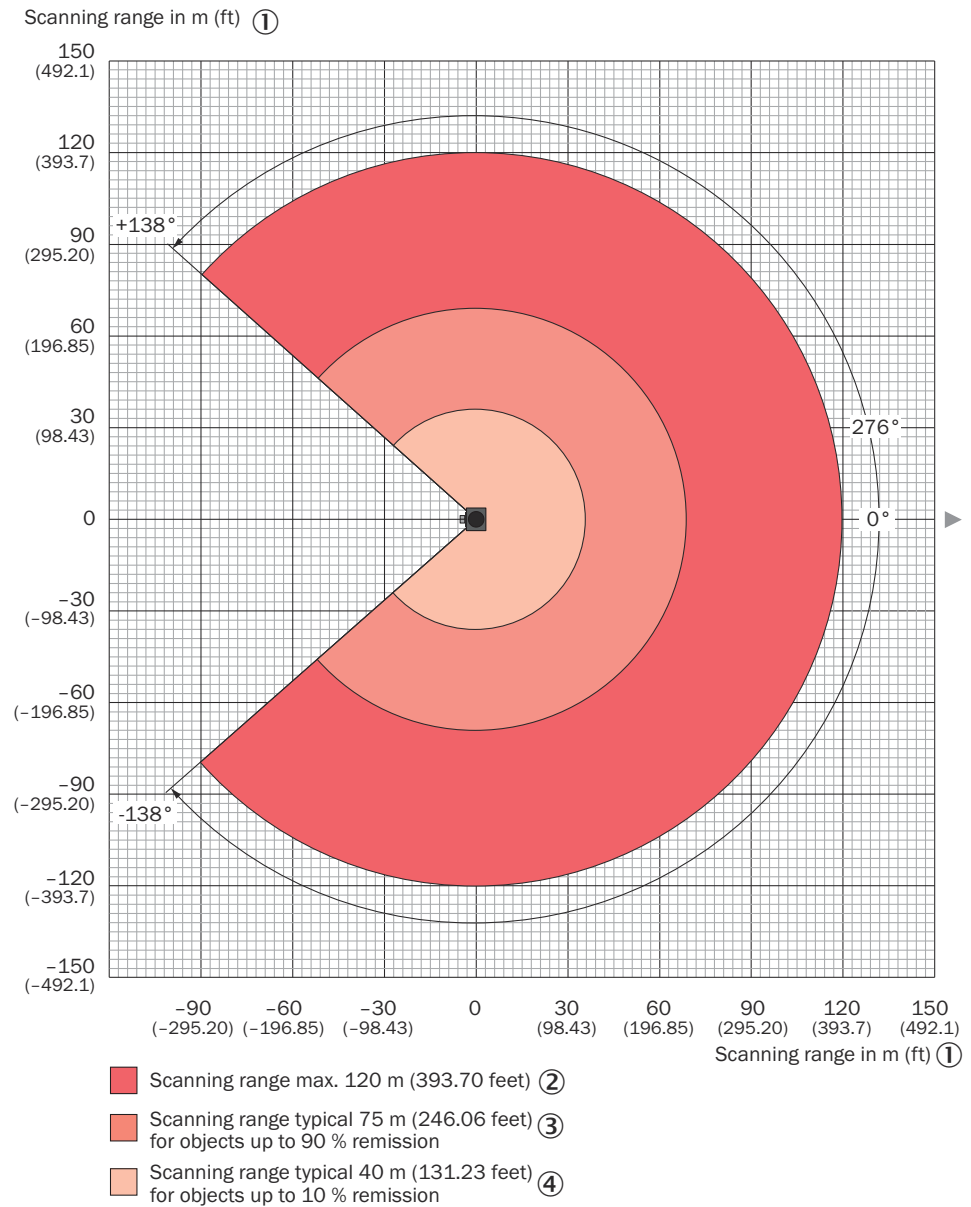


Figure 55: Working range diagram for picoScan Pro

- ① Scanning range in m
- ② Max. scanning range 120 m
- ③ Scanning range typically 75 m for objects with 90 % remission
- ④ Scanning range typically 40 m for objects with 10 % remission

11.2 Mission time

The mission time is the period during which the failure rates can be assumed to be constant. At the end of the mission time, the failure rates gradually increase due to wear. The mission time of the product is influenced by various factors, including operating conditions and user handling.

The typical mission time to be expected under normal operating conditions and proper use can be found in the technical data [see "Data sheet", page 70](#). It is important to note that the actual mission time may vary depending on individual circumstances. Factors such as frequency of use and ambient factors can affect the service life of the product.

According to the IEC/EN 61508-2 standard, a mission time based on general experience can be assumed. In addition, the mission time can be extended by suitable measures taken by the system operator, e.g., operation at a low ambient temperature or avoidance of shocks and vibrations.

Table 21: Example mission times of the product at different ambient temperatures

Continuous use at 10 °C	20 years
Continuous use at 25 °C	12 years
Continuous use rarely above 30 °C	10 years
Continuous use 70% to 30 °C, 30% to 50 °C	7.5 years
Continuous use at 50 °C	4.5 years

11.3 Dimensional drawing

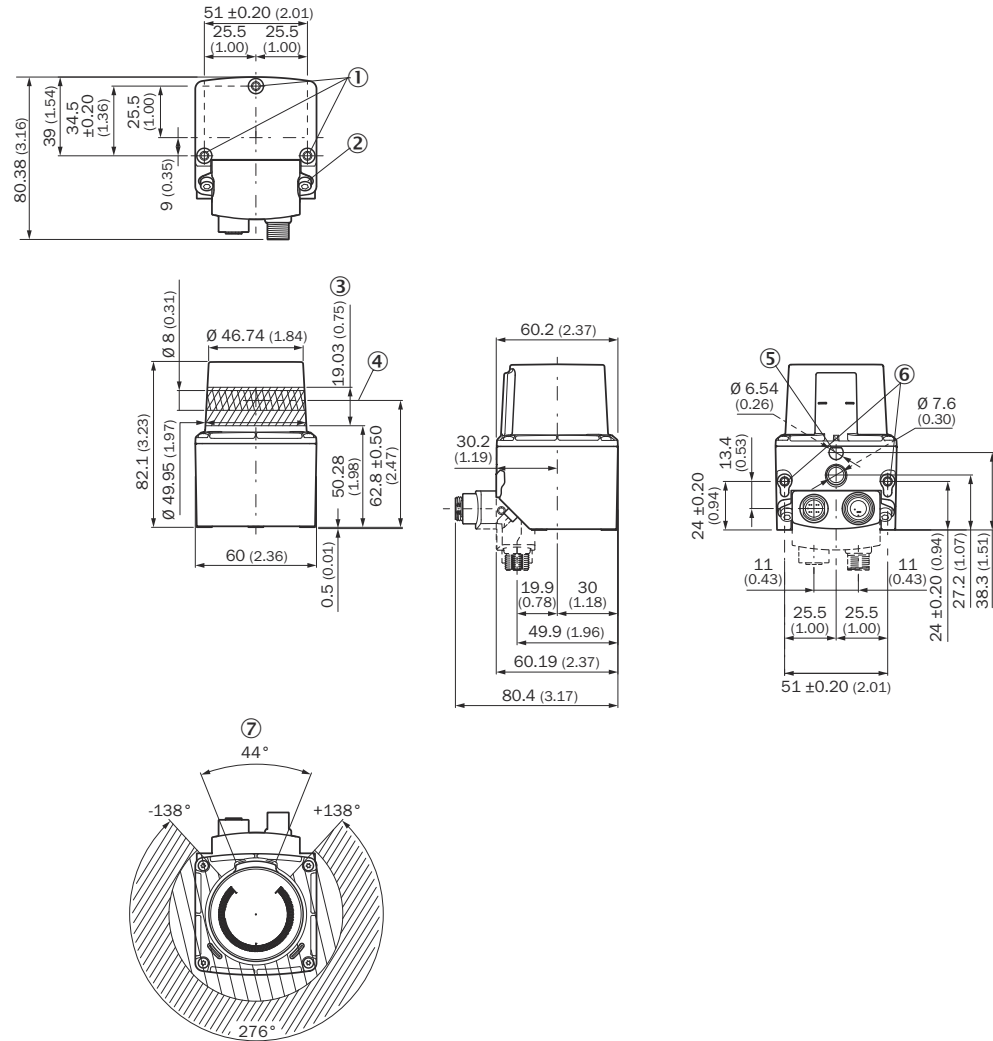


Figure 56: Dimensional drawing, unit of measurement: mm (inch), decimal separator: period

- ① M4 threaded mounting hole; 4.2 mm deep, tightening torque 2 Nm
- ② Tightening torque 2 Nm, screw included in plug unit
- ③ Receiving range
- ④ Transmission axis
- ⑤ Support point
- ⑥ M4 threaded mounting hole; 5.4 mm deep, tightening torque 2 Nm
- ⑦ Area in which no reflective surfaces are permitted when the device is mounted

12 Accessories

Accessories and, if applicable, associated mounting information can be found on the product page.

The call is made via the **SICK Product ID: pid.sick.com/{P/N}/{S/N}**

{P/N} corresponds to the part number of the product, see type label.

{S/N} corresponds to the serial number of the product, see type label (if indicated).

Support Portal



NOTE

In the SICK Support Portal (support.sick.com) you will find, besides useful service and support information for your product, further detailed information on the available accessories and their use.

13 Annex

13.1 Declarations of conformity and certificates

You can download declarations of conformity and certificates via the product page.

The product page can be accessed via the **SICK Product ID: pid.sick.com/{P/N}/{S/N}**

{P/N} corresponds to the part number of the product, see type label.

{S/N} corresponds to the serial number of the product, see type label (if indicated).

13.2 Licenses

SICK uses open source software which is published by the rights holders under a free license. Among others, the following license types are used: GNU General Public License (GPL version 2, GPL version 3), GNU Lesser General Public License (LGPL), MIT license, zlib license and licenses derived from the BSD license.

This program is provided for general use without warranty of any kind. This warranty disclaimer also extends to the implicit assurance of marketability or suitability of the program for a particular purpose.

See the GNU General Public License for more information.

For license texts, see www.sick.com/licensesetexts. Printed copies of the license texts are also available on request.

13.3 Communication interfaces

Communication with the device is possible via CoLa A/B and REST. For more information, see the following English telegram listing.

Under Downloads on the product page, you can download, for example, the Open API file.

The product page can be accessed via the **SICK Product ID: pid.sick.com/{P/N}/{S/N}**

{P/N} corresponds to the part number of the product, see type label.

{S/N} corresponds to the serial number of the product, see type label (if indicated).

13.3.1 HTTPS

Secure communication in accordance with the Cyber Resilience Act (CRA)

The Cyber Resilience Act (CRA) requires the use of secure communication channels for products with digital components. HTTPS (Hypertext Transfer Protocol Secure) is used for communication with the sensor. HTTPS protects data transmission with TLS encryption and ensures the confidentiality of configurations via the REST interface, including SOPASair. Only authorized users have access to the interface. Communication via HTTPS is encrypted, whereas measured data sent via the UDP/TCP protocol is not encrypted.

HTTP remains active by default and must be manually deactivated to prevent unencrypted access.

Certificate requirements for HTTPS

Valid certificates must be stored in the device so that HTTPS can be used. The following requirements must be observed:

- Certificate standard: X.509
- Certificate format: PEM

- Key requirements: RSA $\geq$ 3,072 bits ECDSA $\geq$ 250 bits Curve25519 or Curve448
- Port: 443 Maximum certificate chain: Root → Intermediate1 → Intermediate2 → Leaf
- Recommended certificate chain: Root → Intermediate1 → Leaf

13.4 Integration of the product into mobile platforms

This section provides guidance for customers who want to integrate the product into a mobile application.

Different types of mobile platforms

The integration guide applies to all mobile platforms:

- Mobile platforms in logistics
- AMRs in mobile production
- Service/cleaning robots

LiDAR applications in mobile platforms

The product can perform a variety of tasks in an application. The product can be used for collision avoidance at the front and rear of the platform and for navigation. Protrusion monitoring, detection of pallet pockets or steps and volume measurement is possible when the product is mounted at an angle. The product also offers picking control.

13.4.1 Assignment of integration phases

The following phases are assigned to the listed sections to indicate in which phase the information might be important.

- **Testing:** In this phase, tests are performed to see how the device behaves in the intended target environment.
- **Design-in:** This phase covers the design-in process of the device. It provides information on what to consider when integrating the device into mobile platforms.
- **Production:** In this phase, instructions are given which are to be observed in the production process and in the series installation of the device.
- **Lifecycle:** This phase describes what needs to be considered during the lifecycle of the sensor.

Alignment aid for displaying the laser position [Testing | Design-in]

An alignment aid can help to precisely position the sensor and avoid alignment errors (e.g. part no.: 2101720).

Temperature fluctuations [Testing | Design-in]

When temperatures change rapidly, e.g. when moving from a cold storage facility to a hall at a normal temperature, the optics cover may fog up due to moisture. This can lead to a limited measuring capability.

We recommend using the contamination indication [see "Contamination indication", page 41](#).

Device Ready Status [Testing | Design-in]

The "device ready" status is required for operation of the sensor. If this status is not displayed, an error is present. The status can also be transmitted to all digital outputs, and is easily recognized via the LEDs, [see "Visibility of the LEDs \[Design-in\]", page 84](#).

Error [Testing | Design-in]

So errors can be eliminated, it is necessary to display the errors and the type of error. The errors displayed to the customer are listed under **Diagnostics** in SOPASair [see "Troubleshooting", page 68](#).

Filter types [Testing | Design-in]

By using suitable filters, it is possible to further increase the availability of the sensor in certain applications and also in outdoor area, [see "Filter", page 21](#).

Contamination of the system plug [Testing | Design-in | Lifecycle | Production]

To avoid contamination, it is important that the system plug is always covered or connected. This also applies when a device is replaced. Furthermore, the system plug should be easily accessible so that the device can be replaced without any problems. Mounting and removal must be performed while the device is de-energized.

Permanent inductive charging [Design-in | Testing]

Inductive charging creates a strong magnetic field that can limit the functionality of the device. Particularly at risk are places where several magnetic fields overlap, for example branches in the induction strip. Additional shielding of the device is recommended.

Testing your own targets [Testing | Lifecycle]

The performance of the device also depends on the targets. It is therefore important to test the functionality on the customer's own targets [see "Object sizes", page 28](#) and [see "Scanning range", page 34](#).

Minimum object size [Testing | Lifecycle]

It is important to know what target size can still be detected at what distance. For detailed information, [see "Object sizes", page 28](#).

Note that the operating instructions present the worst case. We therefore recommend in-house testing to verify the performance under the customer's own conditions.

Outdoor [Design-in]

Information on outdoor use is available in the following article in the SICK Support Portal: [picoScan150: Possible challenges and countermeasures in outdoor applications](#).

Scratch protection [Design-in]

Scratches in the optics cover can lead to incorrect values, as this can cause reflection of the beams. The optics cover has an anti-scratch coating for this reason, but the sensor should, if possible, still be mounted in such a way as to protect it from scratches.

Visibility of the LEDs [Design-in]

The LEDs indicate the operational status of the device. It is recommended to make them visible after the device is installed in order to quickly recognize whether the device is switched on or, for example, whether fields have been interrupted. For detailed information, [see "Display and control elements", page 13](#).

The LEDs can be permanently switched off via SOPASair.

Light spot divergence [Design-in]

Due to the light spot divergence that occurs in LiDAR sensors, where the emitted beam spreads over a larger area as the distance between the sensor and the target is increased, a reduction in beam intensity with distance will occur.

The device has a low light spot divergence, but care should still be taken to ensure that the intensity of the beam is sufficient for the application in question, [see "Features", page 70](#).

Vehicle housing [Design-in]

For the product to measure reliably, interference to the laser beam by the vehicle housing must be excluded. Any reflections that occur should also be taken into account. Using a protection hood ensures that the product is reliably protected from external influences such as ambient light and scratches.

Gap width in the housing [Design-in]

The reflected beam is significantly magnified compared to the emitted beam due to the light spot divergence of LiDAR sensors. To generate reliable measured values, this should be taken into account when dimensioning the gap width.

Restrictions in the field of view [Design-in]

In order to exclude irritations caused by reflections of the housing and a resulting object detection, it is recommended to maintain a separation between the scan range and the housing.

Scan field flatness (conical error and tilt) [Design-in]

The scan field flatness describes the production-related vertical deviation of the horizontal scan plane of the sensor. Conical errors and tilt can affect the three-dimensional measurements. This should be taken into consideration to ensure reliable measurement results. Conical errors can only be corrected for a small viewing range.

Tilt errors can be compensated for in many cases by mounting the sensor at a compensating angle. The alignment bracket can be used for this (part number: 2136134).

Scanning behavior behind protective screens [Design-in]

Screens are often used to protect the sensor from, for example, dirt and moisture. It should be noted that optical elements in the transmit and receive paths can influence the measurement behavior. For recommendations on the use of protective screens, [see "Measurement behavior through screens \[Design-in\]", page 85](#).

Measurement behavior through screens [Design-in]

A screen is an optical element and therefore affects the measurement. If the device needs to be protected by a screen, an anti-reflective and thin screen is recommended. It should also be scratch-resistant and have a low refractive index. The screen must be mounted in such a way that a direct reflection does not hit the receiver and no dust cannot settle on it. To this end, it is recommended to install the screen at a slightly inclined angle. We recommend using the multi-echo function and the fog filter.

Mounting on platforms with vibration [Design-in]

If the surface is uneven, the sensor is exposed to vibrations and shock, which can damage the sensor. The resilience of the device can be increased in these cases by using a shock absorber. The device was tested by means of shock and vibration tests, [see "Shock and vibration \[Design-in\]", page 88](#).

Upside down mounting [Design-in]

It is possible to mount the sensor upside down. It should be noted that the laser beam always rotates in the same direction with respect to the sensor coordinate system (counterclockwise), regardless of the installed orientation.

Influence of magnetic fields on the motor of the sensor [Design-in]

Magnetic fields near the motor can cause an imbalance in the internal motor. If this exceeds a permissible limit, the sensor stops. If the sensor is to be used in the vicinity of magnets, a function check-out is recommended. Tests with permanent magnets show that they only have an effect if they are placed in the immediate vicinity of the motor.

Ventilation element [Design-in]

For detailed information, see ["Ventilation element", page 45](#)

Material components of the housing [Design-in]

For detailed information, see ["Mechanics and electronics", page 71](#)

External pressure on the optics cover [Design-in]

The motor of the sensor is located under the optics cover. Make sure that no external pressure is applied to the optics cover.

Temperature effects on the housing [Design-in]

Temperature fluctuations can result in increased tolerances. This effect should be taken into account when mounting the device, e.g., with regard to the spacing of the drill holes.

Reference target for positioning [Design-in | Testing]

Reference targets are often used, for example, for a precise approach to charging or transfer stations, and should be easily recognizable to the sensor. Material that ensures a good measuring capability of the sensor can be purchased using part number 5600079.

Access permission [Design-in]

To prevent the device from being reconfigured by unauthorized persons, it is recommended to change the default password, see ["Managing passwords", page 63](#).

Cybersecurity [Design-in]

Cybersecurity is used to protect against unwanted access by third parties.

A Cybersecurity Hardening Guide can be found at: www.sick.com/8029481.

Software driver [Design-in]

The device is supported by the following drivers:

- ROS1
- ROS2
- C++
- Python

These are available at the following link: https://github.com/SICKAG/sick_scan_xd / <https://github.com/SICKAG/ScanSegmentAPI>

Time stamp [Design-in | Testing]

Time stamps are helpful to evaluate when exactly each distance value was measured. This is of great importance in moving applications. The time stamps are sent to the customer with each data packet. The duration until the customer can access the time stamp depends on the transport time in the network.

Synchronization of devices [Design-in]

Since multiple devices are often used in one application, it makes sense to synchronize them. This is possible with the help of NTP or PTP [see "Time synchronization", page 40](#).

Boot time/power-up time [Design-in]

The power-up time is the time required for the device to be accessible in the network after the supply voltage is applied. The time to full operational readiness ("device ready") is typically longer. At low ambient temperatures, the time until the device is fully operational may also be longer. The device is designed for 24-hour operation and therefore does not necessarily have to be switched off. For the power-up time, [see "Performance", page 72](#).

Voltage supply [Design-in]

With battery-powered AMRs, the supply voltage at the device is not always constant. Limit values must not be exceeded. For the maximum voltage range, [see "Mechanics and electronics", page 71](#).

Energy consumption [Design-in]

The power rating of the device must be considered when sizing the battery, [see "Mechanics and electronics", page 71](#).

Orientation of the system plug [Design-in]

To provide a high level of flexibility during installation, it is possible to mount the system plug in different directions: horizontally to the rear or vertically downwards, [see "Dimensional drawing", page 80](#)

Memory of the system plug [Design-in]

To ensure the network configuration is retained after a system plug is replaced, the configuration is stored in the system plug [see "Data backup", page 64](#).

Heat generation [Design-in]

To avoid overheating of the device, attention should be paid to the heat generation. To dissipate the heat in the best possible way, the device should be thermally connected to the largest possible surface so that it serves as a heat sink. It is best not to install the device in a heat-insulated location.

EMC – Short cable routing [Design-in]

To minimize the EMC effect, avoid coiling cables. Coiling cables can create a larger electromagnetic field. A simple cable routing is therefore recommended.

EMC – Shielded cables [Design-in]

Shielded cables prevent influences or interference from electromagnetic fields.

EMC – Ethernet CAT class [Design-in]

To ensure data transmission, we recommend CAT class 5 for the Ethernet cable.

EMC testing [Design-in]

For the EMC tests performed, [see "Ambient data", page 73.](#)

Horizontal and vertical edge hits [Design-in | Testing]

Edge hits can lead to incorrect distance values. An edge hit occurs when a single beam hits two different objects. This can falsify the distance calculation. The device uses HDDM+ and the multi-echo method, which provides up to three distance values per measuring beam. This applies to both vertical and horizontal edge hits.

Shock and vibration [Design-in]

It is recommended to compare the specified shock and vibration data with the vibration loads in the specific application and to provide a damping system if necessary.

Center of gravity [Design-in]

The center of gravity of the device can be seen in the CAD file. The CAD file can be downloaded from the product page [see "Further information", page 6.](#)

Mutual interference with other light sources [Design-in]

Mutual interference between two or more devices is very unlikely due to the HDDM+ method used.

With the HDDM+ measuring method, several pulses are sent per measurement point. A single measurement point is then calculated from these subpulses. If a subpulse fails, the statistical measurement procedure still has sufficient information to calculate a measurement point. In addition, the emission of the subpulses is randomized in time. The probability of many faulty subpulses is therefore low.

External optical influence [Design-in | Testing]

Common external influences are the sun, arcs from welding, and light from photo-electric sensors. Please determine what sources of ambient light the sensor may be exposed to in the application. A protection hood that shades out ambient light could be helpful.

Ambient light due to AMR illumination [Design-in]

Lights such as LEDs, light strips or the like are often installed on AMRs. Light in the visible range does not affect the device due to a bandpass filter. The wavelength of other light pulse emitting devices must be taken into account (± 30 nm at 905 nm).

Brackets [Design-in]

There are various brackets that can be used for different applications [see "Accessories", page 81.](#) The device can be mounted using the three threads on the bottom or using the two threads on the back.

External reference target [Design-in]

External reference targets are needed if the exact position of the mobile platform is to be determined. It is an easy way to mark specific areas, such as busy hallways or areas with people. Reflectors are well suited as reference targets when RSSI information is used for evaluation. The reflectors must be mounted in the scan field. Reference geometries can also be used.

Screwdriver [Design-in]

To be able to easily tighten the M12 connections and fixing screws, the use of an appropriate tool is recommended. The mounting holes have an M4 thread. A separate torque screwdriver is available for the connections (part no. 2081618).

Update [Design-in]

Updates enable the device to be kept up to date. The manual update is done via SOPASair or SOPAS ET. To update semi-automatically (via batch file), use the SICK AppManager.

Operating temperature range [Design-in]

The device can only measure reliably within a defined temperature range, [see "Ambient data", page 73](#).

It should be noted that cables may have different temperature specifications.

Runoff of liquid [Design-in | Lifecycle]

The device has an IP65/67 protection class rating. Nevertheless, water residues on the device should be avoided. In applications where wetness occurs, a water drain may be useful to keep stagnant liquid away from the device.

Ground hits due to ground unevenness [Design-in | Lifecycle | Production]

An object detection can occur when the device briefly measures the ground due to ground unevenness. If these vehicle pitching movements are unavoidable, a temporal filter can be used (field evaluation).

Water protection testing [Design-in | Lifecycle]

For the use and cleaning of the AMR, it is important to know which IP protection class the device has, [see "Mechanics and electronics", page 71](#).

Detecting optics cover contamination [Design-in | Lifecycle]

If the optics cover is heavily contaminated, measurement inaccuracies may occur due to direct reflection of the emitted light pulse from the contamination.

More information can be found at: [see "Contamination indication", page 41](#)

Tightening torque of screws and connections [Production]

For precise mounting, it is important that the fixing screws and connections are tightened correctly. For the tightening torque of the screws (device and system plug mounting), [see "Dimensional drawing", page 80](#).

The use of special torque screwdrivers is recommended for the cable connections (M12), [see "Screwdriver \[Design-in\]", page 89](#).

Connecting cables [Production]

The pin assignment of the connecting cables can be found in the operating instructions and is printed on the system plug, [see "Connections and pin assignment", page 53](#).

SICK's own connecting cables can be found under the following link: <https://www.sick.com/c/g274507>

The bend radii can be found in the respective specifications.

13.5 Data format description (EN)

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13.5.1.1 Glossary

Designation	Explanation
Azimuth angle	Horizontal angle described by Theta
Beam	With the term beam we denote a beam of light which is emitted by the lidar sensor in a certain direction. There may be multiple reflections of light on that beam which are caused by object which intersect with that beam. These reflections are called echoes.
Elevation angle	Vertical angle described by Phi

Designation	Explanation
Frame	A frame is defined as the data acquired within the entire field of view of a LiDAR sensor (e.g., 360° for rotating LiDAR sensors such as the multiScan100) multiScan136 example: • 1 frame consists of 12 segments • 1 segment comprises n layers = n scans (multiScan136 example: 16 layers = 16 scans) • 1 scan comprises several beams
Layer	A traditional 2D LiDAR has a layer at 0°. Multi-layer scanners can have multiple layers (e.g. multiScan136 => 16 layers).
Module	A module is an object that creates/provides data for different layers
RSSI	RSSI (received signal strength indicator) is defined as an indicator of the strength of the received signal. Context: • Small RSSI value = low signal strength • Large RSSI value = high signal strength The magnitude of the RSSI values is not standardized and can vary from device to device.
Scan	Collection of beams in the azimuth direction of a layer.
Scan segment	A scan segment is defined as a collection of scans that represent a portion of a frame. All scans in a segment have different elevation angles (i.e., they belong to different layers in the case of a multi-layer LiDAR sensor). The angular range in azimuth may (but need not) be different for each scan in a segment. For multi-layer LiDAR sensors, a scan segment is typically used to combine scans that were acquired at the same time or that have the same azimuth range.

13.5.1.2 General information on the transmission of measurement data

The measurement data are transmitted segment by segment, i.e., each transmitted data package (for example a UDP packet or TCP packet) contains a segment (cf. the Segmented data output section in the operating instructions). Each segment can be interpreted separately, i.e., it is not necessary to collect all segments in a frame (=all measurement data recorded in one revolution) to start processing. This makes it possible to reduce the latency between the generation of the measurement data and the processing of that data on the client side.

Two formats are available for the transmission of measurement data, which will be referred to in the following as MSGPACK format and Compact format.

The MSGPACK format encodes the measurement data according to the MSGPACK standard (see also www.msgpack.org), which has the advantage that the data packages can be easily parsed using the standard libraries available for numerous programming languages. The MSGPACK format is self-describing. Each data field is described by a keyword, so it is easy to determine which data field is currently being read without having to know the exact structure of the data.

In the Compact format, the measurement data of the sensor are represented as compactly as possible. The individual fields are not self-describing, but only a string of bytes is transmitted. The structure of the transmitted data package must be known to the user in advance. This has the advantage that as little bandwidth as possible is used on the data line and that a very efficient interpretation of the data is possible by copying the transmitted byte sequence into a structure using a single command (e.g., memcpy in the programming language C/C++).

13.5.1.3 MSPACK format

13.5.1.3.1 Framing

The data packages transmitted in MSGPACK format are enclosed within a frame (see [figure 57, page 92](#)):

The actual MSGPACK payload data is preceded by 4 <STX> characters (hex code 0x02) and the size of the actual payload data (without the checksum at the end) in bytes as a uint32 value. Following the MSGPACK data is a CRC32 checksum calculated on the MSGPACK data only (without the <STX> characters and packet size). The little-endian representation is used for both the size of the payload and the checksum.

4 Bytes	4 Bytes		4 Bytes
\x2\x2\x2\x2	Size of MSGPACK buffer in bytes	MSGPACK buffer	CRC32

Figure 57: Framing in the MSGPACK format

13.5.1.3.2 MSGPACK keywords

To reduce the bandwidth on the transmission line, the keywords used in MSGPACK are encoded as uint8 values. The bandwidth savings from this are significant, but the user must interpret the read uint8 keywords according to the following table. The description of the keywords in the table is for overview purposes only. A detailed description can be found in the sections referred to in the respective table rows. Exactly the same names for the keywords are used there.



NOTE

The data packages can be read using standard MSGPACK parsers after removing the framing. For some parsers, options must be set to allow uint8 values as keywords. For the Python msgpack module, for example, the option `strict_map_key=False` must be set:

```
unpacked = msgpack.unpackb(msgpackValue, strict_map_key=False)
```

Table 22: Used MSGPACK keywords and associated uint8 codes

Keyword name	Uint8 value	Description
classname	0x10	Keyword for the Scan (see "Serialization of the Scan class", page 95) and ScanSegment (see "Serialization of the ScanSegment class", page 94) classes represented in the data
data	0x11	Keyword for the data part, which belongs to the Array, Scan or ScanSegment classes, see "Serialization of the Scan class", page 95 ; see "Serialization of the ScanSegment class", page 94 ; see "Serialization of arrays", page 97 .
numOfElems	0x12	Number of elements in an array, see "Serialization of arrays", page 97
elemSz	0x13	Size of an array element in bytes, see "Serialization of arrays", page 97
endian	0x14	Keyword describing the endianness of the array elements, see "Serialization of arrays", page 97
elemTypes	0x15	Keyword for the type of array elements, see "Serialization of arrays", page 97
Little	0x30	Keyword for endianness "little".
float32	0x31	Data type float32
uint32	0x32	Data type uint32
uint8	0x33	Data type uint8

Keyword name	Uint8 value	Description
uint16	0x34	Data type unit16
ChannelTheta	0x50	Data channel with azimuth angles, see "Serialization of the Scan class", page 95
ChannelPhi	0x51	Data channel with elevation angles, see "Serialization of the Scan class", page 95
DistValues	0x52	Channel with distance values, see "Serialization of the Scan class", page 95
RssiValues	0x53	Channel with RSSI values, see "Serialization of the Scan class", page 95
PropertiesValues	0x54	Channel with further properties of a measurement beam, see "Serialization of the Scan class", page 95 and see "Representation of beam characteristics (Properties)", page 113
Scan	0x70	Keyword for the "Scan" class, see "Serialization of the Scan class", page 95
TimestampStart	0x71	Start time stamp of a scan, see "Serialization of the Scan class", page 95
TimestampStop	0x72	Stop time stamp of a scan, see "Serialization of the Scan class", page 95
ThetaStart	0x73	Azimuth start angle of a scan, see "Serialization of the Scan class", page 95
ThetaStop	0x74	Azimuth stop angle of a scan, see "Serialization of the Scan class", page 95
ScanNumber	0x75	Number of a scan, see "Serialization of the Scan class", page 95
ModuleId	0x76	ModuleID of a scan, see "Serialization of the Scan class", page 95
BeamCount	0x77	Number of beams in a scan, see "Serialization of the Scan class", page 95
EchoCount	0x78	Number of echoes in a scan, see "Serialization of the Scan class", page 95
ScanSegment	0x90	Keyword for the "ScanSegment" class, see "Serialization of the ScanSegment class", page 94
SegmentCounter	0x91	Segment number of a segment, see "Serialization of the ScanSegment class", page 94
FrameNumber	0x92	Frame number of a segment, see "Serialization of the ScanSegment class", page 94
Availability	0x93	Availability of a segment, see "Serialization of the ScanSegment class", page 94
SenderId	0x94	SenderID of a segment, see "Serialization of the ScanSegment class", page 94
SegmentData	0x96	Array with the actual measurement data for each layer, see "Serialization of the ScanSegment class", page 94
LayerId	0xA0	Array with layer IDs, see "Serialization of the ScanSegment class", page 94
TelegramCounter	0xB0	Telegram counter, see "Serialization of the ScanSegment class", page 94

13.5.1.3.3

Serialization of a segment

Each data package transmits one segment enclosed within a frame as per section [section 13.5.1.4.2](#). A segment contains various fields, which are described in [see table 23](#). The actual measurement data is located in the **SegmentData** field, which contains one element of the **Scan** class ([see "Serialization of the Scan class", page 95](#)) for each layer of the sensor ([see figure 58, page 94](#)).

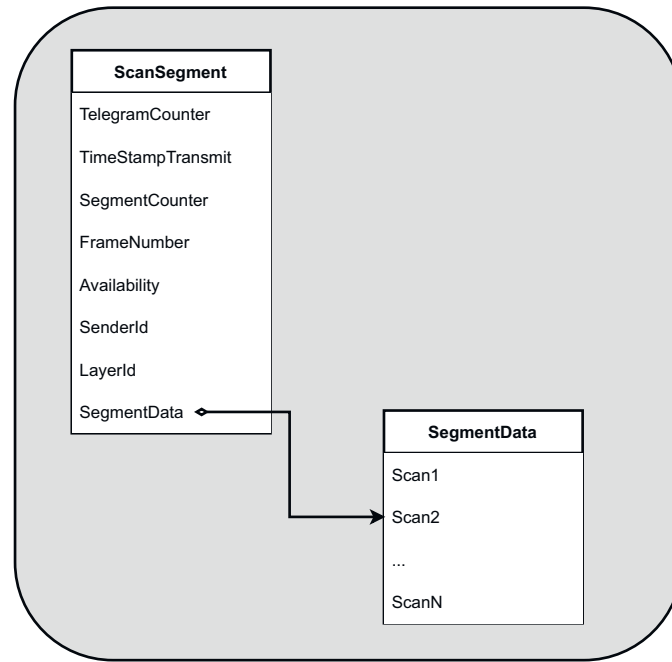


Figure 58: General structure of a segment. In addition to some metadata, the ScanSegment class contains an array named SegmentData that contains an object of type Scan for each layer of the sensor.

The data of a segment are encoded in MSGPACK format. The following exception should be noted: Measurement data in arrays such as distance, RSSI, angle etc. are binary coded here to allow easier serialization (see "Serialization of arrays", page 97).

13.5.1.3.3.1

Notation used

The following notes on the notation used relate to the description of the structures encoded in MSGPACK:

- Msgpack_map_header is used to denote the header of a msgpack map as per this specification: <https://github.com/msgpack/msgpack/blob/master/spec.md#map-format-family>
- The keyword names and other names used in the structures correspond to those from [see table 22, page 92](#).
- Angle brackets are used to specify placeholders for values referred to in the respective declarations, e.g., <numOfElems> for the number of elements of an array.
- Keywords printed in bold refer to substructures of a type, e.g., other types or arrays.
- Although a structure similar to JSON is used for the description in this document, the data is however encoded according to the MSGPACK specification: <https://github.com/msgpack/msgpack/blob/master/spec.md>

13.5.1.3.3.2

Serialization of the ScanSegment class

The **ScanSegment** class is represented as a nesting of MSGPACK maps as follows:

```

msgpack_map_header {
  "classname": ScanSegment,
  "data":
  msgpack_map_header {
    "TelegramCounter": <telegramCounter>,
  
```

```

"TimeStampTransmit": <timeStampTransmit>,
"SegmentCounter": <segmentCounter>,
"FrameNumber": <frameNumber>,
"Availability": <availability>,
"SenderId": <senderId>,
"LayerId": layerIdVector,,
"SegmentData", segmentData
}
}

```

Definition: Definition of the **ScanSegment** class

The meaning of the individual fields is shown in the following table.

Table 23: Description of the attributes of the ScanSegment class

Name	Type	Description
TelegramCounter	MSGPACK int ¹⁾	Counts all telegrams with measurement data sent in MSGPACK format since switching on the device. The counter starts at 1.
TimeStampTransmit	MSGPACK int ¹⁾	Sensor system time in μ s since 1.1.1970 00:00 in UTC.
SegmentCounter	MSGPACK int ¹⁾	Segment counter as described in section.
FrameNumber	MSGPACK int ¹⁾	Counts the number of full revolutions since the device was started.
Reserved		
SenderId	MSGPACK int ¹⁾	Device serial code. It can be used to detect on the recipient which sensor the data was sent from.
LayerId	MSGPACK array ²⁾ of int ¹⁾	Array of layer indices. The layer indices start at 1 and increase with decreasing elevation angle .
SegmentData	MSGPACK array ²⁾ of scans	Array of elements of the Scan class (see "Serialization of the Scan class", page 95) that contain the actual measurement data. The following applies: The scan at position i has the layer number LayerId[i].

¹⁾ github.com/msgpack/msgpack/blob/master/spec.md#int-format-family

²⁾ github.com/msgpack/msgpack/blob/master/spec.md#array-format-family

NOTE | The arrays **LayerId** and **SegmentData** are MSGPACK arrays as per the MSGPACK specification and are not represented like arrays containing measurement data (see "Serialization of arrays", page 97).

13.5.1.3.3.3

Serialization of the Scan class

The **Scan** class is represented as a nesting of MSGPACK maps as follows:

```

msgpack_map_header {
  "classname": Scan,
  "data":
  msgpack_map_header {
    "TimeStampStart": <timeStampStart>,
    "TimeStampStop": <timeStampStop>,
    "ThetaStart": <thetaStart>,
    "ThetaStop": <thetaStop>,

```

```

"ScanNumber": <scanNumber>,
"ModuleId": <moduleId>,
"ChannelTheta": <channelTheta>,
"ChannelPhi": <channelPhi>,
"DistValues": <distValues>,
"RssiValues": <rssiValues>,
"PropertyValues": <propertyValues>,
"BeamCount": <beamCount>,
"EchoCount": <echoCount> }
}

```

Definition: Definition of the **Scan** class. The fields highlighted in gray are optional, i.e., they are not necessarily present in the structure.

The meaning of the individual fields is shown in the following table:

Table 24: Description of the attributes of the **Scan** class

Name	Type	Description
TimeStampStart	MSGPACK int ¹⁾	Acquisition time of the first beam of the scan in μ s. The device's internal time base is used or, if the sensor offers the feature, the time set externally.
TimeStampStop	MSGPACK int ¹⁾	Acquisition time of the last beam of the scan in μ s. The device's internal time base is used or, if the sensor offers the feature, the time set externally.
ThetaStart	MSGPACK int ¹⁾	Azimuth angle of the first beam of the scan in radians.
ThetaStop	MSGPACK int ¹⁾	Azimuth angle of the last beam of the scan in radians.
ScanNumber	MSGPACK int ¹⁾	Not used.
ModuleId	MSGPACK int ¹⁾	Number of the physical module that generated the data. In the case of the multiScan, for example, this is one of the two measuring modules.
ChannelTheta	array of float32	Array of azimuth angles of a specific beam in radians. See also section 13.5.1.5.1 . The encoding of the data in this array is described in section 13.5.1.3.3.4 . The user can configure whether the array is present in the data structure.
ChannelPhi	array of float32	Array with the elevation angle (cf. operating instructions, Coordinate system section) of the scan in radians. For single layer and multi-layer sensors, this array contains only one element. The encoding of the data in this array is described in section 13.5.1.3.3.4 . The user can configure whether the array is present in the data structure.
DistValues	array of (array of float32)	Array containing an array of distance values for each echo of the scan. The number of sub-arrays corresponds to the number of echoes in the scan, see also the EchoCount field below. The distance values are specified in mm. The encoding of the data in this array is described in section 13.5.1.3.3.4 . The user can configure whether the array is present in the data structure. The structure of the nested array is illustrated in figure 59 .

Name	Type	Description
RssiValues	array of (array of uint16)	Array containing an array of RSSI values for each echo of the scan. The number of sub-arrays corresponds to the number of echoes in the scan, see also the EchoCount field below. For the representation of RSSI values, see "Representation of RSSI values", page 112. The encoding of the data in this array is described in section 13.5.1.3.3.4. The user can configure whether the array is present in the data structure. The structure of the nested array is illustrated in figure 59.
PropertyValues	array of uint8	Array with additional properties, for example "reflector", for each beam of a scan. See section 13.5.1.5.4 for details. The encoding of the data in this array is described in section 13.5.1.3.3.4. This array is optional.
BeamCount	MSGPACK int ¹⁾	Number of beams in the current scan.
EchoCount	MSGPACK int ¹⁾	Number of echoes in the current scan.

¹⁾ github.com/msgpack/msgpack/blob/master/spec.md#int-format-family

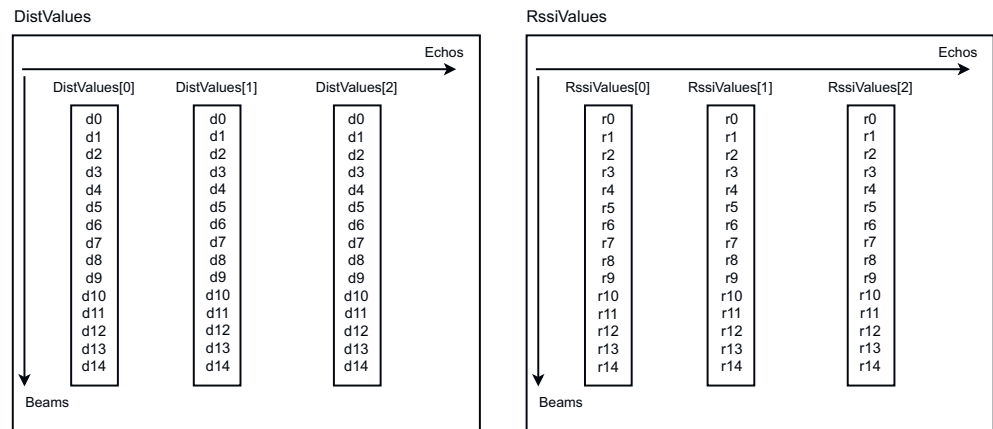


Figure 59: Example of the structure of the DistValues and RssiValues arrays. Shown here are data for a scan with 3 echoes and 15 beams.

13.5.1.3.3.4

Serialization of arrays

Arrays of measurement data (ChannelTheta, ChannelPhi, DistValues, RssiValues and PropertyValues from see table 24, page 96) are encoded as follows:

```
msgpack_map_header {
  "numOfElems": <number of array elements>,
  "elemSz", < size of one array element in bytes>,
  "endian", little,
  "elemTypes", typesArray, ,
  "data": binaryData
}
```

Definition: Definition of the **Array** class

The meaning of the individual fields of the **Array** class are shown in the following table.

Table 25: Description of the attributes of the **Array** class

Name	Description
numOfElems	Number of array elements
elemSz	Size of an array element in bytes, e.g., 4 for float32 or 1 for uint8.
endian	The value is always <code>little</code>
elemTypes	Array of element types. Only arrays with one element are supported here. Example: {float32} for an array of float32 values. To represent an array of tuples, elemTypes would therefore have multiple elements. This is not relevant, however.
data	Actual payload. The data is encoded as a byte array in the MSGPACK int ¹⁾ and can be copied directly into a structure/array after parsing the MSGPACK structure, taking into account the endianness and the data type.

¹⁾ github.com/msgpack/msgpack/blob/master/spec.md#int-format-family

13.5.1.4 Compact format

13.5.1.4.1 Working with Compact

SICK provides the following software tools for working with the Compact data format:

Wireshark Dissector

The Wireshark dissector is a plugin that extends Wireshark with full decoding support for the Compact format, including protocol details and filtering capabilities.

To use the dissector, the following components are required:

- The freely available Wireshark software: www.wireshark.org
- The SICK Compact dissector, including installation instructions, available on the SICK Support Portal support.sick.com (ID KA-08425)

LiDAR SDK

The SICK LiDAR SDK enables fast and seamless integration of SICK LiDAR sensors. It provides functionality for device configuration, handling acyclic device messages, and data transmission when the sensor is configured to use the Compact data format.

The SDK and its documentation are available on GitHub under the project name `sick_perception_sdk` on <https://github.com/SICKAG>.

Additional information and further documentation can be found on the SICK Support Portal: support.sick.com

13.5.1.4.2 Framing

The data packages transmitted in Compact format are enclosed in a frame consisting of a header before the actual payload and a checksum (CRC) after the payload.

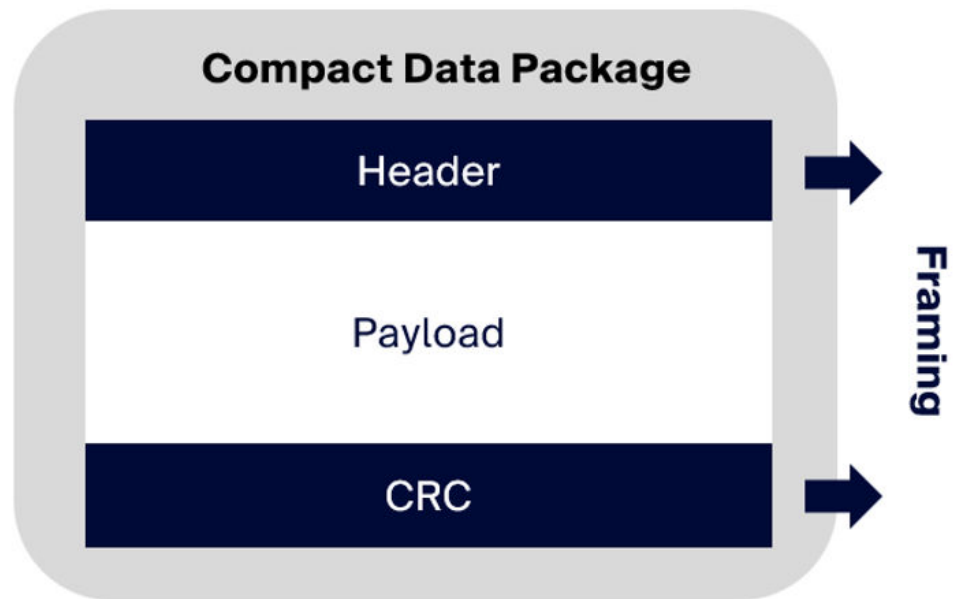


Figure 60: Framing in the Compact format, consisting of a header before the payload and a checksum after the payload.



NOTE

The payload content depends on the **telegramType** field of the frame header. Detailed payload description, deviations from the frame header defined in this chapter as well as **telegramType** specific frame header values are described in the chapter for each **telegramType**.

The structure of the Compact frame header is defined in the table below.

Table 26: Description of the fields of the Compact frame header.

Offset (bytes)	Name	Length (bytes)	Type	Description
0	Start of frame	4	uint32	Four <STX> characters (hex code 0x02): \x2\x2\x2\x2
4	Telegram type	4	uint32	Defines the telegram type and content that follows in the payload section of this frame. This field was named 'commandId' in previous versions of this documentation.
8	Telegram counter	8	uint64	Counts all telegrams sent since the device was switched on. The counter starts at 1.
16	Transmit time-stamp	8	uint64	The sensor system time in microseconds since January 1, 1970, 00:00 (UTC). If a time server is being used, the configured system time is used. This field was named 'timeStampTransmit' in previous versions of this documentation

Offset (bytes)	Name	Length (bytes)	Type	Description
24	Telegram version	4	uint32	Version of the payload for the telegram type defined in the header. The changes of the version are described in the telegram type specific chapters.
28	Payload length	4	uint32	Length of the payload excluding the checksum in bytes. This field was named 'sizeModule0' in previous versions of this documentation
32	Sender Id	4	uint32	Serial number of the sensor sending the data. To be used to identify the origin of the data packet.

The CRC32 checksum that follows the payload is calculated over the entire data package, i.e. over the header and the entire payload. All values in the header and the checksum are encoded as little endian.

13.5.1.4.3 telegramType 1: Primary Data – Spherical Coordinates

13.5.1.4.3.1

Header specifics

The following table describes the specific values and changes to the standard header described in [section 13.5.1.4.2](#) for telegram type 1 **Primary Data – Spherical Coordinate**.

Table 27: Description of the fields of the Compact frame header

Name	Offset (bytes)	Change compared to standard header
Telegram type	4	For telegram type Primary Data – Spherical Coordinates , this is set to 1.
Telegram version	24	For telegram type Primary Data – Spherical Coordinates , this value is set to 4 for the latest version.
Payload length	28	For telegram type Primary Data – Spherical Coordinates , this is the size of the first module.
Sender id	None	Does not exist for this telegram type in the header, following fields are shifted to the previous field without padding.

Table 28: Version history for the telegram version field of the telegram type **Primary Data – Spherical Coordinates**

Version	Note
3	First released version
4	Distance scaling factor was added to the metadata

13.5.1.4.3.2

Payload

The telegram type **Primary Data – Spherical Coordinates** is designed to transport 2D and 3D LiDAR sensor data. To map different sensor specific features, the payload field is separated into different modules that can group multiple layers of the sensor. Every module contains the actual measurement data and metadata describing this module's measurement data.

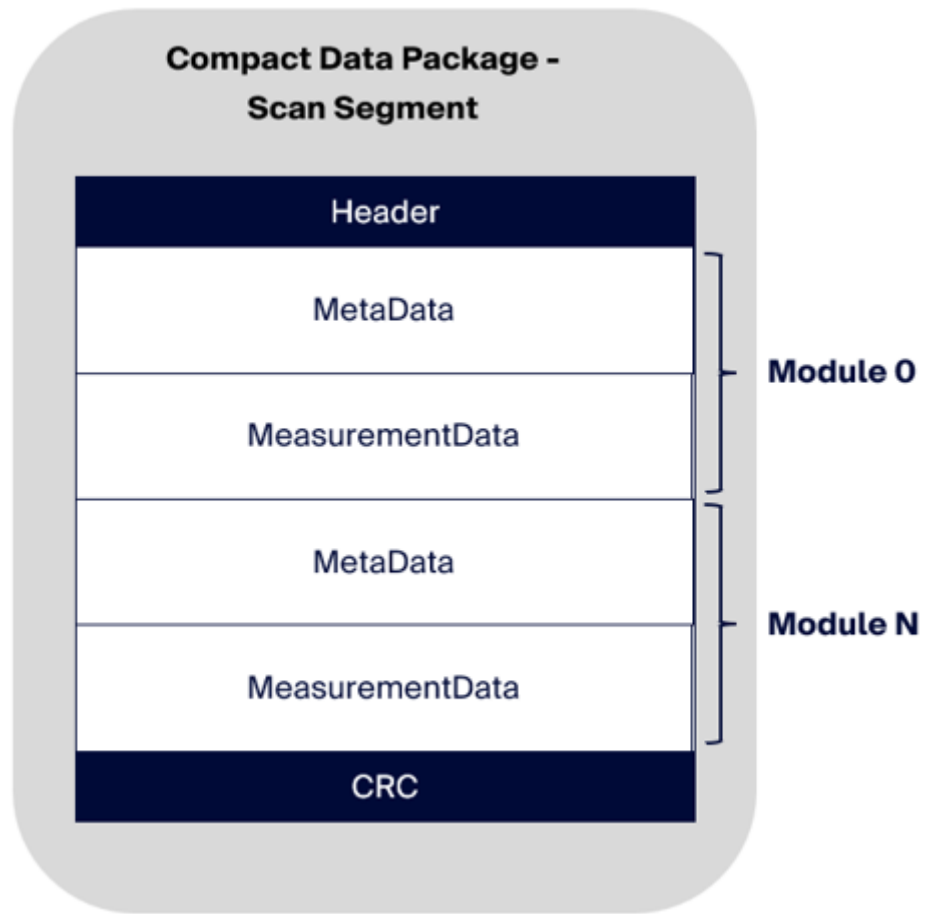


Figure 61: Schematic of the segmentation of the **Primary Data – Spherical Coordinates** payload with a variable number of modules and a Metadata and MeasurementData block per module.

The layers of a sensor are separated in a new module when either the sensor uses multiple physical measurement units or the sensor uses different angular resolutions (e.g. a 3D LiDAR with 0.5° resolution on 0° layer and 1° resolution for other layers would result in 2 modules when only a single physical measurement unit is used).

The following table shows the number of modules depending on the sensor family:

Sensor family	Number of modules
picoScan100	1

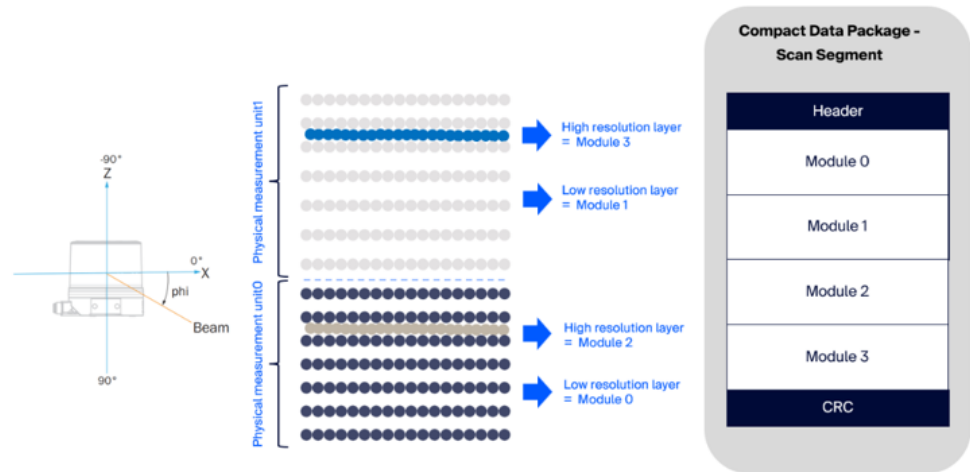


Figure 62: For the multiScan136, a segment consists of four modules, which are shown as colored dots in the figure: For each of the two physical measurement units 0 and 1 there are two modules: One for the layers with 1° angular resolution, and one for the high resolution layers with 1/8° angular resolution.

The modules available in a data package can vary depending on the configuration of the sensor. It is therefore recommended to process module by module when reading the data, as illustrated in the following pseudo code (Example of reading the individual modules from a data package):

```
Read the 32 Byte Header
Set currentModuleSize = payloadLength from the header
As long as currentModuleSize != 0
  Read next module of size currentModuleSize
  Set currentModuleSize = nextModuleSize from the module just read
```

13.5.1.4.3.3

Metadata

This section describes the metadata of a module.

Table 29: Metadata of a module for the Compact format – Version 4

Offset (bytes)	Name	Length (bytes)	Type	Description
32	Segment counter	8	uint64	Segment counter as described in see "Segment counter", page 112
40	Frame number	8	uint64	Counts the number of full revolutions since the device was started.
48	Sender id	4	uint32	Device serial code. It can be used to detect on the receiver which sensor the data was sent from.
52	Number of lines in module	4	uint32	Number of layers contained in one module, figure 65 .
56	Number of beams per scan	4	uint32	Number of beams per scan from one layer, figure 65 . Scans from all layers in a module have the same number of beams.
60	Number of echoes per beam	4	uint32	Number of echoes per beam, figure 64 .

Offset (bytes)	Name	Length (bytes)	Type	Description
64	Time stamp start	Number of elements * 8	array of uint64	Array of acquisition times for the first beam of each scan in the current module. Time base is the sensor system time in microseconds since January 1, 1970, 00:00 (UTC). If a time server is being used, the configured system time is used. The length of the array equals the Number of lines in module .
64 + N * 8 * 1	Time stamp stop	Number of elements * 8	array of uint64	Array of acquisition times for the last beam of each scan in the current module. Time base is the sensor system time in microseconds since January 1, 1970, 00:00 (UTC). If a time server is being used, the configured system time is used. The length of the array equals the Number of lines in module .
64 + N * 8 * 2	Phi	Number of elements * 4	array of float32	Array of elevation angles (cf. operating instructions, Coordinate system section) in radians of each layer in the current module. The length of the array equals the Number of lines in module .
64 + (N * 8) * 2 + (N * 4) * 1	Theta start	Number of elements * 4	array of float32	Array of azimuth angles in radians for the first beam of each scan of a layer in the current module. The length of the array equals the Number of lines in module .
64 + (N * 8) * 2 + (N * 4) * 2	Theta stop	Number of elements * 4	array of float32	Array of azimuth angles in radians for the last beam of each scan of a layer in the current module. The length of the equals the Number of lines in module .
64 + (N * 8) * 2 + (N * 4) * 3	Distance scaling factor	4	float32	This factor is used to scale the distance values in the beam data to allow the display of values over 65,535 mm with 16 bits - Formula: $d_mm_external = DistanceScalingFactor * d$ d is the distance value contained in the beam data. d_mm_external is the distance value in mm which can be derived from a consumer of streaming data from d. The factor can always be rounded to the next whole number.
68 + (N * 8) * 2 + (N * 4) * 3	Next module size	4	uint32	Size of the next module, or 0 if the current module is the last one. This value is important when reading the data using the principle in / figure 63 . For 2D LiDAR always 0.
72 + (N * 8) * 2 + (N * 4) * 3	Reserved	1	uint8	–
73 + (N * 8) * 2 + (N * 4) * 3	Data content echos	1	uint8	The individual bits of this byte describe which data are available in that part of the measurement data that is acquired per echo, e.g., distance or RSSI, table 30 .
74 + (N * 8) * 2 + (N * 4) * 3	Data content beams	1	uint8	The individual bits of this byte describe which data are available in that part of the measurement data that is only acquired once per beam, e.g., azimuth angle or beam properties, table 31 .
75 + (N * 8) * 2 + (N * 4) * 3	Reserved	1	uint8	Reserved field

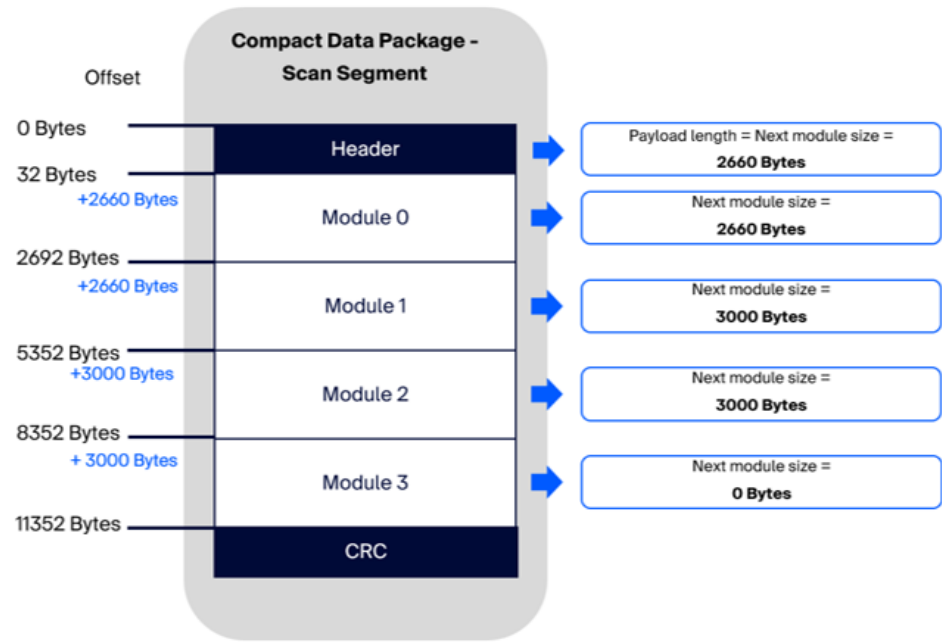


Figure 63: Example of the **NextModuleSize** field for multiScan136. The modules with the 1° layers are 2,660 bytes in size, the modules with the HighRes layers are 3,000 bytes in size. The value for **NextModuleSize** in the header as well as in the first serialized module is therefore 2660 bytes, and in the two subsequent modules 3000 bytes. In the last serialized module the value is 0, because no further module follows.

The bit indices in the **DataContentEchos** and **DataContentBeams** bytes are derived as described in [see "Representation of bit fields", page 112](#).

Table 30: Description of the bits of Data content echos

Bit index	Value
0	1 if distance data is available, otherwise 0.
1	1 if RSSI data is available, otherwise 0.
2	Reserved
3	Reserved
4	Reserved
5	Reserved
6	Reserved
7	Reserved

Table 31: Description of the bits of Data content beams

Bit index	Value
0	1 if additional beam properties are available, otherwise 0.
1	1 if azimuth angles per beam are available, otherwise 0.
2	Reserved
3	Reserved
4	Reserved
5	Reserved
6	Reserved
7	Reserved

13.5.1.4.3.3.1

Measurement data

Each beam of a scan is represented as a tuple whose elements are represented by the bytes (table 30) and **Data content beams** (table 31). The data available per beam (as defined by **Data content beams**) follows. The representation of the respective contents is shown in the following table.

Name	Lenght (bytes)	Type	Description
Distance	2	uint16	The measured distance can be computed as follows: Distance * Distance scaling factor = Distance in mm For "Distance scaling factor" = 1 this value is already the measured distance.
RSSI	2	uint16	see "Representation of RSSI values", page 112
Beam characteristics ("Properties")	1	uint8	see "Representation of beam characteristics (Properties)", page 113
Azimuth angle	2	uint16	uint16 integers, where the following conversion applies: • a_uint: Angle value as integer • a_rad: Angle value in radians. • $a_rad = (a_uint - 16384)/5215$ This conversion ensures that the maximum allowed value range of $[-\pi, 3\pi]$ is fully utilized.

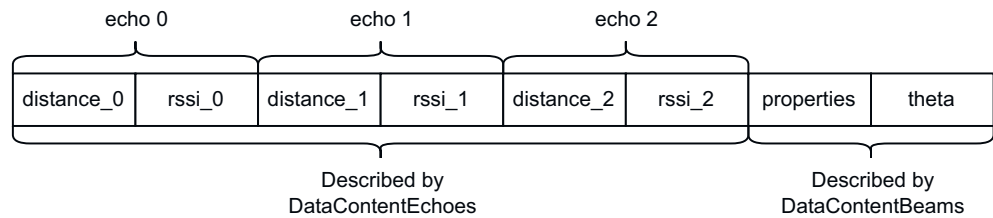
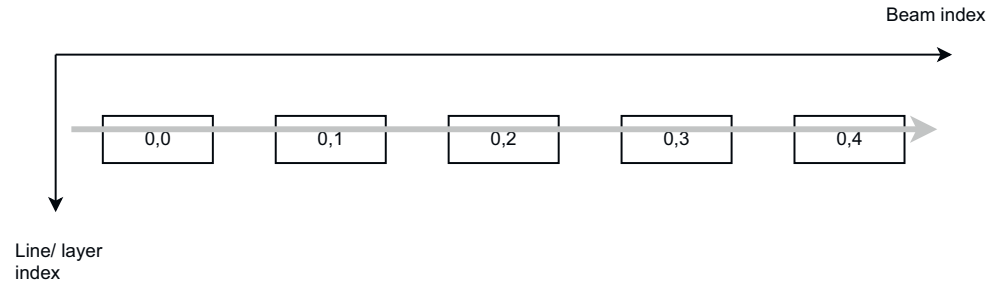


Figure 64: Example representation of a beam as a tuple. There are three echoes available. Both distance and RSSI values exist per echo, i.e., the corresponding bits of DataContentEcho have the value 1. Values for beam properties and azimuth angle are also available, i.e., the corresponding bits of DataContentBeams have the value 1.

The individual tuples are located directly behind each other in the data stream. Their sequence is described in figure 65. Here the data is arranged in a matrix where the individual layers of the current module correspond to the rows, and the columns correspond to the measurement beams of the scans for the individual layers. (Note: This arrangement assumes that all scans of a module have the same length, which is ensured by the division into modules. The order of the beam tuples in the data stream is determined by sweeping the matrix shown in figure 65 column by column, i.e., the tuples of beam index 0 for all layers are incorporated into the data stream first, then the tuples for beam index 1, etc.

Module with one single layer



Module with multiple layers

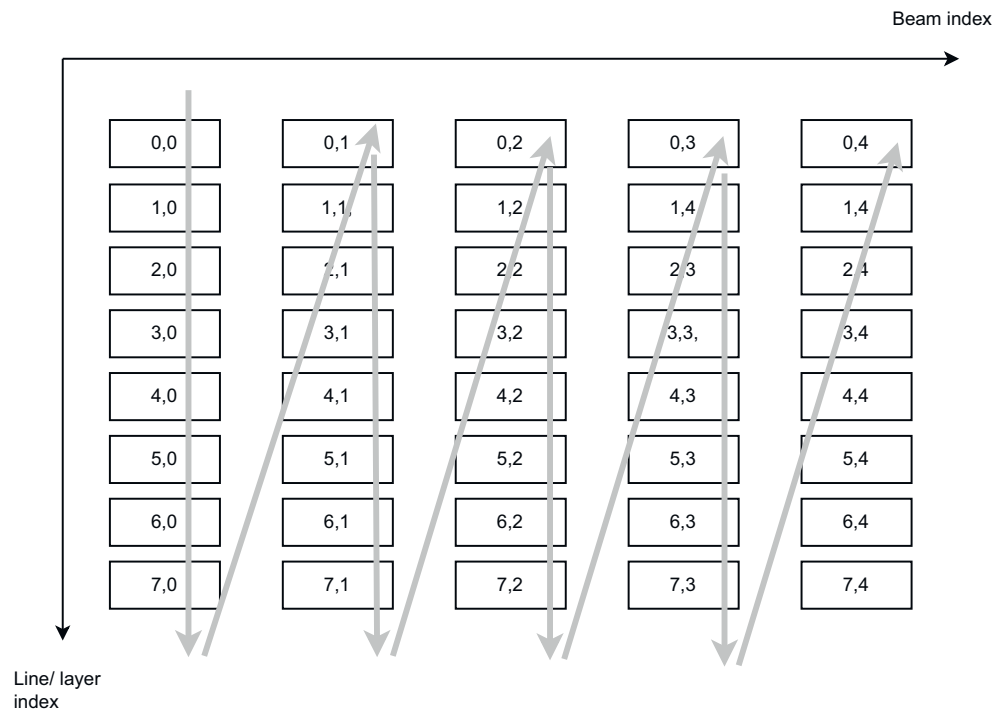


Figure 65: Sequence of data tuples in the memory for the individual beams. Top: Module with only one layer. Bottom: Module with 8 layers. The individual layers of a module correspond to the rows, and the individual measurement beams correspond to the columns. For linear storage of the data in the data package, this matrix is then swept column by column, as indicated by the gray arrows in the figure.

- The elevation angle of the data in row i is $\text{Phi}[i]$ from the metadata.
- The azimuth angle of the first beam in line i is $\text{ThetaStart}[i]$
- The azimuth angle of the last beam in line i is $\text{ThetaStop}[i]$

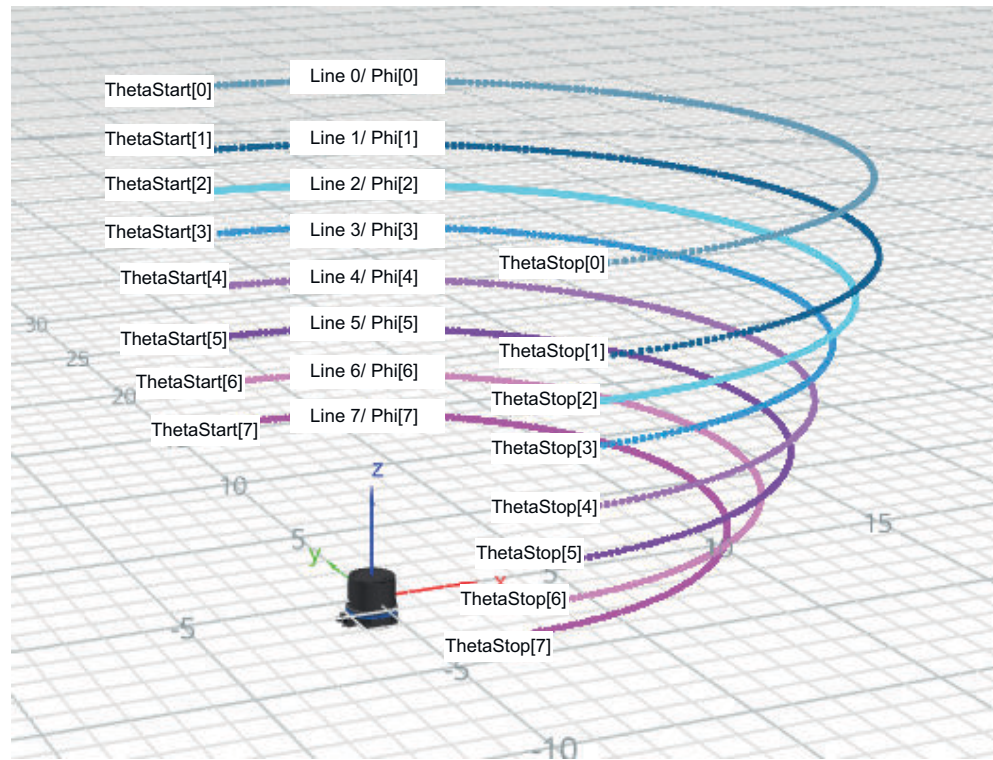


Figure 66: Correspondences between measurement data and metadata using the example of a module with 8 layers. Each line from [figure 65](#) corresponds to a layer in [figure 64](#). The first and last azimuth angles of each layer, and the elevation angles of each layer are in the Phi, ThetaStart and ThetaStop fields.

NOTE | LayerIds, which define the individual layers by their elevation angle as in the MSGPACK format ([see table 23, page 95](#)), are not used in the Compact format. The row index of the matrix with the data relates only to the current module and is therefore generally not the same as the LayerId.

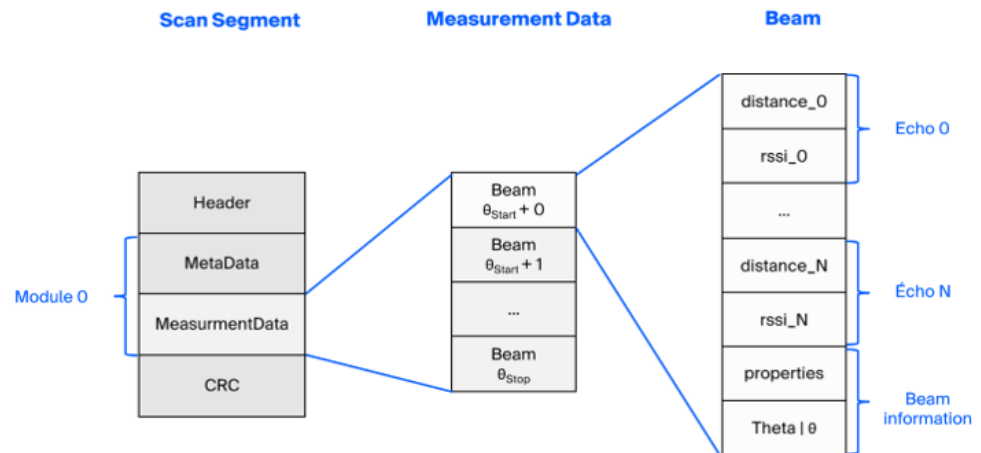


Figure 67: Data structure example for a 2D sensor with a single module

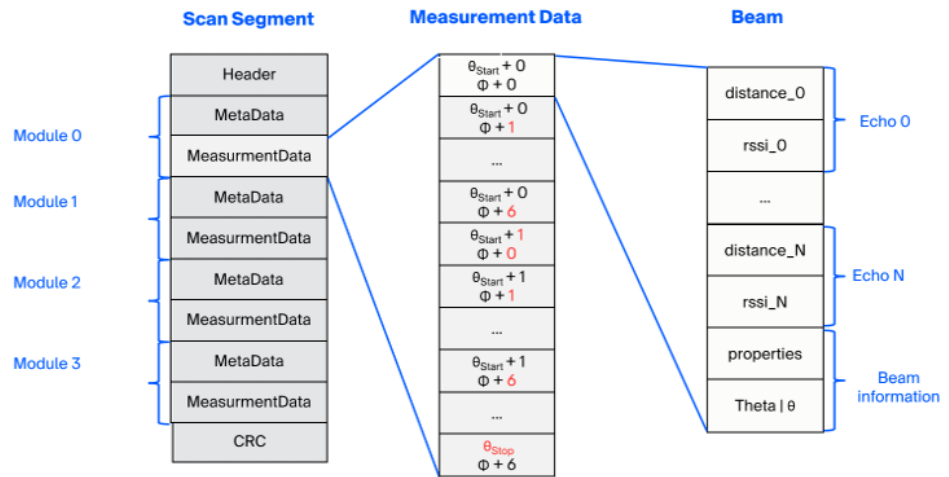


Figure 68: Data structure example for a 3D sensor with a four module

13.5.1.4.4 telegramType2: IMU

13.5.1.4.4.1 Header specifics

The telegram type 2 **IMU** uses a reduced header compared to the standard Header described in [section 13.5.1.4.2](#). The following table describes the specific values and changes to the standard header for telegram type **IMU**.

Table 32: Description of the fields of the Compact frame header

Name	Offset (bytes)	Change compared to standard header
Telegram type	4	For telegram type IMU , this is set to 2.
Telegram counter	None	Does not exist for this telegram type in the header, following fields are shifted to the previous field without padding.
Transmit timestamp	None	Does not exist for this telegram type in the header, following fields are shifted to the previous field without padding.
Telegram version	24	For telegram type: IMU , this value is set to 1 for the latest version.
Payload length	None	Does not exist for this telegram type in the header, following fields are shifted to the previous field without padding.
Sender id	None	Does not exist for this telegram type in the header, following fields are shifted to the previous field without padding.

Table 33: Version history for the telegram version field of the telegram type: **IMU**

Version	Note
1	First released version

The reduced header breaks down to the following structure.

Table 34: Payload of the telegram type 2 **IMU**

Offset (bytes)	Name	Length (bytes)	Type	Description
0	Start of frame	4	uint32	Always four STX characters: \x2\x2\x2\x2.
4	Telegram type	4	uint32	For telegram type: IMU , this is set to 2

Offset (bytes)	Name	Length (bytes)	Type	Description
8	Telegram version	4	uint32	For telegram type: IMU, this value is set to 1 for the latest version

13.5.1.4.4.2

Payload

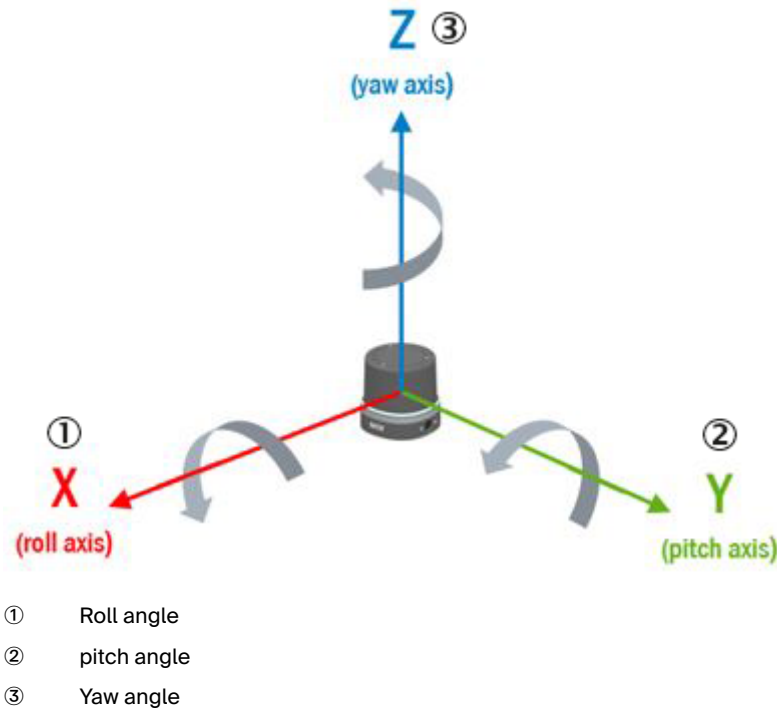
The telegram type **IMU** is designed to efficiently transport device internal IMU information. The reduced standard header described in [section 13.5.1.4.4.1](#) with the payload defined in the table below:

Table 35: Payload of the telegram type 2 IMU- Version 1

Offset (bytes)	Name	Length (bytes)	Type	Description
12	Acceleration x	4	float	Acceleration in m/s ² along the x-axis including gravity; gravity is not subtracted from the data.
16	Acceleration y	4	float	Acceleration in m/s ² along the y-axis including gravity; gravity is not subtracted from the data.
20	Acceleration z	4	float	Acceleration in m/s ² along the z-axis including gravity; gravity is not subtracted from the data.
24	Angular velocity x	4	float	Angular velocity in rad/s along the x-axis.
28	Angular velocity y	4	float	Angular velocity in rad/s along the y-axis.
32	Angular velocity z	4	float	Angular velocity in rad/s along the z-axis.
36	Orientation quaternion w	4	float	Quaternion orientation for w-axis.
40	Orientation quaternion x	4	float	Quaternion orientation for x-axis.
44	Orientation quaternion y	4	float	Quaternion orientation for y-axis.
48	Orientation quaternion z	4	float	Quaternion orientation for z-axis.
52	Sensor time stamp	8	uint64	The sensor system time in microseconds since January 1, 1970, 00:00 (UTC). If a time server is being used, the configured system time is used.

The word size is 32 bits.

The data is specified in the following coordinate system based on the DIN 70000 system: The x-axis lies on the 0° beam of the 0° plane. The y-axis is perpendicular to the x-axis and lies in the 0° plane. The y-values increase in the counter rotation direction (right-handed system). The z-axis is perpendicular to the y-axis and the top of the device points towards increasing z-values.



13.5.1.4.5 telegramType 4: Encoder

13.5.1.4.5.1 Header specifics

The following table describes the specific values and changes to the standard header described in [section 13.5.1.4.2](#) for telegram type 4 **Encoder**.

Table 36: Description of the fields of the Compact frame header

Name	Offset (bytes)	Change compared to standard header
Telegram type	4	For telegram type Encoder , this is set to 4.
Telegram version	24	For telegram type Encoder , this value is set to 1 or 2.

Table 37: Version history for the telegram version field of the telegram type: **Encoder**

Version	Note
1	First released version, Sender id is not part of the payload
2	Sender id is now defined as part of the Compact header. Payload length field for this data type is reduced by 4 bytes. The physical frame length stays unaffected.

13.5.1.4.5.2 Payload

The **Encoder** telegram type is intended for applications in which an external encoder is physically connected to the LiDAR. Its format enables combining the encoder's positional information with the distance data provided by the LiDAR.

The **Frame Number** field within the **Encoder** telegram payload is always synchronized with the **Frame Number** of the **Primary Data – Spherical Coordinates** telegram type contained in the payload metadata. This ensures that LiDAR distance measurements and external encoder information can be merged reliably and with minimal effort.

Table 38: Payload of the telegram type 4 **Encoder** - Version 2

Offset (bytes)	Name	Length (bytes)	Type	Description
36	Frame number	8	uint64	Frame number of the corresponding Primary Data frame where the encoder frame belongs to.
44	Tick counter value	4	uint32	Tick counter value of the encoder from 0 ... $2^{32}-1$
48	Tick counter value reference signal 1	4	uint32	Tick counter value of the encoder reference signal 1 from 0 ... $2^{32}-1$
52	Tick counter value reference signal 2	4	uint32	Tick counter value of the encoder reference signal 2 from 0 ... $2^{32}-1$
56	Speed value	4	float	Moving speed. This value depends on the configured conversion rate per second.
60	Tick counter value time-stamp	8	uint64	Time when the tick value counter last changed. The sensor system time in microseconds since January 1, 1970, 00:00 (UTC). If a time server is being used, the configured system time is used.
68	reference signal 1 time-stamp	8	uint64	Time when the tick value counter last changed. The sensor system time in microseconds since January 1, 1970, 00:00 (UTC). If a time server is being used, the configured system time is used.
76	reference signal 2 timestamp	8	uint64	Time when the tick value counter last changed. The sensor system time in microseconds since January 1, 1970, 00:00 (UTC). If a time server is being used, the configured system time is used.

13.5.1.5 General measurement data definitions

13.5.1.5.1 Azimuth angle

The azimuth angles (also called theta angles in the data formats) of a scan are always monotonically increasing.

For sensors that have a horizontal measuring field of 360° , this means in particular that the angles of a segment can be greater than 180° if the segment exceeds the $+180^\circ/-180^\circ$ limit:

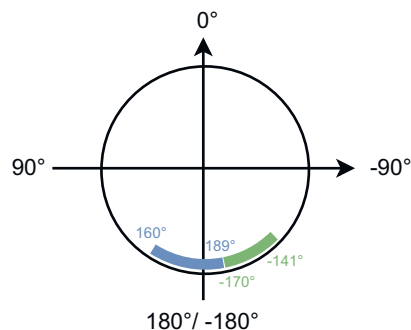


Figure 69: Azimuth angle for different segments. The angular range of the blue segment exceeds the $180^\circ/-180^\circ$ limit. Since the azimuth angles are nevertheless monotonically ascending, angles $> 180^\circ$ are used here. The green segment already starts in the negative angular range, which is why the azimuth angles are negative here.

13.5.1.5.2

Segment counter

The segment counter counts the segments in a frame. A frame is all the data recorded in one revolution. The segment counter is a value between 0 and $< \text{number of segments per revolution} - 1$ and increases with increasing azimuth angle.

An exception to this is shown in [figure 70](#): Here the beams of a segment exceed the $+180^\circ/-180^\circ$ limit so most azimuth angles of this segment have values $> 180^\circ$, [see "Azimuth angle", page 111](#). (This would correspond to negative values close to -180° when normalized to $(-180^\circ, +180^\circ]$.) If the majority of the values of the segment are $> 180^\circ$, this segment is assigned the segment counter 0 again. This only applies to the multiScan100 due to the specific arrangement of the lasers in the measuring module.

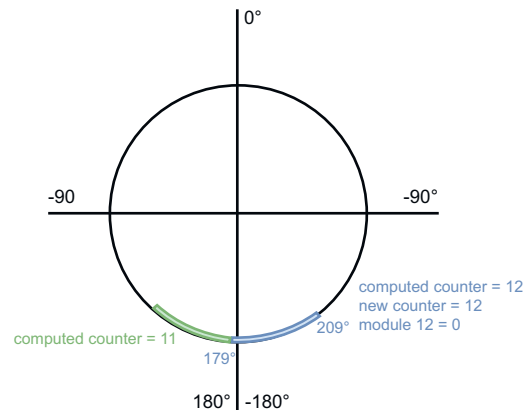


Figure 70: Segment counter for a segment (blue), the majority of whose azimuth angles are $> 180^\circ$. Assuming a segment size of 30° , the segment counter would be 12 if it were calculated only on the basis of the angle values. The majority of the blue segment is already in the next frame, however, which is why it is assigned the segment number 0.

13.5.1.5.3

Representation of RSSI values

RSSI values are represented as a 16-bit integer (uint16). The RSSI is a dimensionless quantity. The values can fall within the complete value range between 0 and $2^{16}-1$, whereby it is possible that the maximum value is rarely or even never reached. The RSSI is generally also not standardized and therefore not exactly comparable between devices.

13.5.1.5.4

Representation of bit fields

For both the DataContentEchos and DataContentBeams values and for the beam properties ([see "Representation of beam characteristics \(Properties\)", page 113](#)), bytes are interpreted as bit fields in which information about the states of the individual bits is encoded.

The relationship between the representation of the byte as a decimal value and its bit indices is as follows:

- Let b_i with $i = 0, \dots, k$ be the bits with index i in the binary representation of a value v with, $b_i \in \{0,1\}$.
- The decimal representation v_{dec} of v is then given as $v_{\text{dec}} = \sum_{i=0..7} b_i \cdot 2^i$

Examples:

Let v be an 8-bit value with decimal representation v_{dec} and hexadecimal representation v_{hex} . Let the bit indices of v be b_0 to b_7 .

Table 39: Examples of the relationship between the bit indices and the decimal representation of a value

Bit number	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Bit value	1	1	1	1	1	1	1	1
Bit position	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
Decimal value	128	64	32	16	8	4	2	1

Table 40: Examples for the decimal value 128.

Bit number	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Bit value	1	0	0	0	0	0	0	0
Bit position	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
Decimal value	128							
Hexadecimal value	0x80							

Table 41: Examples for the decimal value 1.

Bit number	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Bit value	0	0	0	0	0	0	0	1
Bit position	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
Decimal value	1							
Hexadecimal value	0x01							

Table 42: Examples for the decimal value 130.

Bit number	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Bit position	1	0	0	0	0	0	1	0
Bit position	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
Decimal value	130							
Hexadecimal value	0x82							

13.5.1.5.5

Representation of beam characteristics (Properties)

The additional characteristics of a beam are called “properties” here. They are encoded in a bit field according to ["Representation of bit fields", page 112](#). The meaning of the individual bit indices is shown in the following table.

Table 43: Description of the bits of the field for beam characteristics (Properties)

Bit index	Content
0	1 if a reflector was detected for any echo on this beam, otherwise 0.
1	Reserved
2	Reserved
3	Reserved
4	Reserved
5	Reserved
6	Reserved
7	Reserved

If the reflector bit (bit with index 0) is set for a beam, it can be assumed that the last echo measured for this beam came from a reflector. It is virtually impossible physically for a reflector to be measured for an echo and then be followed by another on the same beam.

13.5.1.6 Behavior of serialization for data reduction

Different data reduction options (e.g., limiting the number of available echoes, limiting the azimuth angular range, limiting the layers used) affect the data output in different ways.

The behavior is somewhat different for the MSGPACK format and the Compact format. The data reduction can be configured either via all available configuration paths. These effects are described in this section.

13.5.1.6.1 Behavior in relation to the number of available echoes

Multi-echo capable sensors usually provide the option to define the number of available echoes by means of an echo filter. This then affects the data output as follows:

All echoes setting

If the sensor is configured so that all echoes are available, then all theoretically available echoes are also always serialized, regardless of the number of echoes actually received per beam. If fewer echoes than theoretically available are received on a beam, the measured values for the remaining echoes (distance and RSSI) are padded with 0 (see also the example in [table 44](#)).

Single echo setting

If the sensor is configured so that only a single echo is available (e.g., last or first echo), only the data for this echo is serialized.

Table 44: Example data output for different numbers of available echoes

Angles	All echoes			Last echo
	Distance echo 0	Distance echo 1	Distance echo 2	Distance
42°	3,422 mm	5,022 mm	0 mm	5,022 mm
43°	3,420 mm	0 mm	0 mm	3,420 mm
44°	0 mm	0 mm	0 mm	0 mm

Example:

- The sensor is capable of measuring a maximum of 3 echoes, and it is configured to output all echoes (**All Echoes** columns) or the last echo (**Last Echo** column).
- For the angle 42° the measured distance is 3,422 mm for echo 0 and 5,022mm for echo 1, for the angle 43° the measured distance is 3,420 mm for echo 0, and for angle 44° no valid distance is measured.
- The following distance values are then output (RSSI values are omitted for reasons of clarity, they are treated in the same way as distance values):

13.5.1.6.2 Behavior when restricting the azimuth angular range

If a restricted azimuth angular range is configured for the data output, it is possible for complete segments to lie outside the configured angular range. These segments are not serialized and therefore not transmitted from the device to the client.

For sensors with different physical measuring modules, for example the multiScan1xx, it should be noted that complete segments only lie outside the configured angular range if this applies to the data of both measuring modules. If data from one measuring module only lies within the available angular range, the segment will still be output. The layers of the other measuring module for which the data are outside the available angular range, are then treated as follows:

MSGPACK format

If the data of a layer is completely outside the configured angular range, that layer will not be output, [see "Behavior when reducing the available layers", page 115](#). As soon as at least one beam is within the configured angular range, both the distance value and the RSSI value are set to 0 for the remaining beams of the segment, and the azimuth angle remains unchanged.

Compact format

If the data of a layer are completely outside the configured angular range, then in contrast to the MSGPACK format the distance values and the RSSI values are set to 0 for that layer, and the azimuth value is output unchanged. As soon as at least one beam is within the configured angular range, the procedure is the same as for the MSGPACK format: For the remaining beams of the segment, both the distance value and the RSSI value are set to 0, and the azimuth angle remains unchanged.

The different behavior for the MSGPACK format and the Compact format is due to the fact that in the case of the Compact format, the data structure needs to remain the same for each segment so the data can be easily interpreted (e.g., via memcpy into a structure). This is not necessary with the MSGPACK format, where the data has to be parsed anyway. A more extensive data reduction is therefore possible in this case.

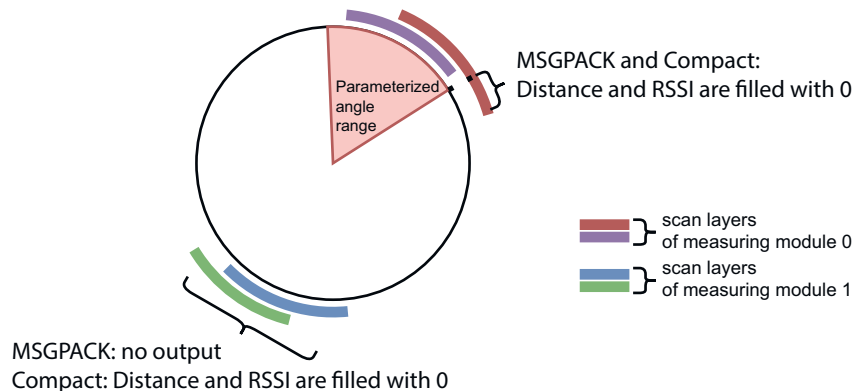


Figure 71: Handling of data that is outside the configured angular range. Data from 4 layers are shown, all belonging to one segment but to different measuring modules. red and violet: measuring module 0, blue and green: measuring module 1. With no restriction of the angular range, the data would be output for all layers. With the angular range restriction shown (red circle segment), the data is output as follows: The blue layer and the green layer are not output in the MSGPACK format, and in the Compact format distance and RSSI are filled with 0. Distance and RSSI values in the part of the red layer outside the configured angular range are padded with 0 in the MSGPACK format and Compact format.

13.5.1.6.3

Behavior when reducing the available layers

For multilayer sensors, for example the multiScan1xx, it is possible for individual layers to fall within the measuring range excluded by data reduction. This can occur, for example, when using a layer filter. The layers for the complete azimuth angular range are excluded. The case where the data for a layer are outside the measuring range only in individual segments is described in ["Behavior when restricting the azimuth angular range", page 114](#).

For the MSGPACK format, this affects the serialization as follows:

- Scan type data for the layers that have been excluded will no longer appear in the SegmentData array ([see table 23, page 95](#)).
- The LayerId array ([see table 23, page 95](#)) is adjusted accordingly, i.e., the IDs of the layers that are no longer present are removed.

For the Compact format, this affects the serialization as follow:

- The data for the layers that were excluded are removed from the measurement data block, i.e., the corresponding rows of the matrix in [figure 65](#) are removed.
- The metadata in the metadata block that are associated with this line will be removed as well. The arrays affected are ThetaStart, ThetaStop, TimeStampStart, TimeStampStop and Phi in [table 29](#).
- If all layers of a module are outside the configured measuring range, the entire module (metadata and measurement data) is not output.

13.6 Telegram listing (EN)

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13.6.1.1 About this document

13.6.1.1.1 Information on the telegram listing



NOTE

In case you prefer to use complete drivers instead of single telegrams, the following options are available:

C++ drivers: https://github.com/SICKAG/sick_perception_sdk

ROS2 drivers: https://github.com/SICKAG/sick_scan_xd



NOTE

Telegrams that are not described in this document for the device should not be implemented as they may either be incompatible or cause undesired effects.



NOTE

CoLa 2 is a SICK specific communication protocol which is used for communication between SICK devices and SICK specific tools and services only.

Please read this chapter carefully before beginning to use the telegram listing.

The telegram listing shows how to send telegrams via a terminal program using the SICK protocol CoLa A (ASCII and hexadecimal values, with TCP port 2111 or 2112) or CoLa B (binary/hexadecimal values, with TCP port 2112 only) to the device. This comprises the query of the current device state or certain parameter values, how to modify parameter values and the way in which the device confirms or responds to commands/telegrams.

The devices generally support automatic IP address discovery.

Default IP address is:

- 192.168.0.1

Subnet mask is 255.255.255.0.

IP ports:

- 2111: CoLa A (fixed)
- 2112: CoLa B (fixed)

Most parameter changes also require certain user levels. Additionally, commands may change during the product lifecycle and development process with a new firmware.

This document is based on the following firmware version (or newer):

- V2.1.0

If commands do not seem to work, please verify that your device version supports this functionality, that the minimum required user level has been selected and check on updates of this documentation.

13.6.1.1.2**Explanation of symbols****NOTE**

... highlights useful tips and recommendations as well as information for efficient and trouble-free operation.



Telegram to device



Telegram from device



Unformatted example to copy and paste

13.6.1.2**Communication format****13.6.1.2.1****Binary telegram (CoLa B)**

The binary telegram is a basic protocol of the scanner (CoLa B). All values are in hexadecimal code and grouped into pairs of two digits (= 1 byte). The string consists of four parts: header, data length, data and checksum (CS). It is highly recommended to use this protocol as the transmitted data amount is only about half as much as with CoLa A).

The header indicates with 4 × STX (02 02 02 02) the start of the telegram.

The data length defines the size of the data part (command part) by indicating the number of digit pairs in the third part. The size of the data length itself is 4 bytes, which means that the data part might have a maximum of $16^8 = 4,294,967,295$ digit pairs (bytes).

The data part comprises the actual command with letters and characters converted to Hex (according to the ASCII chart) and the parameters of either decimal numbers converted to Hex or fixed Hex values with a specific, intrinsic meaning (no conversion). There is always a space (20) between the command and the parameters, but not between the different parameter values.

The checksum finally serves to verify that the telegram has been transferred correctly. The length of the checksum is 1 byte, CRC8. It is calculated with XOR.

Table 45: Example: Binary telegram

02 02 02 02	00 00 00 17	73 4D 4E 20 53 65 74 41 63 63 65 73 73 4D 6F 64 65 20 03 F4 72 47 44	B3
Header	Length	Data	CS

This is an example telegram for setting the user level “Authorized Client”:

- Header = 02 02 02 02
- Length = 23 bytes (17h)
- Data:
 - 73 4D 4E 20 = sMN = start of Sopas command (and space)
 - 53 65 74 41 63 63 65 73 73 4D 6F 64 65 20 = Set Access Mode = the actual command for setting the user level (and space)
 - 03 = fixed Hex value meaning user level “Authorized Client”
 - F4 72 47 44 = fixed Hex value, serving as password for the selected user level “Authorized Client”
- Checksum = B3 from XOR calculation

13.6.1.2.2

ASCII telegram (CoLa A)

The ASCII telegram is an alternative to the binary telegram, suitable especially to parametrize the sensor. However, due to the variable string length of ASCII telegrams, the Binary telegram is still recommended when using scanners with a PLC.

The ASCII telegram has the advantage that commands can be written in plaintext. The string consists only of two parts: the framing and the data part.

The framing indicates with <STX> and <ETX> the start and stop of each telegram.

The data part comprises the actual command with letters and characters (plaintext), parameter values either in decimal (special indicator required) or in hexadecimal (example: a frequency of 25 Hz = +2500 (decimal) = 09C4 (Hex)) and fixed hexadecimal values with a specific, intrinsic meaning.



NOTE

Leading zeros are deleted in ASCII. Therefore a space is always required between all command parts and parameter parts.

As further alternative within CoLa A, depending on the preferences of the user, all values can be written directly in Hex. This means however a 1:1 conversion of all letters and characters including numbers and fixed hexadecimal values via the ASCII chart.



NOTE

The device will confirm parameter values always in hexadecimal code, regardless of the code sent.

Table 46: Example: ASCII telegram

ASCII	<STX>	sMN{SPC}SetAccessMode{SPC}03{SPC}F4724744	<ETX>
Hex	02	73 4D 4E 20 53 65 74 41 63 63 65 73 73 4D 6F 64 65 20 30 33 20 46 34 37 32 34 37 34 34	03
	Start	Data	Stop

This is again an example telegram for setting the user level “Authorized Client”. As only fixed hexadecimal parameter values are needed, the option to use parameter values in decimal code with special indicator cannot be applied here:

- Framing = <STX> = telegram start = 02 (Hex)
- Data:
 - sMN = start of Sopas command (and blank) = 73 4D 4E 20 (Hex)
 - SetAccessMode = the actual command for setting the user level (and blank) = 53 65 74 41 63 63 65 73 73 4D 6F 64 65 20 (Hex)

- 03 = fixed Hex value meaning user level “Authorized Client” (and blank) = 30 33 20 (Hex)
- F4 72 47 44 = fixed Hex value, serving as password for the selected user level “Authorized Client” = 46 34 37 32 34 37 34 34 (Hex)
- Framing = <ETX> = telegram stop = 03 (Hex)

13.6.1.2.3

Variable types

Variable type	Length (byte)	Value range	Sign
Bool_1	1	0 or 1	No
UInt_8	1	0 ... 255	No
Int_8	1	-128 ... +127	Yes
UInt_16	2	0 ... 65,535	No
Int_16	2	-32,768 ... +32,767	Yes
UInt_32	4	0 ... 4,294,967,295	No
Int_32	4	-2,147,483,648 ... +2,147,483,647	Yes
ULInt_64	8	0 ... 18446744073709551616	No
Enum_8	1	Certain values defined in a list of Choices (0 ... 255)	No
Enum_16	2	Certain values defined in a list of Choices (0 ... 65535)	No
String	Context-depend-ent	Strings are not terminated in zeroes	
FlexString	array of visible characters with preceeding current length (UInt lenght) (array of 8 bit)	See description of String and FlexArray	
Real	4	Float according to IEEE754 (see www.h-schmidt.net/FloatConverter/IEEE754de.html)	Yes

Data length is always given in Bytes!

Struct	A structure is a sequence of further types. These types can be of a BasicType, Structs again or an Array.
Array	An Array is a repetition of a type. The length of the array is defined with each Array. The types can be of a BasicType, a Struct or an Array again (n-dimensional).
Flex Array	A FlexArray is a repetition of a type with a variable length. The maximum length of the array is defined with each FlexArray. The current length of the FlexArray is transferred as a UInt preceeding the Array itself. The types can be of a BasicType, a Struct or an Array again (n- dimensional).

13.6.1.2.4

Command basics

SOPAS communication is a index based communication and can be identified with telegram beginning of: sRI, sWI, sMI, sAI, sEI, sSI. Since the parallel usage of one port might be confusing, the usage of separate ports is advised.

Every response telegram starts with a separat framed string:

<STX>sSI 2 1<ETX><STX>“Answer”<ETX>

Description	Value ASCII	Value Hex	Value Binary
Start of text	<STX>	02	02 02 02 02 + given length
End of text	<ETX>	03	Calculated checksum
Read	sRN		73 52 4E
Write	sWN		73 57 4E

Description	Value ASCII	Value Hex	Value Binary
Method	sMN	73 4D 4E	
Event	sEN	73 45 4E	
Answer	sRA	73 52 41	
	sWA	73 57 41	
	sAN	73 41 4E	
	sEA	73 45 41	
	sSN	73 53 4E	
Space	{SPC}	20	20

If values are divided into two parts (e.g. measurement data), they are documented according to LSB 0 (e.g. 00 07), output however is according to MSB (e.g. 07 00).



NOTE

Every write command (sWN) has a read (sRN) counter part even if it may not be described in the Telegram Listing. You can verify the current parameters this way, e.g. read the current IP address via sRN EIIPAddr [see "Read IP address \[sRN EIIPAddr\]", page 231](#).

13.6.1.2.5

Log in: Required user level

Task	Required user level
Change sensor parameters	Authorized Client
Requests or queries (e.g. for measurement data or device state)	None
Manage passwords	Service

In general, every sWN command for changing parameters requires to log in to the device first [see "Log in \[sMN SetAccessMode\]", page 126](#). When being logged in, any desired parameter valid for this user level can be changed. All changes become active only after having logged off again from the device via the sMN Run command [see "Set to run \[sMN Run\]", page 151](#).

In this document, a required, specific user level is indicated in the telegram structure head line.

13.6.1.3 Workflows

13.6.1.3.1 Parameterize the scan

Log in: sMN SetAccessMode [see "Log in \[sMN SetAccessMode\]", page 126](#)

Store parameters: sMN mEEwriteall [see "Save parameters permanently \[sMN mEE-writeall\]", page 150](#)

Log out: sMN Run [see "Set to run \[sMN Run\]", page 151](#)

13.6.1.3.2 Common telegrams

The following telegrams are valid for a wide range of non-safe LiDAR sensors from SICK. Please refer to the telegram listing of the respective device for a detailed description of all valid telegrams.

["Log in \[sMN SetAccessMode\]", page 126](#)

["Start measurement \[sMN LMCstartmeas\]", page 130](#)

["Stop measurement \[sMN LMCstopmeas\]", page 131](#)

["Load factory defaults \[sMN mSCloadfacdef\]", page 132](#)

["Load application defaults \[sMN mSCloadappdef\]", page 133](#)

["Check password \[sMN CheckPassword\]", page 136](#)

"Reboot device [sMN mSCreboot]", page 151
"Save parameters permanently [sMN mEEwriteall]", page 150
"Set to run [sMN Run]", page 151
"Poll one telegram [sRN LMDscandata]", page 153
"Send data permanently [sEN LMDscandata]", page 154
"Set particle filter [sWN LFPparticle]", page 182
"Read state of the inputs [sRN LIDinputstate]", page 206
"Read state of the outputs [sRN LIDoutputstate]", page 207
"Receive outputstate by event [sEN LIDoutputstate]", page 208
"Set output state [sMN mDOSetOutput]", page 209
"Read firmware version [sRN DeviceIdent]", page 211
"Read the device state [sRN SCdevicestate]", page 216
"Read device type [sRN DItyp]", page 219
"Read operating hours [sRN ODoprh]", page 220
"Read power on counter [sRN ODpwr]", page 221
"Set device name [sWN LocationName]", page 223
"Read device name [sRN LocationName]", page 224
"Reset output counter [sMN LIDrstoutpcnt]", page 210
"Set IP address [sWN EllpAddr]", page 230
"Read IP address [sRN EllpAddr]", page 231
"Set Ethernet gateway [sWN Elgate]", page 234
"Read Ethernet gateway [sRN Elgate]", page 235
"Set IP mask [sWN Elmask]", page 237
"Read IP mask [sRN Elmask]", page 238

13.6.1.4 Telegrams

Telegrams listed in this document are described in the following basic structure:

Table 47: Telegram structure: "Command type" "Command"

Telegram structure: "Command type" "Command" (Minimum required user level. If nothing is stated, no user level required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Lists the different parts of the telegram.	Describes the corresponding telegram parts.	Defines the type of the variable.	Defines the length in byte.	Gives further information regarding the values in CoLa A/ CoLa B if necessary.	Defines the value of the telegram part in CoLa A (ASCII).	Defines the value of the telegram part in CoLa B (Binary).



NOTE

Commands are colored blue, parameters orange for further differentiation.

Table 48: Example: "Command type" "Command"

CoLa A	<"Start of text">"Command type value (ASCII)" "space" "Command value (ASCII)" "space" "Parameter value (ASCII)" "space" "Parameter value (ASCII)" <"End of text">				
	Copy example with framing (ASCII)				
	Copy example without framing (ASCII)				
	Copy example with framing (Hex)				
CoLa B	"Start of text and given length" "Command type value (Binary)" "space" "Command value (Binary)" "space" "Parameter value (Binary)" "Parameter value (Binary)" "Calculated checksum"				
	Copy example without framing (Binary)				

Table 49: Telegram structure: "Command type" "Command" (Answer)

Telegram structure: "Command type" "Command"						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Lists the different parts of the telegram.	Describes the corresponding telegram parts.	Defines the type of the variable.	Defines the length in byte.	Gives further information regarding the values in CoLa A/ CoLa B if neccessary.	Defines the value of the telegram part in CoLa A (ASCII).	Defines the value of the telegram part in CoLa B (Binary).

Table 50: Example: "Command type" "Command" (Answer)

CoLa A	<"Start of text">"Command type value (ASCII)" "space" "Command value (ASCII)" "space" "Parameter value (ASCII)" "space" "Parameter value (ASCII)" <"End of text">					
	<"Start of text">"Command type value (Hex)" "space" "Command value (Hex)" "space" "Parameter value (Hex)" "space" "Parameter value (Hex)" <"End of text">					
CoLa B	"Start of text and given length" "Command type value (Binary)" "space" "Command value (Binary)" "space" "Parameter value (Binary)" "Parameter value (Binary)" "Calculated checksum"					

13.6.1.4.1 Log in [sMN SetAccessMode]

A log in to the device is necessary to change parameters. In most cases, the user level 'Authorized client' is needed. Changed parameters will be reset to the previous state via a reboot unless the are saved. To save parameter changes the command "sMN mEEwriteall" (see ["Save parameters permanently \[sMN mEEwriteall\]"](#), [page 150](#)) must be send before log out.

Table 51: Telegram structure: sMN SetAccessMode

Telegram structure: sMN SetAccessMode						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Method	String	3		sMN	73 4D 4E
Command	User level	String	13		SetAccess-Mode	53 65 74 41 63 63 65 73 73 4D 6F 64 65
User level	Select user level	Int_8	1	Maintenance: Authorized client: Service:	2 3 4	02 03 04
Password	Hash value for the selected user level	Uint_32	4	Maintenance: Authorized client: Service:	B21ACE26 F4724744 81BE23AA	B2 1A CE 26 F4 72 47 44 81 BE 23 AA

Table 52: Example: sMN SetAccessMode - Log in as "Authorized client" with password "F4724744"

CoLa A	<STX>sMN{SPC}SetAccessMode{SPC}3{SPC}F4724744<ETX>					
	<STX>sMN SetAccessMode 3 F4724744<ETX>					
	sMN SetAccessMode 3 F4724744					
	02 73 4D 4E 20 53 65 74 41 63 63 65 73 73 4D 6F 64 65 20 30 33 20 46 34 37 32 34 37 34 34 03					
CoLa B	02 02 02 02 00 00 00 17 73 4D 4E 20 53 65 74 41 63 63 65 73 73 4D 6F 64 65 20 03 F4 72 47 44 B3					
	73 4D 4E 20 53 65 74 41 63 63 65 73 73 4D 6F 64 65 20 03 F4 72 47 44					

Table 53: Telegram structure: sAN SetAccessMode

Telegram structure: sAN SetAccessMode						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sAN	73 41 4E
Command	User level	String	13		SetAccess-Mode	53 65 74 41 63 63 65 73 73 4D 6F 64 65
Change user level	Changed level	Bool_1	1	Error: Success:	0 1	00 01

Table 54: Example: sAN SetAccessMode

CoLa A	<STX>sAN{SPC}SetAccessMode{SPC}1<ETX>					
	02 73 41 4E 20 53 65 74 41 63 63 65 73 73 4D 6F 64 65 20 01 03					
CoLa B	02 02 02 02 00 00 00 13 73 41 4E 20 53 65 74 41 63 63 65 73 73 4D 6F 64 65 20 01 38					

13.6.1.4.2 Enable/ disable CoLa user levels [sMN EnableLegacyUserLevel]

This command enables and disables a specific CoLa user level (Maintenance, Authorized Client).

Table 55: Telegram structure: sMN EnableLegacyUserLevel

Telegram structure: sMN EnableLegacyUserLevel						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Method	String	3		sMN	73 4D 4E
Command	Enable/ disable CoLa user level	String	21		EnableLegacyUserLevel	45 6E 61 62 6C 65 4C 65 67 61 63 79 55 73 65 72 4C 65 76 65 6C 20 03 20 01 21
User level	-	Uint_8	1	Maintenance: Authorized Client:	2 3	02 03
Status	Enable/ disable	Bool_1	1	Disable: Enable:	0 1	00 01

Table 56: Example: sMN EnableLegacyUserLevel - Enable User Level Authorized Client

CoLa A	<STX>sMN{SPC}EnableLegacyUserLevel{SPC}3{SPC}1<ETX>					
	<STX>sMN EnableLegacyUserLevel 3 1<ETX>					
	sMN EnableLegacyUserLevel 3 1					
	02 73 4D 4E 20 45 6E 61 62 6C 65 4C 65 67 61 63 79 55 73 65 72 4C 65 76 65 6C 20 33 20 31 03					
	02 02 02 02 00 00 00 1D 73 4D 4E 20 45 6E 61 62 6C 65 4C 65 67 61 63 79 55 73 65 72 4C 65 76 65 6C 20 03 20 01 21					
CoLa B	73 4D 4E 20 45 6E 61 62 6C 65 4C 65 67 61 63 79 55 73 65 72 4C 65 76 65 6C 20 03 20 01					

Table 57: Telegram structure: sAN EnableLegacyUserLevel

Telegram structure: sAN EnableLegacyUserLevel						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sAN	73 41 4E

Telegram structure: sAN EnableLegacyUserLevel						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command	Enable/ disable CoLa user level	String	21		EnableLegacyUserLevel	45 6E 61 62 6C 65 4C 65 67 61 63 79 55 73 65 72 4C 65 76 65 6C
Status code	Confirmation	Bool_1	1	unsuccessful: successful:	0 1	00 01

Table 58: Example: sAN EnableLegacyUserLevel - Enabling/ disabling of the CoLa User Level was successful

CoLa A	<STX>sAN[SPC]EnableLegacyUserLevel[SPC]1<ETX> 02 73 41 4E 20 45 6E 61 62 6C 65 4C 65 67 61 63 79 55 73 65 72 4C 65 76 65 6C 20 31 03
CoLa B	02 02 02 02 00 00 00 1B 73 41 4E 20 45 6E 61 62 6C 65 4C 65 67 61 63 79 55 73 65 72 4C 65 76 65 6C 20 01 0E

13.6.1.4.3 Basic Settings

13.6.1.4.3.1 Read for frequency and angular resolution [sRN LMPscancfg]

Read the scanning frequency, angular resolution and aperture angle of the device. Values show the information of the values before any filters are applied. Therefore aperture angle is always showing the maximum, scanning frequency and angular resolution may also be static depending on device family.

Table 59: Telegram structure: sRN LMPscancfg

Telegram structure: sRN LMPscancfg						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	Info of scan frequency and angular resolution	String	10		LMPscancfg	4C 4D 50 73 63 61 6E 63 66 67

Table 60: Example: sRN LMPscancfg

CoLa A	<STX>sRN[SPC]LMPscancfg<ETX> <STX>sRN LMPscancfg<ETX> sRN LMPscancfg 02 73 52 4E 20 4C 4D 50 73 63 61 6E 63 66 67 03	
CoLa B	02 02 02 02 00 00 00 0E 73 52 4E 20 4C 4D 50 73 63 61 6E 63 66 67 63 73 52 4E 20 4C 4D 50 73 63 61 6E 63 66 67	

Table 61: Telegram structure: sRA LMPscancfg

Telegram structure: sRA LMPscancfg						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	Info of scan frequency and angular resolution	String	10		LMPscancfg	4C 4D 50 73 63 61 6E 63 66 67

Telegram structure: sRA LMPscancfg						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Scan frequency	[1/100 Hz]	Uint_32	4	15 Hz ... 50 Hz: (depends on Dynamic Sensing Profile see "Dynamic Sensing Profiles", page 19)	5DCh ... 1388h	05 DC ... 13 88
Reserved	-	Int_16	2	Always:	1	00 01
Angular resolution	[1/10000°]	Uint_32	4	0.1° ... 1°: (depends on Dynamic Sensing Profile see "Dynamic Sensing Profiles", page 19)	3E8h ... 2710h	03 E8 ... 27 10
Start angle	[1/10000°]	Int_32	4	-48° ... 228°	FFF8AD00h ... 22CA40h	FF F8 AD 00h ... 00 22 CA 40h
Stop angle	[1/10000°]	Int_32	4	-48° ... 228°	FFF8AD00h ... 22CA40h	FF F8 AD 00h ... 00 22 CA 40h

Table 62: Example: sRA LMPscancfg - 30 Hz scan frequency, 0.1° angular resolution, -48° start angle, +138° stop angle

CoLa A	<STX>sRA{SPC}LMPscancfg{SPC}BB8{SPC}1{SPC}3E8{SPC}FFFAF160{SPC}150EA0<ETX> 02 73 52 41 20 4C 4D 50 73 63 61 6E 63 66 67 20 42 42 38 20 31 20 33 45 38 20 46 46 45 41 46 31 36 30 20 31 35 30 45 41 30 03					
CoLa B	02 02 02 02 00 00 00 21 73 52 41 20 4C 4D 50 73 63 61 6E 63 66 67 20 00 00 0B B8 00 01 00 00 03 E8 FF EA F1 60 00 15 0E A0 2A					

13.6.1.4.3.2 Activate standby mode [sMN LMCstandby]

Shut off the laser in order to extend the lifetime of laser diode and reduce motor speed to 25 Hz.

Table 63: Telegram structure: sMN LMCstandby

Telegram structure: sMN LMCstandby (User level 'Authorized Client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Method	String	3		sMN	73 4D 4E
Command	Set device to standby	String	10		LMCstandby	4C 4D 43 73 74 61 6E 64 62 79

Table 64: Example: sMN LMCstandby

CoLa A	<STX>sMN{SPC}LMCstandby<ETX> <STX>sMN LMCstandby<ETX> sMN LMCstandby					
	02 73 4D 4E 20 4C 4D 43 73 74 61 6E 64 62 79 03					
CoLa B	02 02 02 02 00 00 00 0E 73 4D 4E 20 4C 4D 43 73 74 61 6E 64 62 79 65 73 4D 4E 20 4C 4D 43 73 74 61 6E 64 62 79					

Table 65: Telegram structure: sAN LMCstandby

Telegram structure: sAN LMCstandby						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sAN	73 41 4E

Telegram structure: sAN LMCstandby						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command	Set device to standby	String	10		LMCstandby	4C 4D 43 73 74 61 6E 64 62 79
Status code	Accepted when value is 0	Enum_8	1	Success: Not allowed:	0 1	00 01

Table 66: Example: sAN LMCstandby

CoLa A	<STX>sAN{SPC}LMCstandby{SPC}0<ETX> 02 73 41 4E 20 4C 4D 43 73 74 61 6E 64 62 79 20 30 03					
CoLa B	02 02 02 02 00 00 00 10 73 41 4E 20 4C 4D 43 73 74 61 6E 64 62 79 20 00 49					

13.6.1.4.3.3 Start measurement [sMN LMCstartmeas]

Start the laser and (unless in Standby mode) the motor of the device

Table 67: Telegram structure: sMN LMCstartmeas

Telegram structure: sMN LMCstartmeas (User level 'Authorized Client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Method	String	3		sMN	73 4D 4E
Command	Start measurement	String	12		LMCstartmeas	4C 4D 43 73 74 61 72 74 6D 65 61 73

Table 68: Example: sMN LMCstartmeas

CoLa A	<STX>sMN{SPC}LMCstartmeas<ETX>					
	<STX>sMN LMCstartmeas<ETX>					
	sMN LMCstartmeas					
	02 73 4D 4E 20 4C 4D 43 73 74 61 72 74 6D 65 61 73 03					
	02 02 02 02 00 00 00 10 73 4D 4E 20 4C 4D 43 73 74 61 72 74 6D 65 61 73 68					
CoLa B	73 4D 4E 20 4C 4D 43 73 74 61 72 74 6D 65 61 73					

Table 69: Telegram structure: sAN LMCstartmeas

Telegram structure: sAN LMCstartmeas						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sAN	73 41 4E
Command	Start measurement	String	12		LMCstartmeas	4C 4D 43 73 74 61 72 74 6D 65 61 73
Status code	Accepted when value is 0	Enum_8	1	Success: Not allowed:	0 1	00 01

Table 70: Example: sAN LMCstartmeas

CoLa A	<STX>sAN{SPC}LMCstartmeas{SPC}0<ETX> 02 73 41 4E 20 4C 4D 43 73 74 61 72 74 6D 65 61 73 20 30 03					
CoLa B	02 02 02 02 00 00 00 12 73 41 4E 20 4C 4D 43 73 74 61 72 74 6D 65 61 73 20 00 44					

13.6.1.4.3.4 Stop measurement [sMN LMCstopmeas]

Shut off the laser and the motor is running at the set up frequency.

Table 71: Telegram structure: sMN LMCstopmeas

Telegram structure: sMN LMCstopmeas (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Method	String	3		sMN	73 4D 4E
Command	Stop measurement	String	11		LMCstopmeas	4C 4D 43 73 74 6F 70 6D 65 61 73

Table 72: Example: sMN LMCstopmeas

CoLa A	<STX>sMN{SPC}LMCstopmeas<ETX>																
	<STX>sMN LMCstopmeas<ETX>																
	sMN LMCstopmeas																
	02 73 4D 4E 20 4C 4D 43 73 74 6F 70 6D 65 61 73 03																
CoLa B	02 02 02 02 00 00 00 0F 73 4D 4E 20 4C 4D 43 73 74 6F 70 6D 65 61 73 10																
	73 4D 4E 20 4C 4D 43 73 74 6F 70 6D 65 61 73																

Table 73: Telegram structure: sAN LMCstopmeas

Telegram structure: sAN LMCstopmeas						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sAN	73 41 4E
Command	Stop measurement	String	11		LMCstopmeas	4C 4D 43 73 74 6F 70 6D 65 61 73
Status code	Accepted when value is 0	Enum_8	1	No error: Not allowed:	0 1	00 01

Table 74: Example: sAN LMCstopmeas

CoLa A	<STX>sAN{SPC}LMCstopmeas{SPC}0<ETX>				
	02 73 41 4E 20 4C 4D 43 73 74 6F 70 6D 65 61 73 20 30 03				
CoLa B	02 02 02 02 00 00 00 11 73 41 4E 20 4C 4D 43 73 74 6F 70 6D 65 61 73 20 00 3C				

13.6.1.4.3.5 Autostart measurement [sWN LMPautostartmeas]

This parameter defines whether the scanner will start to rotate directly and measure when powering up or remain in idle mode. The changed setting (saved with the command sMN mEEWriteall, [see "Save parameters permanently \[sMN mEEwriteall\]", page 150](#)) will be then be active with the next power-up cycle.

In case autostart is deactivated, the sensor will remain in idle mode with the laser switched off and a motor speed depending on the Dynamic Sensing Profile [see "Dynamic Sensing Profiles", page 19](#).

Table 75: Telegram structure: sWN LMPautostartmeas

Telegram structure: sWN LMPautostartmeas (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E
Command	Autostart measurement	String	16		LMPautostart-meas	4C 4D 50 61 75 74 6F 73 74 61 72 74 6D 65 61 73
Status code	Activate / Deactivate Autostart	Bool_1	1	Autostart off: Autostart on:	0 1	00 01

Table 76: Example: sWN LMPautostartmeas 1

CoLa A	<STX>sWN{SPC}LMPautostartmeas{SPC}1<ETX>					
	<STX>sWN LMPautostartmeas 1<ETX>					
	sWN LMPautostartmeas 1					
	02 73 57 4E 20 4C 4D 50 61 75 74 6F 73 74 61 72 74 6D 65 61 73 20 31 03					
	02 02 02 02 00 00 00 16 73 57 4E 20 4C 4D 50 61 75 74 6F 73 74 61 72 74 6D 65 61 73 20 01 4F					
CoLa B	73 57 4E 20 4C 4D 50 61 75 74 6F 73 74 61 72 74 6D 65 61 73 20 01					

Table 77: Telegram structure: sWA LMDautostartmeas

Telegram structure: sWA LMDautostartmeas						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	Autostart measurement	String	16		LMPautostart-meas	4C 4D 50 61 75 74 6F 73 74 61 72 74 6D 65 61 73

Table 78: Example: sWA LMPautostartmeas

CoLa A	<STX>sWA{SPC}LMPautostartmeas<ETX>					
	02 73 57 41 20 4C 4D 43 73 74 61 72 74 6D 65 61 73 03					
CoLa B	02 02 02 02 00 00 00 15 73 57 41 20 4C 4D 50 61 75 74 6F 73 74 61 72 74 6D 65 61 73 20 41					

13.6.1.4.3.6 Load factory defaults [sMN mSCloadfacdef]



NOTE

The Factory-Reset (Load factory defaults) deletes the entire parametrization of the device. All parameters, settings and system applications will be set to default.

Table 79: Telegram structure: sMN mSCloadfacdef

Telegram structure: sMN mSCloadfacdef (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Method	String	3		sMN	73 4D 4E

Telegram structure: sMN mSCloadfacdef (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command	Load factory defaults	String	13		mSCloadfacdef	6D 53 43 6C 6F 61 64 66 61 63 64 65 66

Table 80: Example: sMN mSCloadfacdef

CoLa A	<STX>sMN{SPC}mSCloadfacdef<ETX>					
	<STX>sMN mSCloadfacdef<ETX>					
	sMN mSCloadfacdef					
CoLa B	02 73 4D 4E 20 6D 53 43 6C 6F 61 64 66 61 63 64 65 66 03					
	02 02 02 02 00 00 00 11 73 4D 4E 20 6D 53 43 6C 6F 61 64 66 61 63 64 65 66 28					
	73 4D 4E 20 6D 53 43 6C 6F 61 64 66 61 63 64 65 66					

Table 81: Telegram structure: sAN mSCloadfacdef

Telegram structure: sAN mSCloadfacdef						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sAN	73 41 4E
Command	Load factory defaults	String	13		mSCloadfacdef	6D 53 43 6C 6F 61 64 66 61 63 64 65 66

Table 82: Example: sAN mSCloadfacdef

CoLa A	<STX>sAN{SPC}mSCloadfacdef<ETX>					
	02 73 4D 4E 20 6D 53 43 6C 6F 61 64 66 61 63 64 65 66 03					
CoLa B	02 02 02 02 00 00 00 12 73 41 4E 20 6D 53 43 6C 6F 61 64 66 61 63 64 65 66 20 04					

13.6.1.4.3.7 Load application defaults [sMN mSCloadappdef]

NOTE The Application-Reset (Load application defaults) deletes only the user parametrization of the Fields, Evaluation cases (EVC) and parameters under the header “Application”. Other parameters like Interface settings, Echo Filter, etc. remain unaffected.

Table 83: Telegram structure: sMN mSCloadappdef

Telegram structure: sMN mSCloadappdef (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Method	String	3		sMN	73 4D 4E
Command	Load application defaults	String	13		mSCloadappdef	6D 53 43 6C 6F 61 64 61 70 70 64 65 66

Table 84: Example: sMN mSCloadappdef

CoLa A	<STX>sMN{SPC}mSCloadappdef<ETX>	
	<STX>sMN mSCloadappdef<ETX>	
	sMN mSCloadappdef	
	02 73 4D 4E 20 6D 53 43 6C 6F 61 64 61 70 70 64 65 66 03	
CoLa B	02 02 02 02 00 00 00 11 73 4D 4E 20 6D 53 43 6C 6F 61 64 61 70 70 64 65 66 2D	
	73 4D 4E 20 6D 53 43 6C 6F 61 64 61 70 70 64 65 66	

Table 85: Telegram structure: sAN mSCloadappdef

Telegram structure: sAN mSCloadappdef						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sAN	73 41 4E
Command	Load application defaults	String	13		mSCloadappdef	6D 53 43 6C 6F 61 64 61 70 70 64 65 66

Table 86: Example: sAN mSCloadappdef

CoLa A	<STX>sAN{SPC}mSCloadappdef<ETX>
	02 73 41 4E 20 6D 53 43 6C 6F 61 64 61 70 70 64 65 66 03
CoLa B	02 02 02 02 00 00 00 12 73 41 4E 20 6D 53 43 6C 6F 61 64 61 70 70 64 65 66 20 01

13.6.1.4.3.8 Read checksum [sRN LFTchecksum]

This command includes the MD5 checksum of multiple software parts to verify their integrity. The checksums help to ensure files have not been corrupted or altered. They are also important for cybersecurity because they detect unauthorized modifications.

Table 87: Telegram structure: sRN LFTchecksum

Telegram structure: sRN LFTchecksum (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	MD5 checksum	String	11		LFTchecksum	4C 46 54 63 68 65 63 6B 73 75 6D

Table 88: Example: sRN LFTchecksum

CoLa A	<STX>sRN{SPC}LFTchecksum<ETX>	
	<STX>sRN LFTchecksum<ETX>	
	sRN LFTchecksum	
	02 73 52 4E 20 4C 46 54 63 68 65 63 6B 73 75 6D 03	
CoLa B	02 02 02 02 00 00 00 1F 73 52 4E 20 4C 46 54 63 68 65 63 6B 73 75 6D 1C	
	73 52 4E 20 4C 46 54 63 68 65 63 6B 73 75 6D	

Table 89: Telegram structure: sRA LFTchecksum

Telegram structure: sRA LFTchecksum						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	MD5 checksum	String	11		LFTchecksum	4C 46 54 63 68 65 63 6B 73 75 6D
Variable Data 1	Checksum validity	Bool_1	1	False: True:	0 1	00 01
Variable Data 2	Length	Uint_16	2		20	32 20
Variable Data 2	Checksum Linux part	Flex-String			<hash>	<hash>
Variable Data 3	Length	Uint_16	2		20	32 20
Variable Data 3	Checksum app part	Flex-String			<hash>	<hash>
Variable Data 4	Length	Uint_16	2		20	32 20
Variable Data 4	Checksum calibration part	Flex-String			<hash>	<hash>
Variable Data 5	Length	Uint_16	2		20	32 20
Variable Data 5	Checksum parameters part	Flex-String			<hash>	<hash>

Table 90: Example: sRA LFTchecksum

CoLa A	<STX>sRA{SPC}LFTcheck-sum{SPC}20{SPC}92b7b9720c40e8b99268229283617dd5{SPC}20{SPC}e002a1c8022b30589ce0e357c87f5a1c{SPC}20{SPC}901736df3fbc807121c46f9eae8ff28{SPC}20{SPC}588e8386c7cd011d1a8369c4377e2a4c<ETX>					
	02 73 52 41 20 4C 46 54 63 68 65 63 6B 73 75 6D 20 31 20 32 30 20 39 32 62 37 62 39 37 32 30 63 34 30 65 38 62 39 39 32 36 38 32 32 39 32 38 33 36 31 37 64 64 35 20 32 30 20 65 30 30 32 61 31 63 38 30 32 32 62 33 30 35 38 39 63 65 30 65 33 35 37 63 38 37 66 35 61 31 63 20 32 30 20 39 30 31 37 33 36 64 66 33 66 62 63 38 30 37 31 32 31 63 34 36 66 39 65 61 65 64 38 66 66 32 38 20 32 30 20 35 38 38 65 38 33 38 36 63 37 63 64 30 31 31 64 31 61 38 33 36 39 63 34 33 37 37 65 32 61 34 63 03					
CoLa B	02 02 02 02 00 00 00 99 73 57 41 20 4C 46 54 63 68 65 63 6B 73 75 6D 20 01 00 20 39 32 62 37 62 39 37 32 30 63 34 30 65 38 62 39 39 32 36 38 32 32 39 32 38 33 36 31 37 64 64 35 00 20 65 30 30 32 61 31 63 38 30 32 32 62 33 30 35 38 39 63 65 30 65 33 35 37 63 38 37 66 35 61 31 63 00 20 39 30 31 37 33 36 64 66 33 66 62 63 38 30 37 31 32 31 63 34 36 66 39 65 61 65 64 38 66 66 32 38 00 20 35 38 38 65 38 33 38 36 63 37 63 64 30 31 31 64 31 61 38 33 36 39 63 34 33 37 37 65 32 61 34 63 6F					

13.6.1.4.3.9 Change password [sMN SetPassword]

Changing the log in password for a specific user level.



NOTE

If logged in with a higher user level you may set the password for lower user levels as well.

Table 91: Telegram structure: sMN SetPassword

Telegram structure: sMN SetPassword (Same user level or higher required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Method	String	3		sMN	73 4D 4E
Command	Set password request	String	13		SetPassword	53 65 74 50 61 73 73 77 6F 72 64
User level	User level that the password will be applied to	Int_8	1	Maintenance: Authorized client: Service:	2 3 4	02 03 04
Password	Hash value of the new password	Uint_32	4		<Hash value>	

Set password for 'Authorized client' to "testtest" (hash value = 1920E4C9).

Table 92: Example: sMN SetPassword

CoLa A	<STX>sMN{SPC}SetPassword{SPC}3{SPC}1920E4C9<ETX>		
	<STX>sMN SetPassword 3 1920E4C9<ETX>		
	sMN SetPassword 3 1920E4C9		
	02 73 4D 4E 20 53 65 74 50 61 73 73 77 6F 72 64 20 33 20 31 39 32 30 45 34 43 39 03 02 02 02 02 00 00 00 15 73 4D 4E 20 53 65 74 50 61 73 73 77 6F 72 64 20 03 19 20 E4 C9 1A		
CoLa B	73 4D 4E 20 53 65 74 50 61 73 73 77 6F 72 64 20 03 19 20 E4 C9		

Calculating the hash value of the password

- Login in SOPASair with user level "Service" to the device.
- Select **Diagnosis > Cybersecurity > Hash calculator**.
- For CoLa A/B take calculated hexadecimal hash value.

Table 93: Telegram structure: sAN SetPassword

Telegram structure: sAN SetPassword						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sAN	73 41 4E
Command	Set password requested	String	13		SetPassword	53 65 74 50 61 73 73 77 6F 72 64
Success	Confirmation	Int_8	1	Failed: Success:	0 1	00 01

Table 94: Example: sAN SetPassword

CoLa A	<STX>sAN{SPC}SetPassword{SPC}1<ETX>	
	02 73 4D 4E 20 53 65 74 50 61 73 73 77 6F 72 64 20 31 03	
CoLa B	02 02 02 02 00 00 00 11 73 41 4E 20 53 65 74 50 61 73 73 77 6F 72 64 20 01 00	

13.6.1.4.3.10 Check password [sMN CheckPassword]

Check the password for a specific user level, e.g. to verify if it has been changed correctly.

Table 95: Telegram structure: sMN CheckPassword

Telegram structure: sMN CheckPassword (Same User level or higher required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Method	String	3		sMN	73 4D 4E
Command	Check password request	String	13		CheckPassword	43 68 65 63 6B 50 61 73 73 77 6F 72 64
User level	User level to check the password for	Int_8	1	Maintenance: Authorized client: Service:	2 3 4	02 03 04
Password	Hash value of the password to be checked	Uint_32	4		<Hash value>	

Check password "testtest" for 'Authorized client'.

Table 96: Example: sMN CheckPassword

CoLa A	<STX>sMN{SPC}CheckPassword{SPC}3{SPC}1920E4C9<ETX>		
	<STX>sMN CheckPassword 3 1920E4C9<ETX>		
	sMN CheckPassword 3 1920E4C9		
	02 73 4D 4E 20 43 68 65 63 6B 50 61 73 73 77 6F 72 64 20 33 20 31 39 32 30 45 34 43 39 03 02 02 02 02 00 00 00 17 73 4D 4E 20 43 68 65 63 6B 50 61 73 73 77 6F 72 64 20 03 19 20 E4 C9 1E		
CoLa B	73 4D 4E 20 43 68 65 63 6B 50 61 73 73 77 6F 72 64 20 03 19 20 E4 C9		

Table 97: Telegram structure: sAN CheckPassword

Telegram structure: sAN CheckPassword						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sAN	73 41 4E
Command	Check password request	String	13		CheckPassword	43 68 65 63 6B 50 61 73 73 77 6F 72 64
Success	Confirmation	Int_8	1	Failed: Success:	0 1	00 01

Table 98: Example: sAN CheckPassword

CoLa A	<STX>sAN{SPC}CheckPassword{SPC}1<ETX>	
	02 73 41 4E 20 43 68 65 63 6B 50 61 73 73 77 6F 72 64 20 31 03	
CoLa B	02 02 02 02 00 00 00 13 73 41 4E 20 43 68 65 63 6B 50 61 73 73 77 6F 72 64 20 01 04	

13.6.1.4.3.11 Set Sensitivity Mode [sWN SensitivityMode]

Define a sensitivity mode for the detection of reflectors [see "Reflector detection", page 32](#).

Table 99: Telegram structure: sWN SensitivityMode

Telegram structure: sWN SensitivityMode (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E

Telegram structure: sWN SensitivityMode (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command	Sensitivity Mode	String	15		SensitivityMode	53 65 6E 73 69 74 69 76 69 74 79 4D 6F 64 65 65
Sensitivity	Choose Sensitivity Mode	Enum_8	1	Maximum robustness: Standard (recommended): Maximum detectivity:	0 1 2	00 01 02

Table 100: Example: sWN SensitivityMode

CoLa A	<STX>sWN{SPC}SensitivityMode{SPC}1<ETX>					
	<STX>sWN SensitivityMode 1<ETX>					
	sWN SensitivityMode 1					
CoLa B	02 73 57 4E 20 53 65 6E 73 69 74 69 76 69 74 79 4D 6F 64 65 20 31 03					
	02 02 02 02 00 00 00 15 73 57 4E 20 53 65 6E 73 69 74 69 76 69 74 79 4D 6F 64 65 20 01 05					
	73 57 4E 20 53 65 6E 73 69 74 69 76 69 74 79 4D 6F 64 65 20 01					

Table 101: Telegram structure: sWA SensitivityMode

Telegram structure: sWA SensitivityMode						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	Sensitivity Mode	String	15		SensitivityMode	53 65 6E 73 69 74 69 76 69 74 79 4D 6F 64 65 65

Table 102: Example: sWA SensitivityMode

CoLa A	<STX>sWA{SPC}SensitivityMode<ETX>					
	02 73 57 41 20 53 65 6E 73 69 74 69 76 69 74 79 4D 6F 64 65 03					
CoLa B	02 02 02 02 00 00 00 14 73 57 41 20 53 65 6E 73 69 74 69 76 69 74 79 4D 6F 64 65 20 0B					

13.6.1.4.3.12 Set contamination indication settings [sWN ContaminationConfig]

Define how and under which conditions the device shall indicate a potential contamination of the optics cover to enable preventive cleaning. (see "Contamination indication", page 41).

Table 103: Telegram structure: sWN ContaminationConfig

Telegram structure: sWN ContaminationConfig (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E
Command		String	19		Contamination-Config	43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 43 6F 6E 66 69 67
Strategy	Strategy code	Enum_8	1	No output: All sectors contaminated: One sector contaminated:	0 1 2	00 01 02

Telegram structure: sWN ContaminationConfig (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Response time	How fast (in seconds) the sensor reacts after a contamination	UInt_16	2	Value range 3 ... 60 Default: 3	+3d ... +60d (03h ... 3Ch)	00 03 ... 00 3C
Threshold warning	Sensitivity	Enum_8	1	High: Medium: Low:	0 1 2	00 01 02
Sector preselection	Selection of either custom sectors or used weather protection hood	Enum_8	1	All sectors: Weather protection hood: Custom sectors:	0 6 255d (FFh)	00 06 FF
Length		UInt_16	2	Always:	7	07
Custom sectors	Enable/ disable individual sectors when custom sectors got selected in Sector preselection	Array	7	Active: 1 Inactive: 0 Default value of all sectors is active	0 ... 1 0 ... 1 0 ... 1 0 ... 1 0 ... 1 0 ... 1 0 ... 1	00 ... 01 00 ... 01 00 ... 01 00 ... 01 00 ... 01 00 ... 01 00 ... 01
Enable Warning	Warning monitoring	Bool_1	1	Off (False): On (True):	0 1	00 01
Enable Error	Error monitoring	Bool_1	1	Off (False): On (True):	0 1	00 01

Table 104: Example: sWN ContaminationConfig One sector contaminated, response time 60 seconds, sensitivity low, weather protection hood, enable warning, enable error

CoLa A	<STX>sWN{SPC}ContaminationConfig{SPC}2{SPC}3C{SPC}2{SPC}6{SPC}7{SPC}1{SPC}1{SPC}1{SPC}1{SPC}1{SPC}1{SPC}1{SPC}1{SPC}1{SPC}1{SPC}1<ETX>	
	<STX>sWN ContaminationConfig 2 3C 2 6 7 1111111111<ETX>	
	sWN ContaminationConfig 2 3C 2 6 7 1111111111	
CoLa B	02 73 57 4E 20 43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 43 6F 6E 66 69 67 20 32 20 33 43 20 32 20 36 20 37 20 31 20 31 20 31 20 31 20 31 20 31 20 31 20 31 20 32 20 33 43 20 32 20 36 20 37 20 31 20 31 20 31 20 31 20 31 20 31 20 31 03	
	02 02 02 02 00 00 00 28 73 57 4E 20 43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 43 6F 6E 66 69 67 20 02 00 3C 02 00 06 07 01 01 01 01 01 01 01 01 01 3C	
CoLa B	73 57 4E 20 43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 43 6F 6E 66 69 67 20 02 00 3C 02 00 06 07 01 01 01 01 01 01 01 01 01 01 01	
	01 01 01	

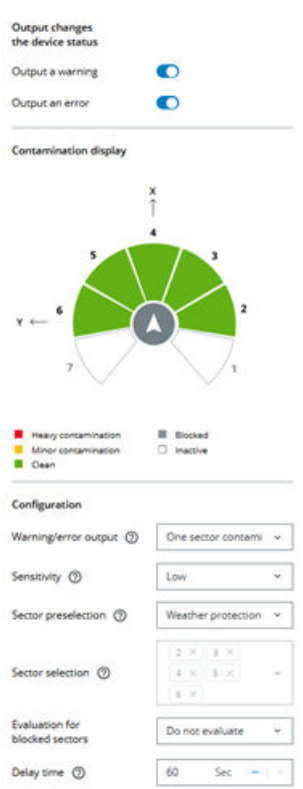


Table 105: Telegram structure: sWA ContaminationConfig

Telegram structure: sWA ContaminationConfig						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command		String	19		Contamination-Config	43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 43 6F 6E 66 69 67 03

Table 106: Example: sWA ContaminationConfig

CoLa A	<STX>sWA{SPC}ContaminationConfig<ETX>
CoLa B	02 02 02 02 00 00 00 18 73 57 41 20 43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 43 6F 6E 66 69 67 20

13.6.1.4.3.13 Read contamination indication settings [sRN ContaminationConfig]

Reads the current contamination settings that define how and under which conditions the device indicates a potential contamination of the optics cover to enable preventive cleaning.

Table 107: Telegram structure: sRN ContaminationConfig

Telegram structure: sRN ContaminationConfig						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E

Telegram structure: sRN ContaminationConfig						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command		String	19		Contamination-Config	43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 43 6F 6E 66 69 67 03

Table 108: Example: sRN ContaminationConfig

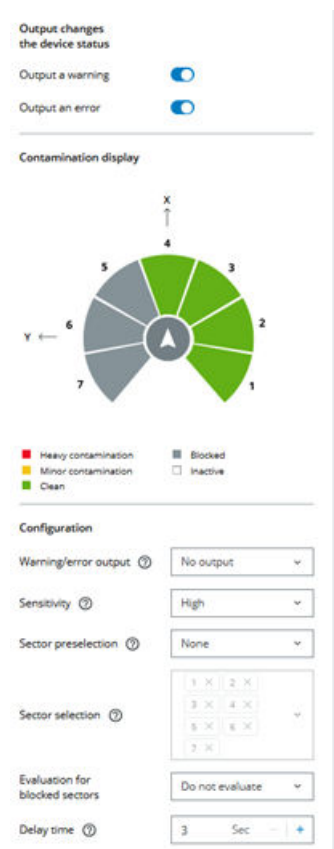
CoLa A	<STX>sRN{SPC}ContaminationConfig<ETX>					
	<STX>sRN ContaminationConfig<ETX>					
	sRN ContaminationConfig					
	02 73 52 4E 20 43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 43 6F 6E 66 69 67 03					
CoLa B	02 02 02 02 00 00 00 17 73 52 4E 20 43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 43 6F 6E 66 69 67 25					
	73 52 4E 20 43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 43 6F 6E 66 69 67 25					

Table 109: Telegram structure: sRA ContaminationConfig

Telegram structure: sRA ContaminationConfig						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command		String	19		Contamination-Config	43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 43 6F 6E 66 69 67 03
Strategy	Strategy code	Enum_8	1	No output: All sectors contaminated: One sector contaminated:	0 1 2	00 01 02
Response time	How fast (in seconds) the sensor reacts after a contamination	Uint_16	2	Value range 3 ... 60 Default: 3	03h ... 3Ch	00 03 ... 00 3C
Threshold warning	Sensitivity	Enum_8	1	High: Medium: Low:	0 1 2	00 01 02
Sector preselection	Selection of either custom sectors or used weather protection hood	Enum_8	1	All sectors: Weather protection hood: Custom sectors:	0 6 FF	00 06 FF
Length		Enum_8	1	Always:	7	07
Custom sectors	Enable/ disable individual sectors when custom sectors got selected in Sector pre-selection	Array	7	Active: 1 Inactive: 0 Default value of all sectors is active	0 ... 1 0 ... 1 0 ... 1 0 ... 1 0 ... 1 0 ... 1 0 ... 1	00 ... 01 00 ... 01 00 ... 01 00 ... 01 00 ... 01 00 ... 01 00 ... 01
Enable Warning	Warning monitoring	Bool_1	1	Off (False): On (True):	0 1	00 01
Enable Error	Error monitoring	Bool_1	1	Off (False): On (True):	0 1	00 01

Table 110: Example: sRA ContaminationConfig No output, response time 3 sec, All sectors active, warning enabled, error enabled

CoLa A	<pre><STX>sRA{SPC}ContaminationCon- fig{SPC}0{SPC}3{SPC}0{SPC}0{SPC}7{SPC}1{SPC}1{SPC}1{SPC}1{SPC}1{SPC}1{SPC}1{SPC}1{SPC}1<ETX></pre> 02 73 52 41 20 43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 43 6F 6E 66 69 67 20 30 20 33 20 30 20 30 20 37 20 31 20 31 20 31 20 31 20 31 20 31 20 31 03
CoLa B	<pre>02 02 02 02 00 00 00 28 73 52 41 20 43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 43 6F 6E 66 69 67 20 00 00 03 00 00 00 07 01 01 01 01 01 01 01 01 01 0F</pre>



13.6.1.4.3.14 Read contamination indication data [sRN ContaminationData]

Get the current contamination indication data.

Table 111: Telegram structure: sRN ContaminationData

Telegram structure: sRN ContaminationData						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	Reading status of all contamination sectors	String	17		Contamination-Data	43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 44 61 74 61

Table 112: Example: sRN ContaminationData

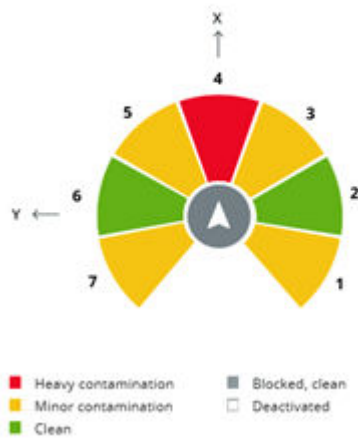
CoLa A	<STX>sRN{SPC}ContaminationData<ETX>
	<STX>sRN ContaminationData<ETX>
	sRN ContaminationData
	02 73 52 4E 20 43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 44 61 74 61 03

	02 02 02 02 00 00 00 15 73 52 4E 20 43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 44 61 74 61 3F	
CoLa B	73 52 4E 20 43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 44 61 74 61	

Table 113: Telegram structure: sRA ContaminationData

Telegram structure: sRA ContaminationData						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	Reading status of all contamination sectors	String	17		Contamination-Data	43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 44 61 74 61
Number of contamination sectors		Uint_16	2	Always:	7	00 07
Contamination data	Status of contamination of each sector	Array of Enum_8	7	Blocked: Clean: Warning: Error:	0 1 2 3	00 01 02 03

Contamination display

**Table 114:** Example: sRA ContaminationData

CoLa A	<STX>sRA{SPC}ContaminationData{SPC}7{SPC}2{SPC}1{SPC}2{SPC}3{SPC}2{SPC}1{SPC}2<ETX>	
	02 73 52 41 20 43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 44 61 74 61 20 37 20 32 20 31 20 32 20 33 20 32 20 31 20 32 03	
CoLa B	02 02 02 02 00 00 00 1F 73 52 41 20 43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 44 61 74 20 00 07 02 01 02 03 02 01 02 14	

13.6.1.4.3.15 Send contamination indication data permanently [sEN ContaminationData]

Provides continuous contamination indication data for the individual sectors of the optical cover. When a change in contamination is detected in any sector, the current contamination status of all sectors will be transmitted

Table 115: Telegram structure: sEN ContaminationData

Telegram structure: sEN ContaminationData						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sEN	73 45 4E
Command	Get the contamination indication data continuously	String	17		Contamination-Data	43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 44 61 74 61
Data	Start/ Stop	Bool_1	1	Stop: Start:	0 1	00 01

Table 116: Example: sEN ContaminationData

CoLa A	<STX>sEN{SPC}ContaminationData{SPC}1<ETX>		
	<STX>sEN ContaminationData 1<ETX>		
	sEN ContaminationData 1		
CoLa B	02 73 45 4E 20 43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 44 61 74 61 20 31 03		
	02 02 02 02 00 00 00 17 73 45 4E 20 43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 44 61 74 61 20 01 09		
	73 45 4E 20 43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 44 61 74 61 20 01		

Table 117: Telegram structure: sSN ContaminationData

Telegram structure: sSN ContaminationData						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sSN	73 53 4E
Command	Get the contamination indication data continuously	String	17		Contamination-Data	43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 44 61 74 61
Number of contamination sectors		Uint_16	2	Always:	7	00 07
Contamination data	Status of contamination of each sector	Array of Enum_8	7	Blocked: Clean: Warning: Error:	0 1 2 3	00 01 02 03

Contamination display



Table 118: Example: sSN ContaminationData

CoLa A	<STX>sSN[SPC]ContaminationData[SPC]7[SPC]0[SPC]0[SPC]0[SPC]2[SPC]2[SPC]1[SPC]1<ETX>
CoLa B	02 73 53 4E 20 43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 44 61 74 61 20 37 20 30 20 30 20 30 20 32 20 32 20 31 20 31 03 01 19

13.6.1.4.3.16 Read contamination indication result [sRN ContaminationResult]

This telegram is intended to read the contamination result. The telegram returns two Boolean. One for warning and one for error. Depending on the strategy of the contamination indication the state changes if one or all sectors are in the specific status (warning or error).

Table 119: Telegram structure: sRN ContaminationResult

Telegram structure: sRN ContaminationResult					 → 	
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command		String	19		Contamination-Result	43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 52 65 73 75 6C 74 03

Table 120: Example: sRN ContaminationResult

CoLa A	<STX>sRN[SPC]ContaminationResult<ETX>	
CoLa B	<STX>sRN ContaminationResult<ETX>	
CoLa B	sRN ContaminationResult	
CoLa A	02 73 52 4E 20 43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 52 65 73 75 6C 74 03	
CoLa B	02 02 02 02 00 00 00 17 73 52 4E 20 43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 52 65 73 75 6C 74 26	

Table 121: Telegram structure: sRA ContaminationResult

Telegram structure: sRA ContaminationResult						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command		String	19		Contamination-Result	43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 52 65 73 75 6C 74
Contamination Result	Warning	Bool_1	1	Not active: Active:	0 1	00 01
	Error	Bool_1	1	Not active: Active:	0 1	00 01

Table 122: Example: sRA ContaminationResult warning inactive error inactive

CoLa A	<STX>sRA{SPC}ContaminationResult{SPC}0{SPC}0<ETX>					
	02 73 52 41 20 43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 52 65 73 75 6C 74 20 30 20 30 03					
CoLa B	02 02 02 02 00 00 00 1A 73 52 41 20 43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 52 65 73 75 6C 74 20 00 00 09					

13.6.1.4.3.17 Send contamination indication result permanently [sEN ContaminationResult]

This telegram is intended to activate the event for read the contamination result permanently. The telegram returns two Boolean. One for warning and one for error. Depending on the strategy of the contamination indication the state changes if one or all sectors are in the specific status (warning or error).

Table 123: Telegram structure: sEN ContaminationResult

Telegram structure: sEN ContaminationResult						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sEN	73 52 4E
Command		String	19		Contamination-Result	43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 52 65 73 75 6C 74
Data	Start/ Stop	Bool_1	1	Stop: Start:	0 1	00 01

Table 124: Example: sEN ContaminationResult

CoLa A	<STX>sEN{SPC}ContaminationResult{SPC}1<ETX>					
	<STX>sEN ContaminationResult 1<ETX>					
	sEN ContaminationResult 1					
	02 73 45 4E 20 43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 52 65 73 75 6C 74 20 31 03					
CoLa B	02 02 02 02 00 00 00 19 73 45 4E 20 43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 52 65 73 75 6C 74 20 01 10					
	73 45 4E 20 43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 52 65 73 75 6C 74 20 01					

Table 125: Telegram structure: sSN ContaminationResult

Telegram structure: sSN ContaminationResult						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sSN	73 53 4E
Contamination Result	Warning	Bool_1	1	Not active: Active:	0 1	00 01
	Error	Bool_1	1	Not active: Active:	0 1	00 01

Table 126: Example: sSN ContaminationResult warning active error inactive

CoLa A	<STX>sRA{SPC}ContaminationResult{SPC}1{SPC}0<ETX>					
	02 73 52 41 20 43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 52 65 73 75 6C 74 20 31 20 30 03					
CoLa B	02 02 02 02 00 00 00 00 1A 73 52 41 20 43 6F 6E 74 61 6D 69 6E 61 74 69 6F 6E 52 65 73 75 6C 74 20 01 00 08					

13.6.1.4.3.18 Set Treat Blocked Sectors As Error Sectors [sWN treatBlockedSectorsAsErrorSectors]

If enabled, blocked sectors get interpreted as error sectors for contamination strategy evaluation.

Table 127: Telegram structure: sWN treatBlockedSectorsAsErrorSectors

Telegram structure: sWN treatBlockedSectorsAsErrorSectors (User level 'Authorized Client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write request	String	3		sWN	73 57 4E
Variable name	treatBlockedSectorsAsErrorSectors	String	33		treatBlocked-SectorsAsError-Sectors	74 72 65 61 74 42 6C 6F 63 6B 65 64 53 65 63 74 6F 72 73 41 73 45 72 72 6F 72 53 65 63 74 6F 72 73
Value	Enable/ disable Treat Blocked Sectors As Error Sectors	Bool_1	1	Disable: Enable:	0 1	00 01

Table 128: Example: sWN treatBlockedSectorsAsErrorSectors

CoLa A	<STX>sWN{SPC}treatBlockedSectorsAsErrorSectors{SPC}1<ETX>					
	<STX>sWN treatBlockedSectorsAsErrorSectors 1<ETX>					
	sWN treatBlockedSectorsAsErrorSectors 1					
	02 73 57 4E 20 74 72 65 61 74 42 6C 6F 63 6B 65 64 53 65 63 74 6F 72 73 41 73 45 72 72 6F 72 53 65 63 74 6F 72 73 20 31 03					
CoLa B	02 02 02 02 00 00 00 00 27 73 57 4E 20 74 72 65 61 74 42 6C 6F 63 6B 65 64 53 65 63 74 6F 72 73 41 73 45 72 72 6F 72 53 65 63 74 6F 72 73 20 01 3F					
	73 57 4E 20 74 72 65 61 74 42 6C 6F 63 6B 65 64 53 65 63 74 6F 72 73 41 73 45 72 72 6F 72 53 65 63 74 6F 72 73 20 01					

Table 129: Telegram structure: sWA treatBlockedSectorsAsErrorSectors

Telegram structure: sWA treatBlockedSectorsAsErrorSectors						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write answer	String	3		sWA	73 57 41
Variable name	Echo of variable name	String	33		treatBlocked-SectorsAsError-Sectors	74 72 65 61 74 42 6C 6F 63 6B 65 64 53 65 63 74 6F 72 73 41 73 45 72 72 6F 72 53 65 63 74 6F 72 73 03

Table 130: Example: sWA treatBlockedSectorsAsErrorSectors

CoLa A	<STX>sWA{SPC}treatBlockedSectorsAsErrorSectors<ETX> 02 73 57 41 20 74 72 65 61 74 42 6C 6F 63 6B 65 64 53 65 63 74 6F 72 73 41 73 45 72 72 6F 72 53 65 63 74 6F 72 73 03
CoLa B	02 02 02 02 00 00 00 25 73 57 41 20 74 72 65 61 74 42 6C 6F 63 6B 65 64 53 65 63 74 6F 72 73 41 73 45 72 72 6F 72 53 65 63 74 6F 72 73 11

13.6.1.4.3.19 Read Treat Blocked Sectors As Error Sectors [sRN treatBlockedSectorsAsErrorSectors]

Returns the state if blocked sectors are treated as error sectors for contamination strategy evaluation.

Table 131: Telegram structure: sRN treatBlockedSectorsAsErrorSectors

Telegram structure: sRN treatBlockedSectorsAsErrorSectors						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read request	String	3		sRN	73 52 4E
Variable name	treatBlockedSectorsAsErrorSectors	String	33		treatBlocked-SectorsAsError-Sectors	74 72 65 61 74 42 6C 6F 63 6B 65 64 53 65 63 74 6F 72 73 41 73 45 72 72 6F 72 53 65 63 74 6F 72 73 03

Table 132: Example: sRN treatBlockedSectorsAsErrorSectors

CoLa A	<STX>sRN{SPC}treatBlockedSectorsAsErrorSectors<ETX>	
	<STX>sRN treatBlockedSectorsAsErrorSectors<ETX>	
	sRN treatBlockedSectorsAsErrorSectors	
	02 73 52 4E 20 74 72 65 61 74 42 6C 6F 63 6B 65 64 53 65 63 74 6F 72 73 41 73 45 72 72 6F 72 53 65 63 74 6F 72 73 03	
CoLa B	02 02 02 02 00 00 00 25 73 52 4E 20 74 72 65 61 74 42 6C 6F 63 6B 65 64 53 65 63 74 6F 72 73 41 73 45 72 72 6F 72 53 65 63 74 6F 72 73 1B	
	73 52 4E 20 74 72 65 61 74 42 6C 6F 63 6B 65 64 53 65 63 74 6F 72 73 41 73 45 72 72 6F 72 53 65 63 74 6F 72 73	

Table 133: Telegram structure: sRA treatBlockedSectorsAsErrorSectors

Telegram structure: sRA treatBlockedSectorsAsErrorSectors						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read answer	String	3		sRA	73 52 41
Variable name	treatBlockedSectorsAsErrorSectors	String	33		treatBlockedSectorsAsErrorSectors	74 72 65 61 74 42 6C 6F 63 6B 65 64 53 65 63 74 6F 72 73 41 73 45 72 72 6F 72 53 65 63 74 6F 72 73
Value	State of Treat Blocked Sectors As Error Sectors (disabled/ enabled)	Bool_1	1	Disabled: Enabled:	0 1	00 01

Table 134: Example: sRA treatBlockedSectorsAsErrorSectors

CoLa A	<STX>sRA{SPC}treatBlockedSectorsAsErrorSectors{SPC}1<ETX> 02 73 52 41 20 74 72 65 61 74 42 6C 6F 63 6B 65 64 53 65 63 74 6F 72 73 41 73 45 72 72 6F 72 53 65 63 74 6F 72 73 20 31 03					
CoLa B	02 02 02 02 00 00 00 27 73 52 41 20 74 72 65 61 74 42 6C 6F 63 6B 65 64 53 65 63 74 6F 72 73 41 73 45 72 72 6F 72 53 65 63 74 6F 72 73 20 01 35					

13.6.1.4.3.20 Set Performance Profile [sWN PerformanceProfileNumber]

Define which Dynamic Sensing Profile shall be used, see "Dynamic Sensing Profiles", page 19.

Table 135: Telegram structure: sWN PerformanceProfileNumber

Telegram structure: sWN PerformanceProfileNumber (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E
Command	Set Dynamic Sensing Profile	String	24		PerformanceProfileNumber	50 65 72 66 6F 72 6D 61 6E 63 65 50 72 6F 66 69 6C 65 4E 75 6D 62 65 72
Performance Profile	Selection of Dynamic Sensing Profile	Enum_8	1	Profile 1: Profile 2: Profile 3: Profile 4: Profile 5: Profile 6: Profile 7: Profile 8: Profile 9: Profile 10: Profile 11:	1 2 3 4 5 6 7 8 9 10 11	01 02 03 04 05 06 07 08 09 0A 0B

Table 136: Example: sWN PerformanceProfileNumber

CoLa A	<STX>sWN{SPC}PerformanceProfileNumber{SPC}2<ETX> <STX>sWN PerformanceProfileNumber 2<ETX> sWN PerformanceProfileNumber 2 02 73 57 4E 20 50 65 72 66 6F 72 6D 61 6E 63 65 50 72 6F 66 69 6C 65 4E 75 6D 62 65 72 20 32 03					
--------	---	--	--	--	--	--

CoLa B	02 02 02 02 00 00 00 00 1E 73 57 4E 20 50 65 72 66 6F 72 6D 61 6E 63 65 50 72 6F 66 69 6C 65 4E 75 6D 62 65 72 20 02 58	
	73 57 4E 20 50 65 72 66 6F 72 6D 61 6E 63 65 50 72 6F 66 69 6C 65 4E 75 6D 62 65 72 20 02	

Table 137: Telegram structure: sWA PerformanceProfileNumber

Telegram structure: sWA PerformanceProfileNumber						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	Set Performance Profile	String	24		Performance-ProfileNumber	50 65 72 66 6F 72 6D 61 6E 63 65 50 72 6F 66 69 6C 65 4E 75 6D 62 65 72

Table 138: Example: sWA PerformanceProfileNumber

CoLa A	<STX>sWA{SPC}PerformanceProfileNumber<ETX>
	02 73 57 41 20 50 65 72 66 6F 72 6D 61 6E 63 65 50 72 6F 66 69 6C 65 4E 75 6D 62 65 72 03
CoLa B	02 02 02 02 00 00 00 00 18 73 57 41 20 50 65 72 66 6F 72 6D 61 6E 63 65 50 72 6F 66 69 6C 65 4E 75 6D 62 65 72 20

13.6.1.4.3.21 Save parameters permanently [sMN mEEwriteall]

Save all parameter changes of the device. Must be sent before logging off and/ or hardware rebooting of the device. Else all changes will be lost.

Table 139: Telegram structure: sMN mEEwriteall

Telegram structure: sMN mEEwriteall (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Method	String	3		sMN	73 4D 4E
Command	Store parameters permanently	String	11		mEEwriteall	6D 45 45 77 72 69 74 65 61 6C 6C

Table 140: Example: sMN mEEwriteall

CoLa A	<STX>sMN{SPC}mEEwriteall<ETX>	
	<STX>sMN mEEwriteall<ETX>	
	sMN mEEwriteall	
	02 73 4D 4E 20 6D 45 45 77 72 69 74 65 61 6C 6C 03	
CoLa B	02 02 02 02 00 00 00 00 0F 73 4D 4E 20 6D 45 45 77 72 69 74 65 61 6C 6C 21	
	73 4D 4E 20 6D 45 45 77 72 69 74 65 61 6C 6C	

Table 141: Telegram structure: sAN mEEwriteall

Telegram structure: sAN mEEwriteall						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sAN	73 41 4E

Telegram structure: sAN mEEwriteall						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command	Store parameters permanently	String	11		mEEwriteall	6D 45 45 77 72 69 74 65 61 6C 6C 6C
Status code	Accepted when value is 1	Bool_1	1	Error: Success:	0 1	00 01

Table 142: Example: sAN mEEwriteall

CoLa A	<STX>sAN{SPC}mEEwriteall{SPC}1<ETX> 02 73 41 4E 20 6D 45 45 77 72 69 74 65 61 6C 6C 20 31 03
CoLa B	02 02 02 02 00 00 00 11 73 41 4E 20 6D 45 45 77 72 69 74 65 61 6C 6C 20 01 0C

13.6.1.4.3.22 Set to run [sMN Run]

Log out from device and activate all parameter changes.

Table 143: Telegram structure: sMN Run

Telegram structure: sMN Run						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Method	String	3		sMN	73 4D 4E
Command	Start the device	String	3		Run	52 75 6E

Table 144: Example: sMN Run

CoLa A	<STX>sMN{SPC}Run<ETX> <STX>sMN Run<ETX> sMN Run 02 73 4D 4E 20 52 75 6E 03	
CoLa B	02 02 02 02 00 00 00 07 73 4D 4E 20 52 75 6E 19 73 4D 4E 20 52 75 6E	

Table 145: Telegram structure: sAN Run

Telegram structure: sAN Run						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sAN	73 41 4E
Command	Start the device	String	3		Run	52 75 6E
Status code	Accepted when value is 1	Bool_1	1	Error: Success:	0 1	00 01

Table 146: Example: sAN Run

CoLa A	<STX>sAN{SPC}Run{SPC}1<ETX> 02 73 41 4E 20 52 75 6E 20 31 03
CoLa B	02 02 02 02 00 00 00 09 73 41 4E 20 52 75 6E 20 01 34

13.6.1.4.3.23 Reboot device [sMN mSCreboot]

This command includes saving all parameters.

Table 147: Telegram structure: sMN mSCreboot

Telegram structure: sMN mSCreboot (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Method	String	3		sMN	73 4D 4E
Command	Reboot device	String	9		mSCreboot	6D 53 43 72 65 62 6F 6F 74

Table 148: Example: sMN mSCreboot

CoLa A	<STX>sMN[SPC]mSCreboot<ETX>					
	<STX>sMN mSCreboot<ETX>					
	sMN mSCreboot					
	02 73 4D 4E 20 6D 53 43 72 65 62 6F 6F 74 03					
CoLa B	02 02 02 02 00 00 00 0D 73 4D 4E 20 6D 53 43 72 65 62 6F 6F 74 2C					
	73 4D 4E 20 6D 53 43 72 65 62 6F 6F 74					

Table 149: Telegram structure: sAN mSCreboot

Telegram structure: sAN mSCreboot						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sAN	73 41 4E
Command	Reboot device	String	9		mSCreboot	6D 53 43 72 65 62 6F 6F 74

Table 150: Example: sAN mSCreboot

CoLa A	<STX>sAN[SPC]mSCreboot<ETX>					
	02 73 41 4E 20 6D 53 43 72 65 62 6F 6F 74 03					
CoLa B	02 02 02 02 00 00 00 0E 73 41 4E 20 6D 53 43 72 65 62 6F 6F 74 20 00					

13.6.1.4.4 Measurement output telegram

13.6.1.4.4.1 Configure the data content for the scan [sWN LMDscandatacfg]

Configuration of the channels (e.g. distance) and information that will be send by the device as measurement data (output data format).

Table 151: Telegram structure: sWN LMDscandatacfg

Telegram structure: sWN LMDscandatacfg (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E
Command	Configure scandata	String	14		LMDscandatacfg	4C 4D 44 73 63 61 6E 64 61 74 61 63 66 67
Reserved	-	Uint_8	2	Always:	0 0	00 00
RSSI	Enable or disable the output of RSSI values	Uint_8	1	RSSI disabled: RSSI enabled:	0 1	00 01
Reserved	-	Enum_8	1	Always:	0	00

**NOTE**

During the bootup phase of the device, there will be no data telegram or answer from the device.

Table 155: Telegram structure: sRN LMDscandata

Telegram structure: sRN LMDscandata						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	Only one telegram	String	11		LMDscandata	4C 4D 44 73 63 61 6E 64 61 74 61

Table 156: Example: sRN LMDscandata

CoLa A	<STX>sRN{SPC}LMDscandata<ETX>					
	<STX>sRN LMDscandata<ETX>					
	sRN LMDscandata					
	02 73 52 4E 20 4C 4D 44 73 63 61 6E 64 61 74 61 03					
CoLa B	02 02 02 02 00 00 00 0F 73 52 4E 20 4C 4D 44 73 63 61 6E 64 61 74 61 05					
	73 52 4E 20 4C 4D 44 73 63 61 6E 64 61 74 61					

Table 157: Telegram structure: sRA LMDscandata

Telegram structure: sRA LMDscandata						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)

Find complete telegram structure of the answer in "[Send data permanently \[sEN LMDscandata\]](#)", page 154.

13.6.1.4.4.3 Send data permanently [sEN LMDscandata]

Start/ stop continuous retrieval of measurement data from device. Data will be transmitted as configured in [see "Set Performance Profile \[sWN PerformanceProfileNumber\]", page 149](#) as soon as measurement data is generated by the device. No data is generated when there is an active log in, the laser is shut off (e.g. in Standby mode), the motor is stopped or in case of certain error modes (e.g. Device not ready).

**NOTE**

Data is transmitted only via LMDscandata when LMDscandata is selected as data output format.

**NOTE**

During the bootup phase of the device, there will be no data telegram or answer from the device.

Table 158: Telegram structure: sEN LMDscandata

Telegram structure: sEN LMDscandata						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Event	String	3		sEN	73 45 4E
Command	Data telegram	String	11		LMDscandata	4C 4D 44 73 63 61 6E 64 61 74 61
Measurement	Start/stop	Enum_8	1	Stop:	0	00
				Start:	1	01

Table 159: Example: sEN LMDscandata

CoLa A	<STX>sEN{SPC}LMDscandata{SPC}1<ETX>	
	<STX>sEN LMDscandata 1<ETX>	
	sEN LMDscandata 1	
	02 73 45 4E 20 4C 4D 44 73 63 61 6E 64 61 74 61 20 31 03	
CoLa B	02 02 02 02 00 00 00 11 73 45 4E 20 4C 4D 44 73 63 61 6E 64 61 74 61 20 01 33	
	73 45 4E 20 4C 4D 44 73 63 61 6E 64 61 74 61 20 01	

Table 160: Telegram structure: sEA LMDscandata

Telegram structure: sEA LMDscandata						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sEA	73 45 41
Command	Data telegram	String	11		LMDscandata	4C 4D 44 73 63 61 6E 64 61 74 61
Measurement	Start/stop	Enum_8	1	Stop: Start:	0 1	00 01

Table 161: Example: Confirmation of sEA LMDscandata

CoLa A	<STX>sEA{SPC}LMDscandata{SPC}1<ETX>
	02 73 45 41 20 4C 4D 44 73 63 61 6E 64 61 74 61 20 31 03
CoLa B	02 02 02 02 00 00 00 11 73 45 41 20 4C 4D 44 73 63 61 6E 64 61 74 61 20 01 3C

Telegram stream

The answer to the telegram will be followed by the scandata:

Table 162: Telegram structure: Datastream of sRA LMDscandata/sSN LMDscandata

Telegram structure: sRA LMDscandata / sSN LMDscandata						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3	Answer to sRN LMDscandata: Answer to sEN LMDscandata:	sRA sSN	73 52 41 73 53 4E
Command	Data telegram	String	11		LMDscandata	4C 4D 44 73 63 61 6E 64 61 74 61
Version number	For detecting format changes by the version.	Uint_16	2		1h ... FFFFh	00 01 ... FF FF
Device						
Device number	Defined with SOPAS	Uint_16	2		0h ... FFFFh	00 00 ... FF FF
Serial number	Production period (year, calendar week, number): YYWWxxxx	Uint_32	4		0h ... FFFFFFFFh	00 00 00 00 ... FF FF FF FF
Device status	(See values column)	Uint_8	2 x 1	Ok: Error:	0 0 1 0	00 00 01 00
Status info						

Telegram structure: sRA LMDscandata / sSN LMDscandata						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Telegram counter	Number of measurement telegrams finished in the scanner and given to the interface. Does not count how many telegrams were really given out; is relevant if not all scans are delivered from the scan core.	Uint_16	2		0h ... FFFFh	00 00 ... FF FF
Scan counter	Number of scans which were created in the device; counts how many scans were really done.	Uint_16	2		0h ... FFFFh	00 00 ... FF FF
Time since start up in μ s	In the output telegram this is the time at the zero index before the measurement itself starts. Time base is the bootup time in μ s since power up, max. 71 min then starting from 0 again.	Uint_32	4		0h ... FFFFFFFFh	00 00 00 00 ... FF FF FF FF
Time of transmission in μ s	In the output telegram this is the time when the complete scan is transmitted to the buffer for data output. Time base is the bootup time in μ s since power up, max. 71 min then starting from 0 again.	Uint_32	4		0h ... FFFFFFFFh	00 00 00 00 ... FF FF FF FF
Status of digital inputs	Low byte represents input 1.	Uint_8	2 x 1	All inputs low: Input 1 high: Input 2 high: Input 1+2 high: All inputs high: (Depends on system plug and port configuration)	0 0 1 0 2 0 3 0 3 F	00 00 01 00 02 00 03 00 03 0F
Status of digital outputs	Low byte represents output 1.	Uint_8	2 x 1	All outputs low: All outputs high:	0 0 3 F	00 00 03 0F
Reserved	-	Uint_16	2	Always:	0	00 00 00 00
Frequencies						
Scan frequency	[1/100 Hz]	Uint_32	4	15 Hz ... 50 Hz: (depends on Dynamic Sensing Profile see "Dynamic Sensing Profiles", page 19)	5DCh ... 1388h	05 DC ... 13 88
Reserved	-	Uint_32	4	Always:	0	00 00 00 00
Encoder						
Amount of encoder	If 0, then next two telegram parts are missing.	Enum_16	2	External encoder	0 1	00 00 00 01
Encoder position	Info in ticks	Uint_32	4	Only filled if parameter is activated	0h ... FFFFFFFFh	00 00 00 00 ... FF FF FF FF

Telegram structure: sRA LMDscandata / sSN LMDscandata						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Encoder speed	mm/sec or milligrad/sec	Uint_16	2	Only filled if parameter is activated	0h ... FFFFh	00 00 ... FF FF
16 bit output channel (Distance)						
Amount of 16 bit channels	Number of 16 bit channels that provide measured data	Enum_16	2	Output channels	1 or 3	01 or 03
Content	Defines the content of the output channel	String	5	Distance values: Distance values: Distance values:	DIST1 DIST2 DIST3	44 49 53 54 31 44 49 53 54 32 44 49 53 54 33
Scale factor	Scale factor or factor of the measurement values	Real as float according to IEEE754	4	Factor x 1: Factor x 2: Factor x 1 (REFL1) (alternative):	3F800000h 40000000h	3F 80 00 00 40 00 00 00
Reserved	-	Real as float according to IEEE754	4	Always:	00000000	00 00 00 00
Start angle	[1/10000°]	Int_32	4	-48° ... 228° (depends on Dynamic Sensing Profile see "Dynamic Sensing Profiles", page 19)	FFF8AD00h ... 22CA40h	FF F8 AD 00h ... 00 22 CA 40h
Size of single angular step	Output format in degree: 1/10000°	Uint_16	2	0.1° ... 1°: (depends on angular resolution of Dynamic Sensing Profile see "Dynamic Sensing Profiles", page 19)	3E8h ... 2710h	03 E8 ... 27 10
Amount of data	Number of transmitted measurement points	Uint_16	2		0h ... FFFFh	00 00 ... FF FF
Data_1 Data_n	Data stream starting Data_1 to Data_n	Uint_16	2		0h ... 61A8h	00 00 00 00 ... 00 00 61 A8
8 bit output channel (RSSI)						
Amount of 8 bit channels	Number of 8 bit channels that provide measured data	Enum_16	2	Output channels:	1 or 3	01 or 03
Content	Defines the content of the output channel (RSSI)	String	5		RSSI1 RSSI2 RSSI3	52 53 53 49 31 52 53 53 49 32 52 53 53 49 33
Scale factor	Scale factor of the RSSI values	Real as float according to IEEE754	4	Factor x 1: Factor x 2:	3F800000h 40000000h	3F 80 00 00 40 00 00 00
Reserved	-	Real as float according to IEEE754	4	Always:	00000000h	00 00 00 00
Start angle	Output format: [1/10000°]	Uint_32	4	-48° ... 228° (depends on Dynamic Sensing Profile see "Dynamic Sensing Profiles", page 19)	FFF8AD00h ... 22CA40h	FF F8 AD 00h ... 00 22 CA 40h

Telegram structure: sRA LMDscandata / sSN LMDscandata						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Size of single angular step	Output format: 1/10000°	Uint_16	2	0.1° ... 1°: (depends on Dynamic Sensing Profile see "Dynamic Sensing Profiles", page 19)	3E8h ... 2710h	03 E8 ... 27 10
Amount of data	Number of transmitted measurement points	Uint_16	2		0h ... FFFFh	00 00 ... FF FF
Data_1 Data_n	Data stream starting Data_1 to Data_n	Uint_8	1		00h ... FFh	00 ... FF
Reserved						
Reserved	-	Enum_16	2	Always:	0	00 00
Reserved						
Reserved	-	Enum_16	2	Always:	0	00 00
Reserved						
Reserved	-	Enum_16	2	Always:	0	00 00
Reserved						
Reserved	-	Enum_16	2	Always:	0	00 00
Reserved						
Reserved	-	Uint_16	2	Always:	0	00 00

LMDscandata - reserved values

Valid distance measurement values are values starting from 16d upwards; everything below has the following meaning:

DIST	RSSI	Description
0d	0d	Invalid measurement value; caused by very low remission (extremely dark object), object distance not within measurement range (too close or too far away) or selected filter settings at device
1d - 15d	0d	Reserved, currently not in use
≥16d	>0h	Valid measurement values
≥16d	254d	Reflector detected
≥16d	255d	Reserved

Higher measurement values will be given out with a zero, that means no measurement value detected.

13.6.1.4.4 Set scan data enable [sWN ScanDataEnable]

Enables/ Disables streaming data output

Table 163: Telegram structure: sWN ScanDataEnable

Telegram structure: sWN ScanDataEnable (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 E4
Command	Enables/ Disables streaming data output.	String	14		ScanDataEnable	53 63 61 6E 44 61 74 61 45 6E 61 62 6C 65
Data		Bool	1	Off: On:	0d (00h) +1d (01h)	00 ... 01

Table 164: Example: sWN ScanDataEnable 0 - Disable the streaming data output

CoLa A	<STX>sWN{SPC}ScanDataEnable{SPC}0<ETX>	
	<STX>sWN ScanDataEnable 0<ETX>	
	sWN ScanDataEnable 0	
	02 73 57 4E 20 53 63 61 6E 44 61 74 61 45 6E 61 62 6C 65 20 30 03	
CoLa B	02 02 02 02 00 00 00 14 73 57 4E 20 53 63 61 6E 44 61 74 61 45 6E 61 62 6C 65 20 00 44	
	73 57 4E 20 53 63 61 6E 44 61 74 61 45 6E 61 62 6C 65 20 00	

Table 165: Telegram structure: sWA ScanDataEnable

Telegram structure: sWA ScanDataEnable (Required User Level authorized clinet)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	Enables/ Disables streaming data output.	String	14		ScanDataEnable	53 63 61 6E 44 61 74 61 45 6E 61 62 6C 65

Table 166: Example: sWA ScanDataEnable

CoLa A	<STX>sWA{SPC}ScanDataEnable<ETX>
	02 73 57 41 20 53 63 61 6E 44 61 74 61 45 6E 61 62 6C 65 03
CoLa B	02 02 02 02 00 00 00 13 73 57 41 20 53 63 61 6E 44 61 74 61 45 6E 61 62 6C 65 20 4B

13.6.1.4.4.5 Set streaming ethernet settings [sWN ScanDataEthSettings]

Ethernet settings for the scan data streaming functionality of the device

Table 167: Telegram structure: sWN ScanDataEthSettings

Telegram structure: sWN ScanDataEthSettings (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E
Command	String	String	19		ScanDataEthSettings	53 63 61 6E 44 61 74 61 45 74 68 53 65 74 74 69 6E 67 73
Protocol	Transport protocol for streaming data	Enum_8	1	UDP (fix):	+1d (01h)	01
IPAddress	IP address of the destination for data receiver	Array	4		0 ...+255d (00...FF) 0 ...+255d (00..FF) 0 ...+255d (00...FF) 0 ...+255d (00...FF)	00 ... FF 00 ... FF 00 ... FF 00 ... FF
Port	Port destination of the data reseiver	Uint_16	2		0 .. +65535d(00 00...FF FF)	00 00 ... FF FF

Example: sWN ScanDataEthSettings +1 +192 +168 +0 +100 +2115

Protocol is set to UDP (1), IPAddress (192.168.0.100), Port (2115)

Table 168: Example: sWN ScanDataEthSettings

CoLa A	<STX>sWN{SPC}ScanDataEthSettings{SPC}+1{SPC}+192{SPC}+168{SPC}+0{SPC}+100{SPC}+2115<ETX>	
	<STX>sWN ScanDataEthSettings +1 +192 +168 +0 +100 +2115<ETX>	
	sWN ScanDataEthSettings +1 +192 +168 +0 +100 +2115	
CoLa B	02 73 57 4E 20 53 63 61 6E 44 61 74 61 45 74 68 53 65 74 74 69 6E 67 73 20 31 20 43 30 20 41 38 20 30 20 36 34 20 38 34 33 03	
	02 02 02 02 00 00 00 1F 73 57 4E 20 53 63 61 6E 44 61 74 61 45 74 68 53 65 74 74 69 6E 67 73 20 01 C0 A8 00 64 08 43 5F	
	73 57 4E 20 53 63 61 6E 44 61 74 61 45 74 68 53 65 74 74 69 6E 67 73 20 01 C0 A8 00 64 08 43	

Table 169: Telegram structure: sWA ScanDataEthSettings

Telegram structure: sWA ScanDataEthSettings						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	String	String	19		ScanDataEth-Settings	53 63 61 6E 44 61 74 61 45 74 68 53 65 74 74 69 6E 67 73

Table 170: Example: sWA ScanDataEthSettings

CoLa A	<STX>sWA{SPC}ScanDataEthSettings<ETX>
	02 73 57 41 20 53 63 61 6E 44 61 74 61 45 74 68 53 65 74 74 69 6E 67 73 03
CoLa B	02 02 02 02 00 00 00 18 73 57 41 20 53 63 61 6E 44 61 74 61 45 74 68 53 65 74 74 69 6E 67 73 20 16

13.6.1.4.4.6 Read streaming ethernet settings [sRN ScanDataEthSettings]

Read ethernet settings for the scan data streaming functionality of the device.

Table 171: Telegram structure: sRN ScanDataEthSettings

Telegram structure: sRN ScanDataEthSettings (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	String	String	19		ScanDataEth-Settings	53 63 61 6E 44 61 74 61 45 74 68 53 65 74 74 69 6E 67 73

Table 172: Example: sRN ScanDataEthSettings

CoLa A	<STX>sWN{SPC}ScanDataEthSettings<ETX>	
	<STX>sRN ScanDataEthSettings<ETX>	
	sRN ScanDataEthSettings	
CoLa B	02 73 52 4E 20 53 63 61 6E 44 61 74 61 45 74 68 53 65 74 74 69 6E 67 73 03	
	02 02 02 02 00 00 00 17 73 52 4E 20 53 63 61 6E 44 61 74 61 45 74 68 53 65 74 74 69 6E 67 73 3C	
	73 57 4E 20 53 63 61 6E 44 61 74 61 45 74 68 53 65 74 74 69 6E 67 73	

Table 173: Telegram structure: sRA ScanDataEthSettings

Telegram structure: sRA ScanDataEthSettings						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	String	String	19		ScanDataEth-Settings	53 63 61 6E 44 61 74 61 45 74 68 53 65 74 74 69 6E 67 73
Protocol	Transport protocol for streaming data	Enum_8	1	UDP: TCP:	+1d (01h) +2d (02h)	01 02
IPAddress	IP address of the destination for data receiver	Array	4		0 ...+255d (00...FF) 0 ...+255d (00..FF) 0 ...+255d (00...FF) 0 ...+255d (00...FF)	00 ... FF 00 ... FF 00 ... FF 00 ... FF
Port	Port destination of the data receiver	Uint_16	2		0 .. +65535d(00 00...FF FF)	00 00 ... FF FF

Example: sRA ScanDataEthSettings +1 +192 +168 +0 +100 +2115

Protocol is set to UDP (1), IPAddress (192.168.0.100), Port (2115)

Table 174: Example: sRA ScanDataEthSettings

CoLa A	<STX>sRA{SPC}ScanDataEthSettings{SPC}1{SPC}C0{SPC}A8{SPC}0{SPC}64{SPC}843<ETX> 02 73 52 41 20 53 63 61 6E 44 61 74 61 45 74 68 53 65 74 74 69 6E 67 73 20 31 20 43 30 20 41 38 20 30 20 36 34 20 38 34 33 03
CoLa B	02 02 02 02 00 00 00 1F 73 52 41 20 53 63 61 6E 44 61 74 61 45 74 68 53 65 74 74 69 6E 67 73 20 01 C0 A8 00 64 08 43 5F

13.6.1.4.4.7 Set Encoder data enable [sWN EncoderDataEnable]

Enables/ Disables streaming Encoder data output.

Table 175: Telegram structure: sWN EncoderDataEnable

Telegram structure: sWN EncoderDataEnable (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E
Command	Set streaming Encoder data output.	String	17		EncoderDataEnable	45 6E 63 6F 64 65 72 44 61 74 61 45 6E 61 62 6C 65
Encoder data stream	Enable/ disable	Bool	1	Disbale: Enable:	0d (00h) +1d (01h)	00 ... 01

Table 176: Example: sWN EncoderDataEnable 0 - Disable the streaming Encoder data output

CoLa A	<STX> sWN {SPC} EncoderDataEnable {SPC}0<ETX>	
	<STX>sWN EncoderDataEnable 0<ETX>	
	sWN EncoderDataEnable 0	
	02 73 57 4E 20 45 6E 63 6F 64 65 72 44 61 74 61 45 6E 61 62 6C 65 20 30 03	
CoLa B	02 02 02 02 00 00 00 17 73 57 4E 20 45 6E 63 6F 64 65 72 44 61 74 61 45 6E 61 62 6C 65 20 00 2F	
	73 57 4E 20 45 6E 63 6F 64 65 72 44 61 74 61 45 6E 61 62 6C 65 20 00	

Table 177: Telegram structure: sWA EncoderDataEnable

Telegram structure: sWA EncoderDataEnable					 	
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	Set streaming Encoder data output.	String	17		EncoderDataEnable	45 6E 63 6F 64 65 72 44 61 74 61 45 6E 61 62 6C 65

Table 178: Example: sWA EncoderDataEnable

CoLa A	<STX> sWA {SPC} EncoderDataEnable <ETX>
	02 73 57 41 20 45 6E 63 6F 64 65 72 44 61 74 61 45 6E 61 62 6C 65 03
CoLa B	02 02 02 02 00 00 00 16 73 57 41 20 45 6E 63 6F 64 65 72 44 61 74 61 45 6E 61 62 6C 65 20 20

13.6.1.4.4.8 Set Encoder data streaming ethernet settings [sWN EncoderDataEthSettings]

Ethernet settings for the encoder data streaming functionality of the device

Table 179: Telegram structure: sWN EncoderDataEthSettings

Telegram structure: sWN EncoderDataEthSettings (User level 'Authorized client' required)					 	
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E
Command	Set Encoder data streaming ethernet settings	String	22		EncoderDataEthSettings	45 6E 63 6F 64 65 72 44 61 74 61 45 74 68 53 65 74 74 69 6E 67 73
Protocol	Transport protocol for streaming IMU data	Enum_8	1	UDP:	+1d (01h)	01
IP address	IP address of the destination for IMU data receiver	Array	4		0 ...+255d (00...FF) 0 ...+255d (00..FF) 0 ...+255d (00...FF) 0 ...+255d (00...FF)	00 ... FF 00 ... FF 00 ... FF 00 ... FF
Port	Port destination of the Encoder data receiver	Uint_16	2		0 ..+65535d(00 00...FF FF)	00 00 ... FF FF

Example: sWN EncoderDataEthSettings +1 +192 +168 +0 +100 +7503

Protocol is set to UPD (1), IPAddress (192.168.0.100), Port (7503)

Table 180: Example: sWN EncoderDataEthSettings

CoLa A	<STX>sWN{SPC}EncoderDataEthSettings{SPC}+1{SPC}+192{SPC}+168{SPC}+0{SPC}+100{SPC}+7503<ETX>	
	<STX>sWN EncoderDataEthSettings +1 +192 +168 +0 +100 +7503<ETX>	
	sWN EncoderDataEthSettings +1 +192 +168 +0 +100 +7503	
	02 73 57 4E 20 45 6E 63 6F 64 65 72 44 61 74 61 45 74 68 53 65 74 74 69 6E 67 73 20 31 20 43 30 20 41 38 20 30 20 36 34 20 31 44 34 46 03	
CoLa B	02 02 02 02 00 00 00 00 22 73 57 4E 20 45 6E 63 6F 64 65 72 44 61 74 61 45 74 68 53 65 74 74 69 6E 67 73 20 01 C0 A8 00 64 1D 4F 2D	
	73 57 4E 20 45 6E 63 6F 64 65 72 44 61 74 61 45 74 68 53 65 74 74 69 6E 67 73 20 01 C0 A8 00 64 1D 4F	

Table 181: Telegram structure: sWA EncoderDataEthSettings

Telegram structure: sWA EncoderDataEthSettings						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	Set Encoder data streaming ethernet settings	String	22		EncoderDataEthSettings	45 6E 63 6F 64 65 72 44 61 74 61 45 74 68 53 65 74 74 69 6E 67 73

Table 182: Example: sWA EncoderDataEthSettings

CoLa A	<STX>sWA{SPC}EncoderDataEthSettings<ETX>
	02 73 57 41 20 45 6E 63 6F 64 65 72 44 61 74 61 45 74 68 53 65 74 74 69 6E 67 73 03
CoLa B	02 02 02 02 00 00 00 00 1B 73 57 41 20 45 6E 63 6F 64 65 72 44 61 74 61 45 74 68 53 65 74 74 69 6E 67 73 20 7D

13.6.1.4.4.9 Set IMU data enable [sWN ImuDataEnable]

Enables/ Disables streaming IMU data output.

Table 183: Telegram structure: sWN ImuDataEnable

Telegram structure: sWN ImuDataEnable (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E
Command	Set streaming IMU data output.	String	13		ImuDataEnable	49 6D 75 44 61 74 61 45 6E 61 62 6C 65
IMU data stream	Enable/ disable	Bool	1	Disbale: Enable:	0d (00h) +1d (01h)	00 ... 01

Table 184: Example: sWN ImuDataEnable 0 - Disable the streaming IMU data output

CoLa A	<STX>sWN{SPC}ImuDataEnable{SPC}0<ETX>	
	<STX>sWN ImuDataEnable 0<ETX>	
	sWN ImuDataEnable 0	
	02 73 57 4E 20 49 6D 75 44 61 74 61 45 6E 61 62 6C 65 20 30 03	
CoLa B	02 02 02 02 00 00 00 00 13 73 57 4E 20 49 6D 75 44 61 74 61 45 6E 61 62 6C 65 20 00 2A	
	73 57 4E 20 49 6D 75 44 61 74 61 45 6E 61 62 6C 65 20 00	

Table 185: Telegram structure: sWA ImuDataEnable

Telegram structure: sWA ImuDataEnable						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	Set streaming IMU data output.	String	13		ImuDataEnable	49 6D 75 44 61 74 61 45 6E 61 62 6C 65

Table 186: Example: sWA ImuDataEnable

CoLa A	<STX>sWA{SPC}ImuDataEnable<ETX>					
	02 73 57 41 20 49 6D 75 44 61 74 61 45 6E 61 62 6C 65 03					
CoLa B	02 02 02 02 00 00 00 12 73 57 41 20 49 6D 75 44 61 74 61 45 6E 61 62 6C 65 20 25					

13.6.1.4.4.10 Set IMU data streaming ethernet settings [sWN ImuDataEthSettings]

Ethernet settings for the IMU data streaming functionality of the device

Table 187: Telegram structure: sWN ImuDataEthSettings

Telegram structure: sWN ImuDataEthSettings (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E
Command	Set IMU data streaming ethernet settings	String	18		ImuDataEthSettings	49 6D 75 44 61 74 61 45 74 68 53 65 74 74 69 6E 67 73
Protocol	Transport protocol for streaming IMU data	Enum_8	1	UDP:	+1d (01h)	01
IP address	IP address of the destination for IMU data receiver	Array	4		0 ...+255d (00...FF) 0 ...+255d (00..FF) 0 ...+255d (00...FF) 0 ...+255d (00...FF)	00 ... FF 00 ... FF 00 ... FF 00 ... FF
Port	Port destination of the IMU data receiver	Uint_16	2		0 ..+65535d(00 00...FF FF)	00 00 ... FF FF

Example: sWN ImuDataEthSettings +1 +192 +168 +0 +100 +7503

Protocol is set to UPD (1), IPAddress (192.168.0.100), Port (7503)

Table 188: Example: sWN ImuDataEthSettings

CoLa A	<STX>sWN{SPC}ImuDataEthSettings{SPC}+1{SPC}+192{SPC}+168{SPC}+0{SPC}+100{SPC}+7503<ETX>					
	<STX>sWN ImuDataEthSettings +1 +192 +168 +0 +100 +7503<ETX>					
	sWN ImuDataEthSettings +1 +192 +168 +0 +100 +7503					
CoLa B	02 73 57 4E 20 49 6D 75 44 61 74 61 45 74 68 53 65 74 74 69 6E 67 73 20 31 20 43 30 20 41 38 20 30 20 36 34 20 31 44 34 46 03					
	02 02 02 02 00 00 00 1E 73 57 4E 20 49 6D 75 44 61 74 61 45 74 68 53 65 74 74 69 6E 67 73 20 01 C0 A8 00 64 1D 4F 28					
	73 57 4E 20 49 6D 75 44 61 74 61 45 74 68 53 65 74 74 69 6E 67 73 20 01 C0 A8 00 64 1D 4F					

Table 189: Telegram structure: sWA ImuDataEthSettings

Telegram structure: sWA ImuDataEthSettings						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	Set IMU data streaming ethernet settings	String	18		ImuDataEthSettings	49 6D 75 44 61 74 61 45 74 68 53 65 74 74 69 6E 67 73

Table 190: Example: sWA ImuDataEthSettings

CoLa A	<STX>sWA{SPC}ImuDataEthSettings<ETX> 02 73 57 41 20 49 6D 75 44 61 74 61 45 74 68 53 65 74 74 69 6E 67 73 03
CoLa B	02 02 02 02 00 00 00 17 73 57 41 20 49 6D 75 44 61 74 61 45 74 68 53 65 74 74 69 6E 67 73 20 78

13.6.1.4.4.11 Read scan data format [sRN ScanDataFormat]

Return of the scan data format

Table 191: Telegram structure: sRN ScanDataFormat

Telegram structure: sRN ScanDataFormat (User Level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	Data serialization format	String	14		ScanDataFormat	53 63 61 6E 44 61 74 61 46 6F 72 6D 61 74

Example: sRN ScanDataFormat

Read of the data serialization format

Table 192: Example: sRN ScanDataFormat

CoLa A	<STX>sRN{SPC}ScanDataFormat<ETX> <STX>sRN ScanDataFormat<ETX> sRN ScanDataFormat 02 73 52 4E 20 53 63 61 6E 44 61 74 61 46 6F 72 6D 61 74 03	
	02 02 02 02 00 00 00 12 73 52 4E 20 53 63 61 6E 44 61 74 61 46 6F 72 6D 61 74 63	
CoLa B	73 52 4E 20 53 63 61 6E 44 61 74 61 46 6F 72 6D 61 74	

Table 193: Telegram structure: sRA ScanDataFormat

Telegram structure: sRA ScanDataFormat						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	Data serialization format	String	14		ScanDataFormat	53 63 61 6E 44 61 74 61 46 6F 72 6D 61 74

Telegram structure: sRA ScanDataFormat						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Data		Enum_8	1	MSGPACK: Compact: LMDscandata: Native ROS2:	1h 2h 3h 4h	01 02 03 04

Example: sRA ScanDataFormat

Scan data format is set to Compact = 2

Table 194: Example: sRA ScanDataFormat

CoLa A	<STX>sRA{SPC}ScanDataFormat{SPC}2<ETX> 02 73 57 41 20 53 63 61 6E 44 61 74 61 46 6F 72 6D 61 74 20 32 03
CoLa B	02 02 02 02 00 00 00 14 73 57 41 20 53 63 61 6E 44 61 74 61 46 6F 72 6D 61 74 20 02 4E

13.6.1.4.12 Set Scan data format [sWN ScanDataFormat]

Set the data serialization format

Table 195: Telegram structure: sWN ScanDataFormat

Telegram structure: sWN ScanDataFormat (User Level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E
Command	Data serialization format	String	14		ScanDataFormat	53 63 61 6E 44 61 74 61 46 6F 72 6D 61 74
Variable Data	Data	Enum_8	1	MSGPACK: Compact (recommended): LMDscandata: Native ROS2:	+1d (1h) +2d (2h) +3d (3h) +4d (4h)	01 02 03 04

Example: sWN ScanDataFormat

Scan data format set to **Compact** format

Table 196: Example: sWN ScanDataFormat

CoLa A	<STX>sWN{SPC}ScanDataFormat{SPC}2<ETX> <STX>sWN ScanDataFormat 2<ETX> sWN ScanDataFormat 2 02 73 57 4E 20 53 63 61 6E 44 61 74 61 46 6F 72 6D 61 74 20 32 03	
	02 02 02 02 00 00 00 14 73 57 4E 20 53 63 61 6E 44 61 74 61 46 6F 72 6D 61 74 20 02 44	
CoLa B	73 57 4E 20 53 63 61 6E 44 61 74 61 46 6F 72 6D 61 74 20 02	

Table 197: Telegram structure: sWA ScanDataFormat

Telegram structure: sWA ScanDataFormat						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41

Telegram structure: sWA ScanDataFormat						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command	Data serialization format	String	14		ScanDataFormat	53 63 61 6E 44 61 74 61 46 6F 72 6D 61 74

Table 198: Example: sWA ScanDataFormat

CoLa A	<STX>sWA{SPC}ScanDataFormat<ETX> 02 73 57 41 20 53 63 61 6E 44 61 74 61 46 6F 72 6D 61 74 03
CoLa B	02 02 02 02 00 00 00 13 73 57 41 20 53 63 61 6E 44 61 74 61 46 6F 72 6D 61 74 20 49

13.6.1.4.13 Set the session timeout for TCP/IP communication [sWN EtherSessionTimeout]

Set the session timeout for TCP/IP communication (e.g. LMDscandata communication). Be careful with low values as they might cause unexpected disconnections. Write access requires user level: Service.

Table 199: Telegram structure: sWN EtherSessionTimeout

Telegram structure: sWN EtherSessionTimeout (User level 'Service' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write request	String	3		sWN	73 57 4E
Variable name	EtherSessionTimeout	String	19		EtherSessionTimeout	45 74 68 65 72 53 65 73 73 69 6F 6E 54 69 6D 65 6F 75 74
Value	Session timeout in ms	UInt_32	4		+15000	00 00 3A 98

Table 200: Example: sWN EtherSessionTimeout

CoLa A	<STX>sWN{SPC}EtherSessionTimeout{SPC}+15000<ETX>	
	<STX>sWN EtherSessionTimeout +15000<ETX>	
	sWN EtherSessionTimeout +15000	
	02 73 57 4E 20 45 74 68 65 72 53 65 73 73 69 6F 6E 54 69 6D 65 6F 75 74 20 2B 31 35 30 30 30 03	
CoLa B	02 02 02 02 00 00 00 1C 73 57 4E 20 45 74 68 65 72 53 65 73 73 69 6F 6E 54 69 6D 65 6F 75 74 20 00 00 3A 98 83	
	73 57 4E 20 45 74 68 65 72 53 65 73 73 69 6F 6E 54 69 6D 65 6F 75 74 20 00 00 3A 98	

Table 201: Telegram structure: sWA EtherSessionTimeout

Telegram structure: sWA EtherSessionTimeout						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write answer	String	3		sWA	73 57 41
Variable name	Echo of variable name	String	19		EtherSessionTimeout	45 74 68 65 72 53 65 73 73 69 6F 6E 54 69 6D 65 6F 75 74

Table 202: Example: sWA EtherSessionTimeout

CoLa A	<STX>sWA{SPC}EtherSessionTimeout<ETX> 02 73 57 41 20 45 74 68 65 72 53 65 73 73 69 6F 6E 54 69 6D 65 6F 75 74 03
CoLa B	02 02 02 02 00 00 00 17 73 57 41 20 45 74 68 65 72 53 65 73 73 69 6F 6E 54 69 6D 65 6F 75 74 0E

13.6.1.4.4.14 Read the session timeout for TCP/IP communication [sRN EtherSessionTimeout]

Returns the session timeout for TCP/IP communication (e.g. LMDscandata communication). Be careful with low values as they might cause unexpected disconnections.

Table 203: Telegram structure: sRN EtherSessionTimeout

Telegram structure: sRN EtherSessionTimeout						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read request	String	3		sRN	73 52 4E
Variable name	EtherSessionTimeout	String	19		EtherSessionTimeout	45 74 68 65 72 53 65 73 73 69 6F 6E 54 69 6D 65 6F 75 74

Table 204: Example: sRN EtherSessionTimeout

CoLa A	<STX>sRN{SPC}EtherSessionTimeout<ETX>					
	<STX>sRN EtherSessionTimeout<ETX>					
	sRN EtherSessionTimeout					
CoLa B	02 73 52 4E 20 45 74 68 65 72 53 65 73 73 69 6F 6E 54 69 6D 65 6F 75 74 03					
	02 02 02 02 00 00 00 17 73 52 4E 20 45 74 68 65 72 53 65 73 73 69 6F 6E 54 69 6D 65 6F 75 74 04					
	73 52 4E 20 45 74 68 65 72 53 65 73 73 69 6F 6E 54 69 6D 65 6F 75 74					

Table 205: Telegram structure: sRA EtherSessionTimeout

Telegram structure: sRA EtherSessionTimeout						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read answer	String	3		sRA	73 52 41
Variable name	EtherSessionTimeout	String	19		EtherSessionTimeout	45 74 68 65 72 53 65 73 73 69 6F 6E 54 69 6D 65 6F 75 74
Value	Session timeout in ms	UInt_32	4		+15000	00 00 3A 98

Table 206: Example: sRA EtherSessionTimeout

CoLa A	<STX>sRA{SPC}EtherSessionTimeout{SPC}+15000<ETX>					
	02 73 52 41 20 45 74 68 65 72 53 65 73 73 69 6F 6E 54 69 6D 65 6F 75 74 20 2B 31 35 30 30 30 03					
CoLa B	02 02 02 02 00 00 00 1C 73 52 41 20 45 74 68 65 72 53 65 73 73 69 6F 6E 54 69 6D 65 6F 75 74 20 00 00 3A 98 89					

13.6.1.4.4.15 Set the content of the Compact format telegram type 1 [sWN compactTelegramType1Content]

Sets the content of the Compact format telegram type 1 (e.g. whether to include the RSSI values).

Table 207: Telegram structure: sWN compactTelegramType1Content

Telegram structure: sWN compactTelegramType1Content (User level 'Authorized Client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write request	String	3		sWN	73 57 4E

Telegram structure: sWN compactTelegramType1Content (User level 'Authorized Client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Variable name	compactTelegramType1Content	String	27		compactTelegramType1Content	63 6F 6D 70 61 63 74 54 65 6C 65 67 72 61 6D 54 79 70 65 31 43 6F 6E 74 65 6E 74
includeRssi	If true, RSSIs are serialized	Bool_1	1	Disable: Enable:	0 1	00 01
includeProperties	If true, properties are serialized	Bool_1	1	Disable: Enable:	0 1	00 01

Table 208: Example: sWN compactTelegramType1Content

CoLa A	<STX>sWN{SPC}compactTelegramType1Content{SPC}1{SPC}1<ETX>					
	<STX>sWN compactTelegramType1Content 1 1<ETX>					
CoLa B	sWN compactTelegramType1Content 1 1					
	02 73 57 4E 20 63 6F 6D 70 61 63 74 54 65 6C 65 67 72 61 6D 54 79 70 65 31 43 6F 6E 74 65 6E 74 20 31 20 31 03					
CoLa B	02 02 02 02 00 00 00 22 73 57 4E 20 63 6F 6D 70 61 63 74 54 65 6C 65 67 72 61 6D 54 79 70 65 31 43 6F 6E 74 65 6E 74 20 01 01 6C					
	73 57 4E 20 63 6F 6D 70 61 63 74 54 65 6C 65 67 72 61 6D 54 79 70 65 31 43 6F 6E 74 65 6E 74 20 01 01					

Table 209: Telegram structure: sWA compactTelegramType1Content

Telegram structure: sWA compactTelegramType1Content						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write answer	String	3		sWA	73 57 41
Variable name	Echo of variable name	String	27		compactTelegramType1Content	63 6F 6D 70 61 63 74 54 65 6C 65 67 72 61 6D 54 79 70 65 31 43 6F 6E 74 65 6E 74

Table 210: Example: sWA compactTelegramType1Content

CoLa A	<STX>sWA{SPC}compactTelegramType1Content<ETX>					
	02 73 57 41 20 63 6F 6D 70 61 63 74 54 65 6C 65 67 72 61 6D 54 79 70 65 31 43 6F 6E 74 65 6E 74 03					
CoLa B	02 02 02 02 00 00 00 1F 73 57 41 20 63 6F 6D 70 61 63 74 54 65 6C 65 67 72 61 6D 54 79 70 65 31 43 6F 6E 74 65 6E 74 43					

13.6.1.4.4.16 Read compactTelegramType1Content [sRN compactTelegramType1Content]

Returns the content of the Compact format telegram type 1 (e.g. whether to include the RSSI values).

Table 211: Telegram structure: sRN compactTelegramType1Content

Telegram structure: sRN compactTelegramType1Content						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read request	String	3		sRN	73 52 4E

Telegram structure: sRN compactTelegramType1Content						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Variable name	compactTelegram-Type1Content	String	27		compactTele-gramType1Con- tent	63 6F 6D 70 61 63 74 54 65 6C 65 67 72 61 6D 54 79 70 65 31 43 6F 6E 74 65 6E 74

Table 212: Example: sRN compactTelegramType1Content

CoLa A	<STX>sRN{SPC}compactTelegramType1Content<ETX>					
	<STX>sRN compactTelegramType1Content<ETX>					
	sRN compactTelegramType1Content					
CoLa B	02 73 52 4E 20 63 6F 6D 70 61 63 74 54 65 6C 65 67 72 61 6D 54 79 70 65 31 43 6F 6E 74 65 6E 74 03					
	02 02 02 02 00 00 00 1F 73 52 4E 20 63 6F 6D 70 61 63 74 54 65 6C 65 67 72 61 6D 54 79 70 65 31 43 6F 6E 74 65 6E 74 49					
	73 52 4E 20 63 6F 6D 70 61 63 74 54 65 6C 65 67 72 61 6D 54 79 70 65 31 43 6F 6E 74 65 6E 74					

Table 213: Telegram structure: sRA compactTelegramType1Content

Telegram structure: sRA compactTelegramType1Content						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read answer	String	3		sRA	73 52 41
Variable name	compactTelegram-Type1Content	String	27		compactTele-gramType1Con- tent	63 6F 6D 70 61 63 74 54 65 6C 65 67 72 61 6D 54 79 70 65 31 43 6F 6E 74 65 6E 74
includerssi	If true, RSSIs are serialized	Bool_1	1	Disabled: Enabled:	0 1	00 01
include-Properties	If true, properties are serialized	Bool_1	1	Disabled: Enabled:	0 1	00 01

Table 214: Example: sRA compactTelegramType1Content

CoLa A	<STX>sRA{SPC}compactTelegramType1Content{SPC}{SPC}<ETX>					
	02 73 52 41 20 63 6F 6D 70 61 63 74 54 65 6C 65 67 72 61 6D 54 79 70 65 31 43 6F 6E 74 65 6E 74 20 31 20 31 03					
CoLa B	02 02 02 02 00 00 00 22 73 52 41 20 63 6F 6D 70 61 63 74 54 65 6C 65 67 72 61 6D 54 79 70 65 31 43 6F 6E 74 65 6E 74 20 01 01 66					

13.6.1.4.4.17 Write ROS parent frame ID [sWN rosParentFrameId]

Set the parent frame ID of the published topics.

Table 215: Telegram structure: sWN rosParentFrameId

Telegram structure: sWN rosParentFrameId (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E

Telegram structure: sWN rosParentFrameld (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command	Set parent frame ID of the published topics	String	16		rosParentFrameld	72 6F 73 50 61 72 65 6E 74 46 72 61 6D 65 49 64
Value	Number of characters of the parent frame id	Uint_16	1		0d ... +16d (0h ... 10h)	00 00 ... 00 10
Value	Configured parent frame ID	Flex-String	64		[Parent frame ID]	[Parent frame ID]

Table 216: Example: sWN rosParentFrameld

CoLa A	<STX>sWN{SPC}rosParentFrameld{SPC}5{SPC}world<ETX>					
	<STX>sWN rosParentFrameld 5 world<ETX>					
	sWN rosFrameld 5 world					
CoLa B	02 73 57 4E 20 72 6F 73 50 61 72 65 6E 74 46 72 61 6D 65 49 64 20 2B 35 20 77 6F 72 6C 64 03					
	02 02 02 02 00 00 00 1D 73 57 4E 20 72 6F 73 50 61 72 65 6E 74 46 72 61 6D 65 49 64 20 00 05 20 77 6F 72 6C 64 0F					
	73 57 4E 20 72 6F 73 50 61 72 65 6E 74 46 72 61 6D 65 49 64 20 00 05 20 77 6F 72 6C 64					

Table 217: Telegram structure: sWA rosParentFrameld

Telegram structure: sWA rosParentFrameld						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	Set parent frame ID of the published topics	String	16		rosParentFrameld	72 6F 73 50 61 72 65 6E 74 46 72 61 6D 65 49 64

Table 218: Example: sWA rosParentFrameld

CoLa A	<STX>sWA{SPC}rosParentFrameld<ETX>					
	02 73 57 41 20 72 6F 73 50 61 72 65 6E 74 46 72 61 6D 65 49 64 03					
CoLa B	02 02 02 02 00 00 00 15 73 57 41 20 72 6F 73 50 61 72 65 6E 74 46 72 61 6D 65 49 64 20 47					

13.6.1.4.4.18 Read ROS parent frame ID [sRN rosParentFrameld]

Returns the parent frame ID of the published topics.

Table 219: Telegram structure: sRN rosParentFrameld

Telegram structure: sRN rosParentFrameld (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	Read parent frame ID of the published topics	String	16		rosParentFrameld	72 6F 73 50 61 72 65 6E 74 46 72 61 6D 65 49 64

Table 220: Example: sRN rosParentFrameld

CoLa A	<STX>sRN{SPC}rosParentFrameld<ETX>	
	<STX>sRN rosParentFrameld<ETX>	
	sRN rosParentFrameld	
	02 73 52 4E 20 72 6F 73 50 61 72 65 6E 74 46 72 61 6D 65 49 64 03	
CoLa B	02 02 02 02 00 00 00 14 73 57 4E 20 72 6F 73 50 61 72 65 6E 74 46 72 61 6D 65 49 64 6D	
	73 57 4E 20 72 6F 73 50 61 72 65 6E 74 46 72 61 6D 65 49 64	

Table 221: Telegram structure: sRA rosParentFrameld

Telegram structure: sRA rosParentFrameld					 	
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	Read parent frame ID of the published topics	String	16		rosParentFrameld	72 6F 73 50 61 72 65 6E 74 46 72 61 6D 65 49 64
Value	Number of characters of the parent frame id	Uint_16	2		0d ... +16d (0h ... 10h)	00 00 ... 00 10
Value	Configured parent frame ID	Flex-String	64		[Parent frame ID]	[Parent frame ID]

Table 222: Example: sRA rosParentFrameld

CoLa A	<STX>sRA{SPC}rosParentFrameld{SPC}5{SPC}world<ETX>
	02 73 52 41 20 72 6F 73 50 61 72 65 6E 74 46 72 61 6D 65 49 64 20 2B 35 20 77 6F 72 6C 64 03
CoLa B	02 02 02 02 00 00 00 1C 73 52 41 20 72 6F 73 50 61 72 65 6E 74 46 72 61 6D 65 49 64 20 00 05 20 77 6F 72 6C 64 61

13.6.1.4.4.19 Write ROS frame ID [sWN rosFrameld]

Set the frame ID of the published topics.

Table 223: Telegram structure: sWN rosFrameld

Telegram structure: sWN rosFrameld (User level 'Authorized client' required)					 	
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E
Command	ROS reference frame to use for data output	String	10		rosFrameld	72 6F 73 46 72 61 6D 65 49 64
Value	Number of characters of the frame id	Uint_16	1		0d ... +16d (0h ... 10h)	00 00 ... 00 10
Value	Configured Frame ID of the device	Flex-String	64		[Frame ID]	[Frame ID]

Table 224: Example: sWN rosFrameld

CoLa A	<STX>sWN{SPC}rosFrameld{SPC}4{SPC}test<ETX>	
	<STX>sWN rosFrameld 4 test<ETX>	
	sWN rosFrameld 4 test	
	02 73 57 4E 20 72 6F 73 46 72 61 6D 65 49 64 20 34 20 74 65 73 74 03	
CoLa B	02 02 02 02 00 00 00 16 73 57 4E 20 72 6F 73 46 72 61 6D 65 49 64 20 00 01 74 65 73 74 43	
	73 57 4E 20 72 6F 73 46 72 61 6D 65 49 64 20 00 01 74 65 73 74	

Table 225: Telegram structure: sWA rosFrameld

Telegram structure: sWA rosFrameld						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	ROS reference frame to use for data output	String	10		rosFrameld	72 6F 73 46 72 61 6D 65 49 64

Table 226: Example: sWA rosFrameld

CoLa A	<STX>sWA{SPC}rosFrameld<ETX> 02 73 57 41 20 72 6F 73 46 72 61 6D 65 49 64 03
CoLa B	02 02 02 02 00 00 00 0F 73 57 41 20 72 6F 73 46 72 61 6D 65 49 64 20 7B

13.6.1.4.4.20 Read ROS frame ID [sRN rosFrameld]

Returns the frame ID of the published topics.

Table 227: Telegram structure: sRN rosFrameld

Telegram structure: sRN rosFrameld (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	ROS reference frame to use for data output	String	10		rosFrameld	72 6F 73 46 72 61 6D 65 49 64

Table 228: Example: sRN rosFrameld

CoLa A	<STX>sRN{SPC}rosFrameld<ETX> <STX>sRN rosFrameld<ETX> sRN rosFrameld 02 73 52 4E 20 72 6F 73 46 72 61 6D 65 49 64 03	
CoLa B	02 02 02 02 00 00 00 0E 73 52 4E 20 72 6F 73 46 72 61 6D 65 49 64 51 73 52 4E 20 72 6F 73 46 72 61 6D 65 49 64	

Table 229: Telegram structure: sRA rosFrameld

Telegram structure: sRA rosFrameld						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	ROS reference frame to use for data output	String	10		rosFrameld	73 57 4E 20 72 6F 73 46 72 61 6D 65 49 64
Value	Number of characters of the frame id	Uint_16	1		0d ... +16d (0h ... 10h)	00 00 ... 00 10
Value	Configured Frame ID of the device	Flex-String	64		[Frame ID]	[Frame ID]

Table 230: Example: sRA rosFrameld - Frame ID is "picoScan_"

CoLa A	<STX>sRA{SPC}rosFrameld{SPC}9{SPC}picoScan_<ETX> 02 73 52 41 20 72 6F 73 46 72 61 6D 65 49 64 20 39 20 70 69 63 6F 53 63 61 6E 5F 03
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CoLa B 02 02 02 02 00 00 00 00 1A 73 52 41 20 72 6F 73 46 72 61 6D 65 49 64 20 00 09 70 69 63 6F 53 63 61 6E 5F 02

13.6.1.4.4.21 Write ROS namespace [sWN rosNamespace]

Adds a prefix to all ROS entities (topics, services, parameters).

Table 231: Telegram structure: sWN rosNamespace

Telegram structure: sWN rosNamespace (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E
Command	Sets the namespace of the published topics	String	10		rosNamespace	72 6F 73 4E 61 6D 65 73 70 61 63 65
Value	Number of characters of the frame id	Uint_16	2		0d ... +16d (0h ... 10h)	00 00 ... 00 10
Value	Name	Flex-String	64		[Namespace]	[Namespace]

Table 232: Example: sWN rosNamespace - Namespace is "rear_lidar"

CoLa A	<STX>sWN{SPC}rosNamespace{SPC}A{SPC}rear_lidar<ETX>					
	<STX>sWN rosNamespace A rear_lidar<ETX>					
	sWN rosNamespace A rear_lidar					
	02 73 57 4E 20 72 6F 73 4E 61 6D 65 73 70 61 63 65 20 41 20 72 65 61 72 5F 6C 69 64 61 72 03					
CoLa B	02 02 02 02 00 00 00 00 1F 73 57 4E 20 72 6F 73 4E 61 6D 65 73 70 61 63 65 20 00 0A 20 72 65 61 72 5F 6C 69 64 61 72 64 20					
	73 57 4E 20 72 6F 73 4E 61 6D 65 73 70 61 63 65 20 00 0A 20 72 65 61 72 5F 6C 69 64 61 72 64					

Table 233: Telegram structure: sWA rosNamespace

Telegram structure: sWA rosNamespace						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	Sets the namespace of the published topics	String	10		rosNamespace	72 6F 73 4E 61 6D 65 73 70 61 63 65

Table 234: Example: sWA rosNamespace

CoLa A	<STX>sWA{SPC}rosNamespace<ETX>					
	02 73 57 41 20 72 6F 73 4E 61 6D 65 73 70 61 63 65 03					
CoLa B	02 02 02 02 00 00 00 00 11 73 57 41 20 72 6F 73 4E 61 6D 65 73 70 61 63 65 20 48					

13.6.1.4.4.22 Read ROS namespace [sRN rosNamespace]

Returns the namespace of the published topics.

Table 235: Telegram structure: sRN rosNamespace

Telegram structure: sRN rosNamespace (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	Reads the namespace of the published topics	String	10		rosNamespace	72 6F 73 4E 61 6D 65 73 70 61 63 65

Table 236: Example: sRN rosNamespace

CoLa A	<STX>sRN{SPC}rosNamespace<ETX>					
	<STX>sRN rosNamespace<ETX>					
	sRN rosNamespace					
	02 73 52 4E 20 72 6F 72 4E 61 6D 65 73 70 61 63 65 03					
CoLa B	02 02 02 02 00 00 00 10 73 52 4E 20 72 6F 73 4E 61 6D 65 73 70 61 63 65 62					
	73 52 4E 20 72 6F 73 4E 61 6D 65 73 70 61 63 65					

Table 237: Telegram structure: sRA rosNamespace

Telegram structure: sRA rosNamespace						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	Reads the namespace of the published topics	String	10		rosNamespace	72 6F 73 4E 61 6D 65 73 70 61 63 65
Value	Number of characters of the frame id	Uint_16	2		0d ... +16d (0h ... 10h)	00 00 ... 00 10
Value	Name	Flex-String	64		[Namespace]	[Namespace]

Table 238: Example: sRA rosNamespace - Namespace is "rear_lidar"

CoLa A	<STX>sRA{SPC}rosNamespace{SPC}A{SPC}rear_lidar<ETX>					
	02 73 52 41 20 72 6F 73 4E 61 6D 65 73 70 61 63 65 20 41 20 72 65 61 72 5F 6C 69 64 61 72 03					
CoLa B	02 02 02 02 00 00 00 1A 73 52 41 20 72 6F 73 4E 61 6D 65 73 70 61 63 65 20 00 09 70 69 63 6F 53 63 61 6E 5F 02					

13.6.1.4.4.23 Configure the ROS domain ID for Native ROS 2 [sWN rosDomainId]

Configure the ROS domain ID for Native ROS 2.

Table 239: Telegram structure: sWN rosDomainId

Telegram structure: sWN rosDomainId (User level 'Authorized Client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write request	String	3		sWN	73 57 4E
Variable name	rosDomainId	String	11		rosDomainId	72 6F 73 44 6F 6D 61 69 6E 49 64
Value	ROS domain ID	UInt_8	1		0	00

Table 240: Example: sWN rosDomainId

CoLa A	<STX>sWN{SPC}rosDomainId{SPC}0<ETX>	
	<STX>sWN rosDomainId 0<ETX>	
	sWN rosDomainId 0	
	02 73 57 4E 20 72 6F 73 44 6F 6D 61 69 6E 49 64 20 30 03	
CoLa B	02 02 02 02 00 00 00 11 73 57 4E 20 72 6F 73 44 6F 6D 61 69 6E 49 64 20 00 09	
	73 57 4E 20 72 6F 73 44 6F 6D 61 69 6E 49 64 20 00	

Table 241: Telegram structure: sWA rosDomainId

Telegram structure: sWA rosDomainId					 	
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write answer	String	3		sWA	73 57 41
Variable name	Echo of variable name	String	11		rosDomainId	72 6F 73 44 6F 6D 61 69 6E 49 64

Table 242: Example: sWA rosDomainId

CoLa A	<STX>sWA{SPC}rosDomainId<ETX>
	02 73 57 41 20 72 6F 73 44 6F 6D 61 69 6E 49 64 03
CoLa B	02 02 02 02 00 00 00 0F 73 57 41 20 72 6F 73 44 6F 6D 61 69 6E 49 64 26

13.6.1.4.4.24 Read the configured ROS domain ID set for Native ROS 2 [sRN rosDomainId]

Returns the configured ROS domain ID set for Native ROS 2.

Table 243: Telegram structure: sRN rosDomainId

Telegram structure: sRN rosDomainId					 	
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read request	String	3		sRN	73 52 4E
Variable name	rosDomainId	String	11		rosDomainId	72 6F 73 44 6F 6D 61 69 6E 49 64

Table 244: Example: sRN rosDomainId

CoLa A	<STX>sRN{SPC}rosDomainId<ETX>	
	<STX>sRN rosDomainId<ETX>	
	sRN rosDomainId	
	02 73 52 4E 20 72 6F 73 44 6F 6D 61 69 6E 49 64 03	
CoLa B	02 02 02 02 00 00 00 0F 73 52 4E 20 72 6F 73 44 6F 6D 61 69 6E 49 64 2C	
	73 52 4E 20 72 6F 73 44 6F 6D 61 69 6E 49 64	

Table 245: Telegram structure: sRA rosDomainId

Telegram structure: sRA rosDomainId					 	
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read answer	String	3		sRA	73 52 41

Telegram structure: sRA rosDomainId						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Variable name	rosDomainId	String	11		rosDomainId	72 6F 73 44 6F 6D 61 69 6E 49 64
Value	ROS domain ID	UInt_8	1		0	00

Table 246: Example: sRA rosDomainId

CoLa A	<STX>sRA{SPC}rosDomainId{SPC}0<ETX> 02 73 52 41 20 72 6F 73 44 6F 6D 61 69 6E 49 64 20 30 03
CoLa B	02 02 02 02 00 00 00 11 73 52 41 20 72 6F 73 44 6F 6D 61 69 6E 49 64 20 00 03

13.6.1.4.5 Time stamp

13.6.1.4.5.1 Set time synchronization [sWN TSCRole]

Set the device synchronization mode.

Table 247: Telegram structure: sWN TSCRole

Telegram structure: sWN TSCRole (User Level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E
Command	Set timestamp role	String	7		TSCRole	54 53 43 52 6F 6C 65
Status	Timestamp role	UInt_8	1	Off: NTP: PTP:	0 1 3	00 01 03

Table 248: Example: sWN TSCRole

CoLa A	<STX>sWN{SPC}TSCRole{SPC}1<ETX> <STX>sWN TSCRole 1<ETX> sWN TSCRole 1 02 73 57 4E 20 54 53 43 52 6F 6C 65 20 31 03	
CoLa B	02 02 02 02 00 00 00 0D 73 57 4E 20 54 53 43 52 6F 6C 65 20 01 1B 73 57 4E 20 54 53 43 52 6F 6C 65 20 01	

Table 249: Telegram structure: sWA TSCRole

Telegram structure: sWA TSCRole						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	Set timestamp role	String	7		TSCRole	54 53 43 52 6F 6C 65

Table 250: Example: sWA TSCRole

CoLa A	<STX>sWA{SPC}TSCRole<ETX> 02 73 57 41 20 54 53 43 52 6F 6C 65 03
--------	---

CoLa B 02 02 02 02 00 00 00 0C 73 57 41 20 54 53 43 52 6F 6C 65 20 15

13.6.1.4.5.2 Set time stamp [sMN LSPsetdatetime]

The data format in the telegram is: +2009{SPC}+7{SPC}+22{SPC}+12{SPC}+0{SPC}+0{SPC}+0.
The numbers represent year, month, day, hour, minute, second, microsecond.

If plus is used up-front the data it is interpreted as an integer decimal number, without the plus it's the scanner reads the data as hex format.

The answer is always in ASCII format.

NOTE
Does only work if NTP- respectively PTP-Client is not active.

NOTE
There is no real time clock inside the device. When the scanner is switched off and after a reboot, the time has to be set again. Therefore, it is recommended to use a NTP or a PTP setting.

Table 251: Telegram structure: sMN LSPsetdatetime

Telegram structure: sMN LSPsetdatetime (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Method	String	3		sMN	73 4D 4E
Command	Set time stamp	String	14		LSPsetdatetime	4C 53 50 73 65 74 64 61 74 65 74 69 6D 65
Year		Uint_16	2		+1970d ... +2099d (07B2h ... 0833h)	07 B2 ... 08 33
Month		Uint_8	1		+1d ... +12d (01h ... 0Ch)	01 ... 0C
Day		Uint_8	1		+1d ... +31d (01h ... 1Fh)	00 ... 1F
Hour		Uint_8	1		+0d ... +23d (00h ... 17h)	00 ... 17
Minute		Uint_8	1		+0d ... +59d (00h ... 3Bh)	00 ... 3B
Second		Uint_8	1		+0d ... +59d (00h ... 3Bh)	00 ... 3B
Microsecond		Uint_32	4		+0d ... +999999d (00000000h ... 000F423Fh)	00 00 00 00 ... 00 0F 42 3F

Table 252: Example 1: sMN LSPsetdatetime Date: 17.02.2009; Time: 16:34

CoLa A	<STX>sMN{SPC}LSPsetdatetime{SPC}7D9{SPC}2{SPC}11{SPC}10{SPC}22{SPC}0{SPC}0<ETX>	
	<STX>sMN LSPsetdatetime 7D9 2 11 10 22 0 0<ETX>	
	sMN LSPsetdatetime 7D9 2 11 10 22 0 0	
	02 73 4D 4E 20 4C 53 50 73 65 74 64 61 74 65 74 69 6D 65 20 37 44 39 20 32 20 31 31 20 31 30 20 32 32 20 30 20 30 03	
CoLa B	02 02 02 02 00 00 00 0E 73 4D 4E 20 4C 53 50 73 65 74 64 61 74 65 74 69 6D 65 20 07 D9 02 11 10 22 00 00 00 00 00 B3	
	73 4D 4E 20 4C 53 50 73 65 74 64 61 74 65 74 69 6D 65 20 07 D9 02 11 10 22 00 00 00 00 00 00	

Table 253: Example 2: sMN LSPsetdatetime Date: 26.01.2010; Time: 10:35

CoLa A	<STX>sMN{SPC}LSPsetdatetime{SPC}+2010{SPC}+01{SPC}+26{SPC}+10{SPC}+35{SPC}0{SPC}0<ETX>					
	<STX>sMN LSPsetdatetime +2010 +01 +26 +10 +35 0 0<ETX>					
	sMN LSPsetdatetime +2010 +01 +26 +10 +35 0 0					
	02 73 4D 4E 20 4C 53 50 73 65 74 64 61 74 65 74 69 6D 65 20 2B 32 30 31 30 20 2B 30 31 20 2B 32 36 20 2B 31 30 20 2B 33 35 20 2B 30 30 20 2B 30 30 30 03					
CoLa B	02 02 02 02 00 00 00 1E 73 4D 4E 20 4C 53 50 73 65 74 64 61 74 65 74 69 6D 65 20 07 DA 01 1A 0A 23 00 00 00 00 00 B3					
	73 4D 4E 20 4C 53 50 73 65 74 64 61 74 65 74 69 6D 65 20 07 DA 01 1A 0A 23 00 00 00 00 00					

Table 254: Telegram structure: sAN LSPsetdatetime

Telegram structure: sAN LSPsetdatetime						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sAN	73 41 4E
Command	Set time stamp	String	14		LSPsetdatetime	4C 53 50 73 65 74 64 61 74 65 74 69 6D 65
Status code	Code number	Enum_8	1	Success:	0	00

Table 255: Example 1, 2: sAN LSPsetdatetime

CoLa A	<STX>sAN{SPC}LSPsetdatetime{SPC}0<ETX>					
	02 73 41 4E 20 4C 53 50 73 65 74 64 61 74 65 74 69 6D 65 20 30 03					
CoLa B	02 02 02 02 00 00 00 14 73 41 4E 20 4C 53 50 73 65 74 64 61 74 65 74 69 6D 65 20 00 50					

Activate time stamp in the output string format or on SOPAS page “data processing”.

13.6.1.4.5.3 Set NTP (Network Time Protocol) parameters

13.6.1.4.5.3.1 Set time server IP address [sWN TSCTCSrvAddr]

Define the IP address from which the device will receive the time synchronization.

Table 256: Telegram structure: sWN TSCTCSrvAddr

Telegram structure: sWN TSCTCSrvAddr (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E
Command	Set time server IP address	String	12		TSCTCSrvAddr	54 53 43 54 43 53 72 76 41 64 64 72
IP address data	Set values	Uint_8	1	First part of IP address	0 ...+255d (00... FF)	00 ... FF
				Second part of IP address	0 ...+255d (00... FF)	00 ... FF
				Third part of IP address	0 ...+255d (00... FF)	00 ... FF
				Fourth part of IP address	0 ...+255d (00... FF)	00 ... FF

Table 257: Example: sWN TSCTCSrvAddr 192.168.0.11

CoLa A	<STX>sWN{SPC}TSCTCSrvAddr{SPC}C0{SPC}A8{SPC}00{SPC}0B<ETX>	
	<STX>sWN TSCTCSrvAddr C0 A8 00 0B<ETX>	
	sWN TSCTCSrvAddr C0 A8 00 0B	
	02 73 57 4E 20 54 53 43 54 43 53 72 76 41 64 64 72 20 43 30 20 41 38 20 30 30 20 30 42 03	
CoLa B	02 02 02 02 00 00 00 15 73 57 4E 20 54 53 43 54 43 53 72 76 41 64 64 72 20 C0 A8 00 0B 3E	
	73 57 4E 20 54 53 43 54 43 53 72 76 41 64 64 72 20 C0 A8 00 0B	

Table 258: Telegram structure: sWA TSCTCSrvAddr

Telegram structure: sWA TSCTCSrvAddr						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	Set time server IP address	String	12		TSCTCSrvAddr	54 53 43 54 43 53 72 76 41 64 64 72

Table 259: Example: sWA TSCTCSrvAddr

CoLa A	<STX>sWA{SPC}TSCTCSrvAddr<ETX>
	02 73 57 41 20 54 53 43 54 43 53 72 76 41 64 64 72 03
CoLa B	02 02 02 02 00 00 00 11 73 57 41 20 54 53 43 54 43 53 72 76 41 64 64 72 20 52

13.6.1.4.5.3.2 Set time zone [sWN TSCTCtimezone]

Table 260: Telegram structure: sWN TSCTCtimezone

Telegram structure: sWN TSCTCtimezone (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E
Command	Set time zone	String	13		TSCTCtimezone	54 53 43 54 43 74 69 6D 65 7A 6F 6E 65
Time zone data	Select the time zone of the client	Enum_8	1	List of time zones see table 261, page 180	+0d ... +104d (00h ... 68h)	00 ... 68

Table 261: Time zone data Values CoLa (ASCII)

0	DATE_LINE_STANDARD	35	MONROVIA_REYKJAVIK	70	MUMBAI_NEUDELHI
1	COORD_WORLD_TIME_11	36	AMSTERDAM_BERLIN_ROM	71	SRI_JAYAWARDENEPURA
2	HAWAII	37	BELGRAD_BUDAPEST_PRAG	72	KATMANDU
3	ALASKA	38	BRUESSEL_MADRID_PARIS	73	ASTANA
4	CALIFORNIA	39	SARAJEVO_WARSCHAU	74	DAKKA
5	USA_CANADA	40	WEST_CENTRAL_AFRICA	75	NOWOSIBIRSK
6	ARIZONA	41	WINDHUK	76	YANGON
7	LA_PAZ	42	AMMAN	77	BANGKOK_HANOI_JAKARTA
8	MOUNTAIN_TIME_USA	43	ATHEN_BUKAREST	78	KRASNOJARSK
9	CENTRAL_TIME_USA	44	BEIRUT	79	IRKUTSK
10	MEXICO_CITY	45	DAMASCUS	80	KUALA_LUMPUR_SINGAPUR
11	MIDDLE_AMERICA	46	HARARE_PRETORIA	81	PEKING_HONGKONG
12	SASKATCHEWAN	47	HELSINKI_KIEW_RIGA	82	PERTH

13	BOGOTA_LIMA	48	ISTANBUL	83	TAIPEH
14	EASTERN_TIME_USA	49	JERUSALEM	84	ULAN_BATOR
15	INDIANA	50	KAIRO	85	JAKUTSK
16	CARACAS	51	KALININGRAD	86	OSAKA_TOKIO
17	ASUNCION	52	EASTERN_EUROPE	87	SEOUL
18	ATLANTIC_KANADA	53	TRIPOLIS	88	ADELAIDE
19	CUIABA	54	BAGDAD	89	DARWIN
20	LAPAZ_SANJUAN	55	KUWAIT_RIAD	90	BRISBANE
21	SANTIAGO	56	MINSK	91	CANBERRA_SYDNEY
22	NEUFUNDLAND	57	MOSKAU_PETERSBURG	92	GUAM_PORT_MORESBY
23	BRASILIA	58	NAIROBI	93	HOBART
24	BUENOS_AIRES	59	TEHERAN	94	MAGADAN
25	CAYENNE_FORTALEZA	60	ABU_DHABI	95	WLADIWOSTOK
26	GROENLAND	61	BAKU	96	SALOMONEN_KALEDONIEN
27	MONTEVIDEO	62	ERIWAN	97	TSCHOKURDACH
28	SALVADOR	63	ISCHEWSK_SAMARA	98	ANADYR
29	COORD_WORLD_TIME_02	64	PORT_LOUIS	99	AUCKLAND_WELLINGTON
30	AZOREN	65	TIFLIS	100	FIDSCHI
31	KAP_VERDE	66	KABUL	101	COORD_WORLD_TIME_12
32	CASABLANCA	67	ASCHGABET_TASCHKENT	102	NAKUALOFA
33	DUBLIN_LISSABON_LONDON	68	ISLAMABAD_KARATSCHI	103	SAMOA
34	COORD_WORLD_TIME	69	JEKATERINBURG	104	KIRITIMATI

Table 262: Example: sWN TSCTCtimezone Amsterdam, Berlin, Rom

CoLa A	<STX>sWN{SPC}TSCTCtimezone{SPC}+36<ETX>	
	<STX>sWN TSCTCtimezone +36<ETX>	
	sWN TSCTCtimezone +36	
	02 73 57 4E 20 54 53 43 54 43 74 69 6D 65 7A 6F 6E 65 20 24 03	
CoLa B	02 02 02 02 00 00 00 13 73 57 4E 20 54 53 43 54 43 74 69 6D 65 7A 6F 6E 65 20 24 16	
	73 57 4E 20 54 53 43 54 43 74 69 6D 65 7A 6F 6E 65 20 24	

Table 263: Telegram structure: sWA TSCTCtimezone

Telegram structure: sWA TSCTCtimezone					 	
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	Set time zone	String	13		TSCTCtimezone	54 53 43 54 43 74 69 6D 65 7A 6F 6E 65

Table 264: Example: sWA TSCTCtimezone

CoLa A	<STX>sWA{SPC}TSCTCtimezone<ETX>
	02 73 57 41 20 54 53 43 54 43 74 69 6D 65 7A 6F 6E 65 03
CoLa B	02 02 02 02 00 00 00 12 73 57 41 20 54 53 43 54 43 74 69 6D 65 7A 6F 6E 65 20 3D

13.6.1.4.5.3.3 Set update time [sWN TSCTCupdatetime]

Define the time period after which the sensor will attempt to get the current system time for the TSC server.

Table 265: Telegram structure: sWN TSCTCupdatetime

Telegram structure: sWN TSCTCupdatetime (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E
Command	Set update time of synchronization	String	15		TSCTCupdatetime	54 53 43 54 43 75 70 64 61 74 65 74 69 6D 65
Update time of synchronization	Set values in seconds	Uint_32	4		+1d ... +3600d (01h ... 0E10h)	00 00 00 00 ... 00 00 0E 10

Table 266: Example: sWN TSCTCupdatetime 600 s

CoLa A	<STX>sWN{SPC}TSCTCupdatetime{SPC}+600<ETX>					
	<STX>sWN TSCTCupdatetime +600<ETX>					
	sWN TSCTCupdatetime +600					
	02 73 57 4E 20 54 53 43 54 43 75 70 64 61 74 65 74 69 6D 65 20 2B 36 30 30 03					
CoLa B	02 02 02 02 00 00 00 18 73 57 4E 20 54 53 43 54 43 75 70 64 61 74 65 74 69 6D 65 20 00 00 02 58 67					
	73 57 4E 20 54 53 43 54 43 75 70 64 61 74 65 74 69 6D 65 20 00 00 02 58					

Table 267: Telegram structure: sWA TSCTCupdatetime

Telegram structure: sWA TSCTCupdatetime						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	Set update time of synchronization	String	15		TSCTCupdatetime	54 53 43 54 43 75 70 64 61 74 65 74 69 6D 65

Table 268: Example: sWA TSCTCupdatetime

CoLa A	<STX>sWA{SPC}TSCTCupdatetime<ETX>					
	02 73 57 41 20 54 53 43 54 43 75 70 64 61 74 65 74 69 6D 65 03					
CoLa B	02 02 02 02 00 00 00 14 73 57 41 20 54 53 43 54 43 75 70 64 61 74 65 74 69 6D 65 20 32					

13.6.1.4.6 Filters

13.6.1.4.6.1 Set particle filter [sWN LFPparticle]

Filter out disturbances in the measurement data caused by particles such as dust, snow flakes or similar [see "Particle filter", page 22](#).

Table 269: Telegram structure: sWN LFPparticle

Telegram structure: sWN LFPparticle (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E

Telegram structure: sWN LFPparticle (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command	Set particle filter	String	11		LFPparticle	4C 46 50 70 61 72 74 69 63 6C 65
Status code	Code number	Bool_1	1	Inactive: Active:	0 1	00 01
Threshold ²⁾	Particle threshold in mm	Uint_16	2	(must be taken)	+500d (1F4h)	01 F4

1) Never change the threshold here, it is taken by the device to handle the particles.

Table 270: Example: sWN LFPparticle

CoLa A	<STX>sWN{SPC}LFPparticle{SPC}1{SPC}+500<ETX>	
	<STX>sWN LFPparticle 1 +500<ETX>	
	sWN LFPparticle 1 +500	
	02 73 57 4E 20 4C 46 50 70 61 72 74 69 63 6C 65 20 31 20 2B 35 30 30 03	
CoLa B	02 02 02 02 00 00 00 13 73 57 4E 20 4C 46 50 70 61 72 74 69 63 6C 65 20 01 01 F4 D0	
	73 57 4E 20 4C 46 50 70 61 72 74 69 63 6C 65 20 01 01 F4	

Table 271: Telegram structure: sWA LFPparticle

Telegram structure: sWA LFPparticle						
Telegram part	Description	Variable	Length	Sensor	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	Set particle filter	String	11		LFPparticle	4C 46 50 70 61 72 74 69 63 6C 65

Table 272: Example: sWA LFPparticle

CoLa A	<STX>sWA{SPC}LFPparticle<ETX>	
	02 73 57 41 20 4C 46 50 70 61 72 74 69 63 6C 65 03	
CoLa B	02 02 02 02 00 00 00 10 73 57 41 20 4C 46 50 70 61 72 74 69 63 6C 65 20 2B	

13.6.1.4.6.2 Set echo filter [sWN FREchoFilter]



NOTE

Only available with firmware versions > V1.10.

Select which measurement value(s) shall be send via LMDscanata, if the meaurement of one angular position returns several distance values (see "Multi-echo analysis", page 18)

Table 273: Telegram structure: sWN FREchoFilter

Telegram structure: sWN FREchoFilter (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E

2) 1)

Telegram structure: sWN FREchoFilter (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command	Set echo filter	String	12		FREchoFilter	46 52 45 63 68 6F 46 69 6C 74 65 72
Status code	Code number	Enum_8	1	First echo: All echos: Last echo:	0 1 2	00 01 02

Table 274: Example: sWN FREchoFilter

CoLa A	<STX>sWN{SPC}FREchoFilter{SPC}1<ETX>		
	<STX>sWN FREchoFilter 1<ETX>		
	sWN FREchoFilter 1		
	02 73 57 4E 20 46 52 45 63 68 6F 46 69 6C 74 65 72 20 31 03		
CoLa B	02 02 02 02 00 00 00 12 73 57 4E 20 46 52 45 63 68 6F 46 69 6C 74 65 72 20 01 7E		
	73 57 4E 20 46 52 45 63 68 6F 46 69 6C 74 65 72 20 01		

Table 275: Telegram structure: sWA FREchoFilter

Telegram structure: sWA FREchoFilter						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	Set echo filter	String	12		FREchoFilter	46 52 45 63 68 6F 46 69 6C 74 65 72

Table 276: Example: sWA FREchoFilter

CoLa A	<STX>sWA{SPC}FREchoFilter<ETX>	
	02 73 57 41 20 46 52 45 63 68 6F 46 69 6C 74 65 72 03	
CoLa B	02 02 02 02 00 00 00 11 73 57 41 20 46 52 45 63 68 6F 46 69 6C 74 65 72 20 70	

13.6.1.4.6.3 Set sensitivity fog filter [sWN MCSenseLevel]

Filter out disturbances in the measurement data caused by fog or steam.

Table 277: Telegram structure: sWN MCSenseLevel

Telegram structure: sWN MCSenseLevel (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E
Command	Sense level	String	12		MCSenseLevel	4D 43 53 65 6E 73 65 4C 65 76 65 6C
Sensitivity level	Enable or disable fog filter and Sense Level	Uint_8	1	Fog Filter off Fog Filter on	0 1	00 01

Table 278: Example: sWN MCSenseLevel

CoLa A	<STX>sWN{SPC}MCSenseLevel{SPC}1<ETX>	
	<STX>sWN MCSenseLevel 1<ETX>	
	sWN MCSenseLevel 1	
	02 73 57 4E 20 4D 43 53 65 6E 73 65 4C 65 76 65 6C 20 31 03	
CoLa B	02 02 02 02 00 00 00 10 73 57 4E 20 4D 43 53 65 6E 73 65 4C 65 76 65 6C 20 01 70	
	73 57 4E 20 4D 43 53 65 6E 73 65 4C 65 76 65 6C 20 01	

Table 279: Telegram structure: sWA MCSenseLevel

Telegram structure: sWA MCSenseLevel					 	
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	Sense level	String	12		MCSenseLevel	4D 43 53 65 6E 73 65 4C 65 76 65 6C

Table 280: Example: sWA MCSenseLevel

CoLa A	<STX>sWA{SPC}MCSenseLevel<ETX>
	02 73 57 41 20 4D 43 53 65 6E 73 65 4C 65 76 65 6C 20 03
CoLa B	02 02 02 02 00 00 00 0F 73 57 41 20 4D 43 53 65 6E 73 65 4C 65 76 65 6C 20 73

13.6.1.4.6.4 Set rectangular filter [sWN LFPcubicareafilter]

Table 281: Telegram structure: sWN LFPcubicareafilter

Telegram structure: sWN LFPcubicareafilter (User level 'Authorized client' required)					 	
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E
Command	CubicAreaFilter limits a polar scan to a axisparallel cube	String	18		LFPcubicareafilter	4C 46 50 63 75 62 69 63 61 72 65 61 66 69 6C 74 65 72
Variable data 1	Enables/Disables the filter.	Bool	1	Off: On:	0d 1d	00 01
Variable Data 2	X min 1/10 mm	Int_32	4		-200000d... +200000d	FF FF B1 E0 ... 00 00 4E 20
Variable Data 3	X max 1/10 mm	Int_32	4		-200000d... +200000d	FF FF B1 E0 ... 00 00 4E 20
Variable Data 4	Y min 1/10 mm	Int_32	4		-200000d... +200000d	FF FF B1 E0 ... 00 00 4E 20
Variable Data 5	Y max 1/10 mm	Int_32	4		-200000d... +200000d	FF FF B1 E0 ... 00 00 4E 20
Variable Data 6	Z min 1/10 mm	Int_32	4	Other values than 0 will be ignored	0	00 00 00 00
Variable Data 7	Z max 1/10 mm	Int_32	4	Other values than 0 will be ignored	0	00 00 00 00

Table 282: Example: sWN LFPcubicAreafilter

	<STX>sWN{SPC}LFPcubicareafilter{SPC}0{SPC}FFFFB1E0{SPC}4E20{SPC}FFFFB1E0{SPC}4E20{SPC}FFFFB1E0{SPC}4E20<ETX>	
CoLa A	<STX>sWN LFPcubicareafilter 0 FFFFB1E0 4E20 FFFFB1E0 4E20 FFFFB1E0 4E20<ETX>	
	sWN LFPcubicareafilter 0 FFFFB1E0 4E20 FFFFB1E0 4E20 FFFFB1E0 4E20	
	02 73 57 4E 20 4C 46 50 63 75 62 69 63 61 72 65 61 66 69 6C 74 65 72 20 30 20 46 46 46 46 42 31 45 30 20 34 45 32 30 20 46 46 46 46 42 31 45 30 20 34 45 32 30 20 46 46 46 46 42 31 45 30 20 34 45 32 30 03	
CoLa B	02 02 02 02 00 00 00 30 73 57 4E 20 4C 46 50 63 75 62 69 63 61 72 65 61 66 69 6C 74 65 72 20 00 FF FF B1 E0 00 00 4E 20 FF FF B1 E0 00 00 4E 20 FF FF B1 E0 00 00 4E 20 66	
	73 57 4E 20 4C 46 50 63 75 62 69 63 61 72 65 61 66 69 6C 74 65 72 20 00 FF FF B1 E0 00 00 4E 20 FF FF B1 E0 00 00 4E 20 FF FF B1 E0 00 00 4E 20	
	00 00 4E 20 FF FF B1 E0 00 00 4E 20	

Table 283: Telegram structure: sWA LFPcubicareafilter

Telegram structure: sWALFPcubicareafilter						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	CubicAreaFilter limits a polar scan to a axisparallel cube	String	18		LFPcubicareafilter	4C 46 50 63 75 62 69 63 61 72 65 61 66 69 6C 74 65 72

Table 284: Example: sWA LFPcubicareafilter

CoLa A	<STX>sWA{SPC}LFPcubicareafilter<ETX>
	02 73 57 41 20 4C 46 50 63 75 62 69 63 61 72 65 61 66 69 6C 74 65 72 03
CoLa B	02 02 02 02 00 00 00 17 73 57 41 20 4C 46 50 63 75 62 69 63 61 72 65 61 66 69 6C 74 65 72 20 56

13.6.1.4.6.5 Activate extended range mode [sWN EnableLongRangeMode]

Extends the maximum scanning distance [see "Extended scanning range", page 40](#).

Table 285: Telegram structure: sWN EnableLongRangeMode

Telegram structure: sWN EnableLongRangeMode (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E
Command	Enable extended range mode	String	19		EnableLongRangeMode	45 6E 61 62 6C 65 4C 6F 6E 67 52 61 6E 67 65 4D 6F 64 65 65
Status code	Activate / deactivate extended range mode	Bool_1	1	Inactive: Active:	0 1	00 01

Table 286: Example: sWN EnableLongRangeMode 1

	<STX>sWN{SPC}EnableLongRangeMode{SPC}1<ETX>	
CoLa A	<STX>sWN EnableLongRangeMode 1<ETX>	
	sWN EnableLongRangeMode 1	
	02 73 57 4E 20 45 6E 61 62 6C 65 4C 6F 6E 67 52 61 6E 67 65 4D 6F 64 65 20 31 03	
CoLa B	02 02 02 02 00 00 00 19 73 57 4E 20 45 6E 61 62 6C 65 4C 6F 6E 67 52 61 6E 67 65 4D 6F 64 65 20 01 1C	
	73 57 4E 20 45 6E 61 62 6C 65 4C 6F 6E 67 52 61 6E 67 65 4D 6F 64 65 20 01	

Table 287: Telegram structure: sWA EnableLongRangeMode

Telegram structure: sWA EnableLongRangeMode						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	Enable extended range mode	String	19		EnableLongRangeMode	45 6E 61 62 6C 65 4C 6F 6E 67 52 61 6E 67 65 4D 6F 64 65

Table 288: Example: sWA EnableLongRangeMode

CoLa A	<STX>sWA{SPC}EnableLongRangeMode<ETX> 02 73 57 41 20 45 6E 61 62 6C 65 4C 6F 6E 67 52 61 6E 67 65 4D 6F 64 65 03
CoLa B	02 02 02 02 00 00 00 18 73 57 41 20 45 6E 61 62 6C 65 4C 6F 6E 67 52 61 6E 67 65 4D 6F 64 65 20 12

13.6.1.4.6.6 Read status of extended range mode [sRN EnableLongRangeMode]

Read if the extended range mode is currently active.

Table 289: Telegram structure: sRN EnableLongRangeMode

Telegram structure: sRN EnableLongRangeMode						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 57 4E
Command	Read status of extended range mode	String	19		EnableLongRangeMode	45 6E 61 62 6C 65 4C 6F 6E 67 52 61 6E 67 65 4D 6F 64 65

Table 290: Example: sRN EnableLongRangeMode

CoLa A	<STX>sRN{SPC}EnableLongRangeMode<ETX> <STX>sRN EnableLongRangeMode<ETX> sRN EnableLongRangeMode 02 73 57 4E 20 45 6E 61 62 6C 65 4C 6F 6E 67 52 61 6E 67 65 4D 6F 64 65 03	
	02 02 02 02 00 00 00 17 73 57 4E 20 45 6E 61 62 6C 65 4C 6F 6E 67 52 61 6E 67 65 4D 6F 64 65 38	
CoLa B	73 57 4E 20 45 6E 61 62 6C 65 4C 6F 6E 67 52 61 6E 67 65 4D 6F 64 65	

Table 291: Telegram structure: sRA EnableLongRangeMode

Telegram structure: sRA EnableLongRangeMode						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	Read status of extended range mode	String	19		EnableLongRangeMode	45 6E 61 62 6C 65 4C 6F 6E 67 52 61 6E 67 65 4D 6F 64 65
Status code	Status of extended range mode	Bool_1	1	Inactive: Active:	0 1	00 01

Table 292: Example: sWA EnableLongrangeMode

CoLa A	<STX>sRA{SPC}EnableLongRangeMode{SPC}1<ETX> 02 73 52 41 20 45 6E 61 62 6C 65 4C 6F 6E 67 52 61 6E 67 65 4D 6F 64 65 20 31 03
CoLa B	02 02 02 02 00 00 00 13 73 52 41 20 45 6E 61 62 6C 65 4C 6F 6E 67 52 61 6E 67 65 4D 6F 64 65 20 01 16

13.6.1.4.6.7 Set angle range filter [sWN LFPangleRangeFilter]

The angle range filter set up the horizontal (theta) and vertical (phi) start- and stop angle in rad.

BeamIncrement = the 'beamIncrement' which is used to subsample the beams within the selected angle range. With a 'beamIncrement' of n only every nth beam from the selected angle range is copied to the output scan, i.e. the angle resolution is reduced by factor n. If the beamIncrement is zero it is set to one.

Table 293: Telegram structure: sWN LFPangleRangeFilter

Telegram structure: sWN LFPangleRangeFilter (User Level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E
Command	filter set up the horizontal (theta) and vertical (phi) start- and stop angle in rad	String	19		LFPangleRangeFilter	4C 46 50 61 6E 67 6C 65 52 61 6E 67 65 46 69 6C 74 65 72
Variable Data 1	Enables/Disables the filter	Bool_1	1	Off: On:	+0d +1d	00 01
Variable Data 2	ThetaStart	Real	4	No impact on sensor setting but needs to be filled (see example)	-1,800,000d ... +1,800,000d (FFE488C0 ... 001B7740h)	FF E4 88 C0 ... 00 1B 77 40
Variable Data 3	ThetaStop	Real	4	No impact on sensor setting but needs to be filled (see example)	-1,800,000d ... +1,800,000d (FFE488C0 ... 001B7740h)	FF E4 88 C0 ... 00 1B 77 40
Variable Data 4	PhiStart	Real	4	No impact on sensor setting but needs to be filled (see example)	-900,000d ... +900,000d (FFF24460 ... 000DBBA0h)	FF F2 44 60 ... 00 0D BB A0
Variable Data 5	PhiStop	Real	4	No impact on sensor setting but needs to be filled (see example)	-900,000d ... +900,000d (FFF24460 ... 000DBBA0h)	FF F2 44 60 ... 00 0D BB A0
Variable Data 6	BeamIncrement	UInt_16	2	No impact on sensor setting but needs to be filled (see example)	1d...+20d	00 01 ... 00 14

Explanation: Enable the angle range filter and set up theta (horizontal) start -90°, theta stop +90°, phi (vertical) start -90°, phi stop +90°, beam increment 1

Table 294: Example: sWN LFPangleRangeFilter

	<STX>sWN{SPC}LFPangleRangeFilter{SPC}1{SPC}-900000{SPC}+900000{SPC}-900000{SPC}+900000{SPC}1<ETX>	
CoLa A	<STX>sWN LFPangleRangeFilter 1 -900000 +900000 -900000 +900000 1<ETX>	
	sWN LFPangleRangeFilter 1 -900000 +900000 -900000 +900000 1	
	02 73 57 4E 4C 46 50 61 6E 67 6C 65 52 61 6E 67 65 46 69 6C 74 65 72 20 31 20 2D 39 30 30 30 30 20 2B 39 30 30 30 30 20 31 03	
CoLa B	02 02 02 02 00 00 00 2B 73 57 4E 20 4C 46 50 61 6E 67 6C 65 52 61 6E 67 65 46 69 6C 74 65 72 20 01 FF F2 44 60 00 0D BB A0 FF F2 44 60 00 0D BB A0 00 01 2E	
	73 57 4E 20 4C 46 50 61 6E 67 6C 65 52 61 6E 67 65 46 69 6C 74 65 72 20 01 FF F2 44 60 00 0D BB A0 FF F2 44 60 00 0D BB A0 00 01	

Table 295: Telegram structure: sWA LFP AngleRangeFilter

Telegram structure: sWA LFPAngleRangeFilter						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	filter set up the horizontal (theta) and vertical (phi) start- and stop angle in rad	String	19		LFPangleRangeFilter	4C 46 50 61 6E 67 6C 65 52 61 6E 67 65 46 69 6C 74 65 72 03

Table 296: Example: sWA LFPAngleRangeFilter

CoLa A	<STX>sWA{SPC}LFPangleRangeFilter<ETX>					
	02 73 57 41 20 4C 46 50 61 6E 67 6C 65 52 61 6E 67 65 46 69 6C 74 65 72 03					
CoLa B	02 02 02 02 00 00 00 18 73 57 41 20 4C 46 50 61 6E 67 6C 65 52 61 6E 67 65 46 69 6C 74 65 72 20 21					

13.6.1.4.6.8 Read angle range filter [sRN LFPAngleRangeFilter]

The angle range filter set up the horizontal (theta) and vertical (phi) start- and stop angle in rad.

BeamIncrement = the 'beamIncrement' which is used to subsample the beams within the selected angle range. With a 'beamIncrement' of n only every nth beam from the selected angle range is copied to the output scan, i.e. the angle resolution is reduced by factor n. If the beamIncrement is zero it is set to one.

Table 297: Telegram structure: sRN LFPAngleRangeFilter

Telegram structure: sRN LFPAngleRangeFilter (User Level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	filter set up the horizontal (theta) and vertical (phi) start- and stop angle in rad	String	19		LFPangleRangeFilter	4C 46 50 61 6E 67 6C 65 52 61 6E 67 65 46 69 6C 74 65 72 03

Table 298: Example: sRN LFPAngleRangeFilter

CoLa A	<STX>sRN{SPC}LFPangleRangeFilter<ETX>					
	02 73 52 4E 20 4C 46 50 61 6E 67 6C 65 52 61 6E 67 65 46 69 6C 74 65 72 03					
CoLa B	02 02 02 02 00 00 00 18 73 52 4E 20 4C 46 50 61 6E 67 6C 65 52 61 6E 67 65 46 69 6C 74 65 72 20 21					

Table 299: Telegram structure: sRA LFP AngleRangeFilter

Telegram structure: sRA LFPAngleRangeFilter						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	filter set up the horizontal (theta) and vertical (phi) start- and stop angle in rad	String	19		LFPangleRangeFilter	4C 46 50 61 6E 67 6C 65 52 61 6E 67 65 46 69 6C 74 65 72 03
Variable Data 1	Enables/Disables the filter	Bool_1	1	Off: On:	+0d +1d	00 01

Telegram structure: sRA LFPangleRangeFilter						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Variable Data 2	ThetaStart	Real	4	No impact on sensor setting	-1,800,000d ... +1,800,000d (FFE488C0 ... 001B7740h)	FF E4 88 C0 ... 00 1B 77 40
Variable Data 3	ThetaStop	Real	4	No impact on sensor setting	-1,800,000d ... +1,800,000d (FFE488C0 ... 001B7740h)	FF E4 88 C0 ... 00 1B 77 40
Variable Data 4	PhiStart	Real	4	No impact on sensor setting	-900,000d ... +900,000d (FFF24460 ... 000DBBA0h)	FF F2 44 60 ... 00 0D BB A0
Variable Data 5	PhiStop	Real	4	No impact on sensor setting	-900,000d ... +900,000d (FFF24460 ... 000DBBA0h)	FF F2 44 60 ... 00 0D BB A0
Variable Data 6	BeamIncrement	UInt_16	2	No impact on sensor setting	1d...+20d	00 01 ... 00 14

Explanation: Angle range filter enabled and set up theta (horizontal) start -90°, theta stop +90°, phi (vertical) start -90°, phi stop +90°, beam increment 1

Table 300: Example: sRA LFPangleRangeFilter

CoLa A	<STX>sRA{SPC}LFPangleRangeFilter{SPC}1{SPC}-900000{SPC}+900000{SPC}-900000{SPC}+900000{SPC}1<ETX>	
	<STX>sRA LFPangleRangeFilter 1 -900000 +900000 -900000 +900000 1<ETX>	
	sRA LFPangleRangeFilter 1 -900000 +900000 -900000 +900000 1	
	02 73 52 41 4C 46 50 61 6E 67 6C 65 52 61 6E 67 65 46 69 6C 74 65 72 20 31 20 2D 39 30 30 30 30 30 20 2B 39 30 30 30 30 30 20 2D 39 30 30 30 30 30 20 2B 39 30 30 30 30 30 20 31 03	
CoLa B	02 02 02 02 00 00 00 2B 73 52 41 20 4C 46 50 61 6E 67 6C 65 52 61 6E 67 65 46 69 6C 74 65 72 20 01 FF F2 44 60 00 0D BB A0 FF F2 44 60 00 0D BB A0 00 01 2E	
	73 52 41 20 4C 46 50 61 6E 67 6C 65 52 61 6E 67 65 46 69 6C 74 65 72 20 01 FF F2 44 60 00 0D BB A0 FF F2 44 60 00 0D BB A0 00 01	

13.6.1.4.6.9 Set interval filter [sWN LFPintervalFilter]

Enables and set up the interval filter. The interval filter reduce the scan output rate by a given factor.

Table 301: Telegram structure: sWN LFPintervalFilter

Telegram structure: sWN LFPintervalFilter (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E
Command	Reduce the scan output rate by a given factor	String	17		LFPintervalFilter	4C 46 50 69 6E 74 65 72 76 61 6C 46 69 6C 74 65 72
Variable data 1	Enables/Disables the filter.	Bool	1	Off: On:	0d (00h) +1d (01h)	00 01

Telegram structure: sWN LFPIntervalFilter (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Variable Data 2	Only every nth scan is output where n is given by the value of uiReductionFactor.	Uint_32	4		1d...+50d (00 00 00 01h ... 00 00 00 32h)	00 00 00 01... 00 00 00 32

Enables the interval filter an set up to the 3rd scan

Table 302: Example: sWN LFPIntervalFilter

CoLa A	<STX>sWN{SPC}LFPIntervalFilter{SPC}1{SPC}3<ETX>					
	<STX>sWN LFPIntervalFilter 1 3<ETX>					
	sWN LFPIntervalFilter 1 3					
	02 73 57 4E 20 4C 46 50 69 6E 74 65 72 76 61 6C 46 69 6C 74 65 72 20 31 20 33 03					
CoLa B	02 02 02 02 00 00 00 1B 73 57 4E 20 4C 46 50 69 6E 74 65 72 76 61 6C 46 69 6C 74 65 72 20 01 00 00 00 03 0E					
	73 57 4E 20 4C 46 50 69 6E 74 65 72 76 61 6C 46 69 6C 74 65 72 20 01 00 00 00 03					

Table 303: Telegram structure: sWA LFPIntervalFilter

Telegram structure: sWA LFPIntervalFilter						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	Reduce the scan output rate by a given factor	String	17		LFPIntervalFilter	4C 46 50 69 6E 74 65 72 76 61 6C 46 69 6C 74 65 72

Table 304: Example: sWA LFPIntervalFilter

CoLa A	<STX>sWA{SPC}LFPIntervalFilter<ETX>					
	02 73 57 41 20 46 50 69 6E 74 65 72 76 61 6C 46 69 6C 74 65 72 03					
CoLa B	02 02 02 02 00 00 00 16 73 57 41 20 4C 46 50 69 6E 74 65 72 76 61 6C 46 69 6C 74 65 72 20 00					

13.6.1.4.6.10 Set moving averaging filter [sWN LFPmovingAveragingFilter]

Enables the moving average filter

Table 305: Telegram structure: sWN LFPmovingAveragingFilter

Telegram structure: sWN LFPmovingAveragingFilter (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E
Command	String	String	24		LFPmovingAveragingFilter	4C 46 50 6D 6F 76 69 6E 67 41 76 65 72 61 67 69 6E 67 46 69 6C 74 65 72
Variable Data 1	Moving averaging is enabled	Bool	1	Off: On:	0d (00h) +1d (01h)	00 01

Telegram structure: sWN LFPmovingAveragingFilter (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Variable Data 2	averaging depth	UInt	2	Minimum: Maximum:	+2d (02h) +4d (0Ah)	00 02 00 04

Disable the moving average filter and set averaging depth to 3

Table 306: Example: sWN LFPmovingAveragingFilter +0 +3

CoLa A	<STX>sWN{SPC}LFPmovingAveragingFilter{SPC}+0{SPC}+3<ETX>					
	<STX>sWN LFPmovingAveragingFilter +0 +3<ETX>					
	sWN LFPmovingAveragingFilter +0 +3					
CoLa B	02 73 57 4E 20 4C 46 50 6D 6F 76 69 6E 67 41 76 65 72 61 67 69 6E 67 46 69 6C 74 65 72 20 2B 30 20 2B 33 03					
	02 02 02 02 00 00 00 20 73 57 4E 20 4C 46 50 6D 6F 76 69 6E 67 41 76 65 72 61 67 69 6E 67 46 69 6C 74 65 72 20 01 00 03					
	73 57 4E 20 4C 46 50 6D 6F 76 69 6E 67 41 76 65 72 61 67 69 6E 67 46 69 6C 74 65 72 20 01 00 03					

Table 307: Telegram structure: sWA LFPmovingAveragingFilter

Telegram structure: sWN LFPmovingAveragingFilter						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	String	String	24		LFPmovingAveragingFilter	4C 46 50 6D 6F 76 69 6E 67 41 76 65 72 61 67 69 6E 67 46 69 6C 74 65 72

Table 308: Example: sWA LFPmovingAveragingFilter

CoLa A	<STX>sWA{SPC}LFPmovingAveragingFilter<ETX>					
	02 73 57 41 20 4C 46 50 6D 6F 76 69 6E 67 41 76 65 72 61 67 69 6E 67 46 69 6C 74 65 72 03					
CoLa B	02 02 02 02 00 00 00 1D 73 57 41 20 4C 46 50 6D 6F 76 69 6E 67 41 76 65 72 61 67 69 6E 67 46 69 6C 74 65 72 20 4D					

13.6.1.4.6.11 Set radial distance range filter [sWN LFPradialDistanceRangeFilter]

Restriction of the scan(s) to a specified distance range.

Table 309: Telegram structure: sWN LFPradialDistanceRangeFilter

Telegram structure: sWN LFPradialDistanceRangeFilter (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E
Command	Restriction of the scan(s) to a specified distance range.	String	28		LFPradialDistanceRangeFilter	4C 46 50 72 61 64 69 61 6C 44 69 73 74 61 6E 63 65 52 61 6E 67 65 46 69 6C 74 65 72
Variable Data 1	Enables/Disables the filter.	Bool_1	1	Off: On:	0d (0h) 1d (1h)	00 01

Telegram structure: sWN LFPradialDistanceRangeFilter (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
DistMin	Lower boundary of the distance range.	Int_32	4	Minimum: Maximum:	+0d (0h) +200000d (30D40h)	00 00 00 00 00 03 0D 40
DistMax	Upper boundary of the distance range.	Int_32	4	Minimum: Maximum:	+0d (0h) +200000d (30D40h)	00 00 00 00 00 03 0D 40

Disable the radial distance range filter and set up the boundaries to min 0mm and max 200000mm.

Table 310: Example: sWN LFPradialDistanceRangeFilter

CoLa A	<STX>sWN{SPC}LFPradialDistanceRangeFilter{SPC}0{SPC}0{SPC}30D40<ETX>					
	<STX>sWN LFPradialDistanceRangeFilter 0 0 30D40<ETX>					
	sWN LFPradialDistanceRangeFilter 0 0 30D40					
CoLa B	02 73 57 4E 20 4C 46 50 72 61 64 69 61 6C 44 69 73 74 61 6E 63 65 52 61 6E 67 65 46 69 6C 74 65 72 20 00 20 00 20 30D40 03					
	02 02 02 02 00 00 00 00 2A 73 57 4E 20 4C 46 50 72 61 64 69 61 6C 44 69 73 74 61 6E 63 65 52 61 6E 67 65 46 69 6C 74 65 72 20 00 00 00 00 03 0D 40 31					
	73 57 4E 20 4C 46 50 72 61 64 69 61 6C 44 69 73 74 61 6E 63 65 52 61 6E 67 65 46 69 6C 74 65 72 20 00 00 00 00 00 00 03 0D 40					

Table 311: Telegram structure: sWA LFPradialDistanceRangeFilter

Telegram structure: sWA LFPradialDistanceRangeFilter						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	Restriction of the scan(s) to a specified distance range.	String	28		LFPradialDistanceRangeFilter	4C 46 50 72 61 64 69 61 6C 44 69 73 74 61 6E 63 65 52 61 6E 67 65 46 69 6C 74 65 72

Table 312: Example: sWA LFPradialDistanceRangeFilter

CoLa A	<STX>sWA{SPC}LFPradialDistanceRangeFilter<ETX>					
	02 73 57 41 20 4C 46 50 72 61 64 69 61 6C 44 69 73 74 61 6E 63 65 52 61 6E 67 65 46 69 6C 74 65 72 03					
CoLa B	02 02 02 02 00 00 00 00 21 73 57 41 20 4C 46 50 72 61 64 69 61 6C 44 69 73 74 61 6E 63 65 52 61 6E 67 65 46 69 6C 74 65 72 20 70					

13.6.1.4.7 Encoder

13.6.1.4.7.1 Set encoder settings [sWN LICencset]

Define if and which type of encoder is connected to the device so that input signals are processed correctly.

Table 313: Telegram structure: sWN LICencset

Telegram structure: sWN LICencset (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E

Telegram structure: sWN LICencset (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command	Encoder settings	String	9		LICencset	4C 49 43 65 6E 63 73 65 74
Encoder setting		Enum_8	1	Off: Single increment/INC1: Direction recognition (phase): Direction recognition (level): Fixed speed:	0 1 2 3 4	00 01 02 03 04

Table 314: Example: sWN LICencset

CoLa A	<STX>sWN{SPC}LICencset{SPC}0<ETX>		
	<STX>sWN LICencset 0<ETX>		
	sWN LICencset 0		
	02 73 57 4E 20 4C 49 43 65 6E 63 73 65 74 20 30 03		
CoLa B	02 02 02 02 00 00 00 0F 73 57 4E 20 4C 49 43 65 6E 63 73 65 74 20 00 26		
	73 57 4E 20 4C 49 43 65 6E 63 73 65 74 20 00		

Table 315: Telegram structure: sWA LICencset

Telegram structure: sWA LICencset						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	Encoder settings	String	9		LICencset	4C 49 43 65 6E 63 73 65 74

Table 316: Example: sWA LICencset

CoLa A	<STX>sWA{SPC}LICencset<ETX>		
	02 73 57 41 20 4C 49 43 65 6E 63 73 65 74 03		
CoLa B	02 02 02 02 00 00 00 0E 73 57 41 20 4C 49 43 65 6E 63 73 65 74 20 29		

13.6.1.4.7.2 Set encoder resolution [sWN LICencres]

Determine the encoder resolution in mm/increment.

Table 317: Telegram structure: sWN LICencres

Telegram structure: sWN LICencres (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E
Command	Set encoder resolution	String	9		LICencres	4C 49 43 65 6E 63 72 65 73
Encoder resolution	Resolution value in mm/Inc as float according to IEEE754	Real	4		+0.001d ... +2000d	3A 83 12 6F ... 44 FA 00 00 (see IEEE 754)

Table 318: Example: sWN LICencres

CoLa A	<STX>sWN{SPC}LICencres{SPC}+1000<ETX>	
	<STX>sWN LICencres +1000<ETX>	
	sWN LICencres +1000	
	02 73 57 4E 20 4C 49 43 65 6E 63 72 65 73 20 2B 31 30 30 03	
CoLa B	02 02 02 02 00 00 00 12 73 57 4E 20 4C 49 43 65 6E 63 72 65 73 20 44 7A 00 00 1E	
	73 57 4E 20 4C 49 43 65 6E 63 72 65 73 20 44 7A 00 00	

Table 319: Telegram structure: sWA LICencres

Telegram structure: sWA LICencres						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	Set encoder resolution	String	9		LICencres	4C 49 43 65 6E 63 72 65 73

Table 320: Example: sWA LICencres

CoLa A	<STX>sWA{SPC}LICencres<ETX>	
	02 73 57 41 20 4C 49 43 65 6E 63 72 65 73 03	
CoLa B	02 02 02 02 00 00 00 0E 73 57 41 20 4C 49 43 65 6E 63 72 65 73 20 2F	

13.6.1.4.7.3 Reset encoder values [sMN LIDrstencoderinc]

Reset current encoder tick value to 0.

Table 321: Telegram structure: sMN LIDrstencoderinc

Telegram structure: sMN LIDrstencoderinc (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Method	String	3		sMN	73 4D 4E
Command	Reset Encoder values	String	16		LIDrstencoderinc	4C 49 44 72 73 74 65 6E 63 6F 64 65 72 69 6E 63

Table 322: Example: sMN LIDrstencoderinc

CoLa A	<STX>sMN{SPC}LIDrstencoderinc<ETX>	
	<STX>sMN LIDrstencoderinc<ETX>	
	sMN LIDrstencoderinc	
	02 73 4D 4E 20 4C 49 44 72 73 74 65 6E 63 6F 64 65 72 69 6E 63 03	
CoLa B	02 02 02 02 00 00 00 14 73 4D 4E 20 4C 49 44 72 73 74 65 6E 63 6F 64 65 72 69 6E 63 74	
	73 4D 4E 20 4C 49 44 72 73 74 65 6E 63 6F 64 65 72 69 6E 63	

Table 323: Telegram structure: sAN LIDrstencoderinc

Telegram structure: sAN LIDrstencoderinc						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sAN	73 41 4E
Command	Reset Encoder values	String	16		LIDrstencoder-inc	4C 49 44 72 73 74 65 6E 63 6F 64 65 72 69 6E 63 63
Reset status		Bool_1	1	Successful Failure/ Insufficient user level	0 1	00 01

Table 324: Example: sAN LIDrstencoderinc

CoLa A	<STX>sAN{SPC}LIDrstencoderinc{SPC}0<ETX> 02 73 41 4E 20 4C 49 44 72 73 74 65 6E 63 6F 64 65 72 69 6E 63 20 30 03					
CoLa B	02 02 02 02 00 00 00 16 73 41 4E 20 4C 49 44 72 73 74 65 6E 63 6F 64 65 72 69 6E 63 20 00 58					

13.6.1.4.7.4 Enable encoder reference input 1 signal [sWN LICpos1enable]

This command enables the encoder reference input signal.

Table 325: Telegram structure: sWN LICpos1enable

Telegram structure: sWN LICpos1enable (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E
Command	Enable encoder reference input signal	String	13		LICpos1enable	4C 49 43 70 6F 73 31 65 6E 61 62 6C 65
Variable Data	Disable/ enable	Bool_1	1	Disable: Enable:	0d (00h) +1d (01h)	00 01

Table 326: Example: sWN LICpos1enable - Enable reference input signal

CoLa A	<STX>sWN{SPC}LICpos1enable{SPC}1<ETX>										
	<STX>sWN LICpos1enable 1<ETX>										
	sWN LICpos1enable 1										
	02 73 57 4E 20 4C 49 43 70 6F 73 31 65 6E 61 62 6C 65 20 31 03										
CoLa B	02 02 02 02 00 00 00 13 73 57 4E 20 4C 49 43 70 6F 73 31 65 6E 61 62 6C 65 20 01 71										
	73 57 4E 20 4C 49 43 70 6F 73 31 65 6E 61 62 6C 65 20 01										

Table 327: Telegram structure: sWA LICpos1enable

Telegram structure: sWA LICpos1enable						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	Enable encoder reference input signal	String	13		LICpos1enable	4C 49 43 70 6F 73 31 65 6E 61 62 6C 65

Table 328: Example: sWA LICpos1enable

CoLa A	<STX>sWA{SPC}LICpos1enable<ETX> 02 73 57 41 20 4C 49 43 70 6F 73 31 65 6E 61 62 6C 65 03
CoLa B	02 02 02 02 00 00 00 12 73 57 41 20 4C 49 43 70 6F 73 31 65 6E 61 62 6C 65 20 7F

13.6.1.4.7.5 Read encoder reference input 1 signal [sRN LICpos1enable]

This command reads the state (enabled/ disabled) of the encoder reference input signal.

Table 329: Telegram structure: sRN LICpos1enable

Telegram structure: sRN LICpos1enable (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	Enable encoder reference input signal	String	13		LICpos1enable	4C 49 43 70 6F 73 31 65 6E 61 62 6C 65

Table 330: Example: sRN LICpos1enable

CoLa A	<STX>sRN{SPC}LICpos1enable<ETX> <STX>sRN LICpos1enable<ETX> sRN LICpos1enable 02 73 52 4E 20 4C 49 43 70 6F 73 31 65 6E 61 62 6C 65 03	
CoLa B	02 02 02 02 00 00 00 11 73 52 4E 20 4C 49 43 70 6F 73 31 65 6E 61 62 6C 65 55 73 52 4E 20 4C 49 43 70 6F 73 31 65 6E 61 62 6C 65	

Table 331: Telegram structure: sRA LICpos1enable

Telegram structure: sRA LICpos1enable						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	Enable encoder reference input signal	String	13		LICpos1enable	4C 49 43 70 6F 73 31 65 6E 61 62 6C 65
Variable Data	Disabled/ enabled	Bool_1	1	Disabled: Enabled:	0d (00h) +1d (01h)	00 01

Table 332: Example: sRA LICpos1enable - Encoder reference input signal is enabled

CoLa A	<STX>sRA{SPC}LICpos1enable{SPC}1<ETX> 02 73 52 41 20 4C 49 43 70 6F 73 31 65 6E 61 62 6C 65 20 31 03
CoLa B	02 02 02 02 00 00 00 13 73 52 41 20 4C 49 43 70 6F 73 31 65 6E 61 62 6C 65 20 01 7B

13.6.1.4.7.6 Enable encoder reference input 2 signal [sWN LICpos2enable]

This command enables the encoder reference input signal.

Table 333: Telegram structure: sWN LICpos2enable

Telegram structure: sWN LICpos2enable (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E
Command	Enable encoder reference input signal	String	13		LICpos2enable	4C 49 43 70 6F 73 32 65 6E 61 62 6C 65
Variable Data	Disable/ enable	Bool_1	1	Disable: Enable:	0d (00h) +1d (01h)	00 01

Table 334: Example: sWN LICpos2enable - Enable reference input signal

CoLa A	<STX>sWN{SPC}LICpos2enable{SPC}1<ETX>		
	<STX>sWN LICpos2enable 1<ETX>		
	sWN LICpos2enable 1		
	02 73 57 4E 20 4C 49 43 70 6F 73 32 65 6E 61 62 6C 65 20 31 03		
CoLa B	02 02 02 02 00 00 00 13 73 57 4E 20 4C 49 43 70 6F 73 32 65 6E 61 62 6C 65 20 01 72		
	73 57 4E 20 4C 49 43 70 6F 73 32 65 6E 61 62 6C 65 20 01		

Table 335: Telegram structure: sWA LICpos2enable

Telegram structure: sWA LICpos2enable						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	Enable encoder reference input signal	String	13		LICpos2enable	4C 49 43 70 6F 73 32 65 6E 61 62 6C 65

Table 336: Example: sWA LICpos2enable

CoLa A	<STX>sWA{SPC}LICpos2enable<ETX>		
	02 73 57 41 20 4C 49 43 70 6F 73 32 65 6E 61 62 6C 65 03		
CoLa B	02 02 02 02 00 00 00 12 73 57 41 20 4C 49 43 70 6F 73 32 65 6E 61 62 6C 65 20 7C		

13.6.1.4.7.7 Read encoder reference input 2 signal [sRN LICpos2enable]

This command reads the state (enabled/ disabled) of the encoder reference input signal.

Table 337: Telegram structure: sRN LICpos2enable

Telegram structure: sRN LICpos2enable (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	Enable encoder reference input signal	String	13		LICpos2enable	4C 49 43 70 6F 73 32 65 6E 61 62 6C 65

Table 338: Example: sRN LICpos2enable

CoLa A	<STX>sRN[SPC]LICpos2enable<ETX>	
	<STX>sRN LICpos2enable<ETX>	
	sRN LICpos2enable	
	02 73 52 4E 20 4C 49 43 70 6F 73 32 65 6E 61 62 6C 65 03	
CoLa B	02 02 02 02 00 00 00 11 73 52 4E 20 4C 49 43 70 6F 73 32 65 6E 61 62 6C 65 56	
	73 52 4E 20 4C 49 43 70 6F 73 32 65 6E 61 62 6C 65	

Table 339: Telegram structure: sRA LICpos2enable

Telegram structure: sRA LICpos2enable					 	
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	Enable encoder reference input signal	String	13		LICpos2enable	4C 49 43 70 6F 73 32 65 6E 61 62 6C 65
Variable Data	Disabled/ enabled	Bool_1	1	Disabled: Enabled:	0d (00h) +1d (01h)	00 01

Table 340: Example: sRA LICpos2enable - Encoder reference input signal is enabled

CoLa A	<STX>sRA[SPC]LICpos2enable[SPC]1<ETX>
	02 73 52 41 20 4C 49 43 70 6F 73 32 65 6E 61 62 6C 65 20 01 03
CoLa B	02 02 02 02 00 00 00 13 73 52 41 20 4C 49 43 70 6F 73 32 65 6E 61 62 6C 65 20 01 7B

13.6.1.4.8 Inputs and Outputs

13.6.1.4.8.1 Read state of the ports [sRN LIDportstate]

LIDportstate has to be available additionally or as successor of the LIDoutputstate telegram.

Valid for all sensors with Ethernet and ports (inputs / outputs).

Table 341: Telegram structure: sRN LIDportstate

Telegram structure: sRN LIDportstate					 	
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	Ask for port configuration	String	12		LIDportstate	4C 49 44 70 6F 72 74 73 74 61 74 65

Table 342: Example: sRN LIDportstate

CoLa A	<STX>sRN[SPC]LIDportstate<ETX>	
	<STX>sRN LIDportstate<ETX>	
	sRN LIDportstate	
	02 73 52 4E 20 4C 49 44 70 6F 72 74 73 74 61 74 65 03	
CoLa B	02 02 02 02 00 00 00 10 73 52 4E 20 4C 49 44 70 6F 72 74 73 74 61 74 65 60	
	73 52 4E 20 4C 49 44 70 6F 72 74 73 74 61 74 65	

Table 343: Telegram structure: sRA LIDportstate

Telegram structure: sRA LIDportstate						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	Port state	String	12		LIDportstate	4C 49 44 70 6F 72 74 73 74 61 74 65
Status code	Version number	Uint_16	2	Current version:	0 ... FFFFh 0	00 01 ... FF FF
	System counter (time in µs since power up max. 71min then starting from 0 again)	Uint_32	4		0 ... FFFFFFFFh	00 00 00 00 ... FF FF FF FF
ARRAY which defines the number of internal ports*	0...n	Uint_16	2	Hex: Not available: Number of ports:	0000 - FFFF 00 01 ... n	00 00 - FF FF
State of the ports and count value in hex	Internal port state	Enum_8	1	Output voltage low: (Relays open) Output voltage high: (Relays closed) Tri-state: Input voltage high (level): Input voltage from low to high (edge) Input voltage low (level): Input voltage high to low (edge)	00 01 02 03 04	00 01 02 03 04
	Internal port counter	Uint_32	4		0 ... FFFFFFFFh	00 00 00 00 ... FF FF FF FF
						
ARRAY which defines the number of external or virtual ports*	0...n	Uint_16	1	Hex: Not available: Nuner of ports:	00 00 - FF FF 00 01 ... n	00 00 - FF FF
State of the ports and count value in hex	External port state	Enum_8	1	Output voltage low: (Relays open) Output voltage high: (Relays closed) Tri-state: Input voltage high (level): Input voltage from low to high (edge) Input voltage low (level): Input voltage high to low (edge)	00 01 02 03 04	00 01 02 03 04
	External port counter	Uint_32	4		0 ... FFFFFFFFh	00 00 00 00 ... FF FF FF FF
Time	States code	Enum_16	1	No time data: Time data:	00 00 00 01	00 00 00 01

Table 347: Telegram structure: sRA PortConfiguration

Telegram structure: sRA PortConfiguration						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	Configuration of all I/Os	String	12		PortConfigura- tion	50 6F 72 74 43 6F 6E 66 69 67 75 72 61 74 69 6F 6E
Start of loop, number of loops = amount of all current and future Inputs and Outputs of device family						
Port Type	Input or Output	Enum_8	1	Input: Output:	0 1	00 01
Port Name	Amount of characters of the following port name	Uint_16	2		0h ... 20h	00 00 ... 00 20
	Port name	String	16 (depend- ing on string length)		[Port name]	[Port name]
Input Settings						
Logic	Logic of the input	Bool_1	1	Active high: Active low:	0 1	00 01
Debounc- ing	Select debouncing time in ms	Uint_16	2	(max. 10,000ms)	0h ... 2710h	00 00 ... 27 10
Reserved	-	Enum_8	1	Always:	0	00
Reserved	Reserved value 1	Uint_16	2		0	00 00
Reserved	Reserved value 2	Uint_16	2		0	00 00
Output Settings						
Logic	Logic of the input	Bool_1	1	Active high: Active low:	0 1	00 01
Output Mode	PNP, NPN or Push-Pull	Enum_8	1	PNP: NPN: Push-Pull:	0 1 2	00 01 02
Restart type	Restart behavior of out-put after event: imme- diatly or after specific time	Enum_8	1	Immediately: Time: Input:	0 1 2	00 01 02
Restart time	[Only with restart type = Time], time in ms	Uint_32	4	(20 ms ... 600,000 ms)	14h ... 927C0h	00 00 00 14 ... 00 09 27 C0
Restart input	[Only with restart type = Input], input for restart	Uint_16	2		1 ... 8	00 01 ... 00 08
Combina- tion	Combining multiple Events and/or Inputs	Enum_8	1	AND: OR: XOR:	0 1 2	00 01 02
Reserved	Reserved value 3	Uint_16	2		0	00 00
Reserved	Reserved value 4	Uint_16	2		0	00 00

	02 02 02 02 00 00 017A 73 52 41 20 50 6F 72 74 43 6F 6E 66 69 67 75 72 6174 69 6F 6E 20 0100 06 49 6E 4F 75 74 31 00 00 0A 00 00 00 00 00 0100 00 00 00 00 C8 00 01 0100 00 00 00 00 01 44 52 44 59 00 00 00 00 00 00 00 00 00 00 01 00 06 49 6E 4F 75 74 32 00 00 0A 0100 00 00 00 00 0100 00 00 00 00 C8 00 01 0100 00 00 00 00 01 44 52 44 59 00 00 00 00 00 00 00 00 00 00 00 00 00 06 49 6E 4F 75 74 33 00 00 0A 0100 00 00 00 00 00 00 00 00 00 C8 00 01 0100 00 00 00 00 00 00 00 00 00 00 00 00 00 06 49 6E 4F 75 74 34 00 00 0A 0100 00 00 00 00 00 00 00 00 00 00 C8 00 01 0100 00 00 00 00 00 00 00 00 00 00 00 00 00 06 49 6E 4F 75 74 35 00 00 0A 0100 00 00 00 00 00 00 00 00 C8 00 01 0100 00 00 00 00 00 00 00 00 00 00 00 00 00 06 49 6E 4F 75 74 36 00 00 0A 0100 00 00 00 00 00 00 00 00 00 C8 00 01 0100 00 00 00 00 00 00 00 00 00 00 00 00 0100 06 49 6E 4F 75 74 37 00 00 0A 0100 00 00 00 00 0100 00 00 00 00 C8 00 01 0100 00 00 00 00 01 44 52 44 59 00 00 00 00 00 00 00 00 00 00 00 0100 06 49 6E 4F 75 74 38 00 00 0A 0100 00 00 00 00 0100 00 00 00 00 C8 00 01 01 00 00 00 00 00 01 44 52 44 59 00 00 00 00 00 00 00 00 00 00 01
CoLa B	73 52 41 20 50 6F 72 74 43 6F 6E 66 69 67 75 72 6174 69 6F 6E 20 0100 06 49 6E 4F 75 74 3100 00 0A 00 00 00 00 00 0100 00 00 00 00 C8 00 01 0100 00 00 00 00 01 44 52 44 59 00 00 00 00 00 00 00 00 00 00 00 00 0100 06 49 6E 4F 75 74 32 00 00 0A 0100 00 00 00 00 0100 00 00 00 00 C8 00 01 0100 00 00 00 00 01 44 52 44 59 00 00 00 00 00 00 00 00 00 00 00 00 06 49 6E 4F 75 74 33 00 00 0A 0100 00 00 00 00 00 00 00 00 00 00 C8 00 01 0100 00 00 00 00 00 00 00 00 00 00 00 00 00 00 06 49 6E 4F 75 74 34 00 00 0A 0100 00 00 00 00 00 00 00 00 00 C8 00 01 0100 00 00 00 00 00 00 00 00 00 00 00 00 00 06 49 6E 4F 75 74 35 00 00 0A 0100 00 00 00 00 00 00 00 00 C8 00 01 0100 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 06 49 6E 4F 75 74 36 00 00 0A 0100 00 00 00 00 00 00 00 00 C8 00 01 0100 00 00 00 00 00 00 00 00 00 00 00 00 00 0100 06 49 6E 4F 75 74 37 00 00 0A 0100 00 00 00 00 0100 00 00 00 00 C8 00 01 0100 00 00 00 00 01 44 52 44 59 00 00 00 00 00 00 00 00 00 0100 06 49 6E 4F 75 74 38 00 00 0A 0100 00 00 00 0100 00 00 00 00 C8 00 01 0100 00 00 00 00 01 44 52 44 59 00 00 00 00 00 00 00 00 00 00

13.6.1.4.8.3 Set port configuration [sWN PortConfiguration]

Configuration of the given ports. Telegram structure represents the configuration of 1 port.

If the device has multiple ports, use the same structure.

Table 349: Telegram structure: sWN PortConfiguration

Telegram structure: sWN PortConfiguration (Required User Level: authorized client)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E
Command	Configuration of the given ports	String	17		PortConfigura- tion	50 6F 72 74 43 6F 6E 66 69 67 75 72 61 74 69 6F 6E
Port Type	Input or Output	Enum_8	1	Input: Output:	+0d (0h) +1d (1h)	00 01
Name	Name of the port	Flex-String	9 (0..32)	Default:	InOut1	00 06 49 6E 4F 75 74 31 00
Input Settings						
Logic	Logic of the input	Enum_8	1	Active high: Active low:	+0d (0h) +1d (1h)	00 01
Debounce	Select debouncing time in ms	Uint_8	1		+0d ... +255d (0h ... FFh)	00 ... FF
Reserved	-	Enum_8	1	Always:	+0d (0h)	00
Reserved1		Uint_16	2		+0d (0h)	00 00
Reserved2		Uint_16	2		+0d (0h)	00 00
Output Settings						
Logic	Definition of the output logic	Enum_8	1	Active high: Active low:	+0d (0h) +1d (1h)	00 01
Output Mode	Set kind of mode for output pin	Enum_8	1	PNP: NPN: Push-Pull:	0 1 2	00 01 02

Table 355: Telegram structure: sRA LIDinputstate

Telegram structure: sRA LIDinputstate						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	Output state	String	14		LIDinputstate	4C 49 44 69 6E 70 75 74 73 74 61 74 65
Status code	Version number	Uint_16	2		0h ... FFFFh	00 00 ... FF FF
	System counter (time in µs since power up max. 71min then starting from 0 again)	Uint_32	4		0 h ... FFFFFFFFh	00 00 00 00 ... FF FF FF FF
State of the inputs 1 ... n	Amount of inputs (n) depending of device family	Enum_8	1	Not active:	0	00
				Active:	1	01
				Input not used:	2	02
Time	States code	Uint_16	2	No time data:	0	00 00
				Time data:	1	00 01
Time Block (sensor-time from the last change of min. one of the out-puts)	Year	Array	2	E. g.	1970	07 B2
	Month		1		1 ... 12	01 ... 0C
	Day		1		1 ... 31	01 ... 1F
	Hour		1		0 ... 23	00 ... 17
	Minute		1		0 ... 59	00 ... 3B
	Second		1		0 ... 59	00 ... 3B
	Microsecond		4		0 ... 999999	00 00 00 00 ... 00 0F 42 3F

Table 356: Example: sRA LIDinputstate In1 not used, In2 inactive, In3 inactive, In4 not used, In5 not used, In6 not used, In7 not used, In8 not used, time: 1970-01-01 0:13 58.443 sec

CoLa A	<STX>sRA{SPC}LIDinput-state{SPC}1{SPC}31F99C10{SPC}2{SPC}0{SPC}0{SPC}2{SPC}2{SPC}2{SPC}2{SPC}2{SPC}1{SPC}7B2{SPC}1{SPC}1{SPC}0{SPC}D{SPC}3A{SPC}6C278<ETX>
	02 73 52 41 20 4C 49 44 69 6E 70 75 74 73 74 61 74 65 20 31 20 39 38 45 37 31 42 44 37 20 32 20 30 20 30 20 32 20 32 20 32 20 32 20 31 20 37 42 32 20 31 20 31 20 30 20 44 20 33 41 20 36 43 32 37 38 03
CoLa B	02 02 02 02 00 00 00 00 2D 73 52 41 20 4C 49 44 69 6E 70 75 74 73 74 61 74 65 20 00 01 31 F9 9C 10 02 00 00 02 02 02 02 02 00 01 07 B2 01 01 00 0D 3A 00 06 C2 78 5A

13.6.1.4.8.5 Read state of the outputs [sRN LIDoutputstate]

Status of all outputs

Table 357: Telegram structure: sRN LIDoutputstate

Telegram structure: sRN LIDoutputstate						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	Output state	String	14		LIDoutputstate	4C 49 44 6F 75 74 70 75 74 73 74 61 74 65

Table 358: Example: sRN LIDoutputstate

CoLa A	<STX>sRN[SPC]LIDoutputstate<ETX>	
	<STX>sRN LIDoutputstate<ETX>	
	sRN LIDoutputstate	
	02 73 52 4E 20 4C 49 44 6F 75 74 70 75 74 73 74 61 74 65 03	
CoLa B	02 02 02 02 00 00 00 12 73 52 4E 20 4C 49 44 6F 75 74 70 75 74 73 74 61 74 65 66	
	73 52 4E 20 4C 49 44 6F 75 74 70 75 74 73 74 61 74 65	

Table 359: Telegram structure: sRA LIDoutputstate

Telegram structure: sRA LIDoutputstate					 ← 	
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Complete telegram structure of the answer see "Receive outputstate by event [sEN LIDoutputstate]", page 208.						

13.6.1.4.8.6 Receive outputstate by event [sEN LIDoutputstate]

Output telegram is sent every time an output state changes.

Table 360: Telegram structure: sEN LIDoutputstate

Telegram structure: sEN LIDoutputstate					 → 	
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Event	String	3		sEN	73 45 4E
Command	Output state	String	14		LIDoutputstate	4C 49 44 6F 75 74 70 75 74 73 74 61 74 65
	Start/stop	Enum_8	1	Start: Stop:	1 0	01 00

Table 361: Example: sEN LIDoutputstate

CoLa A	<STX>sEN[SPC]LIDoutputstate[SPC]1<ETX>	
	<STX>sEN LIDoutputstate 1<ETX>	
	sEN LIDoutputstate 1	
	02 73 45 4E 20 4C 49 44 6F 75 74 70 75 74 73 74 61 74 65 20 31 03	
CoLa B	02 02 02 02 00 00 00 14 73 45 4E 20 4C 49 44 6F 75 74 70 75 74 73 74 61 74 65 20 01 50	
	73 45 4E 20 4C 49 44 6F 75 74 70 75 74 73 74 61 74 65 20 01	

Table 362: Telegram structure: sRA/sSN LIDoutputstate

Telegram structure: sRA/sSN LIDoutputstate					 ← 	
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA / sSN	73 52 41 / 73 53 4E
Command	Output state	String	14		LIDoutputstate	4C 49 44 6F 75 74 70 75 74 73 74 61 74 65

Table 364: Telegram structure: sMN mDOSetOutput

Telegram structure: sMN mDOSetOutput						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Method	String	3		sMN	73 4D 4E
Command	Set output state	String	12		mDOSetOutput	6D 44 4F 53 65 74 4F 75 74 70 75 74
Output number		Uint_8	1	Depends on system plug:	1... 3 1... 6	01... 03 01... 06
Output state		Enum_8	1	Not active: Active:	0 1	00 01

Table 365: Example: sMN mDOSetOutput

CoLa A	<STX>sMN{SPC}mDOSetOutput{SPC}1{SPC}1<ETX>	
	<STX>sMN mDOSetOutput 11<ETX>	
	sMN mDOSetOutput 11	
	02 73 4D 4E 20 6D 44 4F 53 65 74 4F 75 74 70 75 74 20 31 20 31 03	
CoLa B	02 02 02 02 00 00 00 13 73 4D 4E 20 6D 44 4F 53 65 74 4F 75 74 70 75 74 20 01 01 6B	
	73 4D 4E 20 6D 44 4F 53 65 74 4F 75 74 70 75 74 20 01 01	

Table 366: Telegram structure: sAN mDOSetOutput

Telegram structure: sAN mDOSetOutput						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sAN	73 41 4E
Command	Set output state	String	12		mDOSetOutput	6D 44 4F 53 65 74 4F 75 74 70 75 74
Status Code	Status code	Bool_1	1	Error: Success:	0 1	00 01

Table 367: Example: sAN mDOSetOutput

CoLa A	<STX>sAN{SPC}mDOSetOutput{SPC}1<ETX>
	02 73 41 4E 20 6D 44 4F 53 65 74 4F 75 74 70 75 74 20 31 03
CoLa B	02 02 02 02 00 00 00 12 73 41 4E 20 6D 44 4F 53 65 74 4F 75 74 70 75 74 20 01 66

13.6.1.4.8.8 Reset output counter [sMN LIDrstoutpcnt]

Reset the counter which keeps track of how often an digital output has been active (and not active). Information from the counter is included in LIDoutputstate (see ["Receive outputstate by event \[sEN LIDoutputstate\]", page 208](#)).

Table 368: Telegram structure: sMN LIDrstoutpcnt

Telegram structure: sMN LIDrstoutpcnt (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Method	String	3		sMN	73 4D 4E

Telegram structure: sMN LIDrstoutpcnt (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command	Reset output counter	String	13		LIDrstoutpcnt	4C 49 44 72 73 74 6F 75 74 70 63 6E 74

Table 369: Example: sMN LIDrstoutpcnt

CoLa A	<STX>sMN{SPC}LIDrstoutpcnt<ETX>					
	<STX>sMN LIDrstoutpcnt<ETX>					
	sMN LIDrstoutpcnt					
	02 73 4D 4E 20 4C 49 44 72 73 74 6F 75 74 70 63 6E 74 03					
CoLa B	02 02 02 02 00 00 00 11 73 4D 4E 20 4C 49 44 72 73 74 6F 75 74 70 63 6E 74 03					
	73 4D 4E 20 4C 49 44 72 73 74 6F 75 74 70 63 6E 74					

Table 370: Telegram structure: sAN LIDrstoutpcnt

Telegram structure: sAN LIDrstoutpcnt						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sAN	73 41 4E
Command	Reset output counter	String	13		LIDrstoutpcnt	4C 49 44 72 73 74 6F 75 74 70 63 6E 74
Status code	Code number	Bool_1	1	Success:	0	00
				Error:	1	01

Table 371: Example: sAN LIDrstoutpcnt

CoLa A	<STX>sAN{SPC}LIDrstoutpcnt{SPC}0<ETX>					
	02 73 41 4E 20 4C 49 44 72 73 74 6F 75 74 70 63 6E 74 20 30 03					
CoLa B	02 02 02 02 00 00 00 13 73 41 4E 20 4C 49 44 72 73 74 6F 75 74 70 63 6E 74 20 00 2F					

13.6.1.4.9 Status

13.6.1.4.9.1 Read firmware version [sRN Devicelnt]

Table 372: Telegram structure: sRN Devicelnt

Telegram structure: sRN Devicelnt						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	Read firmware version	String	11		Devicelnt	44 65 76 69 63 65 49 64 65 6E 74

Table 373: Example: sRN Devicelnt

CoLa A	<STX>sRN{SPC}Devicelnt<ETX>					
	<STX>sRN Devicelnt<ETX>					
	sRN Devicelnt					
	02 73 52 4E 20 44 65 76 69 63 65 49 64 65 6E 74 03					

	02 02 02 02 00 00 00 0F 73 52 4E 20 44 65 76 69 63 65 49 64 65 6E 74 25	
CoLa B	73 52 4E 20 44 65 76 69 63 65 49 64 65 6E 74	

Table 374: Telegram structure: sRA Devicelident

Telegram structure: sRA Devicelident					 	
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command		String	11		Devicelident	44 65 76 69 63 65 49 64 65 6E 74
Value	Length of firmware designation	Enum_16	2		0 ... 22h	0 ... 22h
Value	Firmware designation for device family	String			(See example)	(See example)
Value	Length of firmware version	Enum_16	2		0 ... 22h	0 ... 22h
Value	Firmware version	String			(See example)	(See example)

Table 375: Example: sRA Devicelident

CoLa A	<STX>sRA{SPC}Devicelident{SPC}8{SPC}picoScan{SPC}8{SPC}1.2.0.0B<ETX>					
	02 73 52 41 20 44 65 76 69 63 65 49 64 65 6E 74 20 38 20 70 69 63 6F 53 63 61 6E 20 38 20 31 2E 32 2E 30 2E 30 42 03					
CoLa B	02 02 02 02 00 00 00 24 73 52 41 20 44 65 76 69 63 65 49 64 65 6E 74 20 00 08 70 69 63 6F 53 63 61 6E 00 08 31 2E 32 2E 30 2E 30 42 4F					

13.6.1.4.9.2 Read version of the application software [sRN FirmwareVersion]

Table 376: Telegram structure: sRN FirmwareVersion

Telegram structure: sRN FirmwareVersion					 	
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	Read version of the application software	String	15		FirmwareVersion	46 69 72 6D 77 61 72 65 56 65 72 73 69 6F 6E 24

Table 377: Example: sRN FirmwareVersion

CoLa A	<STX>sRN{SPC}FirmwareVersion<ETX>					
	<STX>sRN FirmwareVersion<ETX>					
	sRN FirmwareVersion					
	02 73 52 4E 20 46 69 72 6D 77 61 72 65 56 65 72 73 69 6F 6E 03					
CoLa B	02 02 02 02 00 00 00 13 73 52 4E 20 46 69 72 6D 77 61 72 65 56 65 72 73 69 6F 6E 24					
	73 52 4E 20 46 69 72 6D 77 61 72 65 56 65 72 73 69 6F 6E					

Table 378: Telegram structure: sRA FirmwareVersion

Telegram structure: sRA FirmwareVersion						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	Read version of the application software	String	15		FirmwareVersion	46 69 72 6D 77 61 72 65 56 65 72 73 69 6F 6E
Value	Length of version	Uint_16	2		0 ... 28h	0 ... 28h
Value	Version	String	16		(See example)	(See example)

Table 379: Example: sRA FirmwareVersion

CoLa A	<STX>sRA[SPC]FirmwareVersion[SPC]14[SPC]1.2.0-b.0+1225.523ef<ETX> 02 73 52 41 20 46 69 72 6D 77 61 72 65 56 65 72 73 69 6F 6E 20 31 34 20 31 2E 32 2E 30 2D 62 2E 30 2B 31 32 32 35 2E 35 32 33 65 66 03
CoLa B	02 02 02 02 00 00 00 2A 73 52 41 20 46 69 72 6D 77 61 72 65 56 65 72 73 69 6F 6E 20 00 14 31 2E 32 2E 30 2D 62 2E 30 2B 31 32 32 35 2E 35 32 33 65 66 4B

13.6.1.4.9.3 Read the device state [sRN DevSta]

This telegram reads the device state.

Possible status codes:

- Reserved
- Startup
 - This state is used during the startup phase of the device. The state is a temporary state. It tells, that the user shall wait a given time (as specified in the documentation of the device) until the state will change to another value.
- Service mode
 - This state is used during service actions like software update, comprehensive parameter exchange or similar actions. The state does not tell whether the device is able to work properly. But it is a normal non critical state.
- Normal operation
 - This state signals the normal behavior of the device. The startup phase has ended and no errors or warnings occurred so far.
- Suspended operation
 - This state signals that any internal or environmental condition prohibits the adequate measurement.
- Service recommended
 - This state signals that the device is working properly. The measurement function is not affected by any circumstance. Hence the measuring function is as specified. But there is some issue which might be profitable to be handled by a service technician.
- Service required
 - This state signals, that a service technician is required. The device is still able to provide measured values but the results might be slightly affected by any circumstance.
- Recoverable error
 - This state signals a malfunction of the device. The measuring service is most probably disrupted. But the error might be fixed by a service technician.
- Fatal error
 - This state signals a malfunction of the device. The measuring service is most probably disrupted. But the error cannot be fixed by a service technician. The device must be replaced.

Table 380: Telegram structure: sRN DevSta

Telegram structure: sRN DevSta						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	Read the current device state	String	13		DevSta	44 65 76 53 74 61

Table 381: Example: sRN DevSta

CoLa A	<STX>sRN{SPC}DevSta<ETX>										
	<STX>sRN DevSta<ETX>										
	sRN DevSta										
	02 73 52 4E 20 44 65 76 53 74 61 03										
CoLa B	02 02 02 02 00 00 00 0A 73 52 4E 20 44 65 76 53 74 61 5E										
	73 52 4E 20 44 65 76 53 74 61										

Table 382: Telegram structure: sRA DevSta

Telegram structure: sRA DevSta						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	Read the current device state	String	13		DevSta	44 65 76 53 74 61
Status code	Code number	Enum_8	1	Reserved: Startup: Service mode: Normal operation: Suspended operation: Service recommended: Service required: Recoverable error: Fatal error:	0 1 2 3 4 5 6 7 8	00 01 02 03 04 05 06 07 08

Table 383: Example: sRA DevSta

CoLa A	<STX>sRA{SPC}DevSta{SPC}3<ETX>
	02 73 52 41 20 44 65 76 53 74 61 20 33 03
CoLa B	02 02 02 02 00 00 00 0C 73 52 41 20 44 65 76 53 74 61 20 03 72

13.6.1.4.9.4 Receive the device state by event [sEN DevSta]

This telegram reads the device state.

Possible status codes:

- Reserved
- Startup
 - This state is used during the startup phase of the device. The state is a temporary state. It tells, that the user shall wait a given time (as specified in the documentation of the device) until the state will change to another value.
- Service mode

- This state is used during service actions like software update, comprehensive parameter exchange or similar actions. The state does not tell whether the device is able to work properly. But it is a normal non critical state.
- Normal operation
 - This state signals the normal behavior of the device. The startup phase has ended and no errors or warnings occurred so far.
- Suspended operation
 - This state signals that any internal or environmental condition prohibits the adequate measurement.
- Service recommended
 - This state signals that the device is working properly. The measurement function is not affected by any circumstance. Hence the measuring function is as specified. But there is some issue which might be profitable to be handled by a service technician.
- Service required
 - This state signals, that a service technician is required. The device is still able to provide measured values but the results might be slightly affected by any circumstance.
- Recoverable error
 - This state signals a malfunction of the device. The measuring service is most probably disrupted. But the error might be fixed by a service technician.
- Fatal error
 - This state signals a malfunction of the device. The measuring service is most probably disrupted. But the error cannot be fixed by a service technician. The device must be replaced.

Table 384: Telegram structure: sEN DevSta

Telegram structure: sEN DevSta						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Event	String	3		sEN	73 45 4E
Command	Read the current device state	String	13		DevSta	44 65 76 53 74 61

Table 385: Example: sEN DevSta

CoLa A	<STX>sEN{SPC}DevSta<ETX>	
	<STX>sEN DevSta<ETX>	
	sEN DevSta	
	02 73 45 4E 20 44 65 76 53 74 61 03	
	02 02 02 02 00 00 00 0A 73 45 4E 20 44 65 76 53 74 61 5E	
CoLa B	73 45 4E 20 44 65 76 53 74 61	

Table 386: Telegram structure: sSN DevSta

Telegram structure: sSN DevSta						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sSN	73 53 4E
Command	Read the current device state	String	13		DevSta	44 65 76 53 74 61

Telegram structure: sSN DevSta						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Status code	Code number	Enum_8	1	Reserved: Startup: Service mode: Normal operation: Suspended operation: Service recommended: Service required: Recoverable error: Fatal error:	0 1 2 3 4 5 6 7 8	00 01 02 03 04 05 06 07 08

Table 387: Example: sSN DevSta

CoLa A	<STX>sSN{SPC}DevSta{SPC}3<ETX>
	02 73 53 4E 20 44 65 76 53 74 61 20 03 03
CoLa B	02 02 02 02 00 00 00 0C 73 53 4E 20 44 65 76 53 74 61 20 03 72

13.6.1.4.9.5 Read the device state [sRN SCdevicestate]

This telegram reads the general device state.

Table 388: Telegram structure: sRN SCdevicestate

Telegram structure: sRN SCdevicestate						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	Read the device state	String	13		SCdevicestate	53 43 64 65 76 69 63 65 73 74 61 74 65

Table 389: Example: sRN SCdevicestate

CoLa A	<STX>sRN{SPC}SCdevicestate<ETX>
	<STX>sRN SCdevicestate<ETX>
	sRN SCdevicestate
	02 73 52 4E 20 53 43 64 65 76 69 63 65 73 74 61 74 65 03
CoLa B	02 02 02 02 00 00 00 11 73 52 4E 20 53 43 64 65 76 69 63 65 73 74 61 74 65 30
	73 52 4E 20 53 43 64 65 76 69 63 65 73 74 61 74 65

Table 390: Telegram structure: sRA SCdevicestate

Telegram structure: sRA SCdevicestate						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	Read the device state	String	13		SCdevicestate	53 43 64 65 76 69 63 65 73 74 61 74 65
Status code	Code number	Enum_8	1	Busy / logged-in: Ready: Error:	0 1 2	00 01 02

Table 391: Example: sRA SCdevicestate

CoLa A	<STX>sRA{SPC}SCdevicestate{SPC}1<ETX> 02 73 52 41 20 53 43 64 65 76 69 63 65 73 74 61 74 65 20 31 03
CoLa B	02 02 02 02 00 00 00 13 73 52 41 20 53 43 64 65 76 69 63 65 73 74 61 74 65 20 01 1E

13.6.1.4.9.6 Receive the device state by event [sEN SCdevicestate]

This telegram reads the general device state continuously.

Table 392: Telegram structure: sEN SCdevicestate

Telegram structure: sEN SCdevicestate						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Event	String	3		sEN	73 45 4E
Command	Read the device state	String	13		SCdevicestate	53 43 64 65 76 69 63 65 73 74 61 74 65

Table 393: Example: sEN SCdevicestate

CoLa A	<STX>sEN{SPC}SCdevicestate<ETX> <STX>sEN SCdevicestate<ETX> sEN SCdevicestate 02 73 45 4E 20 53 43 64 65 76 69 63 65 73 74 61 74 65 03	
CoLa B	02 02 02 02 00 00 00 11 73 45 4E 20 53 43 64 65 76 69 63 65 73 74 61 74 65 30 73 45 4E 20 53 43 64 65 76 69 63 65 73 74 61 74 65	

Table 394: Telegram structure: sRA SCdevicestate

Telegram structure: sRA SCdevicestate						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sSN	73 53 4E
Command	Read the device state	String	13		SCdevicestate	53 43 64 65 76 69 63 65 73 74 61 74 65
Status code	Code number	Enum_8	1	Busy / logged-in: Ready: Error:	0 1 2	00 01 02

Table 395: Example: sSN SCdevicestate

CoLa A	<STX>sSN{SPC}SCdevicestate{SPC}1<ETX> 02 73 53 4E 20 53 43 64 65 76 69 63 65 73 74 61 74 65 20 31 03
CoLa B	02 02 02 02 00 00 00 13 73 53 4E 20 53 43 64 65 76 69 63 65 73 74 61 74 65 20 01 1E

13.6.1.4.9.7 Read device order number [sRN OrdNum]

This telegram reads the device order number which corresponds to the SICK part number of the device.

Table 396: Telegram structure: sRN OrdNum

Telegram structure: sRN OrdNum						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	Read device order number	String	6		OrdNum	4F 72 64 4E 75 6D

Table 397: Example: sRN OrdNum

CoLa A	<STX>sRN{SPC}OrdNum<ETX>										
	<STX>sRN OrdNum<ETX>										
	sRN OrdNum										
	02 73 52 4E 20 4F 72 64 4E 75 6D 03										
CoLa B	02 02 02 02 00 00 00 0A 73 52 4E 20 4F 72 64 4E 75 6D 40										
	73 52 4E 20 4F 72 64 4E 75 6D										

Table 398: Telegram structure: sRA OrdNum

Telegram structure: sRA OrdNum						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	Read device order number	String	6		OrdNum	4F 72 64 4E 75 6D
Length	Number of characters of the following order number	Uint_16	2		0h ... 20h	00 00 ... 00 20
Order number	Order number in 7 digits	String	7		0000000 ... 9999999	00 00 00 00 00 00 00 00 00 ... FF FF FF FF FF FF FF FF

Example: sRA OrdNum 1134610 (Order Number for picoScan150 Pro-1)**Table 399:** Example for picoScan150 Pro-1: sRA OrdNum

CoLa A	<STX>sRA{SPC}OrdNum{SPC}7{SPC}1134610<ETX>
	02 73 52 41 20 4F 72 64 4E 75 6D 20 37 20 31 31 33 34 36 31 30 03
CoLa B	02 02 02 02 00 00 00 14 73 52 41 20 4F 72 64 4E 75 6D 20 00 07 31 31 33 34 36 31 30 58

13.6.1.4.9.8 Read serial number [sRN SerialNumber]

Read the serial number of the device.

Table 400: Telegram structure: sRN SerialNumber

Telegram structure: sRN SerialNumber						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	Read serial number of the device	String	12		SerialNumber	53 65 72 69 61 6C 4E 75 6D 62 65 72

Table 401: Example: sRN SerialNumber

CoLa A	<STX>sRN{SPC}SerialNumber<ETX>	
	<STX>sRN SerialNumber<ETX>	
	sRN SerialNumber	
	02 73 52 4E 20 53 65 72 69 61 6C 4E 75 6D 62 65 72 03	
CoLa B	02 02 02 02 00 00 00 10 73 52 4E 20 53 65 72 69 61 6C 4E 75 6D 62 65 72 4C	
	73 52 4E 20 53 65 72 69 61 6C 4E 75 6D 62 65 72	

Table 402: Telegram structure: sRA SerialNumber

Telegram structure: sRA SerialNumber					 	
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	Read serial number of the device	String	12		SerialNumber	53 65 72 69 61 6C 4E 75 6D 62 65 72
Length of serial number	Number of characters of the serial number	Uint_16	2		0 ... 8h	00 00 ... 00 08h
Serial number	Production period (year, calendar week, number): YYWWxxxx	String	8		(See example)	(See example)

Table 403: Example: sRA SerialNumber

CoLa A	<STX>sRA{SPC}SerialNumber{SPC}8{SPC}23360024<ETX>
	02 73 52 41 20 53 65 72 69 61 6C 4E 75 6D 62 65 72 20 38 20 32 33 33 36 30 30 32 34 03
CoLa B	02 02 02 02 00 00 00 1B 73 52 41 20 53 65 72 69 61 6C 4E 75 6D 62 65 72 20 00 08 32 33 33 36 30 30 32 34 69

13.6.1.4.9.9 Read device type [sRN Ditype]

This telegram asks for the device type of the product family.

Table 404: Telegram structure: sRN Ditype

Telegram structure: sRN Ditype					 	
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	Ask state	String	6		Ditype	44 49 74 79 70 65

Table 405: Example: sRN Ditype

CoLa A	<STX>sRN{SPC}Ditype<ETX>	
	<STX>sRN Ditype<ETX>	
	sRN Ditype	
	02 73 52 4E 20 44 49 74 79 70 65 03	
CoLa B	02 02 02 02 00 00 00 0A 73 52 4E 20 44 49 74 79 70 65 5A	
	73 52 4E 20 44 49 74 79 70 65	

Table 406: Telegram structure: sRA Dtype

Telegram structure: sRA Dtype						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	Ask state	String	6		Dtype	44 49 74 79 70 65
Length of type key	Number of digits of the following type code length	Uint_8	1		0d ... 255d (0h ... FFh)	00 ... FF
Device type	Type code of the device	String	(var.)		(Device type)	(Device type)

Table 407: sRA Dtype Example for picoScan150:

CoLa A	<STX>sRA{SPC}Dtype{SPC}B{SPC}picoScan150<ETX> 02 73 52 41 20 44 49 74 79 70 65 20 42 20 70 69 63 6F 53 63 61 6E 31 35 30 03					
CoLa B	02 02 02 02 00 00 00 18 73 52 41 20 44 49 74 79 70 65 20 00 0B 70 69 63 6F 53 63 61 6E 31 35 30 60					

13.6.1.4.9.10 Read operating hours [sRN ODoprh]

Views the total number of operating hours during the lifetime of the device.

Table 408: Telegram structure: sRN ODoprh

Telegram structure: sRN ODoprh						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	Read operating hours	String	6		ODoprh	4F 44 6F 70 72 68

Table 409: Example: sRN ODoprh

CoLa A	<STX>sRN{SPC}ODoprh<ETX>					
	<STX>sRN ODoprh<ETX>					
	sRN ODoprh					
CoLa B	02 73 52 4E 20 4F 44 6F 70 72 68 03					
	02 02 02 02 00 00 00 0A 73 52 4E 20 4F 44 6F 70 72 68 41					
	73 52 4E 20 4F 44 6F 70 72 68					

Table 410: Telegram structure: sRA ODoprh

Telegram structure: sRA ODoprh						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	Read operating hours	String	6		ODoprh	4F 44 6F 70 72 68
Value	Operating hours in 1/10 h	Uint_32	4		0h ... FFFFFFFFh	00 00 00 00 ... FF FF FF FF

Table 411: Example: sRA ODoprh

CoLa A	<STX>sRA{SPC}ODoprh{SPC}1B50B<ETX> 02 73 52 41 20 4F 44 6F 70 72 68 20 31 42 35 30 42 03
CoLa B	02 02 02 02 00 00 00 0F 73 52 41 20 4F 44 6F 70 72 68 20 00 01 B5 0B D1

Calculation of the value: 1B50B (hex) → 111883 (dez) × 1/10 h = 11188.3 h

13.6.1.4.9.11 Read operating hours since last power on [sRN ODopdaily]

Shows the runtime duration since the last power on of the device.

Table 412: Telegram structure: sRN ODopdaily

Telegram structure: sRN ODopdaily						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	Read operating hours	String	9		ODopdaily	4F 44 6F 70 64 61 69 6C 79

Table 413: Example: sRN ODopdaily

CoLa A	<STX>sRN{SPC}ODopdaily<ETX> <STX>sRN ODopdaily<ETX> sRN ODopdaily 02 73 52 4E 20 4F 44 6F 70 64 61 69 6C 79 03 02 02 02 02 00 00 00 0D 73 52 4E 20 4F 44 6F 70 64 61 69 6C 79 22	
CoLa B	73 52 4E 20 4F 44 6F 70 64 61 69 6C 79	

Table 414: Telegram structure: sRA ODopdaily

Telegram structure: sRA ODopdaily						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	Read operating hours since last power on	String	9		ODopdaily	4F 44 6F 70 64 61 69 6C 79
Value	Operating hours in 1/10 h	Uint_32	4		0h ... FFFFFFFFh	00 00 00 00 ... FF FF FF FF

Table 415: Example: sRA ODopdaily

CoLa A	<STX>sRA{SPC}ODopdaily{SPC}424772B8<ETX> 02 73 52 41 20 4F 44 6F 70 72 68 20 34 32 34 37 37 32 42 38 03
CoLa B	02 02 02 02 00 00 00 12 73 52 41 20 4F 44 6F 70 72 68 20 42 47 72 B8 D7

13.6.1.4.9.12 Read power on counter [sRN ODpwrcl]

Shows the number of power on cycles.

Table 416: Telegram structure: sRN ODpwrC

Telegram structure: sRN ODpwrC						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	Read power on counter	String	6		ODpwrC	4F 44 70 77 72 63

Table 417: Example: sRN ODpwrC

CoLa A	<STX>sRN{SPC}ODpwrC<ETX>										
	<STX>sRN ODpwrC<ETX>										
	sRN ODpwrC										
	02 73 52 4E 20 4F 44 70 77 72 63 03										
CoLa B	02 02 02 02 00 00 00 0A 73 52 4E 20 4F 44 70 77 72 63 52										
	73 52 4E 20 4F 44 70 77 72 63										

Table 418: Telegram structure: sRA ODpwrC

Telegram structure: sRA ODpwrC						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	Read power on counter	String	6		ODpwrC	4F 44 70 77 72 63
Value	Power on counter	Uint_32	4		0h ... FFFFFFFFh	00 00 00 00 ... FF FF FF FF

Table 419: Example: sRA ODpwrC

CoLa A	<STX>sRA{SPC}ODpwrC{SPC}9A<ETX>
	02 73 52 41 20 4F 44 70 77 72 63 20 39 41 03
CoLa B	02 02 02 02 00 00 00 0F 73 52 41 20 4F 44 70 77 72 63 20 00 00 00 9A E7

13.6.1.4.9.13 Read temperature [sRN OPcurtmpdev]

With this command the internal temperature of the device can be identified. Please note that it does not give an indication of the current ambient temperature.

Table 420: Telegram structure: sRN OPcurtmpdev

Telegram structure: sRN OPcurtmpdev						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	Read temperature of the device	String	11		OPcurtmpdev	4F 50 63 75 72 74 6D 70 64 65 76

Table 421: Example: sRN OPcurtmpdev

CoLa A	<STX>sRN{SPC}OPcurtmpdev<ETX>	
	<STX>sRN OPcurtmpdev<ETX>	
	sRN OPcurtmpdev	
	02 73 52 4E 20 4F 50 63 75 72 74 6D 70 64 65 76 03	
CoLa B	02 02 02 02 00 00 00 0F 73 52 4E 20 4F 50 63 75 72 74 6D 70 64 65 76 2A	
	73 52 4E 20 4F 50 63 75 72 74 6D 70 64 65 76	

Table 422: Telegram structure: sRA OPcurtmpdev

Telegram structure: sRA OPcurtmpdev					 	
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	Read temperature of the device	String	11		OPcurtmpdev	4F 50 63 75 72 74 6D 70 64 65 76
Temperature data	[°C]	Real as float according to IEEE754	4	(-50°C ... +100°C)	C2480000h ... 42C80000h	C2 48 00 00 ... 42 C8 00 00

Example: sRA OPcurtmpdev (35°C)**Table 423:** Example: sRA OPcurtmpdev

CoLa A	<STX>sRA{SPC}OPcurtmpdev{SPC}420C0000<ETX>
	02 73 52 41 20 4F 50 63 75 72 74 6D 70 64 65 76 20 34 32 30 43 30 30 30 30 03
CoLa B	02 02 02 02 00 00 00 14 73 52 41 20 4F 50 63 75 72 74 6D 70 64 65 76 20 42 0C 00 00 4B

13.6.1.4.9.14 Set device name [sWN LocationName]

Give the device a specific description name such as its location.

Table 424: Telegram structure: sWN LocationName

Telegram structure: sWN LocationName (User level 'Maintenance' required)					 	
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E
Command	Set device name	String	12		LocationName	4C 6F 63 61 74 69 6F 6E 4E 61 6D 65
Value	Number of characters of the following device name	Uint_16	2		0d ... +16d (0h ... 10h)	00 00 ... 00 10
Value	Device name	String	32		[Device name]	[Device name]

Table 425: Example: sWN LocationName +9 LongRange

CoLa A	<STX>sWN{SPC}LocationName{SPC}+9{SPC}LongRange<ETX>
	<STX>sWN LocationName +9 LongRange<ETX>
	sWN LocationName +9 LongRange
	02 73 57 4E 20 4C 6F 63 61 74 69 6F 6E 4E 61 6D 65 20 39 20 4C 6F 6E 67 52 61 6E 67 65 03

	02 02 02 02 00 00 00 1D 73 57 4E 20 4C 6F 63 61 74 69 6F 6E 4E 61 6D 65 20 00 09 4C 6F 6E 67 52 61 6E 67 65 2C
CoLa B	73 57 4E 20 4C 6F 63 61 74 69 6F 6E 4E 61 6D 65 20 00 09 4C 6F 6E 67 52 61 6E 67 65

Table 426: Telegram structure: sWA LocationName

Telegram structure: sWA LocationName						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	Set device name	String	12		LocationName	4C 6F 63 61 74 69 6F 6E 4E 61 6D 65

Table 427: Example: sWA LocationName

CoLa A	<STX>sWA{SPC}LocationName<ETX> 02 73 57 41 20 4C 6F 63 61 74 69 6F 6E 4E 61 6D 65 03
CoLa B	02 02 02 02 00 00 00 11 73 57 41 20 4C 6F 63 61 74 69 6F 6E 4E 61 6D 65 20 7F

13.6.1.4.9.15 Read device name [sRN LocationName]

Read the given name of the device (Default is the serial number of the device).

Table 428: Telegram structure: sRN LocationName

Telegram structure: sRN LocationName						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	Read device name	String	12		LocationName	4C 6F 63 61 74 69 6F 6E 4E 61 6D 65

Table 429: Example: sRN LocationName

CoLa A	<STX>sRN{SPC}LocationName<ETX>		
	<STX>sRN LocationName<ETX>		
	sRN LocationName		
	02 73 52 4E 20 4C 6F 63 61 74 69 6F 6E 4E 61 6D 65 03		
CoLa B	02 02 02 02 00 00 00 10 73 52 4E 20 4C 6F 63 61 74 69 6F 6E 4E 61 6D 65 55		
	73 52 4E 20 4C 6F 63 61 74 69 6F 6E 4E 61 6D 65		

Table 430: Telegram structure: sRA LocationName

Telegram structure: sRA LocationName						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)

Find complete telegram structure of the answer in [see table 424, page 223](#)

13.6.1.4.9.16 Initiate an acoustic or visual signal for a defined period of time [sMN FindMe]

This command can be used to make the device easier to find.

Table 431: Telegram structure: sMN FindMe

Telegram structure: sMN FindMe (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Method	String	3		sMN	73 4D 4E
Command	Initiate an acoustic or visual signal	String	11		FindMe	46 69 6E 64 4D 65
Period of time	Duration in seconds	Uint_16	2		0d ... 65535d (0h ... FF FF)	00 ... FF FF

Table 432: Example: sMN FindMe

CoLa A	<STX>sMN{SPC}FindMe{SPC}1<ETX>					
	<STX>sMN FindMe 1<ETX>					
	sMN FindMe 1					
	02 73 4D 4E 20 46 69 6E 64 4D 65 20 31 03					
CoLa B	02 02 02 02 00 00 00 0D 73 4D 4E 20 46 69 6E 64 4D 65 20 01 7C					
	73 4D 4E 20 46 69 6E 64 4D 65 20 01					

Table 433: Telegram structure: sAN FindMe

Telegram structure: sAN FindMe						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sAN	73 41 4E
Command	Initiate an acoustic or visual signal	String	11		FindMe	46 69 6E 64 4D 65

Table 434: Example: sAN FindMe

CoLa A	<STX>sAN{SPC}Findme<ETX>					
	02 73 41 4E 20 46 69 6E 64 4D 65 03					
CoLa B	02 02 02 02 00 00 00 0B 73 41 4E 20 46 69 6E 64 4D 65 20 71					

13.6.1.4.9.17 Read date of last permanent save [sRN Dlpara]

This command reads the date at which the last permanent save (see "Save parameters permanently [sMN mEEwriteall]", page 150) was executed.

Table 435: Telegram structure: sRN Dlpara

Telegram structure: sRN Dlpara						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	Read date of last permanent save	String	6		Dlpara	44 49 70 61 72 61

Table 436: Example: sRN Dlpara

CoLa A	<STX>sRN{SPC}Dlpara<ETX>	
	<STX>sRN Dlpara<ETX>	
	sRN Dlpara	
	02 73 52 4E 20 44 49 70 61 72 61 03	
CoLa B	02 02 02 02 00 00 00 4E 73 52 4E 20 44 49 70 61 72 61 40	
	73 52 4E 20 44 49 70 61 72 61	

Table 437: Telegram structure: sRA Dlpara

Telegram structure: sRA Dlpara						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	Read date of last permanent save	String	6		Dlpara	44 49 70 61 72 61
Reserved	-	Uint_8	1	Always:	Ah	0A
Date of last permanent save	DD.MM.YYYY	Flex-String	10		(see example)	(see example)

Table 438: Example: sRA Dlpara

CoLa A	<STX>sRA{SPC}Dlpara{SPC}A{SPC}09.01.2024<ETX>
	02 73 52 41 20 44 49 70 61 72 61 20 41 20 30 39 2E 30 31 2E 32 30 32 34 03
CoLa B	02 02 02 02 00 00 00 17 73 52 41 20 44 49 70 61 72 61 20 00 0A 30 39 2E 30 31 2E 32 30 32 34 69

13.6.1.4.9.18 Read time of last permanent save [sRN Dlparatm]

This command reads the time at which the last permanent save (see "Save parameters permanently [sMN mEEwri-teal!]", page 150) was executed.

Table 439: Telegram structure: sRN Dlparatm

Telegram structure: sRN Dlparatm						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	Read time of last permanent save	String	8		Dlparatm	44 49 70 61 72 61 74 6D

Table 440: Example: sRN Dlparatm

CoLa A	<STX>sRN{SPC}Dlparatm<ETX>	
	<STX>sRN Dlparatm<ETX>	
	sRN Dlparatm	
	02 73 52 4E 20 44 49 70 61 72 61 74 6D 03	
CoLa B	02 02 02 02 00 00 00 4E 73 52 4E 20 44 49 70 61 72 61 74 6D 59	
	73 52 4E 20 44 49 70 61 72 61 74 6D	

Table 441: Telegram structure: sRA Dparatm

Telegram structure: sRA Dparatm						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	Read time of last permanent save	String	8		Dparatm	44 49 70 61 72 61 74 6D
Reserved	-	Uint_16	2	Always:	5h	00 05
Time of last permanent save	HH:MM	String	5		(see example)	(see example)

Table 442: Example: sRA Dparatm

CoLa A	<STX>sRA{SPC}Dparatm{SPC}5{SPC}12:28<ETX>					
	02 73 52 41 20 44 49 70 61 72 61 74 6D 20 35 20 31 32 3A 32 38 03					
CoLa B	02 02 02 02 00 00 00 14 73 52 41 20 44 49 70 61 72 61 74 6D 20 00 05 31 32 3A 32 38 40					

13.6.1.4.9.19 Read the current device temperature alarm status [sRN temperatureAlarmStatus]

Current device temperature alarm status. If true, the configured temperature alarm is active.

Table 443: Telegram structure: sRN temperatureAlarmStatus

Telegram structure: sRN temperatureAlarmStatus (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	Device temperature alarm status	String	22		temperatureAlarmStatus	74 65 6D 70 65 72 61 74 75 72 65 41 6C 61 72 6D 53 74 61 74 75 73 03

Table 444: Example: sRN temperatureAlarmStatus

CoLa A	<STX>sRN{SPC}temperatureAlarmStatus<ETX>					
	<STX>sRN temperatureAlarmStatus<ETX>					
	sRN temperatureAlarmStatus					
	02 73 52 4E 20 74 65 6D 70 65 72 61 74 75 72 65 41 6C 61 72 6D 53 74 61 74 75 73 03					
CoLa B	02 02 02 02 00 00 00 1A 73 52 4E 20 74 65 6D 70 65 72 61 74 75 72 65 41 6C 61 72 6D 53 74 61 74 75 73 44					
	73 52 4E 20 74 65 6D 70 65 72 61 74 75 72 65 41 6C 61 72 6D 53 74 61 74 75 73					

Table 445: Telegram structure: sRA temperatureAlarmStatus

Telegram structure: sRA temperatureAlarmStatus						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41

Telegram structure: sRA temperatureAlarmStatus						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command	Device temperature alarm status	String	22		temperatureAlarmStatus	74 65 6D 70 65 72 61 74 75 72 65 41 6C 61 72 6D 53 74 61 74 75 73
Data	Alarm state	Bool_1	1	Disbaled: Enabled:	0d (00h) +1d (01h)	00 ... 01

Table 446: Example: sRA temperatureAlarmStatus - Temperature alarm status is off. The device runs in the defined temperature range

CoLa A	<STX>sRA{SPC}temperatureAlarmStatus{SPC}0<ETX> 02 73 52 41 20 74 65 6D 70 65 72 61 74 75 72 65 41 6C 61 72 6D 53 74 61 74 75 73 20 30 03
CoLa B	02 02 02 02 00 00 00 1C 73 52 41 20 74 65 6D 70 65 72 61 74 75 72 65 41 6C 61 72 6D 53 74 61 74 75 73 20 00 6B

13.6.1.4.9.20 Set device temperature alarm configuration [sWN temperatureAlarmConfiguration]

Sets the upper and lower threshold for the temperature alarm status.

Table 447: Telegram structure: sWN temperatureAlarmConfiguration

Telegram structure: sWN temperatureAlarmConfiguration (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E
Command	Device temperature alarm configuration	String	29		temperatureAlarmConfigura-tion	74 65 6D 70 65 72 61 74 75 72 65 41 6C 61 72 6D 43 6F 6E 66 69 67 75 72 61 74 69 6F 6E
Data 1	Upper threshold [°C]	Int_8	1	Default: 55	-128d ... +127d (80h ... 7Fh)	80 ... 7F
Data 2	Lower threshold [°C]	Int_8	1	Default: 50	-128d ... +127d (80h ... 7Fh)	80 ... 7F

Table 448: Example: sWN temperatureAlarmConfiguration - Set the upper threshold to 55 °C and the lower threshold to 50 °C

CoLa A	<STX>sWN{SPC}temperatureAlarmConfiguration{SPC}37{SPC}32<ETX> <STX>sWN temperatureAlarmConfiguration 37 32<ETX> sWN temperatureAlarmConfiguration 37 32	
	02 73 57 4E 20 74 65 6D 70 65 72 61 74 75 72 65 41 6C 61 72 6D 43 6F 6E 66 69 67 75 72 61 74 69 6F 6E 20 33 37 20 33 32 03	
CoLa B	02 02 02 02 00 00 00 24 73 52 4E 20 74 65 6D 70 65 72 61 74 75 72 65 41 6C 61 72 6D 43 6F 6E 66 69 67 75 72 61 74 69 6F 6E 20 37 32 00 73 52 4E 20 74 65 6D 70 65 72 61 74 75 72 65 41 6C 61 72 6D 43 6F 6E 66 69 67 75 72 61 74 69 6F 6E 20 37 32	

Table 449: Telegram structure: sWA temperatureAlarmConfiguration

Telegram structure: sWA temperatureAlarmConfiguration						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	Device temperature alarm configuration	String	29		temperatureAlarmConfiguration	74 65 6D 70 65 72 61 74 75 72 65 41 6C 61 72 6D 43 6F 6E 66 69 67 75 72 61 74 69 6F 6E 03

Table 450: Example: sWA temperatureAlarmConfiguration

CoLa A	<STX>sWA[SPC]temperatureAlarmConfiguration<ETX>					
CoLa B	02 02 02 02 00 00 00 22 73 57 41 20 74 65 6D 70 65 72 61 74 75 72 65 41 6C 61 72 6D 43 6F 6E 66 69 67 75 72 61 74 69 6F 6E 03					

13.6.1.4.9.21 Read device temperature alarm configuration [sRN temperatureAlarmConfiguration]

Returns the upper and lower threshold for the temperature alarm status.

Table 451: Telegram structure: sRN temperatureAlarmConfiguration

Telegram structure: sRN temperatureAlarmConfiguration (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	Device temperature alarm configuration	String	29		temperatureAlarmConfiguration	74 65 6D 70 65 72 61 74 75 72 65 41 6C 61 72 6D 43 6F 6E 66 69 67 75 72 61 74 69 6F 6E

Table 452: Example: sRN temperatureAlarmConfiguration

CoLa A	<STX>sRN[SPC]temperatureAlarmConfiguration<ETX>					
CoLa B	02 02 02 02 00 00 00 21 73 52 4E 20 74 65 6D 70 65 72 61 74 75 72 65 41 6C 61 72 6D 43 6F 6E 66 69 67 75 72 61 74 69 6F 6E 20					

Table 453: Telegram structure: sRA temperatureAlarmConfiguration

Telegram structure: sRA temperatureAlarmConfiguration						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41

Telegram structure: sRA temperatureAlarmConfiguration						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command	Device temperature alarm configuration	String	29		temperatureAlarmConfiguration	74 65 6D 70 65 72 61 74 75 72 65 41 6C 61 72 6D 43 6F 6E 66 69 67 75 72 61 74 69 6F 6E
Data 1	Upper threshold [°C]	Int_8	1	Default: 55	-128d ... +127d (80h ... 7Fh)	80 ... 7F
Data 2	Lower threshold [°C]	Int_8	1	Default: 50	-128d ... +127d (80h ... 7Fh)	80 ... 7F

Table 454: Example: sRA temperatureAlarmConfiguration - 55 °C upper threshold and 50 °C lower threshold

CoLa A	<STX>sRA{SPC}temperatureAlarmConfiguration{SPC}37{SPC}32<ETX> 02 73 52 41 20 74 65 6D 70 65 72 61 74 75 72 65 41 6C 61 72 6D 43 6F 6E 66 69 67 75 72 61 74 69 6F 6E 20 33 37 20 33 32 03					
CoLa B	02 02 02 02 00 00 00 24 73 52 41 20 74 65 6D 70 65 72 61 74 75 72 65 41 6C 61 72 6D 43 6F 6E 66 69 67 75 72 61 74 69 6F 6E 20 37 32 0A					

13.6.1.4.10 Interfaces

13.6.1.4.10.1 Set IP address [sWN EIIpAddr]



NOTE

- Save permanently to set values. Changes will be active after rebooting the device.
- Settings must correspond with network in which scanner is used. Else device cannot be found any more.

Table 455: Telegram structure: sWN EIIpAddr

Telegram structure: sWN EIIpAddr (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E
Command	Set IP address	String	8		EIIpAddr	45 49 49 50 41 64 64 72
IP address	Set values	Uint_8	1	First part of IP address	0 ...+255d (00 ... FF)	00 ... FF
				Second part of IP address	0 ...+255d (00 ... FF)	00 ... FF
				Third part of IP address	0 ...+255d (00 ... FF)	00 ... FF
				Fourth part of IP address	0 ...+255d (00 ... FF)	00 ... FF

Table 456: Example: sWN EIIpAddr 192.168.0.2

CoLa A	<STX>sWN{SPC}EIIpAddr{SPC}C0{SPC}A8{SPC}0{SPC}2<ETX> <STX>sWN EIIpAddr C0 A8 0 2<ETX> sWN EIIpAddr C0 A8 0 2 02 73 57 4E 20 45 49 49 70 41 64 64 72 20 43 30 20 41 38 20 30 20 32 03					
--------	---	--	--	--	--	--

	02 02 02 02 00 00 00 11 73 57 4E 20 45 49 49 70 41 64 64 72 20 C0 A8 00 02 06	
CoLa B	73 57 4E 20 45 49 49 70 41 64 64 72 20 C0 A8 00 02	

Table 457: Telegram structure: sWA EllpAddr

Telegram structure: sWA EllpAddr						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	Set IP address	String	8		EllpAddr	45 49 49 50 41 64 64 72

Table 458: Example: sWA EllpAddr

CoLa A	<STX>sWA{SPC}EllpAddr<ETX> 02 73 57 41 20 45 49 49 70 41 64 64 72 03	
CoLa B	02 02 02 02 00 00 00 0D 73 57 41 20 45 49 49 70 41 64 64 72 20 63	

13.6.1.4.10.2 Read IP address [sRN EllpAddr]

Read the IP address of the device.

Table 459: Telegram structure: sRN EllpAddr

Telegram structure: sRN EllpAddr						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	Read IP address	String	8		EllpAddr	45 49 49 50 41 64 64 72

Table 460: Example: sRN EllpAddr

CoLa A	<STX>sRN{SPC}EllpAddr<ETX> <STX>sRN EllpAddr<ETX> sRN EllpAddr 02 73 52 4E 20 45 49 49 70 41 64 64 72 03	
CoLa B	02 02 02 02 00 00 00 0C 73 52 4E 20 45 49 49 70 41 64 64 72 49 73 52 4E 20 45 49 49 70 41 64 64 72	

Table 461: Telegram structure: sRA EllpAddr

Telegram structure: sRA EllpAddr						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	Read IP address	String	8		EllpAddr	45 49 49 50 41 64 64 72

Telegram structure: sRA EllpAddr						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
IP address	Default: 192.168.0.1	Uint_8	1	First part of IP address	0 ...+255d (00 ... FF)	00 ... FF
				Second part of IP address	0 ...+255d (00 ... FF)	00 ... FF
				Third part of IP address	0 ...+255d (00 ... FF)	00 ... FF
				Fourth part of IP address	0 ...+255d (00 ... FF)	00 ... FF

Table 462: Example: sRA EllpAddr 192.168.0.2

CoLa A	<STX>sRA{SPC}EllpAddr{SPC}C0{SPC}A8{SPC}00{SPC}02<ETX>					
	02 73 57 41 20 45 49 49 70 41 64 64 72 20 C0 20 A8 20 00 20 02 03					
CoLa B	02 02 02 02 00 00 00 11 73 52 41 20 45 49 49 70 41 64 64 72 20 C0 A8 00 02 0C					

13.6.1.4.10.3 Read IP address assigned by DHCP [sRN EllpAddrDHCP]

**NOTE**

DHCP needs to be set as mode for ethernet assignment.

Table 463: Telegram structure: sRN EllpAddrDHCP

Telegram structure: sRN EllpAddrDHCP						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	Read IP address assigned by DHCP	String	12		EllpAddrDHCP	45 49 49 70 41 64 64 72 44 48 43 50

Table 464: Example: srN EllpAddrDHCP

CoLa A	<STX>sRN{SPC}EllpAddrDHCP<ETX>					
	<STX>sRN EllpAddrDHCP<ETX>					
	sRN EllpAddrDHCP					
	02 73 57 4E 20 45 49 49 70 41 64 64 72 44 48 43 50 03					
	02 02 02 02 00 00 00 10 73 52 4E 20 45 49 49 70 41 64 64 72 44 48 43 50 56					
CoLa B	73 52 4E 20 45 49 49 70 41 64 64 72 44 48 43 50					

Table 465: Telegram structure: sRA EllpAddrDHCP

Telegram structure: sRA EllpAddrDHCP						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	Read IP address assigned by DHCP	String	12		EllpAddrDHCP	45 49 49 70 41 64 64 72 44 48 43 50

Telegram structure: sRA EIAddrDHCP						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
IP address	Default: 192.168.0.1	Uint_8	1	First part of IP address	0 ...+255d (00 ... FF)	00 ... FF
				Second part of IP address	0 ...+255d (00 ... FF)	00 ... FF
				Third part of IP address	0 ...+255d (00 ... FF)	00 ... FF
				Fourth part of IP address	0 ...+255d (00 ... FF)	00 ... FF

Table 466: Example: sRA EIAddrDHCP 192.168.0.1

CoLa A	<STX>sRA{SPC}EIAddrDHCP{SPC}C0{SPC}A8{SPC}0{SPC}1<ETX>					
	02 73 52 41 20 45 49 49 70 41 64 64 72 44 48 43 50 20 43 30 20 41 38 20 30 20 31 03					
CoLa B	02 02 02 02 00 00 00 00 15 73 52 41 20 45 49 49 70 41 64 64 72 44 48 43 50 20 C0 A8 00 01 10					

13.6.1.4.10.4 Set mode for ethernet address assignment [sWN EIAddrMode]

This Command determines the mode for the ethernet address assignment.

Table 467: Telegram structure: sWN EIAddrMode

Telegram structure: sWN EIAddrMode (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E
Command	Set mode for ethernet address assignment	String	10		EIAddrMode	45 49 41 64 64 72 4D 6F 64 65
Ethernet address assignment	Static IP address / DHCP	Enum_8	1	Static: DHCP:	0 1	00 01

Table 468: Example: sWN EIAddrMode

CoLa A	<STX>sWN{SPC}EIAddrMode{SPC}1<ETX>					
	<STX>sWN EIAddrMode 1<ETX>					
	sWN EIAddrMode 1					
	02 73 57 4E 20 45 49 41 64 64 72 4D 6F 64 65 20 31 03					
CoLa B	02 02 02 02 00 00 00 00 4E 73 57 4E 20 45 49 41 64 64 72 4D 6F 64 65 20 01 76					
	73 57 4E 20 45 49 41 64 64 72 4D 6F 64 65 20 01					

Table 469: Telegram structure: sWA EIAddrMode

Telegram structure: sWA EIAddrMode						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	Set mode for ethernet address assignment	String	10		EIAddrMode	45 49 41 64 64 72 4D 6F 64 65

Table 470: Example: sWA ElAddrMode

CoLa A	<STX>sWA{SPC}ElAddrMode<ETX> 02 73 57 41 20 45 49 41 64 64 72 4D 6F 64 65 03
CoLa B	02 02 02 02 00 00 00 0F 73 57 41 20 45 49 41 64 64 72 4D 6F 64 65 20 79

13.6.1.4.10.5 Set fallback for DHCP [sWN EIDHCPFallback]

This Command determines the fallback when DHCP is not successful.

Table 471: Telegram structure: sWN EIDHCPFallback

Telegram structure: sWN EIDHCPFallback (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E
Command	Set fallback for DHCP	String	14		EIDHCPFallback	45 49 44 48 43 50 46 61 6C 6C 62 61 63 6B
Fallback ethernet address assignment	Use Static IP address / Retry DHCP	Enum_8	1	Static IP address: DHCP retry:	0 1	00 01

Table 472: Example: sWN EIDHCPFallback

CoLa A	<STX>sWN{SPC}EIDHCPFallback{SPC}1<ETX> <STX>sWN EIDHCPFallback 1<ETX> sWN EIDHCPFallback 1 02 73 57 4E 20 45 49 44 48 43 50 46 61 6C 6C 62 61 63 6B 20 31 03	
CoLa B	02 02 02 02 00 00 00 14 73 57 4E 20 45 49 44 48 43 50 46 61 6C 6C 62 61 63 6B 20 01 54 73 57 4E 20 45 49 44 48 43 50 46 61 6C 6C 62 61 63 6B 20 01	

Table 473: Telegram structure: sWA EIDHCPFallback

Telegram structure: sWA EIDHCPFallback						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	Set fallback for DHCP	String	14		EIDHCPFallback	45 49 44 48 43 50 46 61 6C 6C 62 61 63 6B

Table 474: Example: sWA EIDHCPFallback

CoLa A	<STX>sWA{SPC}EIDHCPFallback<ETX> 02 73 57 41 20 45 49 44 48 43 50 46 61 6C 6C 62 61 63 6B 03
CoLa B	02 02 02 02 00 00 00 13 73 57 41 20 45 49 44 48 43 50 46 61 6C 6C 62 61 63 6B 20 5A

13.6.1.4.10.6 Set Ethernet gateway [sWN Elgate]

Change Ethernet gateway IP address (TCP/IP)

**NOTE**

- Save permanently to set values. Changes will be active after rebooting the device.
- Settings must correspond with network in which scanner is used. Else device cannot be found any more.

Table 475: Telegram structure: sWN Elgate

Telegram structure: sWN Elgate (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E
Command	Set gateway adress	String	6		Elgate	45 49 67 61 74 65
Gateway address	Set values	Uint_8	1	First part of gateway address	0 ...+255d (00... FF)	00 ...FF
				Second part of gateway address	0 ...+255d (00... FF)	00 ...FF
				Third part of gateway address	0 ...+255d (00... FF)	00 ...FF
				Fourth part of gateway address	0 ...+255d (00... FF)	00 ...FF

Table 476: Example: sWN Elgate 192.168.0.1

CoLa A	<STX>sWN{SPC}Elgate{SPC}C0{SPC}A8{SPC}00{SPC}01<ETX>					
	<STX>sWN Elgate C0 A8 00 01<ETX>					
	sWN Elgate C0 A8 00 01					
	02 73 57 4E 20 45 49 67 61 74 65 20 43 30 20 41 38 20 30 30 20 30 31 03					
CoLa B	02 02 02 02 00 00 00 0F 73 57 4E 20 45 49 67 61 74 65 20 C0 A8 00 01 18					
	73 57 4E 20 45 49 67 61 74 65 20 C0 A8 00 01					

Table 477: Telegram structure: sWA Elgate

Telegram structure: sWA Elgate						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	Set gateway adress	String	6		Elgate	45 49 67 61 74 65

Table 478: Example: sWA Elgate

CoLa A	<STX>sWA{SPC}Elgate<ETX>					
	02 73 57 41 20 45 49 67 61 74 65 03					
CoLa B	02 02 02 02 00 00 00 0B 73 57 41 20 45 49 67 61 74 65 20 7E					

13.6.1.4.10.7 Read Ethernet gateway [sRN Elgate]

Read for the Ethernet gateway (TCP/IP)

Table 479: Telegram structure: sRN Elgate

Telegram structure: sRN Elgate						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	Read gateway address	String	6		Elgate	45 49 67 61 74 65

Table 480: Example: sRN Elgate

CoLa A	<STX>sRN{SPC}Elgate<ETX>																
	<STX>sRN Elgate<ETX>																
	sRN Elgate																
	02 73 52 4E 20 45 49 67 61 74 65 03																
CoLa B	02 02 02 02 00 00 00 0A 73 52 4E 20 45 49 67 61 74 65 54																
	73 52 4E 20 45 49 67 61 74 65																

Table 481: Telegram structure: sRA Elgate

Telegram structure: sRA Elgate						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	Read gateway address	String	6		Elgate	45 49 67 61 74 65
Gateway address	Default: 0.0.0.0	Uint_8	1	First part of gateway address	0 ...+255d (00... FF)	00 ... FF
				Second part of gateway address	0 ...+255d (00... FF)	00 ... FF
				Third part of gateway address	0 ...+255d (00... FF)	00 ... FF
				Fourth part of gateway address	0 ...+255d (00... FF)	00 ... FF

Table 482: Example: sRA Elgate 192.168.0.1

CoLa A	<STX>sRA[SPC]Elgate[SPC]C0[SPC]A8[SPC]00[SPC]01<ETX>				
	02 73 52 41 20 45 49 67 61 74 65 20 C0 A8 00 01 03				
CoLa B	02 02 02 02 00 00 00 0F 73 52 41 20 45 49 67 61 74 65 20 C0 A8 00 01 12				

13.6.1.4.10.8 Read ethernet gateway IP address assigned by DHCP [sRN ElgateDHCP]



NOTE

DHCP needs to be set as mode for ethernet assignment.

Read for the ethernet gateway IP address which was assigned by DHCP.

Table 483: Telegram structure: sRN ElgateDHCP

Telegram structure: sRN ElgateDHCP						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	Read ethernet gateway IP address assigned by DHCP	String	6		ElgateDHCP	45 49 67 61 74 65 44 48 43 50

Table 484: Example: sRN Elgate

CoLa A	<STX>sRN{SPC}ElgateDHCP<ETX>					
	<STX>sRN ElgateDHCP<ETX>					
	sRN ElgateDHCP					
	02 73 52 4E 20 45 49 67 61 74 65 44 48 43 50 03					
CoLa B	02 02 02 02 00 00 00 0E 73 52 4E 20 45 49 67 61 74 65 44 48 43 50 4B					
	73 52 4E 20 45 49 67 61 74 65 44 48 43 50					

Table 485: Telegram structure: sRA ElgateDHCP

Telegram structure: sRA ElgateDHCP						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	Read ethernet gateway IP address assigned by DHCP	String	6		ElgateDHCP	45 49 67 61 74 65 44 48 43 50
Gateway IP address	Default: 0.0.0.0	Uint_8	1	First part of gateway IP address	0 ...+255d (00... FF)	00 ... FF
				Second part of gateway IP address	0 ...+255d (00... FF)	00 ... FF
				Third part of gateway IP address	0 ...+255d (00... FF)	00 ... FF
				Fourth part of gateway IP address	0 ...+255d (00... FF)	00 ... FF

Table 486: Example: sRA ElgateDHCP 0.0.0.0

CoLa A	<STX>sRA{SPC}ElgateDHCP{SPC}0{SPC}0{SPC}0<ETX>					
	02 73 52 41 20 45 49 67 61 74 65 44 48 43 50 20 30 20 30 20 30 20 30 03					
CoLa B	02 02 02 02 00 00 00 13 73 52 41 20 45 49 67 61 74 65 44 48 43 50 20 00 00 00 00 64					

13.6.1.4.10.9 Set IP mask [sWN Elmask]

Define the subnet mask of the device.

NOTE

- Save permanently to set values. Changes will be active after rebooting the device.
- Settings must correspond with network in which scanner is used. Else device cannot be found any more.

Table 487: Telegram structure: sWN Elmask

Telegram structure: sWN Elmask (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E
Command	Set IP mask	String	6		Elmask	45 49 6D 61 73 6B
IP mask	Set values	Uint_8	1	First part of IP mask	0 ...+255d (00...FF)	00 ...FF
				Second part of IP mask	0 ...+255d (00...FF)	00 ...FF
				Third part of IP mask	0 ...+255d (00...FF)	00 ...FF
				Fourth part of IP mask	0 ...+255d (00...FF)	00 ...FF

Table 488: Example: sWN Elmask 255.255.254.0

CoLa A	<STX>sWN{SPC}Elmask{SPC}FF{SPC}FF{SPC}FE{SPC}00<ETX>					
	<STX>sWN Elmask FF FF FE 00<ETX>					
	sWN Elmask FF FF FE 00					
	02 73 57 4E 20 45 49 6D 61 73 6B 20 46 46 20 46 46 20 46 45 20 30 30 03					
CoLa B	02 02 02 02 00 00 00 0F 73 57 4E 20 45 49 6D 61 73 6B 20 FF FF FE 00 8C					
	73 57 4E 20 45 49 6D 61 73 6B 20 FF FF FE 00					

Table 489: Telegram structure: sWA Elmask

Telegram structure: sWA Elmask						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	Set IP mask	String	6		Elmask	45 49 6D 61 73 6B

Table 490: Example: sWA Elmask

CoLa A	<STX>sWA{SPC}Elmask<ETX>					
	02 73 57 41 20 45 49 6D 61 73 6B 03					
CoLa B	02 02 02 02 00 00 00 0B 73 57 41 20 45 49 6D 61 73 6B 20 7D					

13.6.1.4.10.10 Read IP mask [sRN Elmask]

Read the subnet mask of the device.

Table 491: Telegram structure: sRN Elmask

Telegram structure: sRN Elmask						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	Read IP mask	String	6		Elmask	45 49 6D 61 73 6B

Table 492: Example: sRN Elmask

CoLa A	<STX>sRN{SPC}Elmask<ETX>	
	<STX>sRN Elmask<ETX>	
	sRN Elmask	
	02 73 52 4E 20 45 49 6D 61 73 6B 03	
CoLa B	02 02 02 02 00 00 00 0A 73 52 4E 20 45 49 6D 61 73 6B 57	
	73 52 4E 20 45 49 6D 61 73 6B	

Table 493: Telegram structure: sRA Elmask

Telegram structure: sRA Elmask						
Telegram part	Description	Variable	Length	Sensor	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	Read IP mask	String	6		Elmask	45 49 6D 61 73 6B
IP mask	Default: 255.255.255.0	Uint_8	1	First part of IP mask	0 ...+255d (00... FF)	00 ... FF
				Second part of IP mask	0 ...+255d (00... FF)	00 ... FF
				Third part of IP mask	0 ...+255d (00... FF)	00 ... FF
				Fourth part of IP mask	0 ...+255d (00... FF)	00 ... FF

Table 494: Example: sRA Elmask 255.255.254.0

CoLa A	<STX>sRA{SPC}Elmask{SPC}FF{SPC}FF{SPC}FE{SPC}00<ETX>
	02 73 52 41 20 45 49 6D 61 73 6B 20 45 49 6D 61 73 6B 03
CoLa B	02 02 02 02 00 00 00 0F 73 52 41 20 45 49 6D 61 73 6B 20 FF FF FE 00 86

13.6.1.4.10.11 Read IP mask assigned by DHCP [sRN ElmaskDHCP]

**NOTE**

DHCP needs to be set as mode for ethernet assignment.

Read for the IP mask which was assigned by DHCP.

Table 495: Telegram structure: sRN ElmaskDHCP

Telegram structure: sRN ElmaskDHCP						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	Read IP mask assigned by DHCP	String	10		ElmaskDHCP	45 49 6D 61 73 6B 44 48 43 50

Table 496: Example: sRN ElmaskDHCP

CoLa A	<STX>sRN{SPC}ElmaskDHCP<ETX>	
	<STX>sRN ElmaskDHCP<ETX>	
	sRN ElmaskDHCP	
	02 73 52 4E 20 45 49 6D 61 73 6B 44 48 43 50 03	

	02 02 02 02 00 00 00 0E 73 52 4E 20 45 49 6D 61 73 6B 44 48 43 50 4B
CoLa B	73 52 4E 20 45 49 6D 61 73 6B 44 48 43 50

Table 497: Telegram structure: sRA ElmaskDHCP

Telegram structure: sRA ElmaskDHCP						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	Read IP mask assigned by DHCP	String	10		ElmaskDHCP	45 49 6D 61 73 6B 44 48 43 50
IP mask	Default: 255.255.255.0	Uint_8	1	First part of IP mask	0 ...+255d (00... FF)	00 ... FF
				Second part of IP mask	0 ...+255d (00... FF)	00 ... FF
				Third part of IP mask	0 ...+255d (00... FF)	00 ... FF
				Fourth part of IP mask	0 ...+255d (00... FF)	00 ... FF

Table 498: Example: sRA ElmaskDHCP 255.255.255.0

CoLa A	<STX>sRA{SPC}ElgateDHCP{SPC}FF{SPC}FF{SPC}FF{SPC}0<ETX>
	02 73 52 41 20 45 49 6D 61 73 6B 44 48 43 50 20 46 46 20 46 46 20 46 46 20 30 03
CoLa B	02 02 02 02 00 00 00 13 73 52 41 20 45 49 6D 61 73 6B 44 48 43 50 20 FF FF FF 00 98

13.6.1.4.10.12 Read MAC address [sRN EIMacAdr]

Read the MAC address of the device.

Table 499: Telegram structure: sRN EIMacAdr

Telegram structure: sRN EIMacAdr						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	Read MAC address of the device	String	8		EIMacAdr	45 49 4D 61 63 41 64 72

Table 500: Example: sRN EIMacAdr

CoLa A	<STX>sRN{SPC}EIMacAdr<ETX>
	<STX>sRN EIMacAdr<ETX>
	sRN EIMacAdr
	02 73 57 4E 20 45 49 4D 61 63 41 64 72 03
CoLa B	02 02 02 02 00 00 00 0C 73 52 4E 20 45 49 4D 61 63 41 64 72 5B
	73 52 4E 20 45 49 4D 61 63 41 64 72

Table 501: Telegram structure: sRA EIMacAdr

Telegram structure: sRA EIMacAdr						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	Read MAC address of the device	String	8		EIMacAdr	45 49 4D 61 63 41 64 72
MAC address	Values	Uint_8	1	First part of MAC address	0 ...+255d (00... FF)	00 ... FF
		Uint_8	1	Second part of MAC address	0 ...+255d (00... FF)	00 ... FF
		Uint_8	1	Third part of MAC address	0 ...+255d (00... FF)	00 ... FF
		Uint_8	1	Fourth part of MAC address	0 ...+255d (00... FF)	00 ... FF
		Uint_8	1	Fifth part of MAC address	0 ...+255d (00... FF)	00 ... FF
		Uint_8	1	Sixth part of MAC address	0 ...+255d (00... FF)	00 ... FF

Table 502: Example: sRA EIMacAdr 00:06:77:22:40:EA

CoLa A	<STX>sRA{SPC}EIMacAdr{SPC}0{SPC}6{SPC}77{SPC}22{SPC}40{SPC}EA<ETX>					
	02 73 52 41 20 45 49 4D 61 63 41 64 72 20 30 20 36 20 37 37 20 32 32 20 34 30 20 45 41 03					
CoLa B	02 02 02 02 00 00 00 13 73 52 41 20 45 49 4D 61 63 41 64 72 20 00 06 77 22 40 EA 8D					

13.6.1.4.10.13 Set device search mode [sWN EtherColaScanMode]

This command configures the Auto IP functionalities (Device search by SICK Engineering Tools) and the changeability of the IP address. The functionality can reduce the number of open ports to a minimum.

Table 503: Telegram structure: sWN EtherCoLaScanMode

Telegram structure: sWN EtherCoLaScanMode (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E
Command	Set CoLa Scan / AutoIP search settings	String	17		EtherCoLaScanMode	45 74 68 65 72 43 6F 4C 61 53 63 61 6E 4D 6F 64 65
CoLa Scan / AutoIP search settings	Choose setting	Uint_8	1	IP address can be found/ changed without login:	0	00
				IP address can be found/ changed with login:	1	01
				Disable the Auto IP search/ IP change:	2	02
				IP address can be found/ changed for 1 min:	3	03

Table 504: Example: sWN EtherCoLaScanMode 2 - Disable the Auto IP search/ IP change

CoLa A	<STX>sWN{SPC}EtherCoLaScanMode{SPC}2<ETX>					
	<STX>sWN EtherCoLaScanMode 2<ETX>					
	sWN EtherCoLaScanMode 2					
	02 73 57 4E 20 45 74 68 65 72 43 6F 4C 61 53 63 61 6E 4D 6F 64 65 20 32 03					

	02 02 02 02 00 00 00 14 73 57 4E 20 45 74 68 65 72 43 6F 4C 61 53 63 61 6E 4D 6F 64 65 20 02 3B	
CoLa B	73 57 4E 20 45 74 68 65 72 43 6F 4C 61 53 63 61 6E 4D 6F 64 65 20 02	

Table 505: Telegram structure: sWA EtherCoLaScanMode

Telegram structure: sWA EtherCoLaScanMode					 	
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	Set CoLa Scan / AutoIP search settings	String	17		EtherCoLaScanMode	45 74 68 65 72 43 6F 4C 61 53 63 61 6E 4D 6F 64 65

Table 506: Example: sWA EtherCoLaScanMode

CoLa A	<STX> sWA {SPC} EtherCoLaScanMode <ETX>
	02 73 57 41 20 45 74 68 65 72 43 6F 4C 61 53 63 61 6E 4D 6F 64 65 03
CoLa B	02 02 02 02 00 00 00 16 73 57 41 20 45 74 68 65 72 43 6F 4C 61 53 63 61 6E 4D 6F 64 65 20 36

13.6.1.4.10.14 Read device search mode [sRN EtherColaScanMode]

This command polls the Auto IP functionalities (Device search by SICK Engineering Tools) and the IP address changeability settings.

Table 507: Telegram structure: sRN EtherCoLaScanMode

Telegram structure: sRN EtherCoLaScanMode (User level 'Authorized client' required)					 	
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sRN	73 52 4E
Command	Read CoLa Scan / AutoIP search settings	String	17		EtherCoLaScanMode	45 74 68 65 72 43 6F 4C 61 53 63 61 6E 4D 6F 64 65

Table 508: Example: sRN EtherCoLaScanMode

	<STX> sRN {SPC} EtherCoLaScanMode <ETX>	
CoLa A	<STX>sRN EtherCoLaScanMode<ETX>	
	sRN EtherCoLaScanMode	
	02 73 52 4E 20 45 74 68 65 72 43 6F 4C 61 53 63 61 6E 4D 6F 64 65 03	
	02 02 02 02 00 00 00 15 73 52 4E 20 45 74 68 65 72 43 6F 4C 61 53 63 61 6E 4D 6F 64 65 1C	
CoLa B	73 52 4E 20 45 74 68 65 72 43 6F 4C 61 53 63 61 6E 4D 6F 64 65	

Table 509: Telegram structure: sRA EtherCoLaScanMode

Telegram structure: sRA EtherCoLaScanMode					 	
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41

Telegram structure: sRA EtherCoLaScanMode						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command	Read CoLa Scan / AutoIP search settings	String	17		EtherCoLaScanMode	45 74 68 65 72 43 6F 4C 61 53 63 61 6E 4D 6F 64 65
CoLa Scan / AutoIP search settings	Poll setting	Uint_8	1	IP address can be found/changed without login:	0	00
				IP address can be found/changed with login:	1	01
				Auto IP search/ IP change is disabled:	2	02
				IP address can be found/changed for 1 min:	3	03

Table 510: Example: sRA EtherCoLaScanMode 3 - IP address can be found/changed for 1 min

CoLa A	<STX>sRA{SPC}EtherCoLaScanMode{SPC}3<ETX>					
	02 73 52 41 20 45 74 68 65 72 43 6F 4C 61 53 63 61 6E 4D 6F 64 65 20 33 03					
CoLa B	02 02 02 02 00 00 00 17 73 52 41 20 45 74 68 65 72 43 6F 4C 61 53 63 61 6E 4D 6F 64 65 20 03 30					

13.6.1.4.10.15 Enable/ disable CoLa1 interface [sWN EIAuxEnable]

After enabling the CoLa1 interface, use port 2111 for CoLa A and port 2112 for CoLa B.

Table 511: Telegram structure: sWN EIAuxEnable

Telegram structure: sWN EIAuxEnable (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write	String	3		sWN	73 57 4E
Command	Set CoLa1 interface	String	11		EIAuxEnable	45 49 41 75 78 45 6E 61 62 6C 65
CoLa1 interface	Enable/ disable	Bool_1	1	Disable:	0	00
				Enable:	1	01

Table 512: Example: sWN EIAuxEnable

CoLa A	<STX>sWN{SPC}EIAuxEnable{SPC}1<ETX>					
	<STX>sWN EIAuxEnable 1<ETX>					
	sWN EIAuxEnable 1					
	02 73 57 4E 20 45 49 41 75 78 45 6E 61 62 6C 65 20 31 03					
CoLa B	02 02 02 02 00 00 00 11 73 57 4E 20 45 49 41 75 78 45 6E 61 62 6C 65 20 01 0A					
	73 57 4E 20 45 49 41 75 78 45 6E 61 62 6C 65 20 01					

Table 513: Telegram structure: sWA EIAuxEnable

Telegram structure: sWA EIAuxEnable						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41

Telegram structure: sWA EIAuxEnable						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command	Set CoLa1 interface	String	11		EIAuxEnable	45 49 41 75 78 45 6E 61 62 6C 65

Table 514: Example: sWA EIAuxEnable

CoLa A	<STX>sWA{SPC}EIAuxEnable<ETX> 02 73 57 41 20 45 49 41 75 78 45 6E 61 62 6C 65 03
CoLa B	02 02 02 02 00 00 00 10 73 57 41 20 45 49 41 75 78 45 6E 61 62 6C 65 20 04

13.6.1.4.10.16 Set Webserver state [sMN SetWebserverEnabled]

This command enables/ disables the Webserver. Port 80 will not be opened after a reboot.

Table 515: Telegram structure: sMN SetWebserverEnabled

Telegram structure: sMN SetWebserverEnabled (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Method	String	3		sMN	73 4D 4E
Command	Set Webserver state	String	19		SetWebserver-Enabled	53 65 74 57 65 62 73 65 72 76 65 72 45 6E 61 62 6C 65 64
State	Enable/ disable	Bool_1	1	Disable: Enable:	0 1	00 01

Table 516: Example: sMN SetWebserverEnabled

CoLa A	<STX>sMN{SPC}SetWebserverEnabled{SPC}1<ETX> <STX>sMN SetWebserverEnabled 1<ETX> sMN SetWebserverEnabled 1 02 73 4D 4E 20 47 65 74 57 65 62 73 65 72 76 65 72 45 6E 61 62 6C 65 64 20 31 03
CoLa B	02 02 02 02 00 00 00 19 73 4D 4E 20 53 65 74 57 65 62 73 65 72 76 65 72 45 6E 61 62 6C 65 64 20 01 23 73 4D 4E 20 53 65 74 57 65 62 73 65 72 76 65 72 45 6E 61 62 6C 65 64 20 01

Table 517: Telegram structure: sAN SetWebserverEnabled

Telegram structure: sAN SetWebserverEnabled						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sAN	73 41 4E
Command	Set Webserver state	String	19		SetWebserver-Enabled	53 65 74 57 65 62 73 65 72 76 65 72 45 6E 61 62 6C 65 64

Table 518: Example: sAN SetWebserverEnabled

CoLa A	<STX>sAN{SPC}SetWebserverEnabled<ETX> 02 73 41 4E 20 47 65 74 57 65 62 73 65 72 76 65 72 45 6E 61 62 6C 65 64 03
CoLa B	02 02 02 02 00 00 00 18 73 41 4E 20 47 65 74 57 65 62 73 65 72 76 65 72 45 6E 61 62 6C 65 64 2E

13.6.1.4.10.17 Read Webserver state [sMN GetWebserverEnabled]

Returns state if Webserver is enabled.

Table 519: Telegram structure: sMN GetWebserverEnabled

Telegram structure: sMN GetWebserverEnabled (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Method	String	3		sMN	73 4D 4E
Command	Webserver state	String	19		GetWebserver-Enabled	47 65 74 57 65 62 73 65 72 76 65 72 45 6E 61 62 6C 65 64

Table 520: Example: sMN GetWebserverEnabled

CoLa A	<STX>sMN{SPC}GetWebserverEnabled<ETX>					
	<STX>sMN GetWebserverEnabled<ETX>					
	sMN GetWebserverEnabled					
	02 73 4D 4E 20 47 65 74 57 65 62 73 65 72 76 65 72 45 6E 61 62 6C 65 64 03					
	02 02 02 02 00 00 00 17 73 4D 4E 20 47 65 74 57 65 62 73 65 72 76 65 72 45 6E 61 62 6C 65 64 16					
CoLa B	73 4D 4E 20 47 65 74 57 65 62 73 65 72 76 65 72 45 6E 61 62 6C 65 64					

Table 521: Telegram structure: sAN GetWebserverEnabled

Telegram structure: sAN GetWebserverEnabled						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sAN	73 41 4E
Command	Webserver state	String	19		GetWebserver-Enabled	47 65 74 57 65 62 73 65 72 76 65 72 45 6E 61 62 6C 65 64
State	Status of the webserver	Bool_1	1	Disabled: Enabled:	0 1	00 01

Table 522: Example: sAN GetWebserverEnabled

CoLa A	<STX>sAN{SPC}GetWebserverEnabled{SPC}1<ETX>					
	02 73 41 4E 20 47 65 74 57 65 62 73 65 72 76 65 72 45 6E 61 62 6C 65 64 20 31 03					
CoLa B	02 02 02 02 00 00 00 19 73 41 4E 20 47 65 74 57 65 62 73 65 72 76 65 72 45 6E 61 62 6C 65 64 20 01 3B					

13.6.1.4.10.18 Apply configured ethernet settings [sMN mEthUpdt]

Apply the configured ethernet settings.

Table 523: Telegram structure: sMN mEthUpdt

Telegram structure: sMN mEthUpdt (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Method	String	3		sMN	73 4D 4E
Command	Apply ethernet settings	String	8		mEthUpdt	6D 45 74 68 55 70 64 74

Table 524: Example: sMN mEthUpdt

CoLa A	<STX> sMN {SPC} mEthUpdt <ETX>	
	<STX>sMN mEthUpdt<ETX>	
	sMN mEthUpdt	
	02 73 4D 4E 20 6D 45 74 68 55 70 64 74 03	
CoLa B	02 02 02 02 00 00 00 0C 73 4D 4E 20 6D 45 74 68 55 70 64 74 51	
	73 4D 4E 20 6D 45 74 68 55 70 64 74	

Table 525: Telegram structure: sAN mEthUpdt

Telegram structure: sAN mEthUpdt						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sAN	73 41 4E
Command	Apply ethernet settings	String	8		mEthUpdt	6D 45 74 68 55 70 64 74

Table 526: Example: sAN mEthUpdt

CoLa A	<STX> sAN {SPC} mEthUpdt <ETX>
	02 73 41 4E 20 6D 45 74 68 55 70 64 74 03
CoLa B	02 02 02 02 00 00 00 0D 73 41 4E 20 6D 45 74 68 55 70 64 74 20 7D

13.6.1.4.10.19 Enable/ disable LEDs [sWN LEDEnable]

This command enables/ disables the LEDs of the device.

Table 527: Telegram structure: sWN LEDEnable

Telegram structure: sWN LEDEnable (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Method	String	3		sWN	73 57 4E
Command	Set LEDs	String	9		LEDEnable	4C 45 44 45 6E 61 62 6C 65
Status	Enable/ disable	Bool_1	1	Off: On:	0 1	00 01

Table 528: Example: sWN LEDEnable

CoLa A	<STX> sWN {SPC} LEDEnable {SPC} 1 <ETX>	
	<STX>sWN LEDEnable 1<ETX>	
	sWN LEDEnable 1	
	02 73 57 4E 20 4C 45 44 45 6E 61 62 6C 65 20 31 03	
CoLa B	02 02 02 02 00 00 00 0F 73 57 4E 20 4C 45 44 45 6E 61 62 6C 65 20 01 07	
	73 57 4E 20 4C 45 44 45 6E 61 62 6C 65 20 01	

Table 529: Telegram structure: sWA LEDEnable

Telegram structure: sWA LEDEnable						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	Set LEDs	String	9		LEDenable	4C 45 44 45 6E 61 62 6C 65

Table 530: Example: sWA LEDEnable

CoLa A	<STX>sWA{SPC}LEDenable<ETX> 02 73 57 41 20 4C 45 44 45 6E 61 62 6C 65 03					
CoLa B	02 02 02 02 00 00 00 0E 73 57 41 20 73 57 41 20 4C 45 44 45 6E 61 62 6C 65 20 09					

13.6.1.4.10.20 Read state of LEDs [sRN LEDState]

Read the current state of the LEDs.

Table 531: Telegram structure: sRN LEDState

Telegram structure: sRN LEDState						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Method	String	3		sRN	73 52 4E
Command	Read state of LEDs	String	8		LEDState	4C 45 44 53 74 61 74 65

Table 532: Example: sRN LEDState

CoLa A	<STX>sRN{SPC}LEDState<ETX>					
	<STX>sRN LEDState<ETX>					
	sRN LEDState 02 73 52 4E 20 4C 45 44 53 74 61 74 65 03					
CoLa B	02 02 02 02 00 00 00 0C 73 52 4E 20 4C 45 44 53 74 61 74 65 55					
	73 52 4E 20 4C 45 44 53 74 61 74 65					

Table 533: Telegram structure: sRA LEDState

Telegram structure: sRA LEDState						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	State of LEDs	String	8		LEDState	4C 45 44 53 74 61 74 65
LED Color	-	Enum_8	1	Green: Yellow: Red:	0 1 2	00 01 02

Telegram structure: sRA LEDState						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
LED behavior	-	Enum_8	1	On: Off: Blinking: Blinking fast: Blinking delayed: Find me active (see "Initiate an acoustic or visual signal for a defined period of time [sMN FindMe]", page 224):	0 1 2 3 4 5	00 01 02 03 04 05
LED ID	Name of the LED	String	8		LED1 LED2	4C 45 44 31 4C 45 44 32

Table 534: Example: sRA LEDState

CoLa A	<STX>sRA{SPC}LED-State{SPC}2{SPC}0{SPC}0{SPC}LED2{SPC}0{SPC}0{SPC}0{SPC}0{SPC}1{SPC}0{SPC}LED1{SPC}0{SPC}0{SPC}0{SPC}0<ETX>					
CoLa B	02 73 52 41 20 4C 45 44 53 74 61 74 65 20 32 20 30 20 30 20 4C 45 44 32 00 00 00 00 20 31 20 30 20 4C 45 44 31 00 00 00 00 03					
CoLa B	02 02 02 02 00 00 00 23 73 52 41 20 4C 45 44 53 74 61 74 65 20 00 02 00 00 4C 45 44 32 00 00 00 00 01 00 4C 45 44 31 00 00 00 00 7A					

13.6.1.4.10.21 Set the port number for CoLa ASCII TCP/IP [sWN EIAuxPort]

Set the port number for CoLa ASCII TCP/IP (e.g. LMDscandata).

Table 535: Telegram structure: sWN EIAuxPort

Telegram structure: sWN EIAuxPort (User level 'Authorized Client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write request	String	3		sWN	73 57 4E
Variable name	EIAuxPort	String	9		EIAuxPort	45 49 41 75 78 50 6F 72 74
Value	TCP port for CoLa ASCII communication	UInt_16	2		+2111	08 3F

Table 536: Example: sWN EIAuxPort

CoLa A	<STX>sWN{SPC}EIAuxPort{SPC}+2111<ETX>					
CoLa A	<STX>sWN EIAuxPort +2111<ETX>					
CoLa A	sWN EIAuxPort +2111					
CoLa B	02 73 57 4E 20 45 49 41 75 78 50 6F 72 74 20 2B 32 31 31 31 03					
CoLa B	02 02 02 02 00 00 00 10 73 57 4E 20 45 49 41 75 78 50 6F 72 74 20 08 3F 24					
CoLa B	73 57 4E 20 45 49 41 75 78 50 6F 72 74 20 08 3F					

Table 537: Telegram structure: sWA EIAuxPort

Telegram structure: sWA EIAuxPort						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write answer	String	3		sWA	73 57 41

Telegram structure: sWA EIAuxPort						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Variable name	Echo of variable name	String	9		EIAuxPort	45 49 41 75 78 50 6F 72 74

Table 538: Example: sWA EIAuxPort

CoLa A	<STX>sWA{SPC}EIAuxPort<ETX>					
	02 73 57 41 20 45 49 41 75 78 50 6F 72 74 03					
CoLa B	02 02 02 02 00 00 00 0D 73 57 41 20 45 49 41 75 78 50 6F 72 74 3C					

13.6.1.4.10.22 Read the port number for CoLa ASCII TCP/IP [sRN EIAuxPort]

Returns the port number for CoLa ASCII TCP/IP (e.g. LMDscandata).

Table 539: Telegram structure: sRN EIAuxPort

Telegram structure: sRN EIAuxPort						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read request	String	3		sRN	73 52 4E
Variable name	EIAuxPort	String	9		EIAuxPort	45 49 41 75 78 50 6F 72 74

Table 540: Example: sRN EIAuxPort

CoLa A	<STX>sRN{SPC}EIAuxPort<ETX>					
	<STX>sRN EIAuxPort<ETX>					
	sRN EIAuxPort					
	02 73 52 4E 20 45 49 41 75 78 50 6F 72 74 03					
	02 02 02 02 00 00 00 0D 73 52 4E 20 45 49 41 75 78 50 6F 72 74 36					
CoLa B	73 52 4E 20 45 49 41 75 78 50 6F 72 74					

Table 541: Telegram structure: sRA EIAuxPort

Telegram structure: sRA EIAuxPort						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read answer	String	3		sRA	73 52 41
Variable name	EIAuxPort	String	9		EIAuxPort	45 49 41 75 78 50 6F 72 74
Value	TCP port for CoLa ASCII communication	UInt_16	2		+2111	08 3F

Table 542: Example: sRA EIAuxPort

CoLa A	<STX>sRA{SPC}EIAuxPort{SPC}+2111<ETX>					
	02 73 52 41 20 45 49 41 75 78 50 6F 72 74 20 2B 32 31 31 03					
CoLa B	02 02 02 02 00 00 00 10 73 52 41 20 45 49 41 75 78 50 6F 72 74 20 08 3F 2E					

13.6.1.4.10.23 Set the port number for CoLa Binary TCP/IP [sWN EIHstPort]

Set the port number for CoLa Binary TCP/IP (e.g. LMDscandata).

Table 543: Telegram structure: sWN EIHstPort

Telegram structure: sWN EIHstPort (User level 'Authorized Client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write request	String	3		sWN	73 57 4E
Variable name	EIHstPort	String	9		EIHstPort	45 49 48 73 74 50 6F 72 74
Value	TCP port for CoLa Binary communication	UInt_16	2		+2112	08 40

Table 544: Example: sWN EIHstPort

CoLa A	<STX>sWN{SPC}EIHstPort{SPC}+2112<ETX>					
	<STX>sWN EIHstPort +2112<ETX>					
	sWN EIHstPort +2112					
	02 73 57 4E 20 45 49 48 73 74 50 6F 72 74 20 2B 32 31 31 32 03					
CoLa B	02 02 02 02 00 00 00 10 73 57 4E 20 45 49 48 73 74 50 6F 72 74 20 08 40 58					
	73 57 4E 20 45 49 48 73 74 50 6F 72 74 20 08 40					

Table 545: Telegram structure: sWA EIHstPort

Telegram structure: sWA EIHstPort						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write answer	String	3		sWA	73 57 41
Variable name	Echo of variable name	String	9		EIHstPort	45 49 48 73 74 50 6F 72 74

Table 546: Example: sWA EIHstPort

CoLa A	<STX>sWA{SPC}EIHstPort<ETX>					
	02 73 57 41 20 45 49 48 73 74 50 6F 72 74 03					
CoLa B	02 02 02 02 00 00 00 0D 73 57 41 20 45 49 48 73 74 50 6F 72 74 3F					

13.6.1.4.10.24 Read the port number for CoLa Binary TCP/IP [sRN EIHstPort]

Returns the port number for CoLa Binary TCP/IP (e.g. LMDscandata).

Table 547: Telegram structure: sRN EIHstPort

Telegram structure: sRN EIHstPort						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read request	String	3		sRN	73 52 4E
Variable name	EIHstPort	String	9		EIHstPort	45 49 48 73 74 50 6F 72 74

Table 548: Example: sRN EIHstPort

CoLa A	<STX>sRN{SPC}EIHstPort<ETX>					
	<STX>sRN EIHstPort<ETX>					
	sRN EIHstPort					
	02 73 52 4E 20 45 49 48 73 74 50 6F 72 74 03					

	02 02 02 02 00 00 00 0D 73 52 4E 20 45 49 48 73 74 50 6F 72 74 35	
CoLa B	73 52 4E 20 45 49 48 73 74 50 6F 72 74	

Table 549: Telegram structure: sRA EIHstPort

Telegram structure: sRA EIHstPort						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read answer	String	3		sRA	73 52 41
Variable name	EIHstPort	String	9		EIHstPort	45 49 48 73 74 50 6F 72 74
Value	TCP port for CoLa Binary communication	UInt_16	2		+2112	08 40

Table 550: Example: sRA EIHstPort

CoLa	<STX>sRA{SPC}EIHstPort{SPC}+2112<ETX>
A	02 73 52 41 20 45 49 48 73 74 50 6F 72 74 20 2B 32 31 31 32 03
CoLa B	02 02 02 02 00 00 00 10 73 52 41 20 45 49 48 73 74 50 6F 72 74 20 08 40 52

13.6.1.4.10.25 Enable cloning plug functionality [sWN enableCloningPlug]

Enables or disables the cloning plug functionality.

Table 551: Telegram structure: sWN enableCloningPlug

Telegram structure: sWN enableCloningPlug (User level 'Service' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write request	String	3		sWN	73 57 4E
Variable name	enableCloningPlug	String	17		enableCloning-Plug	65 6E 61 62 6C 65 43 6C 6F 6E 69 6E 67 50 6C 75 67
Value	Enable/ disable cloning plug functionality	Bool_1	1	Disable: Enable:	0 1	00 01

Table 552: Example: sWN enableCloningPlug

CoLa	<STX>sWN{SPC}enableCloningPlug{SPC}1<ETX>	
A	<STX>sWN enableCloningPlug 1<ETX>	
	sWN enableCloningPlug 1	
	02 73 57 4E 20 65 6E 61 62 6C 65 43 6C 6F 6E 69 6E 67 50 6C 75 67 20 31 03	
	02 02 02 02 00 00 00 17 73 57 4E 20 65 6E 61 62 6C 65 43 6C 6F 6E 69 6E 67 50 6C 75 67 20 01 0A	
CoLa B	73 57 4E 20 65 6E 61 62 6C 65 43 6C 6F 6E 69 6E 67 50 6C 75 67 20 01	

Table 553: Telegram structure: sWA enableCloningPlug

Telegram structure: sWA enableCloningPlug						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Write answer	String	3		sWA	73 57 41

Telegram structure: sWA enableCloningPlug						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Variable name	Echo of variable name	String	17		enableCloning-Plug	65 6E 61 62 6C 65 43 6C 6F 6E 69 6E 67 50 6C 75 67

Table 554: Example: sWA enableCloningPlug

CoLa A	<STX>sWA{SPC}enableCloningPlug<ETX> 02 73 57 41 20 65 6E 61 62 6C 65 43 6C 6F 6E 69 6E 67 50 6C 75 67 03					
CoLa B	02 02 02 02 00 00 00 15 73 57 41 20 65 6E 61 62 6C 65 43 6C 6F 6E 69 6E 67 50 6C 75 67 24					

13.6.1.4.10.26 Read cloning plug functionality [sRN enableCloningPlug]

Check if the cloning plug functionality is enabled.

Table 555: Telegram structure: sRN enableCloningPlug

Telegram structure: sRN enableCloningPlug						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read request	String	3		sRN	73 52 4E
Variable name	enableCloningPlug	String	17		enableCloning-Plug	65 6E 61 62 6C 65 43 6C 6F 6E 69 6E 67 50 6C 75 67

Table 556: Example: sRN enableCloningPlug

CoLa A	<STX>sRN{SPC}enableCloningPlug<ETX> <STX>sRN enableCloningPlug<ETX> sRN enableCloningPlug					
	02 73 52 4E 20 65 6E 61 62 6C 65 43 6C 6F 6E 69 6E 67 50 6C 75 67 03					
	02 02 02 02 00 00 00 15 73 52 4E 20 65 6E 61 62 6C 65 43 6C 6F 6E 69 6E 67 50 6C 75 67 2E					
CoLa B	73 52 4E 20 65 6E 61 62 6C 65 43 6C 6F 6E 69 6E 67 50 6C 75 67					

Table 557: Telegram structure: sRA enableCloningPlug

Telegram structure: sRA enableCloningPlug						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read answer	String	3		sRA	73 52 41
Variable name	enableCloningPlug	String	17		enableCloning-Plug	65 6E 61 62 6C 65 43 6C 6F 6E 69 6E 67 50 6C 75 67
Value	Cloning plug functionality disabled / enabled	Bool_1	1	Disabled: Enabled:	0 1	00 01

Table 558: Example: sRA enableCloningPlug

CoLa A	<STX>sRA{SPC}enableCloningPlug{SPC}1<ETX> 02 73 52 41 20 65 6E 61 62 6C 65 43 6C 6F 6E 69 6E 67 50 6C 75 67 20 31 03					
CoLa B	02 02 02 02 00 00 00 17 73 52 41 20 65 6E 61 62 6C 65 43 6C 6F 6E 69 6E 67 50 6C 75 67 20 01 00					

13.6.1.4.11 Application

13.6.1.4.11.1 Field Evaluation

13.6.1.4.11.1.1 Set activation of evaluation group [sMN ActivateEvaluationGroup]

The telegram is intended to activate or deactivate groups via telegram. The group activation needs to be changed from always to telegram in advance.

Table 559: Telegram structure: sMN ActivateEvaluationGroup

Telegram structure: sMN ActivateEvaluationGroup (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Method	String	3		sMN	73 4D 4E
Command	Activate / deactivate evaluation group	String	23		ActivateEvaluationGroup	41 63 74 69 76 61 74 65 45 76 61 6C 75 61 74 69 6F 6E 47 72 6F 75 70
Amount of evaluation groups to activate/deactivate	Array	UInt_16	2	Each array element contains two parameters: - Evaluation group number - Deactivation / activation of evaluation group	1d ... 48d (1 ... 30h)	00 01 ... 00 30
Evaluation group number	Evaluation group 1	UInt_16	2		1	00 01
Deactivation / activation of evaluation group	Activate / deactivate evaluation group 1	Bool_1	1	Deactivate: Activate:	0 1	00 01
Evaluation group number	Evaluation group 2	UInt_16	2		2	00 02
Activate/deactivate evaluation group	Activate / deactivate evaluation group 2	Bool_1	1	Deactivate: Activate:	0 1	00 01
...						
Evaluation group number	Evaluation group 48	UInt_16	2		48d (30h)	00 30
Activate/deactivate evaluation group	Activate / deactivate evaluation group 48	Bool_1	1	Deactivate: Activate:	0 1	00 01

Table 560: Example1: sMN ActivateEvaluationGroup – Deactivate evaluation group 1

CoLa A	<STX>sMN{SPC}ActivateEvaluationGroup{SPC}1{SPC}1{SPC}0<ETX>	
	<STX>sMN ActivateEvaluationGroup 1 1 0<ETX>	
	sMN ActivateEvaluationGroup 1 1 0	
	02 73 4D 4E 20 41 63 74 69 76 61 74 65 45 76 61 6C 75 61 74 69 6F 6E 47 72 6F 75 70 20 31 20 31 20 30 03	

CoLa B	02 02 02 02 00 00 00 21 73 4D 4E 20 41 63 74 69 76 61 74 65 45 76 61 6C 75 61 74 69 6F 6E 47 72 6F 75 70 20 00 01 00 01 00 20	
	73 4D 4E 20 41 63 74 69 76 61 74 65 45 76 61 6C 75 61 74 69 6F 6E 47 72 6F 75 70 20 00 01 00 01 00	

Table 561: Example2: sMN ActivateEvaluationGroup – Deactivate evaluation group 1, activate evaluation group 2 and 3

CoLa A	<STX>sMN{SPC}ActivateEvaluationGroup{SPC}3{SPC}1{SPC}0{SPC}2{SPC}1{SPC}3{SPC}1<ETX>	
	<STX>sMN ActivateEvaluationGroup 3 1 0 2 1 3 1<ETX>	
CoLa B	sMN ActivateEvaluationGroup 3 1 0 2 1 3 1	
	02 73 4D 4E 20 41 63 74 69 76 61 74 65 45 76 61 6C 75 61 74 69 6F 6E 47 72 6F 75 70 20 33 20 31 20 30 20 32 20 31 20 33 20 31 03	

Table 562: Telegram structure: sAN ActivateEvaluation

Telegram structure: sAN ActivateEvaluation					 	
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sAN	73 52 41
Command	Activate / deactivate evaluation group	String	23		ActivateEvaluationGroup	41 63 74 69 76 61 74 65 45 76 61 6C 75 61 74 69 6F 6E 47 72 6F 75 70
Amount of activated/deactivated evaluation groups	-	UInt_16	2		1h... 30h	00 01 ... 00 30
State of activation / deactivation	-	Bool_1	1	Activation/Deactivation failed: Successfully activated/deactivated:	0 1	00 01

Table 563: Example 1: sAN ActivateEvaluation – Successfully deactivated evaluation group 1

CoLa A	<STX>sAN{SPC}ActivateEvaluation{SPC}1{SPC}1<ETX>
CoLa B	02 73 41 4E 20 41 63 74 69 76 61 74 65 45 76 61 6C 75 61 74 69 6F 6E 47 72 6F 75 70 20 31 20 31 03

Table 564: Example 2: sAN ActivateEvaluation – Successfully deactivated evaluation group 1, successfully activated evaluation group 2, failed activation of evaluation group 3

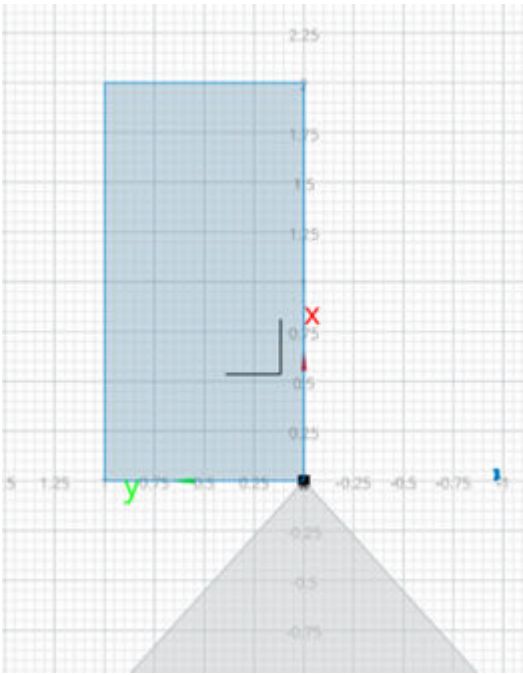
CoLa A	<STX>sAN{SPC}ActivateEvaluation{SPC}3{SPC}1{SPC}1{SPC}0<ETX>
CoLa B	02 73 41 4E 20 41 63 74 69 76 61 74 65 45 76 61 6C 75 61 74 69 6F 6E 47 72 6F 75 70 20 31 20 31 20 30 03

13.6.1.4.11.1.2 Read evaluation group type [sRN EvaluationGroupType]

Array with the type of each group, either perpendicular distance or field evaluation.

Table 569: Telegram structure: sMN SetFieldEvaluationContour

Telegram structure: sMN SetFieldEvaluationContour (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Method	String	3		sMN	73 4D 4E
Command	Set the region of interest of an object detection evaluation	String	25		SetFieldEvaluationContour	53 65 74 46 69 65 6C 64 45 76 61 6C 75 61 74 69 6F 6E 43 6F 6E 74 6F 75 72
Evaluation ID	Specific ID of the evaluation that is to be changed	UInt_16	2		+1d ... +48d (1h ... 30h)	00 01 ... 00 30
Amount of polygon vertices	E. g. triangle = 3, square = 4	Array	3 ... 100		+3d ... +800d (3h ... 320h)	00 03... 03 20
Polygon vertex	Coordinates of first polygon vertex in mm	Dint_32	4	X coordinate:	-120,000d ... +120,000d (FFFE2B40h ... 1D4C0h)	FF FE 2B 40 ... 00 01 D4 C0
		Dint_32	4	Y coordinate:	-120,000d ... +120,000d (FFFE2B40h ... 1D4C0h)	FF FE 2B 40 ... 00 01 D4 C0
Polygon vertex	Coordinates of second polygon vertex in mm	Dint_32	4	X coordinate:	-120,000d ... +120,000d (FFFE2B40h ... 1D4C0h)	FF FE 2B 40 ... 00 01 D4 C0
		Dint_32	4	Y coordinate:	-120,000d ... +120,000d (FFFE2B40h ... 1D4C0h)	FF FE 2B 40 ... 00 01 D4 C0
...						
Polygon vertex	Coordinates of last polygon vertex in mm	Dint_32	4	X coordinate:	-120,000d ... +120,000d (FFFE2B40h ... 1D4C0h)	FF FE 2B 40 ... 00 01 D4 C0
		Dint_32	4	Y coordinate:	-120,000d ... +120,000d (FFFE2B40h ... 1D4C0h)	FF FE 2B 40 ... 00 01 D4 C0
Lower Z limit	Lower bound of extrusion in Z direction	Dint_32	4	Must be: (Other values than 0 will lead to an error)	0	00 00 00 00
Upper Z limit	Upper bound of extrusion in Z direction	Dint_32	4	Must be: (Other values than 0 will lead to an error)	0	00 00 00 00

**Table 570:** Example: sMN SetFieldEvaluationContour

CoLa A	<STX>sMN{SPC}SetFieldEvaluationContour{SPC}1{SPC}4{SPC}+2000{SPC}+1000{SPC}+0{SPC}+1000{SPC}+0{SPC}+0{SPC}+2000{SPC}+0{SPC}+0{SPC}+0<ETX>	
	<STX>sMN SetFieldEvaluationContour 1 4 +2000 +1000 +0 +1000 +0 +0 +2000 +0 +0 +0<ETX>	
	sMN SetFieldEvaluationContour 1 4 +2000 +1000 +0 +1000 +0 +0 +2000 +0 +0 +0	
	02 73 4D 4E 20 53 65 74 46 69 65 6C 64 45 76 61 6C 75 61 74 69 6F 6E 43 6F 6E 74 6F 75 72 20 31 20 34 20 2B 32 30 30 30 20 2B 31 30 30 30 20 2B 30 20 2B 31 30 30 30 20 2B 30 20 2B 30 20 2B 32 30 30 30 20 2B 30 20 2B 30 03	
CoLa B	02 02 02 02 00 00 00 4A 73 4D 4E 20 53 65 74 46 69 65 6C 64 45 76 61 6C 75 61 74 69 6F 6E 43 6F 6E 74 6F 75 72 20 00 01 00 04 00 00 07 D0 00 00 03 E8 00 00 00 00 00 03 E8 00 00 00 00 00 00 00 00 00 07 D0 00 00 00 00 00 00 00 00 00 00 00 00 1D	
	73 4D 4E 20 53 65 74 46 69 65 6C 64 45 76 61 6C 75 61 74 69 6F 6E 43 6F 6E 74 6F 75 72 20 00 01 00 04 00 00 07 D0 00 00 03 E8 00 00 00 00 00 03 E8 00 00 00 00 00 00 00 00 00 07 D0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	

Table 571: Telegram structure: sAN SetFieldEvaluationContour

Telegram structure: sAN SetFieldEvaluationContour						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sAN	73 52 41
Command	Set the region of interest of an object detection evaluation	String	25		SetFieldEvaluationContour	53 65 74 46 69 65 6C 64 45 76 61 6C 75 61 74 69 6F 6E 43 6F 6E 74 6F 75 72
State of evaluation conversion	-	Enum_8	1	Conversion successful: Invalid Evaluation: Invalid Polygon: Invalid Z limit:	0 1 2 3	00 01 02 03

Table 572: Example: sAN SetFieldEvaluationContour – Evaluation conversion successful

CoLa A	<STX>sAN{SPC}SetFieldEvaluationContour{SPC}0<ETX>	
	02 73 41 4E 20 53 65 74 46 69 65 6C 64 45 76 61 6C 75 61 74 69 6F 6E 43 6F 6E 74 6F 75 72 20 30 03	

CoLa B 02 02 02 02 00 00 00 1F 73 41 4E 20 53 65 74 46 69 65 6C 64 45 76 61 6C 75 61 74 69 6F 6E 43 6F 6E 74 6F 75 72 20 00 14

13.6.1.4.11.1.4 Get field evaluation contour [sMN GetFieldEvaluationContour]

This telegram can be used to read the contour/ polygon of an already defined field.

Table 573: Telegram structure: sMN GetFieldEvaluationContour

Telegram structure: sMN GetFieldEvaluationContour (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Method	String	3		sMN	73 4D 4E
Command	Get the region of interest of an object detection evaluation	String	25		GetFieldEvaluationContour	47 65 74 46 69 65 6C 64 45 76 61 6C 75 61 74 69 6F 6E 43 6F 6E 74 6F 75 72
Evaluation ID	Specific ID of the evaluation that is to be changed	UInt_16	2		+1d ... +48d (1h ... 30h)	00 01 ... 00 30

Table 574: Example: sMN GetFieldEvaluationContour

CoLa A	<STX>sMN{SPC}GetFieldEvaluationContour{SPC}1<ETX>					
	<STX>sMN GetFieldEvaluationContour 1<ETX>					
	sMN GetFieldEvaluationContour 1					
CoLa B	02 73 4D 4E 20 47 65 74 46 69 65 6C 64 45 76 61 6C 75 61 74 69 6F 6E 43 6F 6E 74 6F 75 72 20 31 03					
	02 02 02 02 00 00 00 20 73 4D 4E 20 47 65 74 46 69 65 6C 64 45 76 61 6C 75 61 74 69 6F 6E 43 6F 6E 74 6F 75 72 20 01 0D					
	73 4D 4E 20 47 65 74 46 69 65 6C 64 45 76 61 6C 75 61 74 69 6F 6E 43 6F 6E 74 6F 75 72 20 01					

Table 575: Telegram structure: sAN GetFieldEvaluationContour

Telegram structure: sAN GetFieldEvaluationContour						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sAN	73 52 41
Command	Get the region of interest of an object detection evaluation	String	25		GetFieldEvaluationContour	47 65 74 46 69 65 6C 64 45 76 61 6C 75 61 74 69 6F 6E 43 6F 6E 74 6F 75 72
Length of array		UInt_8	1	Always:	1	00 01
Evaluation ID	Specific ID of the evaluation that is to be changed	UInt_16	2		1h ... 30h	00 01 ... 00 30
Amount of polygon vertices	E. g. triangle = 3, square = 4	Array	3 ... 100		3h ... 320h	00 03... 03 20
Polygon vertex	Coordinates of first polygon vertex in mm	Dint_32	4	X coordinate:	FFFE2B40h ... 1D4C0h	FF FE 2B 40 ... 00 01 D4 C0
		Dint_32	4	Y coordinate:	FFFE2B40h ... 1D4C0h	FF FE 2B 40 ... 00 01 D4 C0

Telegram structure: sAN GetFieldEvaluationContour						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Polygon vertex	Coordinates of second polygon vertex in mm	Dint_32	4	X coordinate:	FFFE2B40h ... 1D4C0h	FF FE 2B 40 ... 00 01 D4 C0
		Dint_32	4	Y coordinate:	FFFE2B40h ... 1D4C0h	FF FE 2B 40 ... 00 01 D4 C0
...						
Polygon vertex	Coordinates of last polygon vertex in mm	Dint_32	4	X coordinate:	FFFE2B40h ... 1D4C0h	FF FE 2B 40 ... 00 01 D4 C0
		Dint_32	4	Y coordinate:	FFFE2B40h ... 1D4C0h	FF FE 2B 40 ... 00 01 D4 C0
Lower Z limit	Lower bound of extrusion in Z direction	Dint_32	4	Always:	0	00 00 00 00
Upper Z limit	Upper bound of extrusion in Z direction	Dint_32	4	Always:	0	00 00 00 00

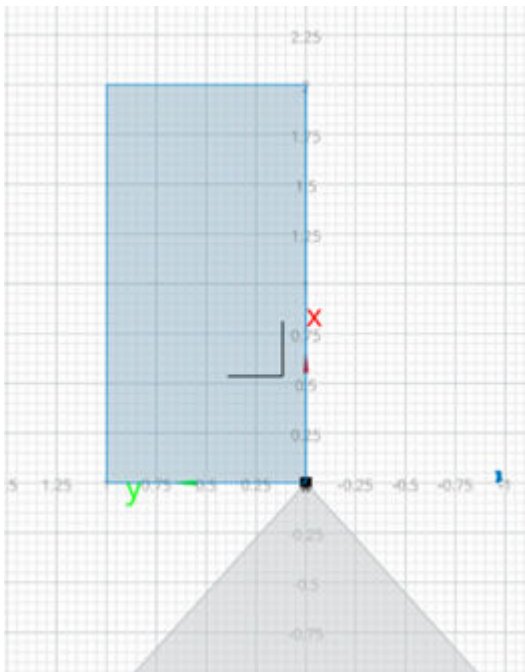


Table 576: Example: sAN GetFieldEvaluationContour

CoLa A	<STX>sAN{SPC}GetFieldEvaluationContour{SPC}{i}{SPC}4{SPC}7D0{SPC}3E8{SPC}0{SPC}3E8{SPC}0{SPC}0{SPC}7D0{SPC}0{SPC}0{SPC}0<ETX>
	02 73 41 4E 20 47 65 74 46 69 65 6C 64 45 76 61 6C 75 61 74 69 6F 6E 43 6F 6E 74 6F 75 72 20 31 20 31 20 34 20 37 44 30 20 33 45 38 20 30 20 33 45 38 20 30 20 30 20 37 44 30 20 30 20 30 20 30 03
CoLa B	02 02 02 02 00 00 00 4C 73 41 4E 20 47 65 74 46 69 65 6C 64 45 76 61 6C 75 61 74 69 6F 6E 43 6F 6E 74 6F 75 72 20 00 01 00 01 00 04 00 00 07 D0 00 00 03 E8 00 00 00 00 00 00 03 E8 00 00 00 00 00 00 00 00 00 00 07 D0 00 00 00 00 00 00 00 00 00 04

13.6.1.4.11.5 Read field evaluation result [sRN FieldEvaluationResult]

This telegram returns the status information of all evaluations.

Possible status codes:

- Not configured
 - This evaluation has not been configured.
- Inactive

- This evaluation has been configured but is currently not evaluated.
- Free
 - The field is not infringed.
- Infringed
 - The field is infringed.
- Detecting infringed
 - An infringement has been detected in the field but the response time has not been reached yet.
- Detecting free
 - The field has been detected as free but the restart time has not been reached yet.

Table 577: Telegram structure: sRN FieldEvaluationResult

Telegram structure: sRN FieldEvaluationResult						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	Info on field evaluation result	String	21		FieldEvaluation-Result	46 69 65 6C 64 45 76 61 6C 75 61 74 69 6F 6E 52 65 73 75 6C 74

Table 578: Example: sRN FieldEvaluationResult

CoLa A	<STX>sRN{SPC}FieldEvaluationResult<ETX>					
	<STX>sRN FieldEvaluationResult<ETX>					
	sRN FieldEvaluationResult					
CoLa B	02 73 52 4E 20 46 69 65 6C 64 45 76 61 6C 75 61 74 69 6F 6E 52 65 73 75 6C 74 03					
	02 02 02 02 00 00 00 19 73 52 4E 20 46 69 65 6C 64 45 76 61 6C 75 61 74 69 6F 6E 52 65 73 75 6C 74 12					
73 52 4E 20 46 69 65 6C 64 45 76 61 6C 75 61 74 69 6F 6E 52 65 73 75 6C 74						

Table 579: Telegram structure: sRA FieldEvaluationResult

Telegram structure: sRA FieldEvaluationResult						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	Info on field evaluation result	String	21		FieldEvaluation-Result	46 69 65 6C 64 45 76 61 6C 75 61 74 69 6F 6E 52 65 73 75 6C 74
Version	Version number	UInt_16	2		0 ... FFFFh	00 ... FF FF
Time-stamp	Time of the last field infringement. Time base is the sensor system time in milliseconds since January 1, 1970, 00:00 (UTC). If a time server is being used, the configured system time is used.	ULInt_64	8		0 ... FFFFFFFFh	00 00 00 00 00 00 00 00 ... FF FF FF FF FF FF FF FF

Table 580: Example: sRA FieldEvaluationResult - field 1 is infringed, field 2 is free

13.6.1.4.11.1.6 Receive field evaluation result by event [sEN FieldEvaluationResult]

Possible status codes:

- Not configured
 - This evaluation has not been configured.
- Inactive
 - This evaluation has been configured but is currently not evaluated.

- Free
 - The field is not infringed.
- Infringed
 - The field is infringed.
- Detecting infringed
 - An infringement has been detected in the field but the response time has not been reached yet.
- Detecting free
 - The field has been detected as free but the restart time has not been reached yet.

Table 581: Telegram structure: sEN FieldEvaluationResult

Telegram structure: sEN FieldEvaluationResult						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Event	String	3		sEN	73 45 4E
Command	Info on field evaluation result	String	21		FieldEvaluation-Result	46 69 65 6C 64 45 76 61 6C 75 61 74 69 6F 6E 52 65 73 75 6C 74
Reporting	Start/ stop	Enum_8	1	Stop: Start:	0 1	00 01

Table 582: Example: sEN FieldEvaluationResult

CoLa A	<STX>sEN{SPC}FieldEvaluationResult{SPC}1<ETX>	
	<STX>sEN FieldEvaluationResult 1<ETX>	
	sEN FieldEvaluationResult 1	
	02 73 45 4E 20 46 69 65 6C 64 45 76 61 6C 75 61 74 69 6F 6E 52 65 73 75 6C 74 20 31 03	
CoLa B	02 02 02 02 00 00 00 1B 73 45 4E 20 46 69 65 6C 64 45 76 61 6C 75 61 74 69 6F 6E 52 65 73 75 6C 74 20 01 24	
	73 45 4E 20 46 69 65 6C 64 45 76 61 6C 75 61 74 69 6F 6E 52 65 73 75 6C 74 20 01	

Table 583: Telegram structure: sSN FieldEvaluationResult

Telegram structure: sSN FieldEvaluationResult						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sSN	73 53 4E
Command	Info on field evaluation result	String	21		FieldEvaluation-Result	46 69 65 6C 64 45 76 61 6C 75 61 74 69 6F 6E 52 65 73 75 6C 74
Version	Version number	UInt_16	2		0 ... FFFFh	00 ... FF FF
Time-stamp	Time of the latest field infringement. Time base is the sensor system time in milliseconds since January 1, 1970, 00:00 (UTC). If a time server is being used, the configured system time is used.	ULInt_64	8		0 ... FFFFFFFFh	00 00 00 00 00 00 00 00 ... FF FF FF FF FF FF

Table 584: Example: sSN FieldEvaluationResult - field 1 is infringed, field 2 is free

Start teach-in of the environment.

Table 585: Telegram structure: sMN StartTeachIn

Telegram structure: sMN StartTeachIn (User level 'Authorized Client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Method invocation	String	3		sMN	73 4D 4E
Method name	StartTeachIn	String	12		StartTeachIn	53 74 61 72 74 54 65 61 63 68 49 6E

Table 586: Example: sMN StartTeachIn

CoLa A	<STX>sMN[SPC]StartTeachIn<ETX>					
	<STX>sMN StartTeachIn<ETX>					
	sMN StartTeachIn					
	02 73 4D 4E 20 53 74 61 72 74 54 65 61 63 68 49 6E 03					
CoLa B	02 02 02 02 00 00 00 10 73 4D 4E 20 53 74 61 72 74 54 65 61 63 68 49 6E 6C					
	73 4D 4E 20 53 74 61 72 74 54 65 61 63 68 49 6E					

Table 587: Telegram structure: sAN StartTeachIn

Telegram structure: sAN StartTeachIn						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Method answer	String	3		sAN	73 41 4E
Method name	Echo of method name	String	12		StartTeachIn	53 74 61 72 74 54 65 61 63 68 49 6E

Table 588: Example: sAN StartTeachIn

CoLa A	<STX>sAN[SPC]StartTeachIn<ETX>					
	02 73 41 4E 20 53 74 61 72 74 54 65 61 63 68 49 6E 03					
CoLa B	02 02 02 02 00 00 00 11 73 41 4E 20 53 74 61 72 74 54 65 61 63 68 49 6E 20 40					

13.6.1.4.11.1.8 Stop Teach-In [sMN StopTeachIn]

Stop teach-in of the environment.

Table 589: Telegram structure: sMN StopTeachIn

Telegram structure: sMN StopTeachIn (User level 'Authorized Client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Method invocation	String	3		sMN	73 4D 4E
Method name	StopTeachIn	String	11		StopTeachIn	53 74 6F 70 54 65 61 63 68 49 6E

Table 590: Example: sMN StopTeachIn

CoLa A	<STX>sMN[SPC]StopTeachIn<ETX>					
	<STX>sMN StopTeachIn<ETX>					
	sMN StopTeachIn					
	02 73 4D 4E 20 53 74 6F 70 54 65 61 63 68 49 6E 03					

	02 02 02 02 00 00 00 0F 73 4D 4E 20 53 74 6F 70 54 65 61 63 68 49 6E 14
CoLa B	73 4D 4E 20 53 74 6F 70 54 65 61 63 68 49 6E

Table 591: Telegram structure: sAN StopTeachIn

Telegram structure: sAN StopTeachIn						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Method answer	String	3		sAN	73 41 4E
Method name	Echo of method name	String	11		StopTeachIn	53 74 6F 70 54 65 61 63 68 49 6E

Table 592: Example: sAN StopTeachIn

CoLa A	<STX>sAN{SPC}StopTeachIn<ETX>
	02 73 41 4E 20 53 74 6F 70 54 65 61 63 68 49 6E 03
CoLa B	02 02 02 02 00 00 00 10 73 41 4E 20 53 74 6F 70 54 65 61 63 68 49 6E 20 38

13.6.1.4.11.2 Perpendicular Distance

13.6.1.4.11.2.1 Request perpendicular distance once [sRN perpendicularDistanceResult]

This telegram can be used to request the perpendicular distance. The shortest and furthest measuring point in a range to a reference plane will be transmitted. A field and a reference plane must be defined in the user interface under **Application > Perpendicular distance measurement**.

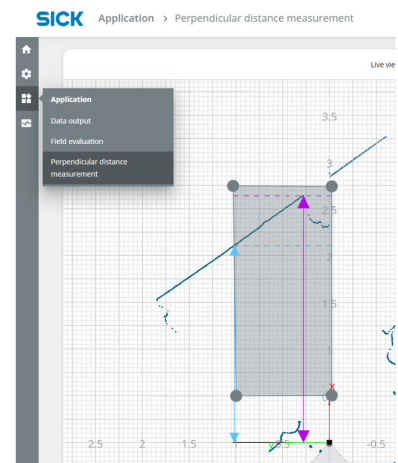


Table 593: Telegram structure: sRN perpendicularDistanceResult

Telegram structure: sRN perpendicularDistanceResult (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read	String	3		sRN	73 52 4E
Command	Perpendicular distance values	String	27		perpendicular-DistanceResult	70 65 72 70 65 6E 64 69 63 75 6C 61 72 44 69 73 74 61 6E 63 65 52 65 73 75 6C 74

Table 594: Example: sRN perpendicularDistanceResult

CoLa A	<STX>sRN{SPC}perpendicularDistanceResult<ETX>	
	<STX>sRN perpendicularDistanceResult<ETX>	
	sRN perpendicularDistanceResult	
	02 73 52 4E 20 70 65 72 70 65 6E 64 69 63 75 6C 61 72 44 69 73 74 61 6E 63 65 52 65 73 75 6C 74 03	
CoLa B	02 02 02 02 00 00 00 00 1F 73 52 4E 20 70 65 72 70 65 6E 64 69 63 75 6C 61 72 44 69 73 74 61 6E 63 65 52 65 73 75 6C 74 3D	
	73 52 4E 20 70 65 72 70 65 6E 64 69 63 75 6C 61 72 44 69 73 74 61 6E 63 65 52 65 73 75 6C 74	

Table 595: Telegram structure: sRA perpendicularDistanceResult

Telegram structure: sRA perpendicularDistanceResult					  	
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	Perpendicular distance values	String	27		perpendicular-DistanceResult	70 65 72 70 65 6E 64 69 63 75 6C 61 72 44 69 73 74 61 6E 63 65 52 65 73 75 6C 74
Time-stamp	Time of the latest object detection in the defined fields. Time base is the sensor system time in milliseconds since January 1, 1970, 00:00 (UTC). If a time server is being used, the configured system time is used.	Uint_32	8		0 ... FFFFFFFFh	00 00 00 00 00 00 00 00... FF FF FF FF FF FF FF FF
Amount of channels	Amount of perpendicular distance evaluation channels/fields	Enum_8	2		1 ... 20d (1 ... 14h)	00 01 ... 00 14

Telegram structure: sRA perpendicularDistanceResult						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Evaluation ID X	ID of perpendicular distance field	Enum_8	2		1 ... 20d (1 ... 14h)	00 01 ... 00 14
	Minimum distance [mm]	Int_32	4		-120000d ... +120000d (FFFE2B3F ... 1D4C0h)	FF FE 2B 3F ... 00 01 D4 C0
	Maximum distance [mm]	Int_32	4		-120000d ... +120000d (FFFE2B3F ... 1D4C0h)	FF FE 2B 3F ... 00 01 D4 C0
	X minimum distance [mm]	Int_32	4		-120000d ... +120000d (FFFE2B3F ... 1D4C0h)	FF FE 2B 3F ... 00 01 D4 C0
	Y minimum distance [mm]	Int_32	4		-120000d ... +120000d (FFFE2B3F ... 1D4C0h)	FF FE 2B 3F ... 00 01 D4 C0
	Reserved	Int_32	4	Always:	0	00 00 00 00
	X maximum distance [mm]	Int_32	4		-120000d ... +120000d (FFFE2B3F ... 1D4C0h)	FF FE 2B 3F ... 00 01 D4 C0
	Y maximum distance [mm]	Int_32	4		-120000d ... +120000d (FFFE2B3F ... 1D4C0h)	FF FE 2B 3F ... 00 01 D4 C0
	Reserved	Int_32	4	Always:	0	00 00 00 00
Evaluation ID ...	...	...	...	...	...	...

Table 596: Example: sRA perpendicularDistanceResult - 1 configured field

CoLa A	<STX>sRA{SPC}perpendicularDistanceResult{SPC}138D305E8{SPC}1{SPC}1{SPC}9D{SPC}7EA{SPC}9D{SPC}FFFFFFC67{SPC}0{SPC}7EA{SPC}FFFFFFD70{SPC}0<ETX>
	02 73 52 41 20 70 65 72 70 65 6E 64 69 63 75 6C 61 72 44 69 73 74 61 6E 63 65 52 65 73 75 6C 74 20 31 33 38 44 33 30 35 45 38 20 31 20 31 20 39 44 20 37 45 41 20 39 44 20 46 46 46 46 46 43 36 37 20 30 20 37 45 41 20 46 46 46 46 46 44 37 30 20 30 03
CoLa B	02 02 02 02 00 00 00 4C 73 52 41 20 70 65 72 70 65 6E 64 69 63 75 6C 61 72 44 69 73 74 61 6E 63 65 52 65 73 75 6C 74 20 00 00 00 01 31 80 92 88 00 01 00 01 00 00 00 9F 00 00 07 EB 00 00 00 9F FF FF FC 8C 00 00 00 00 00 00 07 EB FF FF FD 70 00 00 00 00 45

13.6.1.4.11.2.2 Request perpendicular distance continuously on event [sEN perpendicularDistanceResult]

This telegram can be used to request the perpendicular distance continuously on event. The shortest and furthest measuring point in a range to a reference plane will be transmitted. A field and a reference plane must be defined in the user interface under **Application > Perpendicular distance measurement**.

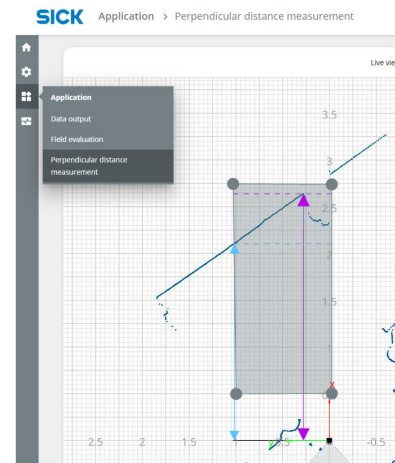


Table 597: Telegram structure: sEN perpendicularDistanceResult

Telegram structure: sEN perpendicularDistanceResult (User level 'Authorized client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Event	String	3		sEN	73 45 4E
Command	Perpendicular distance values	String	27		perpendicularDistanceResult	70 65 72 70 65 6E 64 69 63 75 6C 61 72 44 69 73 74 61 6E 63 65 52 65 73 75 6C 74
Start/ Stop	-	Enum_8	1	Stop: Start:	0 1	00 01

Table 598: Example: sEN perpendicularDistanceResult

CoLa A	<STX>sEN[SPC]perpendicularDistanceResult[SPC]1<ETX>		
	<STX>sEN perpendicularDistanceResult 1<ETX>		
	sEN perpendicularDistanceResult 1		
CoLa B	02 73 45 4E 20 70 65 72 70 65 6E 64 69 63 75 6C 61 72 44 69 73 74 61 6E 63 65 52 65 73 75 6C 74 20 31 03		
	02 02 02 02 00 00 00 21 73 45 4E 20 70 65 72 70 65 6E 64 69 63 75 6C 61 72 44 69 73 74 61 6E 63 65 52 65 73 75 6C 74 20 01 0B		
	73 45 4E 20 70 65 72 70 65 6E 64 69 63 75 6C 61 72 44 69 73 74 61 6E 63 65 52 65 73 75 6C 74 20 01		

Table 599: Telegram structure: sSN perpendicularDistanceResult

Telegram structure: sSN perpendicularDistanceResult						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sSN	73 53 4E
Command	Perpendicular distance values	String	27		perpendicularDistanceResult	70 65 72 70 65 6E 64 69 63 75 6C 61 72 44 69 73 74 61 6E 63 65 52 65 73 75 6C 74

Telegram structure: sSN perpendicularDistanceResult						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Time-stamp	Time of the latest object detection in the defined fields. Time base is the sensor system time in milliseconds since January 1, 1970, 00:00 (UTC). If a time server is being used, the configured system time is used.	Uint_32	8		0 ... FFFFFFFFh	00 00 00 00 00 00 00 00 00 00 00... FF FF FF FF FF FF FF FF
Amount of channels	Amount of perpendicular distance evaluation channels/fields	Enum_8	2		1 ... 20d (1 ... 14h)	00 01 ... 00 14
Evaluation ID X	ID of perpendicular distance field	Enum_8	2		1 ... 20d (1 ... 14h)	00 01 ... 00 14
	Minimum distance [mm]	Int_32	4		-120000d ... +120000d (FFFE2B3F ... 1D4C0h)	FF FE 2B 3F ... 00 01 D4 C0
	Maximum distance [mm]	Int_32	4		-120000d ... +120000d (FFFE2B3F ... 1D4C0h)	FF FE 2B 3F ... 00 01 D4 C0
	X minimum distance [mm]	Int_32	4		-120000d ... +120000d (FFFE2B3F ... 1D4C0h)	FF FE 2B 3F ... 00 01 D4 C0
	Y minimum distance [mm]	Int_32	4		-120000d ... +120000d (FFFE2B3F ... 1D4C0h)	FF FE 2B 3F ... 00 01 D4 C0
	Reserved	Int_32	4	Always:	0	00 00 00 00
	X maximum distance [mm]	Int_32	4		-120000d ... +120000d (FFFE2B3F ... 1D4C0h)	FF FE 2B 3F ... 00 01 D4 C0
	Y maximum distance [mm]	Int_32	4		-120000d ... +120000d (FFFE2B3F ... 1D4C0h)	FF FE 2B 3F ... 00 01 D4 C0
	Reserved	Int_32	4	Always:	0	00 00 00 00
Evaluation ID ...	...	...	...	...	...	...

Table 600: Example: sSN perpendicularDistanceResult - 1 configured field

CoLa A	<STX>sSN{SPC}perpendicularDistanceResult{SPC}D5BD2DA8{SPC}1{SPC}1{SPC}174{SPC}7CE{SPC}174{SPC}FFFFFFC{SPC}0{SPC}7CE{SPC}FFFFFFD13{SPC}0<ETX>
	02 73 53 4E 20 70 65 72 70 65 6E 64 69 63 75 6C 61 72 44 69 73 74 61 6E 63 65 52 65 73 75 6C 74 20 44 35 42 44 32 44 41 38 20 31 20 31 20 31 37 34 20 37 43 45 20 31 37 34 20 46 46 46 46 46 46 43 20 30 20 37 43 45 20 46 46 46 46 44 31 33 20 30 03
CoLa B	02 02 02 02 00 00 00 4C 73 53 4E 20 70 65 72 70 65 6E 64 69 63 75 6C 61 72 44 69 73 74 61 6E 63 65 52 65 73 75 6C 74 20 00 00 00 00 F9 B6 D5 88 00 01 00 00 00 9D 00 00 07 D0 00 00 00 9D FF FF FC D0 00 00 00 00 00 00 07 D0 FF FF FD 1F 00 00 00 00 C0

13.6.1.4.11.3

Detection History

13.6.1.4.11.3.1 Enable Detection History Logging [sWN EnableDetectionHistory]

Enables or disables the detection history logging.

Table 601: Telegram structure: sWN EnableDetectionHistory

Telegram structure: sWN EnableDetectionHistory						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type		String	3		sWN	73 57 4E
Command	EnableDetectionHistory	String	22		EnableDetectionHistory	45 6E 61 62 6C 65 44 65 74 65 63 74 69 6F 6E 48 69 73 74 6F 72 79
Detection history logging	Enable/ Disable	Bool_1	1	Disable: Enable:	0 1	00 01

Table 602: Example: sWN EnableDetectionHistory

CoLa A	<STX>sWN{SPC}EnableDetectionHistory{SPC}1<ETX>					
	<STX>sWN EnableDetectionHistory 1<ETX>					
	sWN EnableDetectionHistory 1					
	02 73 57 4E 20 45 6E 61 62 6C 65 44 65 74 65 63 74 69 6F 6E 48 69 73 74 6F 72 79 20 31 03					
CoLa B	02 02 02 02 00 00 00 1C 73 52 4E 20 45 6E 61 62 6C 65 44 65 74 65 63 74 69 6F 6E 48 69 73 74 6F 72 79 20 01 47					
	73 52 4E 20 45 6E 61 62 6C 65 44 65 74 65 63 74 69 6F 6E 48 69 73 74 6F 72 79 20 01					

Table 603: Telegram structure: sWA EnableDetectionHistory

Telegram structure: sWA EnableDetectionHistory						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sWA	73 57 41
Command	EnableDetectionHistory	String	22		EnableDetectionHistory	45 6E 61 62 6C 65 44 65 74 65 63 74 69 6F 6E 48 69 73 74 6F 72 79

Table 604: Example: sWA EnableDetectionHistory

CoLa A	<STX>sWA{SPC}EnableDetectionHistory<ETX>					
	02 73 57 41 20 45 6E 61 62 6C 65 44 65 74 65 63 74 69 6F 6E 48 69 73 74 6F 72 79 03					
CoLa B	02 02 02 02 00 00 00 1A 73 52 41 20 45 6E 61 62 6C 65 44 65 74 65 63 74 69 6F 6E 48 69 73 74 6F 72 79 69					

13.6.1.4.11.3.2 Read Detection History Logging state [sRN EnableDetectionHistory]

Reads whether the detection history logging is enabled or disabled.

Table 605: Telegram structure: sRN EnableDetectionHistory

Telegram structure: sRN EnableDetectionHistory						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type		String	3		sRN	73 52 4E

Telegram structure: sRN EnableDetectionHistory						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command	EnableDetectionHistory	String	22		EnableDetectionHistory	45 6E 61 62 6C 65 44 65 74 65 63 74 69 6F 6E 48 69 73 74 6F 72 79

Table 606: Example: sRN EnableDetectionHistory

CoLa A	<STX>sRN{SPC}EnableDetectionHistory<ETX>					
	<STX>sRN EnableDetectionHistory<ETX>					
	sRN EnableDetectionHistory					
	02 73 52 4E 20 45 6E 61 62 6C 65 44 65 74 65 63 74 69 6F 6E 48 69 73 74 6F 72 79 03					
CoLa B	02 02 02 02 00 00 00 00 1A 73 52 4E 20 45 6E 61 62 6C 65 44 65 74 65 63 74 69 6F 6E 48 69 73 74 6F 72 79 63					
	73 52 4E 20 45 6E 61 62 6C 65 44 65 74 65 63 74 69 6F 6E 48 69 73 74 6F 72 79					

Table 607: Telegram structure: sRA EnableDetectionHistory

Telegram structure: sRA EnableDetectionHistory						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Answer	String	3		sRA	73 52 41
Command	EnableDetectionHistory	String	22		EnableDetectionHistory	45 6E 61 62 6C 65 44 65 74 65 63 74 69 6F 6E 48 69 73 74 6F 72 79
Detection history logging	Enabled/ Disabled	Bool_1	1	Disabled: Enabled:	0 1	00 01

Table 608: Example: sRA EnableDetectionHistory

CoLa A	<STX>sRA{SPC}EnableDetectionHistory{SPC}1<ETX>					
	02 73 52 41 20 45 6E 61 62 6C 65 44 65 74 65 63 74 69 6F 6E 48 69 73 74 6F 72 79 20 31 03					
CoLa B	02 02 02 02 00 00 00 00 1C 73 52 41 20 45 6E 61 62 6C 65 44 65 74 65 63 74 69 6F 6E 48 69 73 74 6F 72 79 20 01 4D					

13.6.1.4.11.3.3 Reset Detection History Logging [sMN ResetDetectionHistory]

Deletes all entries in the field evaluation logging list.

Table 609: Telegram structure: sMN ResetDetectionHistory

Telegram structure: sMN ResetDetectionHistory (User level 'Authorized Client' required)						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Method invocation	String	3		sMN	73 4D 4E
Method name	ResetDetectionHistory	String	21		ResetDetectionHistory	52 65 73 65 74 44 65 74 65 63 74 69 6F 6E 48 69 73 74 6F 72 79

Table 610: Example: sMN ResetDetectionHistory

CoLa A	<STX>sMN{SPC}ResetDetectionHistory<ETX>	
	<STX>sMN ResetDetectionHistory<ETX>	
	sMN ResetDetectionHistory	
	02 73 4D 4E 20 52 65 73 65 74 44 65 74 65 63 74 69 6F 6E 48 69 73 74 6F 72 79 03	
CoLa B	02 02 02 02 00 00 00 19 73 4D 4E 20 52 65 73 65 74 44 65 74 65 63 74 69 6F 6E 48 69 73 74 6F 72 79 08	
	73 4D 4E 20 52 65 73 65 74 44 65 74 65 63 74 69 6F 6E 48 69 73 74 6F 72 79	

Table 611: Telegram structure: sAN ResetDetectionHistory

Telegram structure: sAN ResetDetectionHistory						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Method answer	String	3		sAN	73 41 4E
Method name	Echo of method name	String	21		ResetDetectionHistory	52 65 73 65 74 44 65 74 65 63 74 69 6F 6E 48 69 73 74 6F 72 79

Table 612: Example: sAN ResetDetectionHistory

CoLa A	<STX>sAN{SPC}ResetDetectionHistory<ETX>
	02 73 41 4E 20 52 65 73 65 74 44 65 74 65 63 74 69 6F 6E 48 69 73 74 6F 72 79 03
CoLa B	02 02 02 02 00 00 00 19 73 41 4E 20 52 65 73 65 74 44 65 74 65 63 74 69 6F 6E 48 69 73 74 6F 72 79 04

13.6.1.5 Diagnostics

13.6.1.5.1 Read laser type [sRN laserType]

Reads the laser type of the device.

Table 613: Telegram structure: sRN laserType

Telegram structure: sRN laserType						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read request	String	3		sRN	73 52 4E
Variable name	laserType	String	9		laserType	6C 61 73 65 72 54 79 70 65

Table 614: Example: sRN laserType

CoLa A	<STX>sRN{SPC}laserType<ETX>	
	<STX>sRN laserType<ETX>	
	sRN laserType	
	02 73 52 4E 20 6C 61 73 65 72 54 79 70 65 03	
CoLa B	02 02 02 02 00 00 00 0D 73 52 4E 20 6C 61 73 65 72 54 79 70 65 1E	
	73 52 4E 20 6C 61 73 65 72 54 79 70 65	

Table 615: Telegram structure: sRA laserType

Telegram structure: sRA laserType						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read answer	String	3		sRA	73 52 41
Variable name	laserType	String	9		laserType	6C 61 73 65 72 54 79 70 65
Value	Laser type identifier	UInt_8	1		0	00

Table 616: Example: sRA laserType

CoLa A	<STX>sRA[SPC]laserType[SPC]0<ETX> 02 73 52 41 20 6C 61 73 65 72 54 79 70 65 20 30 03					
CoLa B	02 02 02 02 00 00 00 0F 73 52 41 20 6C 61 73 65 72 54 79 70 65 20 00 31					

13.6.1.5.2 Read the SiPM type of the device [sRN sipmType]

Return the SiPM type of the device.

Table 617: Telegram structure: sRN sipmType

Telegram structure: sRN sipmType						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read request	String	3		sRN	73 52 4E
Variable name	sipmType	String	8		sipmType	73 69 70 6D 54 79 70 65

Table 618: Example: sRN sipmType

CoLa A	<STX>sRN[SPC]sipmType<ETX>					
	<STX>sRN sipmType<ETX>					
	sRN sipmType 02 73 52 4E 20 73 69 70 6D 54 79 70 65 03					
CoLa B	02 02 02 02 00 00 00 0C 73 52 4E 20 73 69 70 6D 54 79 70 65 70					
	73 52 4E 20 73 69 70 6D 54 79 70 65					

Table 619: Telegram structure: sRA sipmType

Telegram structure: sRA sipmType						
Telegram part	Description	Variable	Length	Additional details	Values CoLa A (ASCII)	Values CoLa B (Binary)
Command type	Read answer	String	3		sRA	73 52 41
Variable name	sipmType	String	8		sipmType	73 69 70 6D 54 79 70 65
Value	SiPM sensor type identifier	UInt_8	1		0	00

Table 620: Example: sRA sipmType

CoLa A	<STX>sRA[SPC]sipmType[SPC]0<ETX> 02 73 52 41 20 73 69 70 6D 54 79 70 65 20 30 03					
CoLa B	02 02 02 02 00 00 00 0E 73 52 41 20 73 69 70 6D 54 79 70 65 20 00 5F					

13.6.1.5.3 SOPAS error codes

sFA ErrorCode

Table 621: SOPAS error codes

Telegram structure: sFA ErrorCode			
Error code	Description	Dec.	Hex.
Sopas_Ok	No error	0	0
Sopas_Error_METHODIN_ACCESSDENIED	Wrong userlevel, access to method not allowed	1	1
Sopas_Error_METHODIN_UNKNOWNINDEX	Trying to access a method with an unknown Sopas index	2	2
Sopas_Error_VARIABLE_UNKNOWNINDEX	Trying to access a variable with an unknown Sopas index	3	3
Sopas_Error_LOCALCONDITIONFAILED	Local condition violated, e.g. giving a value that exceeds the minimum or maximum allowed value for this variable	4	4
Sopas_Error_INVALID_DATA	Invalid data given for variable, this errorcode is deprecated (is not used anymore).	5	5
Sopas_Error_UNKNOWN_ERROR	An error with unknown reason occurred, this errorcode is deprecated.	6	6
Sopas_Error_BUFFER_OVERFLOW	The communication buffer was too small for the amount of data that should be serialised.	7	7
Sopas_Error_BUFFER_UNDERFLOW	More data was expected, the allocated buffer could not be filled.	8	8
Sopas_Error_ERROR_UNKNOWN_TYPE	The variable that shall be serialised has an unknown type. This can only happen when there are variables in the firmware of the device that do not exist in the released description of the device. This should never happen.	9	9
Sopas_Error_VARIABLE_WRITE_ACCESSDENIED	It is not allowed to write values to this variable. Probably the variable is defined as read-only.	10	A
Sopas_Error_UNKNOWN_CMD_FOR_NAMESERVER	When using names instead of indices, a command was issued that the nameserver does not understand.	11	B
Sopas_Error_UNKNOWN_COLA_COMMAND	The CoLa protocol specification does not define the given command, command is unknown.	12	C
Sopas_Error_METHODIN_SERVER_BUSY	It is not possible to issue more than one command at a time to an SRT device.	13	D
Sopas_Error_FLEX_OUT_OF_BOUNDS	An array was accessed over its maximum length.	14	E
Sopas_Error_EVENTREG_UNKNOWNINDEX	The event you wanted to register for does not exist, the index is unknown.	15	F
Sopas_Error_COLA_A_VALUE_OVERFLOW	The value does not fit into the value field, it is too large.	16	10
Sopas_Error_COLA_A_INVALID_CHARACTER	Character is unknown, probably not alphanumeric.	17	11
Sopas_Error_OSAI_NO_MESSAGE	Only when using SRTOS in the firmware and distributed variables this error can occur. It is an indication that no operating system message could be created. This happens when trying to GET a variable.	18	12
Sopas_Error_OSAI_NO_ANSWER_MESSAGE	This is the same as <code>Sopas_Error_OSAI_NO_MESSAGE</code> with the difference that it is thrown when trying to PUT a variable.	19	13
Sopas_Error_INTERNAL	Internal error in the firmware, probably a pointer to a parameter was null.	20	14
Sopas_Error_HubAddressCorrupted	The Sopas Hubaddress is either too short or too long.	21	15
Sopas_Error_HubAddressDecoding	The Sopas Hubaddress is invalid, it can not be decoded (Syntax).	22	16

Telegram structure: sFA ErrorCode			
Error code	Description	Dec.	Hex.
Sopas_Error_HubAddressAddressExceeded	Too many hubs in the address	23	17
Sopas_Error_HubAddressBlankExpected	When parsing a HubAddress an expected blank was not found. The HubAddress is not valid.	24	18
Sopas_Error_AsyncMethodsAreSuppressed	An asynchronous method call was made although the device was built with "AsyncMethodsSuppressed". This is an internal error that should never happen in a released device.	25	19
Sopas_Error_ComplexArraysNotSupported	Device was built with „ComplexArraysSuppressed“ because the compiler does not allow recursions. But now a complex array was found. This is an internal error that should never happen in a released device.	26	1A

Table 622: Example: sFA ErrorCode Wrong userlevel

CoLa A	<STX>sFA{SPC}i<ETX>
	02 73 46 41 20 31 03
CoLa B	02 02 02 02 00 00 00 05 73 46 41 20 00 01 75

SICK AT A GLANCE

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